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EXPERIMENTAL REALIZATION OF A TWO-PHOTON RAMAN LASER

by

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Dr. John E. Thomas

Dissertation submitted in partial fulfillment of the
requirements for the degree of Doctor of Philosophy
in the Department of Physics
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ABSTRACT

(Physics)

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Abstract

This thesis describes the development of a novel quantum oscillator known as a two-photon Raman laser. It is based on two-photon stimulated emission in strongly driven potassium atoms. Two-photon lasers were first proposed in the 1960's, but only recently have such devices been experimentally realized. The two-photon Raman laser is an important step forward as it provides the first opportunity to study the turn-on behavior, instabilities, and noise properties of a pure two-photon optical device.

The necessary ingredients for a two-photon laser are a medium displaying two-photon laser beam amplification and an optical resonator. In my two-photon Raman laser the amplification arises from a multi-photon process in which state-prepared potassium atoms undergo two-photon Raman transitions. This amplification process was studied using a high-density, small-Doppler-width potassium atomic beam that was driven by a strong pump laser and probed by a weak probe laser. I observed two-photon Raman amplification for a range of pump laser frequencies, atomic beam number densities, and probe beam powers. The two-photon Raman gain is linearly dependent on the input probe power as expected for a two-photon process. This gain mechanism is also spectrally isolated from other mechanisms occurring in strongly driven potassium atoms.

The optical resonator consists of a sub-confocal high finesse cavity. The cavity is constructed so that the two-photon Raman process will lase while all other processes are suppressed. The cavity buildup is sufficient to support lasing given the maximum two-photon Raman gain observed in the amplification experiment.

Using this apparatus I have observed two-photon lasing. In agreement with theoretical predictions, an external photon source is required to initiate two-photon lasing.

I initiated the two-photon laser using an externally injected pulse of light and using a frequency degenerate one-photon process. The two-photon Raman laser threshold was mapped as a function of the potassium atomic beam number density.

Polarization instabilities were observed in the output light of the two-photon Raman laser. These instabilities were present for all experimental parameters we used and were a function of the magnetic field strength in the gain medium. These instabilities may be due to competition between various polarization pathways for the two-photon Raman process in potassium.

An operating two-photon Raman laser opens the door to the study of the noise properties and photon statistics of a two-photon device.

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Chapter 1

The two-photon laser

1.1 Introduction

The goal of my research is the experimental realization of a new type of quantum oscillator known as the two-photon laser. First suggested by Prokhorov [1] in his Nobel Prize lecture and independently by Sorokin and Braslau [2] shortly thereafter, a two-photon laser possesses many of the characteristics of other (*i.e.*, one-photon) lasers. However, the two-photon laser is a fundamentally new type of device with novel properties that are of great scientific interest. Experimentalists have been trying to build two-photon lasers since they were first conceptualized. These efforts were unsuccessful until recent theoretical and technological developments overcame the many obstacles. The realization of a robust two-photon laser is an important step forward in experimental quantum optics.

In order to show where two-photon lasers fit into the research field of quantum optics, I first describe the important ideas behind one-photon lasers. Building on these general ideas of lasers, I then explain the specifics of the two-photon laser. The motivation for the interest in the two-photon laser is contained in the following section. I then describe the hurdles to building a two-photon laser and the solutions to these problems. This chapter closes with an overview of the remainder of the thesis.

1.2 Lasers

Lasers are a class of devices that combine a method for amplifying light and an optical resonator to produce intense, spectrally pure, coherent light. Conceived by Townes and Schawlow [3] in the mid-1950's, lasers have quickly progressed from experimental novelty to an integral part of our technology. Applications for lasers include bar-code scanning in supermarkets, industrial cutting and welding, medical surgery, and information storage and retrieval. As practical uses were being found for lasers in many arenas, new types of lasers were being built in the lab. Recent research has produced lasers without inversion, single-atom lasers, and two-photon lasers.

To understand lasers, we begin with the three fundamental light-atom interactions: absorption, spontaneous emission and stimulated emission. These processes were first suggested by Einstein in a 1916 paper [4], in an attempt to reconcile Planck's equation for blackbody radiation with Bohr's hypothesis that electrons "orbiting" a nucleus can only occupy certain discrete energy levels [5]. This was an extension of Einstein's earlier work on blackbody radiation where he first developed the idea that light exists in discrete quanta; what we now call photons. An excellent discussion of this early period of quantum mechanics can be found in Whitaker's **Einstein, Bohr and the Quantum Dilemma** [6].

Einstein started by considering only two electronic energy levels of an atom interacting with a resonant electromagnetic field, as can be seen in Figure 1.1. Absorption occurs when an atom initially in the ground state undergoes a transition to the excited state and a photon from the field is annihilated. The absorption rate is given by

$$W_{absorption} = B_{absorption} \cdot N_{ground} \cdot q , \quad (1.1)$$

where N_{ground} is the number of atoms in the ground state, q is the number of photons,

and $B_{absorption}$ is the Einstein absorption coefficient. The process of spontaneous emission occurs when an atom in the excited state drops spontaneously to the ground state and emits a photon. It occurs at a rate given by

$$W_{spontaneous} = A_{spontaneous} \cdot N_{excited} , \quad (1.2)$$

where $N_{excited}$ is the number of atoms with electrons in the excited state and $A_{spontaneous}$ is the Einstein spontaneous emission coefficient. The third process is stimulated emission, which occurs when an atom in the excited state interacts with a photon whose energy is equal to the transition. This interaction causes the atom to drop to the ground state and emit a second photon identical to the initial, stimulating photon. The rate for stimulated emission is given by

$$W_{stimulated} = B_{stimulated} \cdot N_{excited} \cdot q , \quad (1.3)$$

where $B_{stimulated}$ is the Einstein stimulated emission coefficient. Notice that in each of these processes, the photons have the same energy as the energy difference between the two electronic energy levels of the atom. For later comparison, these processes are referred to as one-photon absorption, one-photon spontaneous emission, and one-photon stimulated emission. All three processes occur simultaneously when light interacts with matter. For example, a photon given off from an atom by spontaneous emission may be absorbed by a neighboring atom.

The important process for creating a laser is stimulated emission. The second photon emitted when the electron drops to the ground state is identical in frequency, polarization, and direction of propagation to the stimulating photon. Thus, stimulated emission allows photons to be replicated. The challenge in designing a laser is to perform this replication process over and over until there are not two, but billions of photons; all alike and all doing the same thing. An operating laser produces a beam of light which is both spectrally pure and well collimated.

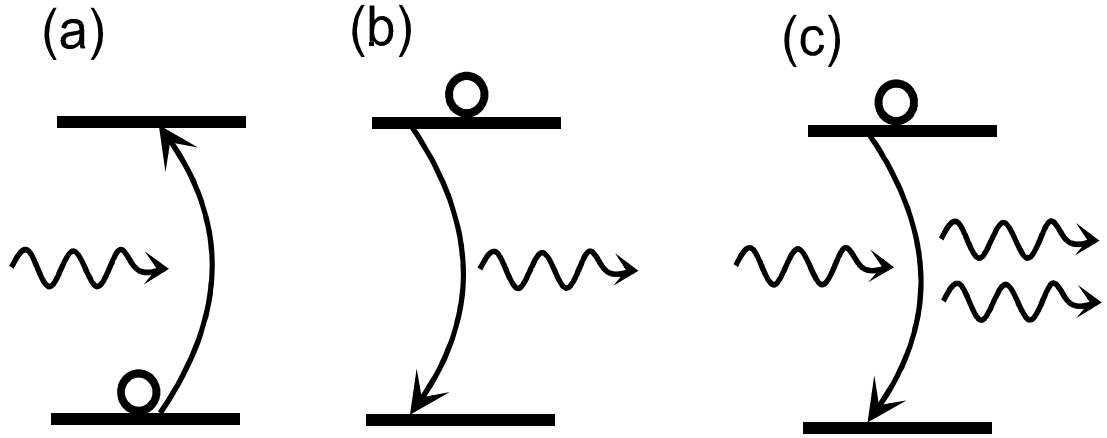


Figure 1.1: The three fundamental light-atom interactions are (a) absorption, where an atom absorbs a photon and the electron moves from a lower energy level to an higher one, (b) spontaneous emission, where an electron drops from an excited state to a lower energy state and the atom emits a photon, and (c) stimulated emission, where a photon induces an atom to make a transition, giving off a second identical photon in the process. The wavy lines represent photons, the circles are electrons, and the arrows symbolize electron transitions.

Absorption and spontaneous emission both compete with stimulated emission since absorption removes photons from the laser beam and spontaneous emission emits photons in all directions rather than into the laser beam direction. Thus to realize a laser, one needs a system where stimulated emission dominates the other two processes. **Since $B_{\text{absorption}} = B_{\text{stimulated}}$, stimulated emission will happen more often than absorption if there are more atoms with electrons in the excited state than in the ground state. This condition is referred to as a population inversion.** Additionally, stimulated emission will happen more often than spontaneous emission if the excited state has a long lifetime, i.e., the rate of spontaneous emission is small. Many systems with these criteria have been discovered or designed. Such a system is referred to as a gain medium or optical amplifier, since light passing through the medium gains additional photons. More

explicitly, one may write a differential equation for a beam with intensity I passing through a slice of a gain medium as

$$\frac{dI}{dz} = \alpha \cdot I \quad (1.4)$$

where

$$\alpha = \frac{h}{\lambda} (B_{stimulated} \cdot N_{excited} - B_{absorption} \cdot N_{ground}) , \quad (1.5)$$

is the linear gain coefficient, h is Planck's constant, and λ is the optical wavelength. Since all quantities are positive, this shows that one must have $N_{excited} > N_{ground}$ for the optical beam to experience gain.

In addition to the population inversion, an optical resonator (also known as a optical cavity, see illustration in Figure 1.2) is needed to select a spatial mode of the electromagnetic field and feedback a particular wavelength. A cavity is formed by placing two mirrors parallel to each other. Light bounces back and forth between the mirrors along a single path, which is called the cavity axis. **Adding a gain medium with more round trip gain than round trip loss will result in lasing.** This laser output builds up from spontaneous emission as the light in the cavity increases with each round trip until some steady-state is reached. The cavity can be thought of as a selective feedback mechanism that chooses which photons will be replicated: since the electromagnetic field must vanish at the mirrors, only certain frequencies of light will be resonant within the cavity. Typically, one of the mirrors is a partial reflector, thus allowing some of the laser light to exit the cavity. These fundamental ideas underlying the normal one-photon laser can be expanded to describe the two-photon laser.

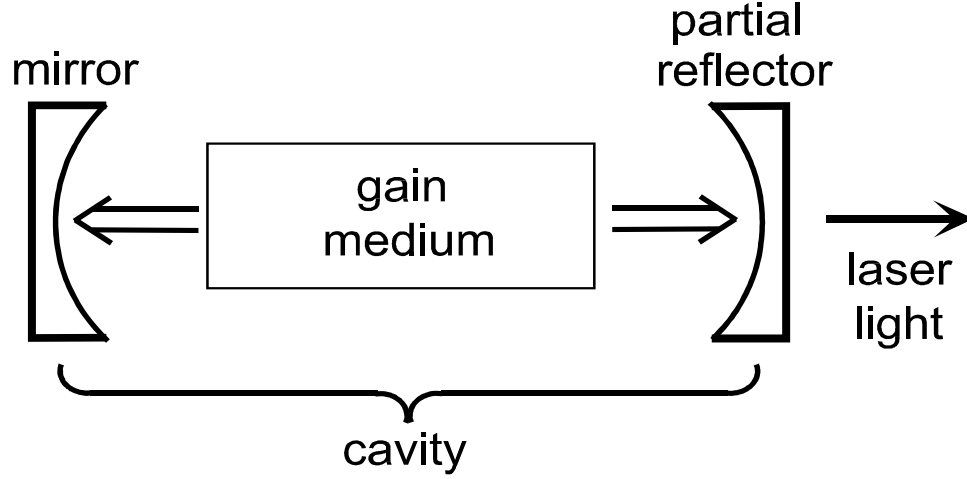


Figure 1.2: Pieces of the laser include the gain medium which amplifies the light passing through it and the cavity formed by the mirrors which selects the particular spatial mode and frequency which lases.

1.3 Two-photon processes

The insight of Prohkorov [1] and Sorokin and Braslau [2] was to realize that there is no restriction on the number of photons participating in the stimulated emission process. The only criteria are that the sum of the energies of the photon is equal to the energy difference between the electronic energy levels and that the transition is not forbidden. Figure 1.3 contains the diagrams for the three fundamental two-photon processes, analogous to the one-photon processes shown in Figure 1.1. Figure 1.3(a) shows two-photon absorption where an atom in the ground state undergoes a transition to the excited state via a virtual intermediate state and two photons from the field are annihilated. Assuming the degenerate case, *i.e.*, that the two photons have the same energy, the absorption rate for this process is given by

$$W_{2\gamma \text{ absorption}} = B_{2\gamma \text{ absorption}} \cdot N_{\text{ground}} \cdot q^2, \quad (1.6)$$

where $B_{2\gamma \text{ absorption}}$ is the Einstein two-photon absorption coefficient. Two-photon spontaneous emission occurs when an electron drops to the ground state and the atom emits two photons and the rate is given by

$$W_{2\gamma \text{ spontaneous}} = A_{2\gamma \text{ spontaneous}} \cdot N_{\text{excited}} , \quad (1.7)$$

where $A_{2\gamma \text{ spontaneous}}$ is the Einstein two-photon spontaneous emission coefficient. Finally two-photon stimulated emission occurs when an atom in the excited state interacts with two-photons of the suitable energy sum and the electron drops to the ground state. The rate for two-photon stimulated emission is given by

$$W_{2\gamma \text{ stimulated}} = B_{2\gamma \text{ stimulated}} \cdot N_{\text{excited}} \cdot q^2 \quad (1.8)$$

where $B_{2\gamma \text{ stimulated}}$ is the Einstein two-photon stimulated emission coefficient. There is no restriction on the energy of the two photons, as long as the sum of these energies matches the energy difference of the electronic energy levels. The majority of the work to date has occurred in systems where the two photons are degenerate. All of the work described in this thesis involves degenerate two-photon processes, so, unless stated, degenerate operation can be assumed anytime a two-photon process is mentioned.

In a similar fashion to Equation 1.4 one may write a gain equation for a light beam passing through a two-photon gain medium. Again we have

$$\frac{dI}{dz} = \alpha \cdot I \quad (1.9)$$

but now

$$\alpha = \frac{h}{c\lambda} (B_{2\gamma \text{ stimulated}} \cdot N_{\text{excited}} - B_{2\gamma \text{ absorption}} \cdot N_{\text{ground}}) \cdot I . \quad (1.10)$$

We still have the condition that $N_{\text{excited}} > N_{\text{ground}}$ in order for the beam to experience gain, but now the gain coefficient α is a linear function of the beam intensity I . That is, the higher the intensity, the larger the gain.

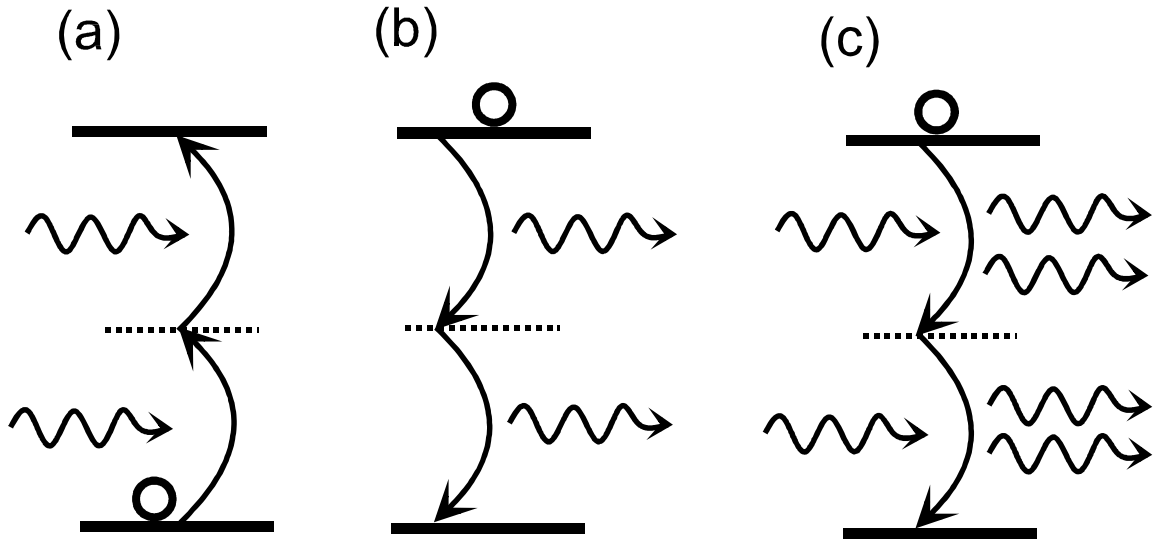


Figure 1.3: The three fundamental two-photon interactions are (a) two-photon absorption, where an atom absorbs two photons and the electron jumps from a lower energy level to an higher one, (b) two-photon spontaneous emission, where an electron drops from an excited state to a lower state and the atom emits two photons, and (c) two-photon stimulated emission, where two photons induces an atom to make a transition giving off two additional photons. The wavy lines represent photons, the circles are electrons, and the arrows symbolize electron transitions. The solid horizontal lines are energy levels and the dotted horizontal lines are virtual energy levels.

A two-photon laser is a device that uses two-photon stimulated emission as the replication method for the photons. Just like the one-photon laser, the challenge is to find a physical system that can be inverted (more population in the excited state than in the ground state) so that two-photon stimulated emission is the dominant process. Using such a gain system in conjunction with a cavity will result in a two-photon laser, provided the gain per round trip is greater than the loss per round trip. This description overlooks the substantial difficulties involved in finding a suitable inversion and building a cavity. I shall discuss these difficulties and their solutions briefly later in this chapter and in detail in Chapter 2. Given the idea of what a two-photon laser is, I will now describe why it is an interesting problem to study.

1.4 Motivation

The experimental realization of a two-photon laser represents a substantial scientific and technical breakthrough that will provide solid evidence to support or contradict the two-photon laser theories to date. Furthermore it will provide a testbed for exploring nonlinear dynamics of optical systems, new types of photon statistics and coherences, and light-atom interactions in the presence of a cavity. On the practical side, an operational two-photon laser opens the door to exploration of applications such as low-noise optical communication techniques and switches for optical computation.

Initially in the 1960's, there was substantial excitement about two-photon lasers because two-photon processes offered relief from the constraints of atomic energy levels. Scientists were exploring possibilities for building lasers at new wavelengths and lasers that might be tunable. Two-photon transitions promised a breakthrough because the conservation of energy requirement is that the sum of the energies of the two-photons matches the transition energy. This permits the possibility of tuning the wavelengths of the two photons while just making sure that the sum remains fixed. In addition, since the gain is proportional to the intensity, it was hypothesized that two-photon lasers might operate at very high intensities and might make excellent pulsed lasers.

These early motivations are no longer the driving interests in two-photon lasers. Since the 1960's, one-photon lasers have progressed to the point that much of the visible, infrared and ultraviolet wavelengths are now covered by Ti:Sapphire lasers, dye lasers, and diode lasers, just to name a few. Extremely bright lasers have been built and pulsed lasers with pulse widths on the order of a few femtoseconds are now standard. Thus, the two-photon laser is not sought as a tool for generating new

wavelengths or high intensities, but rather as a fundamentally new type of device that will teach us about light-matter interactions.

The experimental realization of a two-photon laser will help to settle many theoretical questions. Numerous theoretical treatments of the two-photon laser have been undertaken; however there is substantial variation in the conclusions drawn by these theories. Two-photon lasers are sufficiently complex that varying results are obtained depending on where the theory starts and what assumptions and approximations are made. A working two-photon laser will help resolve these issues and provide impetus for new theoretical work, since experimental comparisons will now be possible. Some of these issues include the nonlinear nature of the dynamics of a two-photon laser, the novel photon statistics of a two-photon laser, and the highly quantum nature of a two-photon device.

Since the mechanism for two-photon stimulated emission is nonlinear, the dynamics of a two-photon laser are also expected to be nonlinear. In all lasers, the condition for lasing (*i.e.*, the laser threshold) is that round trip gain equals round trip loss. In a one-photon laser, the laser will turn on smoothly when a sufficient inversion is attained via pumping and the laser output is proportional to the pumping until saturation is reached. For a two-photon laser there is the additional condition that a minimum number of photons are required in the cavity, since the gain is proportional to the intensity. As a result, a field must be injected into the cavity for the two-photon laser to start [1, 2, 7, 8]. Once the laser starts, the intensity of the field circulating in the cavity will be equal to or greater than the two-photon saturation intensity. Thus, the two-photon laser undergoes a discontinuous transition from off to on and exhibits bistable behavior; one hallmark of a nonlinear system. In addition, any spatial variation in the number of photons will result in spatial variation in the gain which may lead to spatio-temporal chaos within the two-photon laser [9–13].

The statistics of the light (*i.e.*, the statistics of the intervals between the arrivals of successive photons) generated by a two-photon laser are expected to be novel. Before lasers, most light sources were thermal sources, which are characterized by very short coherence lengths and photon bunching. Lasers opened the door to light sources with very long coherence lengths, hundreds of meters or more, and photon statistics which are Poisson or random. Two-photon lasers promise yet another type of light source, one that may naturally exhibit novel photon statistics and even phase-squeezing. Squeezing occurs when the noise in one quadrature is below the point where the noise in the two quadratures is equal and at the Heisenberg uncertainty limit.

Another area of interest is the fundamental quantum mechanical nature of the two-photon interaction. Extremely high finesse cavities are required for two-photon lasers, for example, our two-photon Raman laser operates with only a few thousand photons and a few hundred thousand atoms in the cavity. This is in contrast to typical lasers which have billions of photons and several thousand trillion atoms in the cavity at any one time. Thus the two-photon Raman laser is in the area between classical devices and quantum devices, providing an opportunity to explore the quantum nature of light-matter interactions.

There has been tremendous interest in building a two-photon laser since it was first suggested, but experimental success has happened only recently. Theorists can easily envision what they think a two-photon laser should be like, but experimental reality is not so simple. The numerous challenges and the solutions that ultimately resolved these challenges are the next topic.

1.5 Challenges to building the two-photon laser

All of the challenges to building the two-photon laser come from the question of how to make two-photon stimulated emission the dominant process in the laser. In this section I work through some of the competing mechanisms and how they are suppressed. The first step is to find an atomic system which favors two-photon stimulated emission. As equations 1.6, 1.7, and 1.8 (the rate equations) showed, both the two-photon absorption and two-photon stimulated emissions rates are proportional to the square of the number of available photons. For two-photon stimulated emission to happen at a faster rate than absorption, there must be more atoms in the excited state than in the ground state (an inversion). Once there are atoms in the excited state, one-photon spontaneous emission will occur, robbing photons from the inversion. Unfortunately, one-photon spontaneous emission is present in any atomic system. Since the two-photon stimulated emission rate is proportional to the square of the photon number, it would seem that we can increase the rate by increasing the number of photons. However, large intensities increase the rates of other competing nonlinear optical processes such as dressed-state processes, parametric wave-mixing, etc., which also have rates proportional to the square of the photon number. Thus, the goal is to find systems that may be inverted and where $B_{2\gamma \text{ stimulated}}$ is as large as possible. To actually achieve two-photon lasing, the system must still meet the requirements that the gain exceeds the cavity losses and the photon number is insufficient for other competing nonlinear mechanisms.

To develop this intuition further, consider the two-photon stimulated emission rate coefficient for the simple three-level atomic system shown in Figure 1.4. The two-photon rate coefficient for this system is

$$B_{2\gamma \text{ stimulated}} = \frac{32\pi^2 |\boldsymbol{\mu}_{\text{ei}} \cdot \boldsymbol{\epsilon}|^2 |\boldsymbol{\mu}_{\text{ig}} \cdot \boldsymbol{\epsilon}|^2 \omega^2}{V_c^2 \hbar^2 \Delta_{ig}^2 \Gamma_{eg}} \quad (1.11)$$

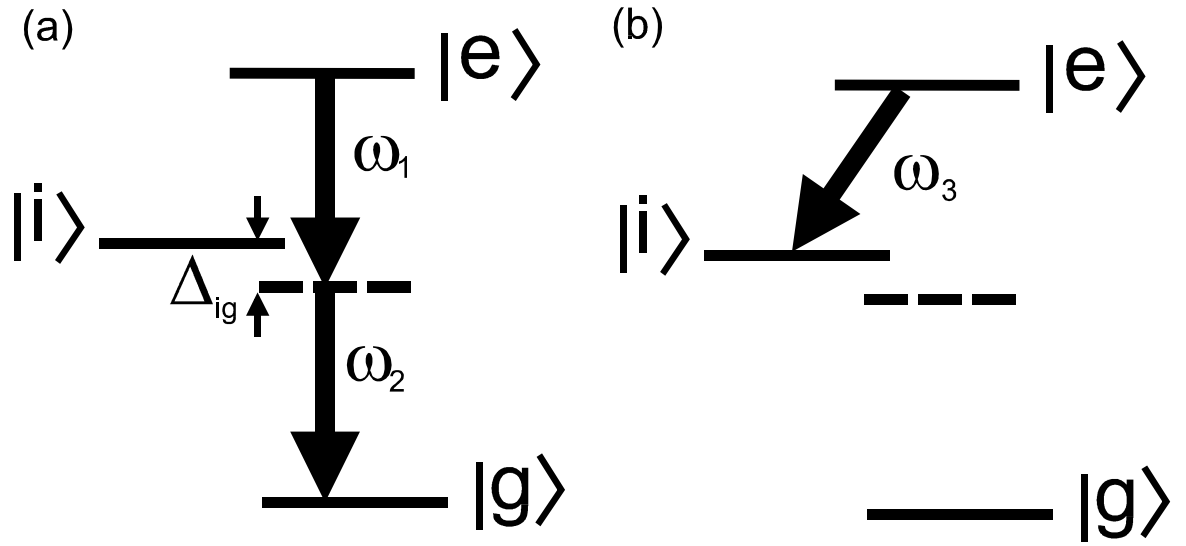


Figure 1.4: (a) Two-photon stimulated emission occurs when two photons cause a transition between an excited and a ground state via a virtual intermediate state. There is an intermediate level ($|i\rangle$) which resonantly enhances the transition since $|e\rangle \rightarrow |i\rangle$ and $|i\rangle \rightarrow |g\rangle$ are strong dipole allowed transitions. (b) One photon transition between the excited and intermediate states. This transition is detuned from the two-photon transition by Δ_{ig} .

where μ is a dipole matrix element, V_c is the mode volume, and Γ_{eg} is the coherence dephasing rate of the two-photon transition [14]. This coefficient is proportional to the square of the dipole matrix elements between the ground and intermediate state and the intermediate and excited state, so large dipole matrix elements are advantageous for two-photon stimulated emission. Additionally, the rate coefficient is inversely proportional to the square of the detuning of the virtual level (the dashed line in Figure 1.4) from the real intermediate level. The smaller this detuning, the larger the two-photon stimulated emission rate. However, if this detuning is too small, transitions will occur from the excited state to the intermediate state and finally to the ground state, *i.e.*, there will be two stepwise one-photon transitions instead of a single two-photon transition (see Figure 1.4(b)). This is where the proper choice of the cavity comes into play.

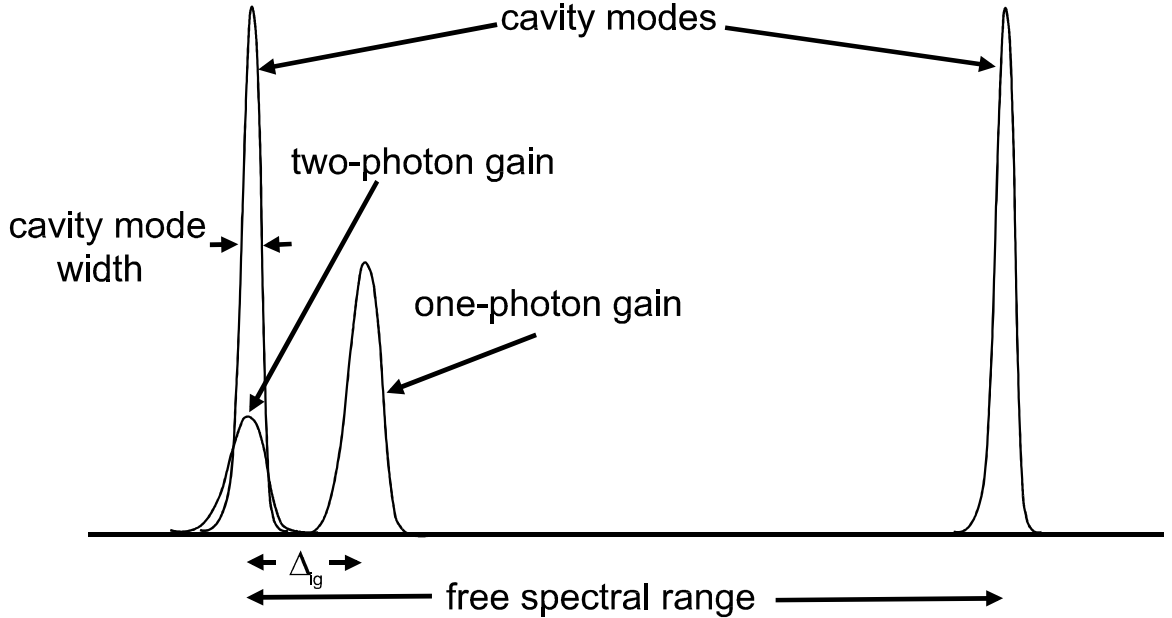


Figure 1.5: A short cavity (one with a large free spectral range) with a high finesse (i.e., a narrow cavity resonance) allows the two-photon gain to be enhanced while avoiding other features such as the nearby one-photon gain.

There will always be other optical processes occurring in a system exhibiting two-photon gain. Once we have chosen an atomic system that minimizes these other processes, the next step is to use a cavity to selectively enhance the two-photon gain. As is shown in Figure 1.5 there are two properties of the cavity that help enhance two-photon gain at the expense of other processes. First, by using a relatively short cavity the longitudinal modes¹ are spaced far enough away in frequency that the adjacent mode frequencies do not match any other gain processes in the system. Next, by using a high finesse cavity, the cavity resonance will be very sharp (*i.e.*, the frequency width of the cavity resonance will be small), so features close to the two-photon gain will not be enhanced. For example, in the simple three-level system again, we know that there is a one photon process which is detuned from the two-photon gain by Δ_{ig} , as shown in Figure 1.5 (b), and we want to suppress this one-photon process while not affecting the two-photon process. The sharper the resonance, the smaller the detuning that can be used, thus increasing the two-photon gain.

In summary, to increase the two-photon stimulated emission rate we need a system with a large population inversion and a large two-photon stimulated emission coefficient. The two-photon stimulated emission coefficient may be maximized by employing an atom with large dipole matrix elements and a small detuning between the real and virtual intermediate states. Once there is sufficient two-photon gain, other processes can be suppressed by using a short cavity with high finesse.

Using these ideas Haroche's group was able to successfully build a two-photon maser. Haroche realized that the Rydberg atomic states are an excellent system for two-photon gain because of the large electric dipole moment between nearby

¹Longitudinal modes refer to the standing electromagnetic waves that vanish at the cavity mirrors. Each mode is made up of an integral number of halfwavelengths of the field and are separated in frequency by the free spectral range of the optical cavity which is given by $c/(2nl)$, where c is the speed of light, l is the length of the cavity and n is the index of refraction inside the cavity.

states [16]. In addition Δ_{ig} can be minimized by searching for a near degeneracy between pairs of Rydberg transitions. Haroche was able to find a set of three Rydberg that are nearly ideal, assisted by the plethora of Rydberg states and nearly degenerate spacings between adjacent set of levels. An extremely high finesse cavity was used to selectively enhanced the two-photon masing while suppressing competing processes. Finding such a system is a fluke of nature and does not occur readily.

There are no known optical transitions possessing a near resonant intermediate level and that can be inverted easily. However, Mossberg realized that such a system can be artificially created by strongly driving a two-level atom with a near resonant laser beam of frequency ω_d [17–19]. This composite atom + laser system has many desirable characteristics, similar to the Rydberg atom states. The detuning from the intermediate state is <1 GHz, which is very small on the scale of optical frequencies of tens of terahertz, and can be easily controlled. Also the dipole matrix elements are the full allowed ground to excited state transition matrix elements.

Figure 1.6(a) shows how this two-photon dressed-state process works in the low-field limit. Three pump photons (the solid arrows in Figure 1.6(a)) and two probe photons (the dotted arrows in Figure 1.6(a)) stimulate a transition from $|g\rangle$ to $|e\rangle$. Four identical probe photons are emitted as a result of this transition.

In the dressed-state picture one views the two-level atom and the strong driving field as a system, as shown in Figure 1.6(b). The energy levels of this atom+strong laser system consist of a infinite ladder of pairs of levels (see Figure 1.6) where the pairs are separated by ω_d and the levels within the pair are separated by the generalized Rabi frequency Ω' . Since the strong pump is detuned from the transition there is more population in $|-\rangle$ than in $|+\rangle$, providing the inversion necessary for gain. Two-photon transitions occur between the $|-, n+1\rangle$ level and the $|+, n-1\rangle$ level. The frequency of the two-photon transition is $\omega_d - \Omega'/2$ and the $|+/-, n\rangle$ levels act

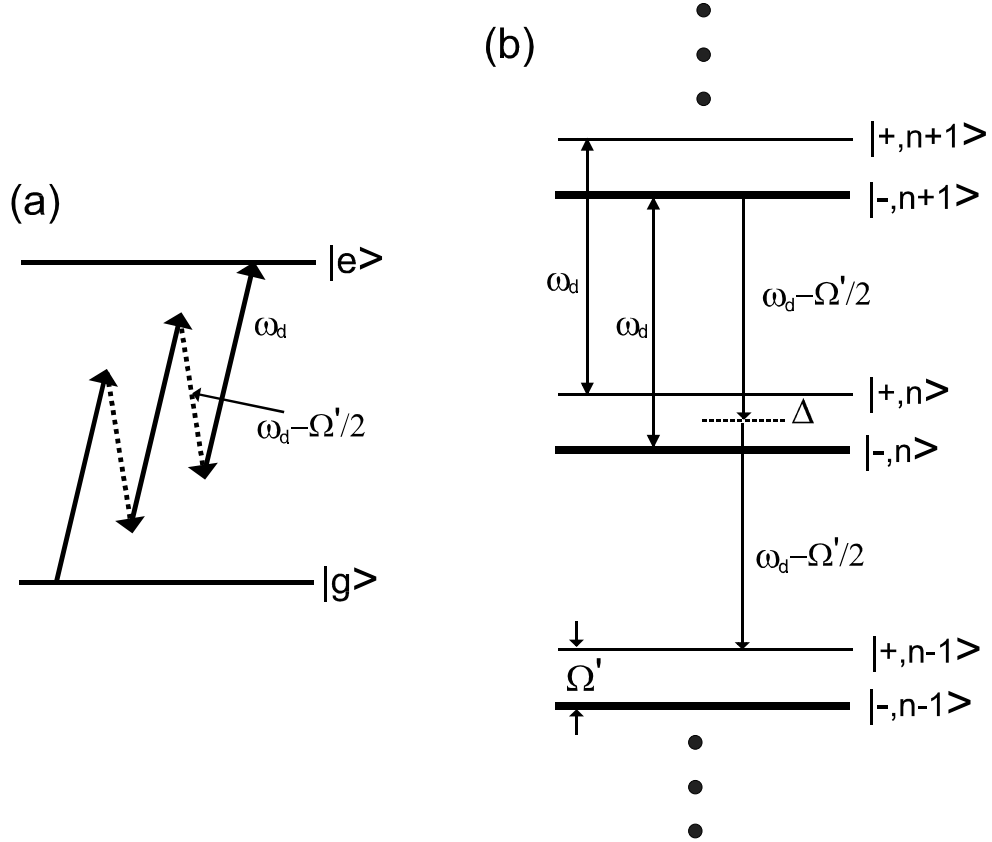


Figure 1.6: (a) Two-photon transition using three pump photons (up arrows) and two probe photons (dotted down arrows). (b) Same transition in the pump dressed-atom picture. There exists an infinite ladder of pairs of levels and the thick lines represent the state with the higher population. Using this picture it is easy to see the two-photon inversion between states $|-, n+1\rangle$ and $|+, n-1\rangle$, with resonant frequency $\omega - \Omega/2$. Note that the levels $|-, n\rangle$ and $|+, n\rangle$ provide the nearly resonant intermediate level for the two-photon process.

as the near resonant intermediate levels.

Using these techniques a two-photon dressed-state laser was built in Mossberg's group [20]. While a remarkable achievement, this laser had several shortcomings. It was not a pure two-photon laser, in that there was both one-photon and two-photon stimulated emission contributing to the gain. In addition, the laser was very unstable and only lased for short periods of time.

The two-photon laser described in this thesis builds on the idea of the two-photon dressed-state laser. Instead of dressed-state gain however, two-photon Raman gain is used. Raman transitions occur between nearby ground states and use the excited state as the near resonant intermediate level. The two-photon Raman process is lower order than two-photon dressed-state processes and thus proceeds with fewer pump photons. Excellent spectral resolution of the one- and two-photon gain is possible by careful selection of the ground state splitting. Furthermore since the full dipole matrix element is used there is a large transition rate for the two-photon Raman process.

Two-photon Raman gain had already been observed in this lab [21], however it was unsuitable for a two-photon laser. We discovered a new type of two-photon Raman scattering which employs a novel beam geometry. In this geometry the pump beams are orthogonal to the probe beam thus suppressing any competing wave-mixing effects, which require a small angle between the pump and probe, and substantially simplifying the design and construction of the two-photon laser cavity.

A bare atom level diagram for this two-photon Raman process in potassium (occurring on the D1 line) is shown in Figure 1.7. The population inversion is created by two circularly polarized optical pumping beams (with σ^+ or right circular polarization), which move all the population from the $F=1$ and $F=2$ hyperfine levels into the $F=2$, $m_f=2$ magnetic sublevel (see Figure 1.7(a)). Two probe photons (with z or

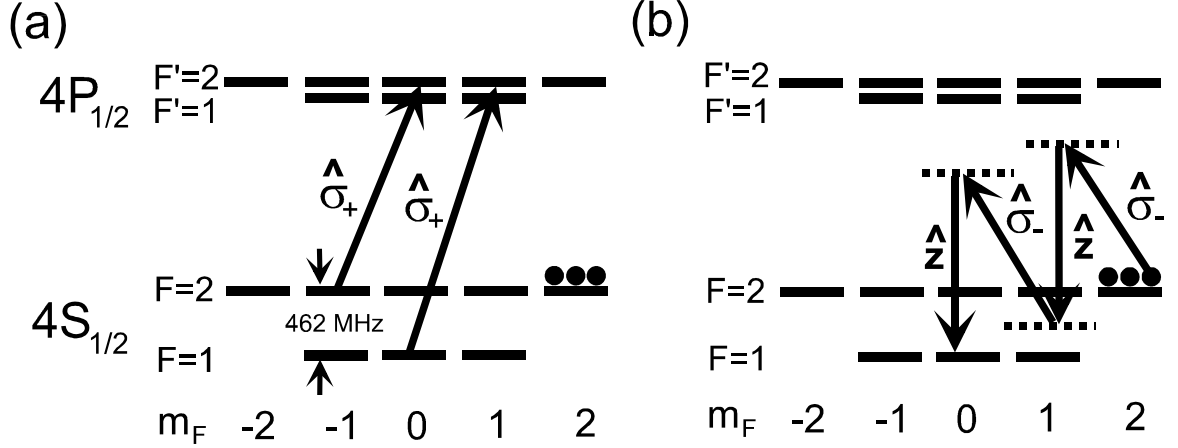


Figure 1.7: Bare atom picture for two-photon Raman gain. Right circularly polarized photons move all the electrons to the $F=2$, $m_f = 2$ magnetic sublevel. From this level, two left circularly polarized pump photons and two linearly $\hat{z}$ polarized probe (or laser) photons stimulate a transition to another level.

vertical polarization) then stimulate a transition back the ground state annihilating two pump photons (with σ^- or left circular polarization) in the process. In Figure 1.7(b) the probe photons are represented by the arrows pointing down and the Raman pump photons are represented by the up-left arrows. A dressed-state picture of this process is shown in Chapter 3.

1.6 Overview of thesis

This section provides an outline of the rest of the chapters in my thesis. In Chapter 2, previous work on the two-photon laser is presented. The work done in this lab by former students and the new ideas that constitute my contribution are outlined in Chapter 3. Chapter 4 documents the two-photon gain measurements made using our atomic beam apparatus. Chapter 5 describes the design and construction of the laser cavity. Chapter 6 covers the experimental results including the observation of

two-photon Raman lasing. Finally, Chapter 7 has the conclusions and outlines the future directions for this experiment.

In Chapter 2, I present an overview of the two-photon laser. Relevant background information is discussed, including the difficulties faced in building a two-photon laser or maser and the properties of such a device. Further explanation is given for the particular characteristics of the two-photon laser that I studied experimentally. The chapter closes with a section on the experimental realizations of a two-photon maser by Haroche and collaborators and a two-photon dressed-state laser by Mossberg, Gauthier and collaborators.

Chapter 3 begins with the previous work done in this lab, which included the experimental observation of 30% two-photon gain. When I began working on the two-photon laser experiment, this record gain had already been observed but there were several competing mechanisms that needed to be suppressed or eliminated before a laser could be built. I spent substantial time trying various methods to suppress this features, including the introduction of various buffer gases, but was unsuccessful in these efforts. Reflecting on these failures led us to the discovery of a new type of two-photon Raman process which employs a novel beam geometry. A simple dressed-state theory for the atomic beam system is presented. This theory predicts what gain features will be present and at what frequency they will occur.

Chapter 4 covers the experimental search for two-photon Raman gain in an atomic beam system. The two-photon Raman gain occurs in a beam of potassium atoms driven by optical pumping beams and a Raman pump beam. Orthogonal to the atomic beam and the pump beams is the probe beam that experiences two-photon gain. An explanation of the entire apparatus is given including the vacuum system, the atomic beam system, and all of the lasers. Experimental results are given for various pump frequencies and probe powers, documenting our search for the maxi-

mum two-photon gain and the location with the best spectral isolation. Numerous experimental spectra are displayed, as well as plots showing the probe power dependent gain, which is a signature of a two-photon process (see Figure 1.8). Chapter 4 closes with a discussion of these results.

In Chapter 5, I cover the design and development of the high finesse cavity used for the two-photon Raman laser. This begins with some general background information on optical cavities and then moves on to the particular requirements for our optical cavity. All mechanical pieces of the cavity are described in-depth. The last section contains the experimental verification of the cavity operation, including cavity alignment and the measurement of the cavity finesse.

The results for the two-photon laser experiment are presented in Chapter 6. I describe the procedures employed to search for two-photon lasing using our high finesse optical cavity. I observed two-photon Raman lasing initiated by two distinct mechanisms; an externally injected pulse of laser light (see Figure 1.9) and by choosing the pump detuning so that the two-photon Raman process was degenerate in frequency with a one-photon process. The number density threshold for the two-photon Raman laser was mapped out. I also observed polarization instabilities in the output of the two-photon laser. These instabilities are a function of the magnetic field strength, however they are always present for our experimental parameters. Two-photon dressed-state lasing was also observed. A discussion of these significant results concludes Chapter 6.

This experiment is just the beginning. A reliable two-photon Raman laser of sufficient power opens the door to a wide variety of further experiments, some of which are outlined in Chapter 7. These include further study of the turn-on behavior of the two-photon laser and several experiments concerning the quantum statistics of the light generated by a two-photon laser. There is also the possibility of obtaining

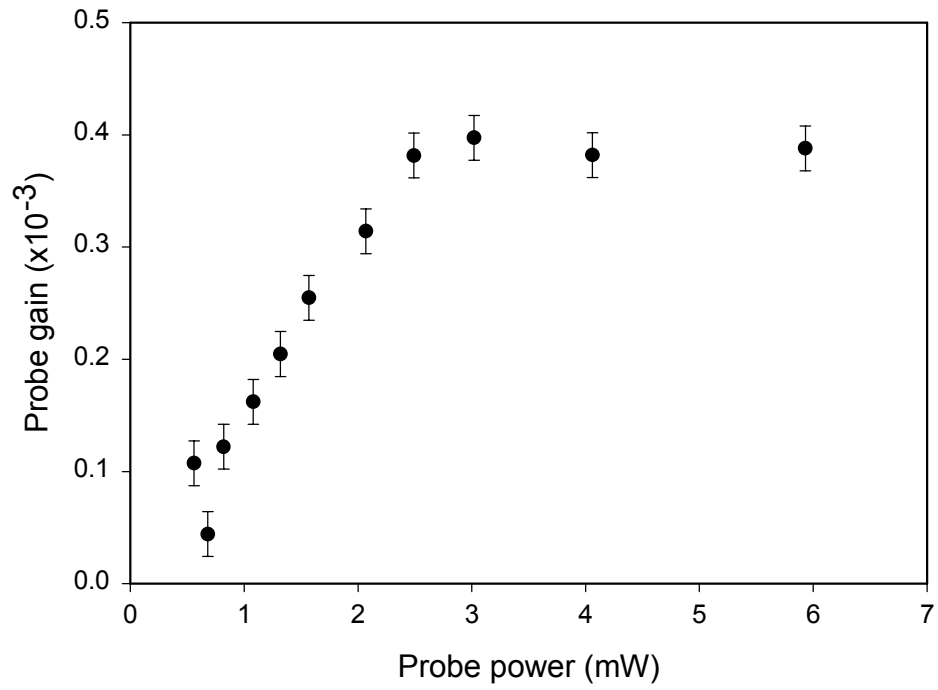


Figure 1.8: Gain versus probe power. This graph shows the linear relationship between probe power and gain before saturation. After saturation the gain slowly decreases with increases probe power.

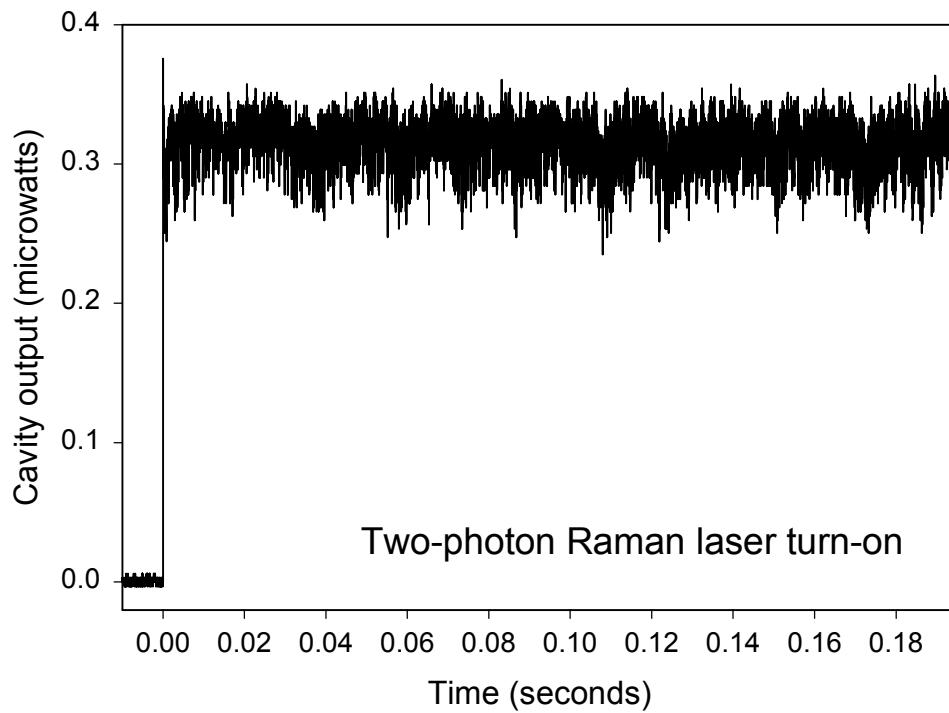


Figure 1.9: Turn-on behavior for the two-photon Raman laser. A 100 ns. optical pulse is injected into the cavity, initiating the two-photon laser, which then remains on for almost 1/5th of a second after the pulse is turned off.

three-photon lasing with this experimental apparatus. In the more distant future, experiments exploring spatio-temporal chaos in a two-photon laser might be possible, as well as a two-photon laser in a magneto-optical trap.

Chapter 2

Background of the two-photon laser

This chapter covers some of the background material relevant to the two-photon Raman laser. Included are some of the interesting properties of two-photon lasers with a more in-depth discussion of the properties related to the observations I made. This is also a description of the other experimental two-photon devices that have been built to date.

Two-photon stimulated emission was first suggested by Prokhorov [1] and Sorokin and Braslau [2] in 1964. The idea of two-photon absorption had already been put forth, but this was the first time that anyone thought of using two-photon stimulated emission to build an oscillator. In his Nobel Prize acceptance speech, Prokhorov suggested the two-photon laser as a device where the photon energy is not equal to the spacing of the electronic energy levels, thus allowing laser oscillation at frequencies other than the resonance frequencies of atoms [1]. Prokhorov also noted that such a device would require an injected trigger to start, a prediction born out in the dressed-state two-photon laser [20]. Sorokin and Braslau independently and simultaneously proposed the two-photon laser and suggested that it could be used to obtain “giant pulses” of laser light [2]. These scientists assumed that it would only be a matter of a few years before a two-photon laser was actually built, however such was not to be the case.

Experimental realization of a two-photon device was not to happen for another 23 years, when Haroche and collaborators finally built a two-photon maser. This was not for a lack of trying, but rather due to the many obstacles. First, scientists had to find a system where it is possible to create a two-photon inversion. Given the

requirement of a near resonant intermediate level, necessary to resonantly enhance the two-photon transition, few naturally occurring systems are available. On top of that, there must not be a one-photon process that occurs at the same frequency. It bears repeating that this is a difficult problem which many scientists have attempted to understand.

Early experimental work focused on the standard systems and techniques with which scientists were already familiar in connection to their work on one-photon lasers. In 1965, Hall observed photoionization, via two-photon absorption, of atomic cesium and negative ions of atomic iodine [22–24]. Shortly after this, Lipeles *et al.* [25–27] observed two-photon spontaneous emission from the $2^2S_{1/2}$ state of singly ionized helium. Some time later, Bräunlich and Lambropoulos [28] experimentally explored singly stimulated two-photon emission in deuterium atoms¹. In 1978, the first two-photon stimulated emission was observed by Loy [29–31] using a transient two-photon inversion in NH_3 . Even though these were important experimental steps in understanding two-photon process, none of these systems meet the criteria for building a two-photon laser.

Despite the slow experimental progress, theorists have made numerous predictions about how a two-photon laser should work [7, 32–37]. Very early on it was realized that a two-photon laser will not start from spontaneous emission, but will require an external pulse to initiate lasing [1]. Further work indicated that the two-photon laser may be unstable for some, or even all, experimental conditions. It was also recognized that the light emitted by the two-photon laser will have novel properties. The photon statistics of the light may be different from the Poissonian statistics of one-photon lasers [38–45]. Also the noise properties will not be the same; the two-

¹Singly stimulated two-photon emission occurs when one of the two nondegenerate frequencies is stimulated by an external laser resulting in one photon generated by stimulated emission and one photon generated by spontaneous emission.

photon laser may exhibit squeezing [46–56]. Further explanation of these ideas can be found in H. Concannon’s thesis Chapter 6 [57] and in Teich and Saleh’s review article on squeezed and antibunched light [58]. I will go into further depth on concerning the characteristics which I studied experimentally, the turn-on dynamics and instabilities of the two-photon laser.

2.1 Turn-on dynamics

Novel turn-on dynamics are expected from the two-photon laser. To contrast, I will first describe one-photon laser dynamics. The one-photon laser smoothly transitions from spontaneous emission to lasing as the pump rate (proportional to the gain in the system) is increased (see Figure 2.2). Dynamically, this is a transcritical bifurcation that occurs when the gain in the amplifier overcomes the losses in the cavity and lasing begins. The photons necessary to initiate lasing come from spontaneous emission due to the inversion between the excited and ground states in the amplifying medium.

2.1.1 Rate equation model for turn-on dynamics

To help our understanding of the turn-on dynamics of both the one- and two-photon laser it is instructive to consider a simple rate equation model of a laser. This system is an expansion of the rate equation model presented in H. Concannon’s doctoral thesis, Chapter 7 [14]. I have created a model which includes both a one-photon process and a two-photon process, resulting in a system which may exhibit pure one-photon lasing (by turning off the two-photon mechanism), two-photon lasing (by turning off the one-photon mechanism) or a mixture of both one-photon and two-photon lasing. This mixture of lasing mechanisms is observed in our experiment for certain experimental parameters. I shall first present the entire model and then consider the threshold behavior and turn-on dynamics for a one-photon laser, a two-

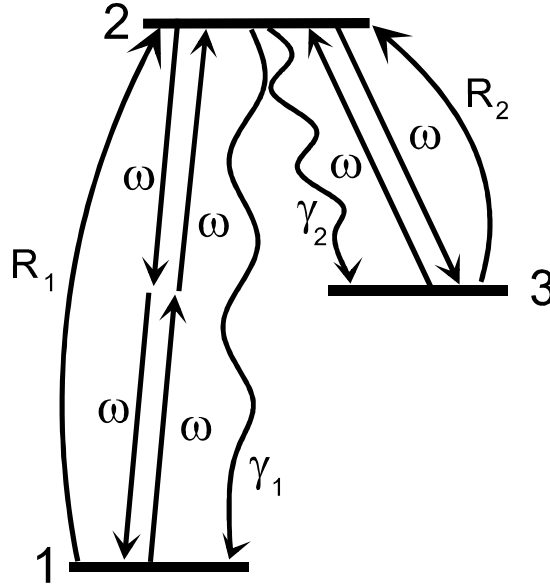


Figure 2.1: Level diagram for the simple rate equation model. A two-photon process is incorporated between levels 1 and 2, while a one-photon process occurs between levels 2 and 3. The straight lines represent laser photons, the oscillating lines indicate spontaneous emission and the curved lines are the incoherent pump mechanisms.

photon laser initiated by an external pulse, and a two-photon laser initiated by a frequency degenerate one-photon process.

Figure 2.1 shows a level diagram for a three level system which supports a one-photon process on the $2 \longleftrightarrow 3$ transition and a two-photon process on the $2 \longleftrightarrow 1$ transition. The frequency of the photons is the same for both the one- and two-photon processes, *i.e.* they are degenerate in frequency. Each process has a pump rate (R_1 and R_2) and a decay rate due to spontaneous emission (γ_1 and γ_2). The rate equation for the number of photons in the cavity q is

$$\frac{dq}{dt} = V_a B^{(2)} q^2 (N_2 - N_1) + V_a B^{(1)} (q + 1) (N_2 - N_3) - \gamma_c (q - q_{injected}) \quad (2.1)$$

where V_a is the volume of the cavity, $B^{(2)}$ is the two-photon absorption and stimulated emission rate, $B^{(1)}$ is the one-photon absorption and stimulated emission rate, γ_c is

the decay rate of photons out of the cavity, and $q_{injected}$ is the number of photons injected into the cavity by an external source. Note that the second term in the right hand side of Eq. 2.1 is proportional to $q + 1$; the 1 is added in to account for spontaneous emission from level 2 to level 3, as there is roughly one photon in the cavity at any given time due to one-photon spontaneous emission. The rate equations for the three atomic energy levels are given by

$$\frac{dN_2}{dt} = R_1 N_3 - \gamma_1 N_2 - B^{(1)} q N_2 + B^{(1)} q N_3 + R_2 N_1 - \gamma_2 N_2 - B^{(2)} q^2 N_2 + B^{(2)} q^2 N_1 , \quad (2.2)$$

$$\frac{dN_1}{dt} = -R_2 N_1 + \gamma_2 N_2 + B^{(2)} q^2 N_2 - B^{(2)} q^2 N_1 , \quad (2.3)$$

and

$$\frac{dN_3}{dt} = -R_1 N_3 + \gamma_1 N_2 + B^{(1)} q N_2 - B^{(1)} q N_3 . \quad (2.4)$$

These equations may be solved explicitly, but for simplicity I solved them numerically using Mathematica. In addition I performed a linear stability analysis on the solutions to find out which solutions were stable and which were unstable.

It should be noted that this model is incomplete, as there are other processes that can occur. For example, there is a singly stimulated two-photon process that can occur between the upper state and the ground state, where one photon is created by stimulated emission and one photon is created by spontaneous emission. Well above laser threshold this singly stimulated emission process should be unimportant, but near threshold it may be significant. Since we were building the simplest qualitative model we could, we ignored this singly stimulated two-photon process.

2.1.2 One-photon laser turn-on dynamics

I begin with the model for one-photon laser, which obtained by setting $B^{(2)} = 0$. In this case the results are identical to the model described in H. Concannon's thesis in section 7.1. Figure 2.2 shows the steady-state solution of this model, plotted as the

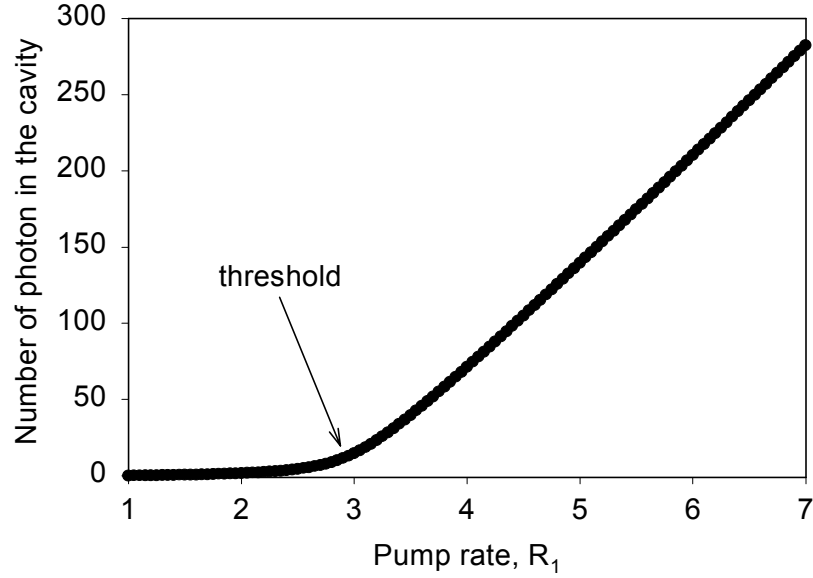


Figure 2.2: Turn-on behavior for a one-photon laser. Note the sharp corner as lasing starts. Parameters are $\gamma_1 = 1$, $\gamma_2 = 1$, $\gamma_c = 0.5$, $V = 50$, $q_{injected} = 0$, $B^{(1)} = 0.04$, and $B^{(2)} = 0$.

number of photons in the cavity as a function of the pump rate. The one-photon laser smoothly turns on as the pump rate is increased. This turn-on occurs when the round trip gain due to the pumping mechanism just exceeds the losses in the cavity. Note that there is only one solution for each value of the pump rate and that all the solutions are stable. This is a second-order phase transition since there is a smooth turn-on as a function of the pump rate. Dynamically, the turn-on behavior is a transcritical bifurcation [15].

2.1.3 Two-photon laser turn-on via injected field

For the two-photon laser to turn on, two conditions must be met. The round trip gain must exceed the round trip loss and there must be a sufficient number of photons in the cavity [1]. There are several ways to obtain the necessary photons to initiate lasing. For two-photon masers there are sufficient photons from spontaneous emission to initiate two-photon masing [16,59–61], however such is not the case for optical two-photon lasers. One possible source of photons is to inject a laser pulse. A simple model for an injected beam is given in H. Concannon's thesis, section 7.2 [14, 57]. Figure 2.3 was generated using my model with $B^{(1)} = 0$. In Figure 2.3(a) the solutions for the number of photons in the cavity as a function of the pump rate are displayed; the dark solid lines are stable solutions and the light dotted lines are unstable solutions. The solution $q = 0$ is always stable, thus demonstrating that the two-photon laser will not turn-on without an external source of photons. For higher pump rates, there is a second branch of stable solutions; if a pulse of light is injected into the cavity it is possible for the laser to jump to this second solution, *i.e.* the two-photon laser initiates. It is also possible to inject a constant number of photons from an external source, the graph of intracavity photon number versus pump rate for this case is shown in 2.3(b). As the pump rate is increased the photon number remains on the lower branch until the pump rate is ≈ 4 at which point the photon number jumps to the higher branch. If the pump rate is now smoothly decreased, the two-photon laser will remain on until the pump rate drops below ≈ 2.5 . Thus there is a bistability or hysteresis in the photon number for this two-photon laser model. This behavior is a first-order phase transition as was noted by Haken in 1984 [7]. Likewise from a nonlinear dynamics perspective, this is an imperfect subcritical pitchfork bifurcation, or a backward bifurcation [15].

To show what happens dynamically when a pulse of light is injected into the

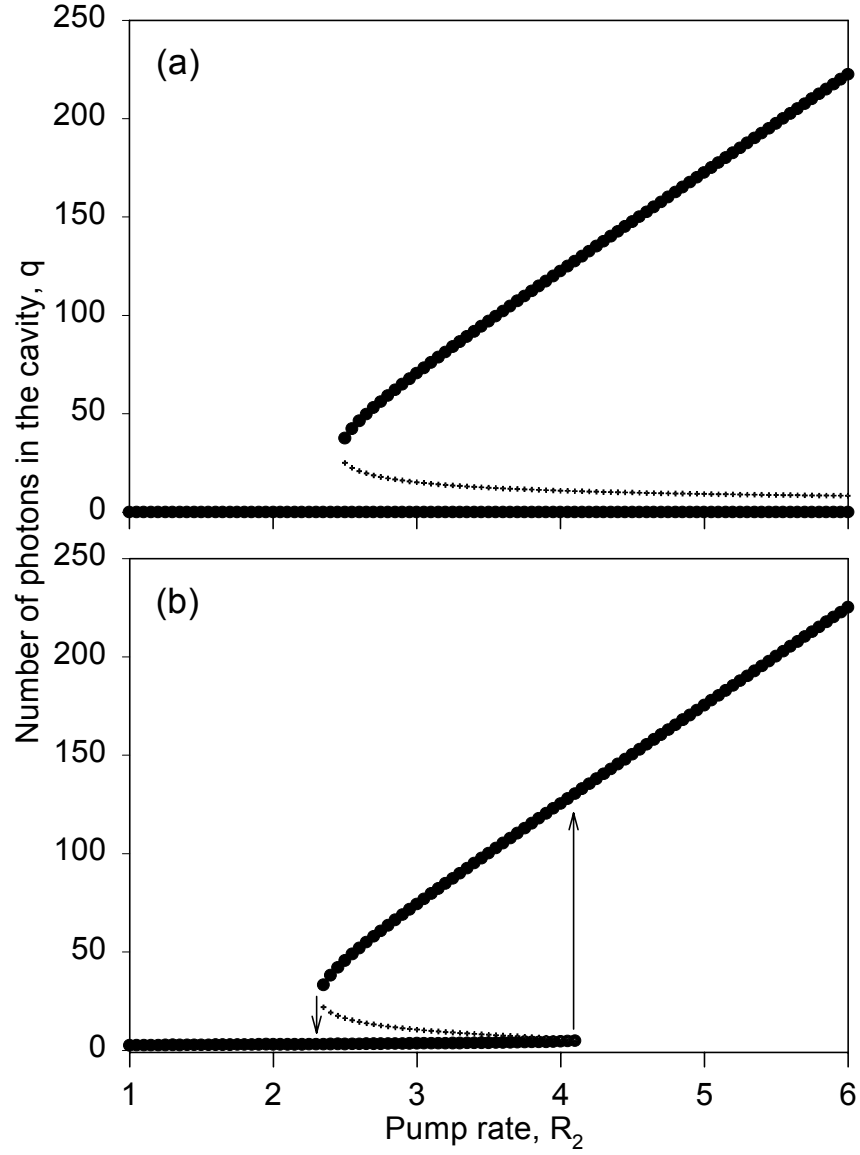


Figure 2.3: Number of photons in the cavity versus the pump rate for the two-photon laser. For (a) $q_{injected} = 0$ and for (b) $q_{injected} = 2.5$, the other parameters are $\gamma_1 = 1$, $\gamma_2 = 1$, $\gamma_c = 0.5$, $V = 50$, $B^{(1)} = 0$, and $B^{(2)} = 0.0002$.

cavity, I integrated the rate equations in time. Figure 2.4 shows the number of photons in the cavity as a function of time for four different injected pulse powers. For all four plots the vertical lines mark the beginning and the end of the injected pulse. Figure 2.4(a) is a pulse well below threshold and the cavity photon number quickly decays away after the pulse is turned off. In Figure 2.4(b) the pulse power is just below threshold and cavity photon number takes a significant amount of time to decay back to zero. Figure 2.4(c) shows a pulse power just above threshold and the photon number builds up to a steady-state value even after the pulse is turned off. Finally, Figure 2.4(d) is for a pulse power well above threshold and we see a spike that decays back to the steady-state laser power after the pulse is turned off.

Even though this is a very simple model we see qualitatively similar experimental results as will be described in Chapter 6. In particular, as we map the two-photon laser behavior as a function of the input probe power we obtain time series that look much like Figures 2.4 (a), (b), and (c). For other experimental conditions we see large spikes at threshold similar to Figure 2.4 (d)

2.1.4 Two-photon laser turn-on via one-photon process

Yet another way to initiate a two-photon laser is by using photons generated from a frequency degenerate one-photon process. The one-photon process will begin lasing from spontaneous emission and can then provide enough photons to start the two-photon laser. We observed this turn-on mechanism for the two-photon Raman laser.

This behavior can be studied with the rate-equation model by using nonzero values for $B^{(1)}$ and $B^{(2)}$. Figure 2.5 shows the curve of photon number as a function of the pump rate for three different values of the one-photon stimulated emission rate, $B^{(1)}$. As before, the dark lines indicate stable solutions to the rate equation model, the light dotted lines are unstable solutions. In Figure 2.5(a) we see that

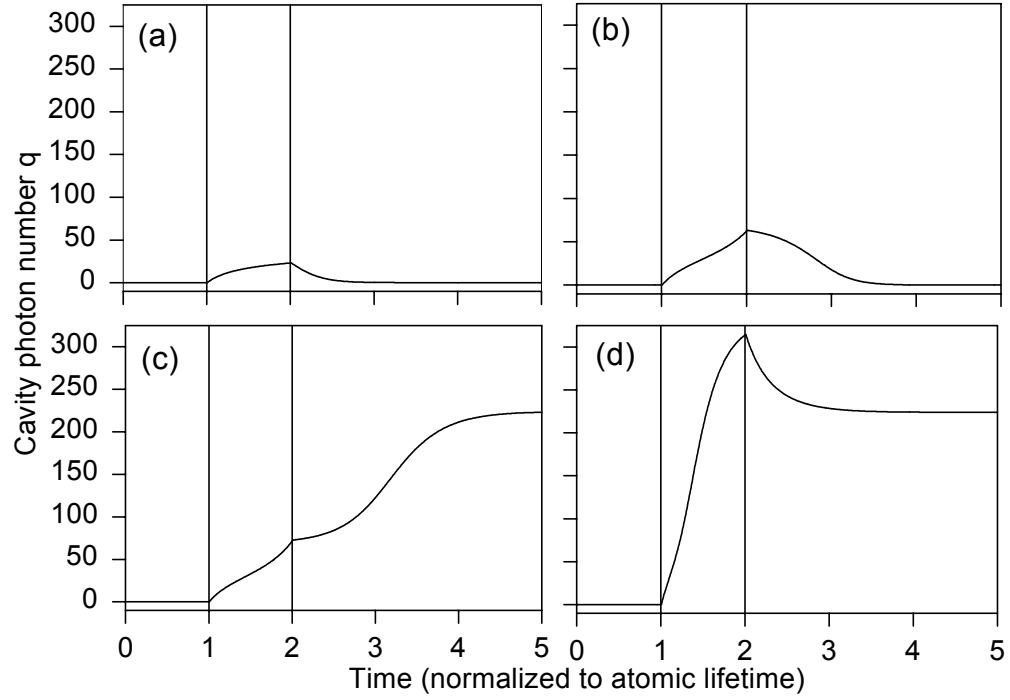


Figure 2.4: Turn-on dynamics of the two-photon laser for four different injected pulse powers. The vertical lines represent the beginning and end of the injected pulse. The four plots are: (a) pulse power well below threshold, $q_{injected} = 15$, (b) pulse power just below threshold, $q_{injected} = 23$, (c) pulse power just about threshold, $q_{injected} = 24$, and (d) pulse power well above threshold, $q_{injected} = 75$. The other parameters are $\gamma_1 = 1$, $\gamma_2 = 1$, $\gamma_c = 5$, $V = 50$, $B^{(1)} = 0$, and $B^{(2)} = 0.002$

there is a large range of the pump rate for which three solutions exist, two of which are stable. As the pump rate is increased to large values the system will jump from the lower branch to the upper branch, just as in the case of a constant injected field. If the pump rate is then decreased, the laser will remain on the upper branch (*i.e.* lasing) until the pump rate reaches the point that only one solution exists. This is the same kind of bistability or hysteresis that we see in the model with an injected field. Unfortunately, we were not able to observe this hysteresis experimentally with the current configuration. If the one-photon stimulated emission rate is increased we obtain the curve in Figure 2.5(b), where for each pump rate there is only one solution, but there is a large jump in the photon number as the pump rate is increased. Further increasing the one-photon stimulated emission rate produces the curve in Figure 2.5(c), which looks much like pure one-photon turn-on except for a small kink. Experimentally we do observe a large jump in the two-photon laser output as the pump rate is increased, indicating that we are in the small one-photon stimulated emission rate regime.

As in the case of the two-photon laser initiated by an injected pulse, we are interested in the dynamical behavior of the two-photon laser initiated by a frequency degenerate one-photon process. We can perform this experiment in the lab by quickly turning on the strong Raman pump beam, in the model we can simulate this by quickly turning on the pump rate. Figure 2.6 shows the number of photons in the cavity as a function of time for the same three sets of one-photon and two-photon stimulated emission rates as in Figure 2.5. In all three plots the two pump rates go from 0 to 60 at time $t=1$. In Figure 2.6(a) the photon number begins to build up very slowly in time. For Figures 2.6(b) and (c) the two-photon laser smoothly turns on and the larger the one-photon stimulated emission rate the faster the laser initiates.

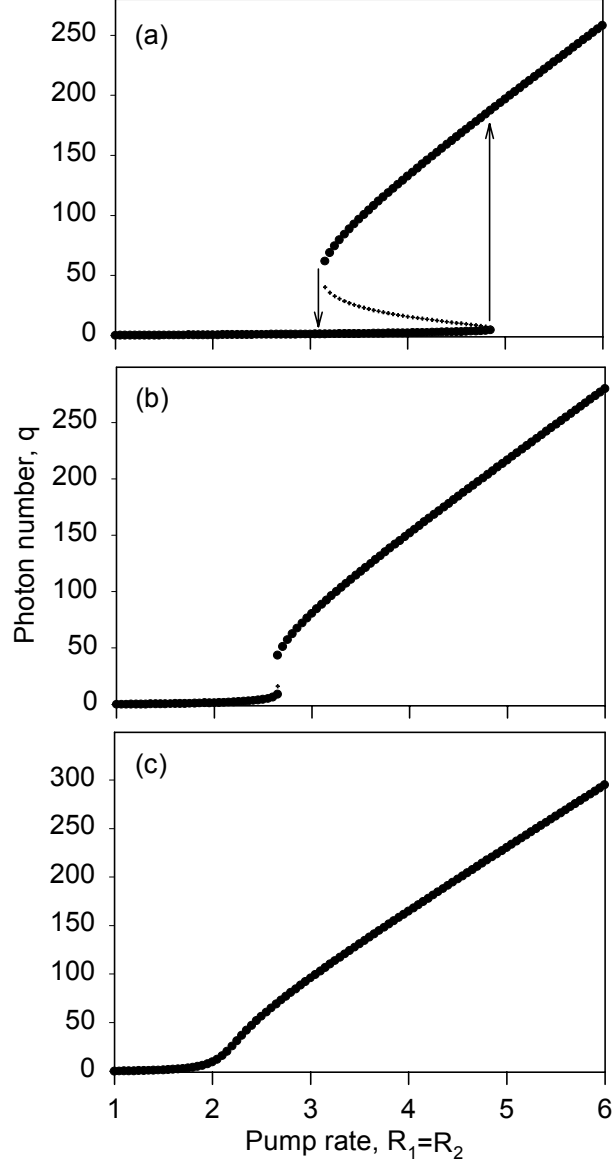


Figure 2.5: Graphs showing the photon number q as a function of the pump rate for a system with both one-photon and two-photon gain. The one-photon and two-photon stimulated emission rates are (a) $B^{(1)} = 0.013$, and $B^{(2)} = 0.0005$, (b) $B^{(1)} = 0.024$, and $B^{(2)} = 0.0005$, and (c) $B^{(1)} = 0.04$, and $B^{(2)} = 0.0005$. The other parameters are $\gamma_1 = 1$, $\gamma_2 = 1$, $\gamma_c = 0.5$, $V = 50$, and $q_{injected} = 0$.

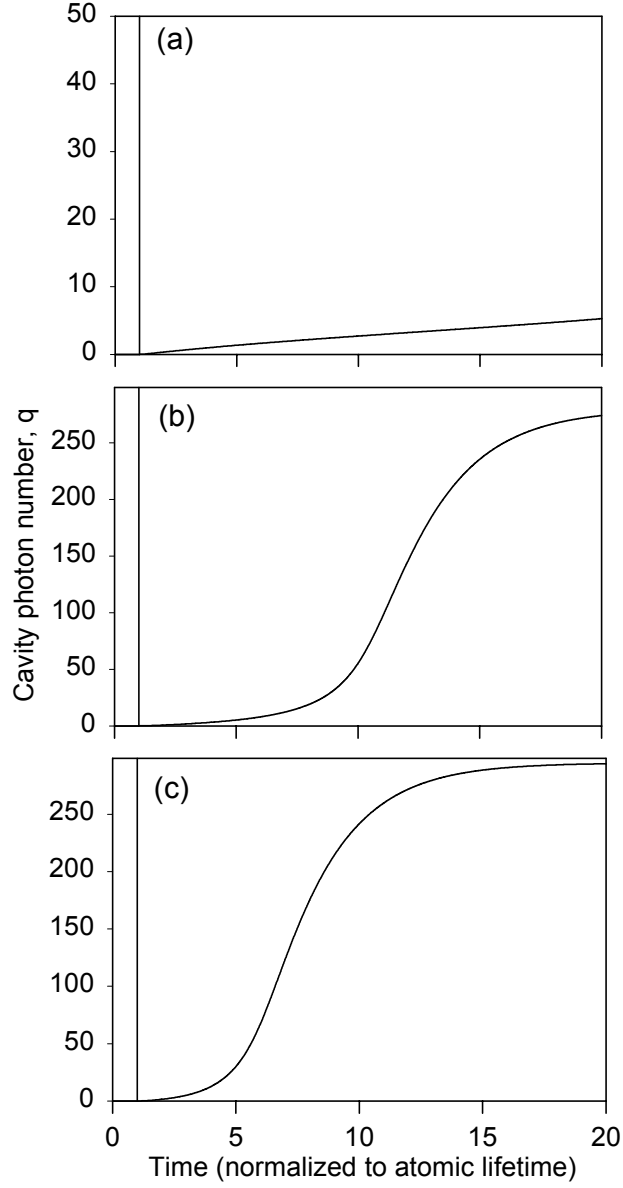


Figure 2.6: Dynamical behavior for a two-photon laser initiated by a frequency degenerate one-photon laser. At time $t=1$ the pump rate is turned on. The one-photon and two-photon rates are (a) $B^{(1)} = 0.013$, and $B^{(2)} = 0.0005$, (b) $B^{(1)} = 0.024$, and $B^{(2)} = 0.0005$, and (c) $B^{(1)} = 0.04$, and $B^{(2)} = 0.0005$. The other parameters are $\gamma_1 = 1$, $\gamma_2 = 1$, $\gamma_c = 5$, $V = 50$, $q_{injected} = 0$, and $R_1 = R_2 = 60$.

I have already noted that there are mathematical solutions to the two-photon laser equations which are unstable. This indicates that this is not an experimentally realizable output of the device. If you start the device at an unstable state, it will quickly evolve to a stable state. However, there are other types of instabilities in lasers and in the two-photon laser in particular.

2.2 Instabilities in the two-photon laser

The term instabilities indicates that a device is no longer running in a stable fashion. For lasers, instabilities typically take the form of spiking, periodic oscillation, or aperiodic (chaotic) oscillation in the intensity, phase, or polarization of the output light.

Instabilities have been observed since the earliest laser. The ruby laser produces large spikes in the output intensity which are still not entirely understood [62]. Instabilities have been observed in other lasers including far infrared gas laser [63], CO₂ lasers, and diode lasers [64]. The source of these instabilities is the nonlinear nature of the laser itself. Lasers can be modeled as a system with a driving source (the pump mechanism), dissipative mechanisms (spontaneous emission and cavity losses), and feedback (the laser cavity), which is a natural recipe for instabilities. These conditions are only exacerbated by the nonlinear gain mechanism in the two-photon laser.

Much of the two-photon laser theory has dealt with the stability of the system. Even the simple rate equation model presented in the previous section shows that there are solutions to the system of equations which are not stable in time. Many much more sophisticated models have supported the idea that two-photon lasers are inherently unstable [9–12, 65].

To help understand the highly nonlinear nature of the two-photon laser, consider

the following line of argument [66]. The polarization of the medium can be written as the susceptibility times the electric field

$$P(\omega) = \chi(\omega)E(\omega) . \quad (2.5)$$

In a typical one-photon laser, the susceptibility is proportional to the inversion

$$\chi(\omega) \propto \Delta N \quad (2.6)$$

which is in turn proportional to

$$\Delta N \propto \frac{1}{1 + I/I_{sat}} \quad (2.7)$$

where I is the intensity in the laser and I_{sat} is the saturation intensity. The vast majority of laser operate will below the saturation intensity ($I \ll I_{sat}$) so a Taylor series expansion can be made

$$\frac{1}{1 + I/I_{sat}} \approx (1 - I/I_{sat} + \dots) \quad (2.8)$$

and we have

$$P(\omega) \approx \chi^{(1)}(\omega)E(\omega) + \chi^{(3)}(\omega)E^3(\omega) + \dots \quad (2.9)$$

where $\chi^{(3)} \ll \chi^{(1)}$. However in the two-photon laser one has

$$\Delta N \propto \frac{1}{1 + (I/I_{sat})^2} \quad (2.10)$$

and the laser naturally operates with $I \approx I_{sat}$ as shown in H. Concannon's thesis p.224 [57]. Thus one must keep all orders in the expansion and

$$P(\omega) \approx \sum_{n=0}^{\infty} \chi^{(4n+1)}(\omega)E^{(4n+1)}(\omega) . \quad (2.11)$$

Given that one photon lasers show instabilities and yet may be accurately modeled with $\chi^{(3)}$ theory, it is to be expected that two-photon lasers will be extremely susceptible to instabilities.

There are several types of instabilities that occur in one-photon lasers, the ones most relevant here are intensity instabilities, multimode instabilities, and polarization instabilities. Several laser models predict intensity instabilities in one-photon lasers. In 1975 Haken [67] showed that the single mode laser model is isomorphic with the Lorenz [68] equations, opening the door to instabilities and even chaos in lasers. This Haken-Lorentz model predicts unstable oscillations if the cavity linewidth is larger than the sum of the atomic relaxation rates and the unsaturated gain exceeds the laser threshold by at least a factor of ten. These are difficult conditions to achieve for most lasers, typically the gain exceeds threshold by only a factor of two or three. Our one-photon Raman laser could satisfy these conditions. For the current cavity the gain is ~ 250 times the threshold and the cavity linewidth is $\sim 1/5$ of the relaxation rates. A different cavity configuration could easily satisfy both of these conditions, opening the door to the possibility of observing chaotic behavior in the one-photon Raman laser.

Instabilities can also occur between different transverse modes of a laser. Theoretically, the transverse modes are orthogonal and thus do not overlap spatially, but experimentally such is not the case. This can lead to gain competition between different spatial modes, which in turn can lead to instabilities in the laser output [69]. Our laser system is an excellent test bed for such multimode instabilities because of the high gain (relative to threshold) in the one-photon Raman processes and the highly nonlinear gain in two-photon Raman processes. It is not however, our intention to study these type of instabilities at this time. In fact, we have placed apertures in the cavity expressly to suppress lasing on any higher order transverse modes.

It was not until we began operating the two-photon laser that we realized there was even the possibility of polarization instabilities. Typically, polarization instabil-

ities occur when there are multiple quantum pathways within the atom that involve different light polarizations. Recently, Vilaseca and colleagues have studied theoretically polarization instabilities in cascade laser² systems [70–72]. They find that asymmetries due to the coherent pumping will lead to polarization instabilities in the output [73, 74]. While not identical to the cascade system, the two-photon laser is similar in many respects and this lends theoretical support to our experimental observations. Polarization instabilities have also been considered in diode laser systems with an externally applied magnetic field [75].

Polarization instabilities have also been observed in counterpropagating beams in atomic sodium vapor [76, 77]. These instabilities occurred in very intense beams tuned near the $3S_{1/2} \rightarrow 3P_{1/2}$ transition in sodium. As the power of the beams is increased, the output polarization goes from constant to periodic to chaotic. Our two-photon laser will experience similar conditions since there are two intense counterpropagating beams inside the two-photon laser cavity.

In summary, two-photon lasers are highly nonlinear devices that exhibit novel turn-on behavior, noise properties, photon statistics, and are highly susceptible to instabilities. With this in mind, I will discuss the previous experimental realizations of two-photon devices. These experiments opened the door, but did not answer all the questions about how and why two-photon devices work.

2.3 Experimental realization of two-photon stimulated emission devices

The first reported observation of two-photon lasing occurred in 1981, when Toschek’s group claimed to have seen two-photon lasing in laser pumped lithium vapor [78].

²Cascade laser systems involve multiple connected transitions where the ground state of the upper system is the excited state of the lower system. Thus the upper system feeds population to the lower system.

However subsequent work by Jackson and Wayne [79] suggested that this was actually a parametric oscillator based on six-wave mixing, not a two-photon laser. Such missteps underscore the difficulty in finding two-photon processes and the necessity of carefully checking any results and claims.

2.3.1 Two-photon maser

Experimental realization of a two-photon oscillator did not occur until the late 1980's. After considering the two-photon stimulated emission rate coefficient, Haroche realized that Rydberg atoms might provide a system with a very large coefficient. Recalling from Chapter 1, the two-photon stimulated emission rate coefficient for a simple three-level system is given by

$$B_{2\gamma \text{ stimulated}} = \frac{32\pi^2 |\boldsymbol{\mu}_{\text{ei}} \cdot \boldsymbol{\epsilon}|^2 |\boldsymbol{\mu}_{\text{ig}} \cdot \boldsymbol{\epsilon}|^2 \omega^2}{V_c^2 \hbar^2 \Delta_{ig}^2 \Gamma_{eg}} . \quad (2.12)$$

The dipole matrix elements $\boldsymbol{\mu}$ are very large in the Rydberg levels, in excess of 1500 *a.u.* for Haroche's maser. The Rydberg levels are numerous and closely spaced, allowing Haroche to choose two levels with an intermediate level nearly halfway between the ground and excited state, thus making Δ_{ig} very small. In addition, a very high finesse cavity was used so that the number of photons circulating in the cavity was relatively small and the cavity resonances were very narrow, thus suppressing competing gain mechanisms.

This two-photon maser was first operated in 1987 [16, 59–61]. The two-photon transition occurred between the $40S_{1/2}$ (the upper level) and the $39S_{1/2}$ (the lower level of the laser transition) Rydberg levels in rubidium. The $39P_{3/2}$ level acted as the near-resonant intermediate level, resulting in a Δ_{ig} of 39 MHz, as illustrated in Figure 2.7. The rubidium atoms from the atomic beam were pumped to the upper state using light from three diode lasers and microwaves from an X-band

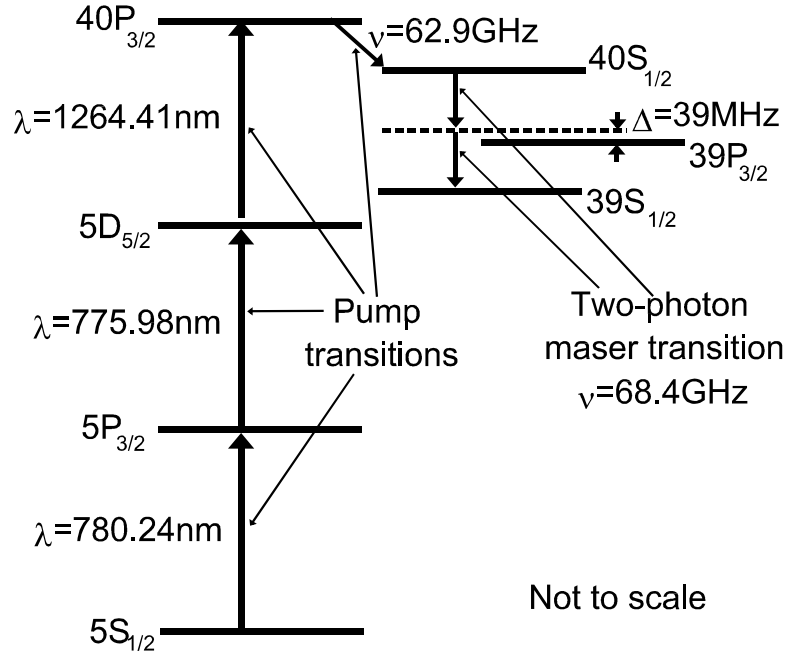


Figure 2.7: The atomic system used to build the first two-photon micromaser. Three diode lasers pump the atom into the $40P_{3/2}$ Rydberg level. A microwave field stimulates the atom to the $40S_{1/2}$ level, resulting in a two-photon inversion between the $40S_{1/2}$ level and the $39S_{1/2}$ level. The $39P_{3/2}$ level is the near resonant intermediate level for the stimulated two-photon transition.

klystron. These excited atoms then passed through a high finesse cavity, which had a Q (quality factor, which is a measurement of the finesse used for microwave cavities) of approximately 10^8 . Masing could be achieved with approximately 40 photons and five atoms in the cavity. Since the intracavity intensity was very low, competing nonlinear processes such as multiwave-mixing were insignificant. The output power from this two-photon maser was less than 2×10^{-17} W, making detection difficult and experiments concerning photon statistics or squeezing nearly impossible. This two-photon micromaser is a substantial experimental step, but only adds to the interest in building a two-photon optical laser.

The technology that made the two-photon maser possible was the development of

a super cavity with extremely low losses and correspondingly high finesse (or quality factor Q). For the two-photon maser, Haroche used a niobium cavity with a length of 7.5 mm and a diameter of 7.7 mm, which was cooled to reduce the number of thermal photons in the cavity. The cavity was chilled to 1.7 K using liquid helium. At this temperature the cavity finesse was $Q \simeq 10^8$; if the temperature is increased to 2.5K, the Q drops to 3×10^7 . Without this super cavity, the two-photon maser would not have been possible. For a lower Q cavity, losses would have robbed photons from the two-photon maser and there would have been one-photon transitions instead of the two-photon transition. High finesse cavities are used in the optical two-photon lasers for similar reasons.

Several other amazing experiments have recently been conducted using extremely high finesse cavities. All of these experiments, like the two-photon laser, explore the quantum mechanical interaction of light and atoms. For example, using a high finesse cavity, Haroche has been able to observe quantum Rabi oscillations which provide a direct measurement of the number of photons in the cavity [80]. Even more recently, Haroche used a high Q cavity as a quantum memory device with a single photon in the cavity as the “bit” of information [81]. In a similar vein is Feld’s single-atom laser [82, 83] where mirrors with a reflectivity of 99.9999% enable a photon to remain in the cavity for a quarter of a million rounds trips. Passing excited barium atoms through the cavity results in a laser with an efficiency of $>50\%$ and an average of less than one atom in the cavity at any given moment. This is in contrast to typical lasers with billions of photons and several thousand trillion atoms in the cavity at one time.

2.3.2 Two-photon dressed-state laser

The search for experimental systems has yet to find an atom or molecule with optical transitions suitable for a two-photon laser. However, shortly after Haroche built the two-photon maser, Mossberg realized that it might be possible to engineer an optical system that had properties similar to the Rydberg levels. This system is a dressed atom, where an atom is driven by a very strong pump laser which is tuned near a strong electric-dipole allowed transition. The energy levels of the combined atom plus driving laser system are called dressed-state levels since the strong pump laser is altering or dressing the atomic energy levels. This dressed-state system looks much like the Rydberg levels with the same advantageous of easily created inversions and near resonant intermediate levels.

Dressed-state levels were first explored by Cohen-Tannoudji and Reynaud in 1977 [84, 85]. The atom itself has energy levels; for simplicity, consider a two level atom with just a ground and excited state. In addition there are the energy levels of the strong laser beam, consisting of a ladder of states corresponding to the number of photons in the beam. The atom and the laser are coupled via the electric dipole interaction. The resulting system has a system of levels which is a mixture of the atom levels and the field levels, see Figure 2.8. Photons are added or removed by moving up or down the rungs of the ladder. In this approach, spontaneous emission is neglected; instead resonance fluorescence appears as spontaneous emission in the atom+laser system. Stimulated emission happens when a second field (usually referred to as the probe) is added that stimulates transitions between the dressed states.

Mossberg realized that this system is ideal for two-photon stimulated emission. As can be seen in Figure 2.8 there is an inversion created by the strong dressing laser

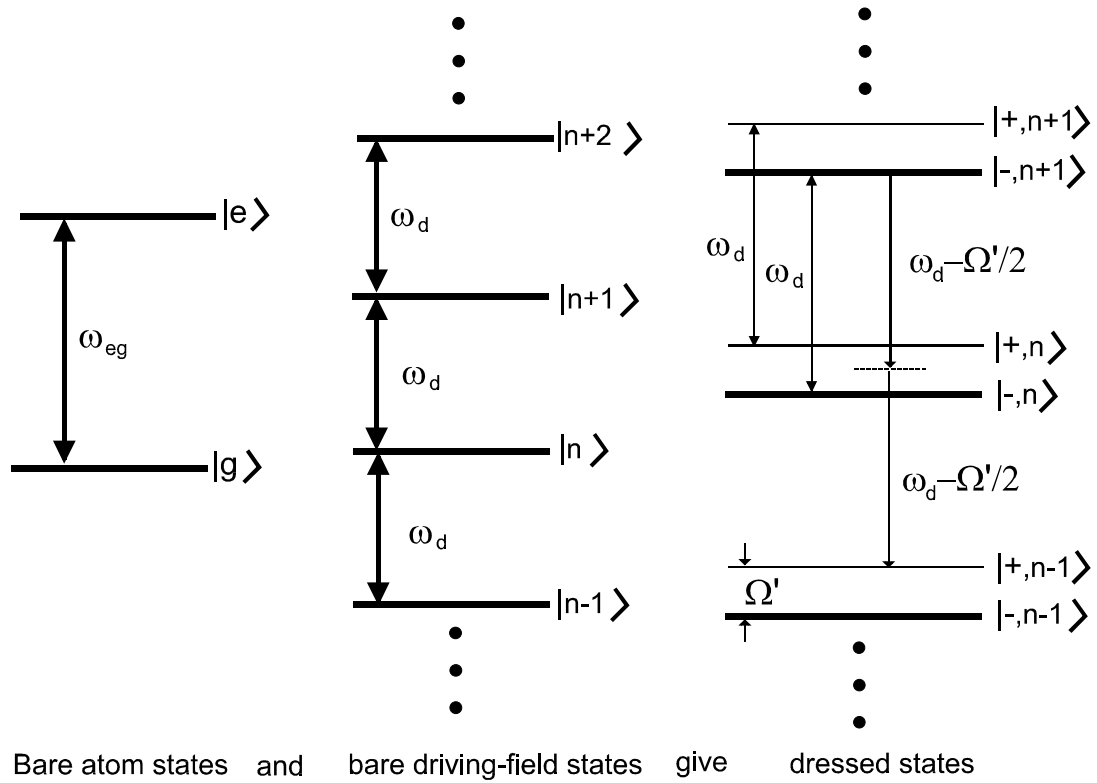


Figure 2.8: Dressed-atom picture of the atom+laser system for a two-level atom driven by a strong pump. The ground and excited states of the two-level atom interact with the ladder of states representing the number of photon in the field to give the dressed states. Using this picture it is easy to see the two-photon inversion between states $|-,n+1\rangle$ and $|+,n-1\rangle$, with resonant frequency $\omega - \Omega/2$. Note that the levels $|-,n\rangle$ and $|+,n\rangle$ provide the nearly resonant intermediate level for the two-photon process.

(the wide lines in the dressed states represent the states with greater population). Additionally there is a near resonant intermediate state which will enhance the two-photon stimulated emission rate. This detuning can be easily controlled since Ω' (the generalized Rabi frequency) depends on the strength and detuning of the pump laser. Finally, the full ground to excited state dipole matrix elements are involved, thus maximizing the possible two-photon stimulated emission rate.

In a series of three papers in 1991, Zabrewski, Lewenstein, and Mossberg explored theoretically the behavior of dressed state lasers. The first paper [17] shows that a strongly dressed two-level atom will exhibit one-photon and two-photon dressed-state gain and possibly higher orders. Additionally, they explore the regions of stability as a function of the drive laser Rabi frequency and detuning, finding large areas of stable operation for both one- and two-photon dressed state lasing. The second paper [18] considers the effects of phase diffusion and the possibility of squeezing in the laser light emitted from dressed-state lasers. Interestingly, both the one- and two-photon dressed state lasers exhibit squeezing theoretically. Unfortunately, squeezing only exists for short time measurements, as phase diffusion effects wash out any squeezing in the long time limit. The third and final paper [19] in the series goes on to consider pump depletion effects. Results indicate that these effects reduce the amount of squeezing in the output of the dressed state laser, but, at the same time, introduce a small amount of squeezing into the depleted pump laser field.

Using the idea of two-photon dressed state gain, a two-photon laser was built in Mossberg's lab in 1992. The first step was observing two-photon gain, which was done by Zhu *et al.* using a laser pumped beam of barium atoms [86]. Gauthier *et al.* [20] then managed to build a laser using this gain mechanism. The apparatus consists of an barium atomic beam interacting with two orthogonal laser beams, one pump and one probe, as illustrated in Figure 2.9. For the laser, a cavity is added

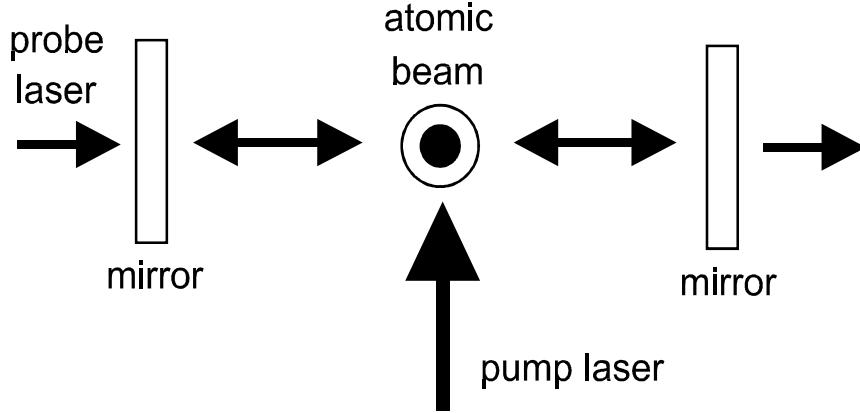


Figure 2.9: Schematic for two-photon laser based on dressed-state gain using a laser driven barium atomic beam. The atomic beam is in the direction perpendicular to the page and interacts with a strong pump beam and an orthogonally positioned probe beam.

in the direction of the probe laser. As predicted, an external trigger pulse, with intensity greater than threshold, is required to start two-photon lasing.

Even though lasing was observed in this experiment there were several shortcomings. The first was that both one and two photon lasing were present because the Doppler width was on the order of the frequency separation between the two gain mechanisms. The one-photon gain accounted for $\approx 35\%$ of the gain in the system. Using a higher finesse cavity would help in this situation because a narrower Doppler width could be employed. In addition, the two-photon laser only remained on for $\approx 50 \mu\text{sec}$ before external perturbation disrupted lasing. Thus the only characteristic of two-photon lasers that could be really studied was turn-on properties. Photon statistics, noise measurements, and long-time effects awaited the advent of a stable two-photon laser.

2.3.3 Other possible two-photon gain sources

Several other two-photon gain sources have been suggested or observed. One theoretical idea for a two-photon gain mechanism came from Ironside in 1992 [87]. He suggested using a semiconductor media for two-photon gain. This system is advantageous since two-photon effects are well understood in semiconductors and there is already significant technical know-how for dealing with semiconductors. Unfortunately, other nonlinear processes will present significant competition to the two-photon gain. Ironside explores theoretically the electrically pumped GaAs double heterostructure laser and optically pumped InSb structures; in both cases two-photon gain is predicted at low temperatures (77 K).

In an experimental observation in 1994, Hänsch *et al.* [88] reported multiphoton Raman transitions between vibrational levels of an optical lattice of rubidium atoms. They created a 3-D optical lattice of rubidium 85 atoms, first by cooling the atoms using a magneto-optical trap, and then by trapping the atoms in the anti-nodes formed by standing waves created in each dimension. Once the atoms are cooled, the MOT fields are turned off. Pump-probe spectra are taken using one of the standing waves as the pump and an additional probe beam. Transmission spectra of the probe beam reveal a Rayleigh resonance and Raman resonances utilizing one-photon through four-photon transitions. These Raman resonances occur between vibrational levels of the trapped atoms. The one-photon resonance is 162 kHz from the Rayleigh resonance and has a width of ≈ 40 kHz, a two-photon resonance occurs at 82 kHz and there is even a three-photon resonance at 56 kHz. However these resonances all appear as bumps on the side of a spectrally broader feature, which Hänsch attributes to Raman transitions employing two vibrational quanta. This inability to spectrally separate the various resonances render this mechanism unsuitable as a source of two-photon gain. Additionally the trap must be reloaded once the atoms are used, so a

two-photon laser built around this mechanism would have to be pulsed.

These experiments set the stage for the two-photon laser research carried on in this lab. My work builds on the idea of creating a suitable system for two-photon gain by using a strong pump laser. The challenge has been to find a system with sufficient spectrally isolated two-photon gain and no competing gain mechanisms. As I will show, laser driven potassium atoms provide such a system and ultimately enable a two-photon Raman laser.

Chapter 3

Laser driven alkali atoms as a source of two-photon gain

3.1 Introduction

Having covered some of the historical ideas and experiments, I now discuss the previous work done in this lab that leads into my work. As discussed in section 2.3.2, the only optical two-photon laser to date employed dressed-state gain in a beam of barium atoms. Two-photon lasing was observed, but one-photon lasing was present at the same frequency and the resulting device was therefore not a pure two-photon laser. Work in this lab has been focused on finding other mechanisms and atomic systems that would produce enough spectrally isolated two-photon gain to build a pure two-photon laser. It is important to have multiple mechanisms for building two-photon lasers so that we can sort out which effects are due to two-photon lasers in general and which are results of the specific mechanism.

The first part of this chapter covers the work done by previous students in this lab. Most significantly, 30% two-photon gain was observed in a vapor cell experiment. This gain is 100 times larger than any other two-photon gain observed to date. I arrived shortly after this observation was made and my task was to build a laser employing this gain mechanism. Unfortunately, for reasons which I will outline later, it proved impossible to build a two-photon laser using the vapor cell two-photon Raman gain. The next phase was to take a step back and consider what systems might be better suited for our needs. Ultimately, we developed a novel two-photon Raman process based on laser driven potassium atoms generated by an atomic beam

apparatus. The final section of the chapter covers a simple theory we developed to predict what gain (and hence lasing) features would exist in the experiment.

3.2 Previous experimental work with the vapor cell system

When Dr. Gauthier came to Duke, his first goal was to find two-photon Raman gain in the alkali atoms. In general, Raman processes are multi-photon transitions between closely spaced atomic levels. Figure 3.1(a) shows a one-photon Raman transition between two ground states in a three-level atom. The upward arrow represents a photon from the pump beam and the downward arrow represents a photon from the probe beam. The result of the interaction is the annihilation of one pump photon, the addition of a photon to the probe beam and an atomic transition from level $|a\rangle$ to level $|c\rangle$. In a similar fashion, two pump photons and two probe photons can also induce a transition from $|a\rangle$ to $|c\rangle$. This process is referred to as a two-photon Raman transition, see Figure 3.1(b).

Two-photon Raman gain should have several advantages to the previously observed two-photon dressed-state gain. Only two pump photons are needed for two-photon Raman transition instead of the three pump photons needed for the two-photon dressed-state transition, so, all else being equal, the two-photon Raman process will have a larger scattering rate than the two-photon dressed-state process. The detuning between the one-photon Raman process and the two-photon Raman process is set by the ground state splitting of the atom, so it is possible to obtain sufficient frequency separation between the features by proper selection of the atom. Finally, as previously noted, it is important to have multiple two-photon gain mechanisms so that we may discern what are actually fundamental features of the two-photon process.

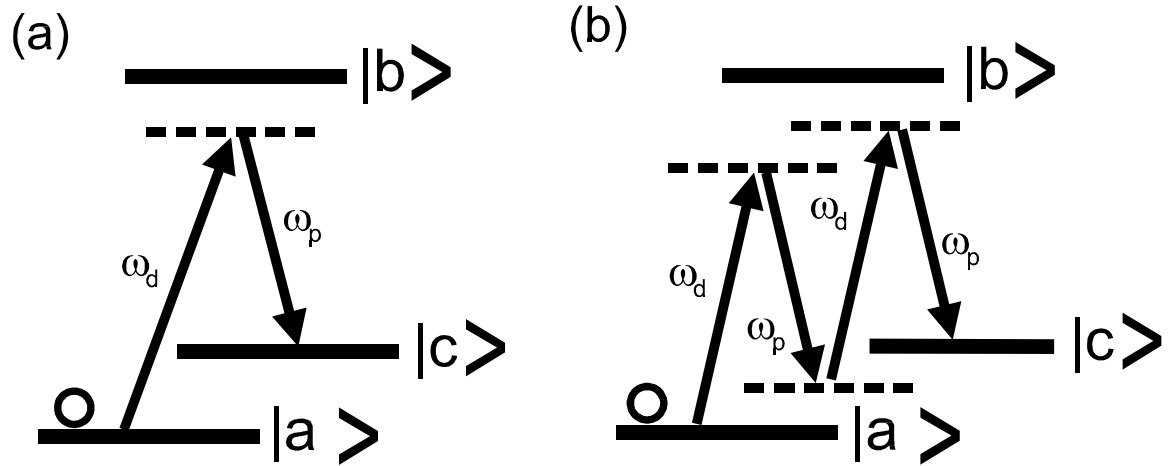


Figure 3.1: Scattering diagrams for Raman transitions, the solid lines are atomic energy levels and the dashed lines are virtual intermediate levels. In (a) one pump photon (the up arrow) and one probe photon (the down arrow) stimulate a transition from state $|a\rangle$ to state $|c\rangle$. Similarly in (b) two pump photons and two probe photons stimulate the transition. All orders are possible for the process (*i.e.* three-photon, four-photon, etc.).

The first experiments conducted in this lab were focused on finding and characterizing two-photon Raman gain in a buffer cell. The experimental configuration consisted of a vacuum cell with a cold finger filled with an alkali metal. When the cold finger was heated, the alkali formed a vapor with atomic number densities on the order of 10^{13} atoms/cm³. Passing through this vapor cell were the strong pump beam (up to 1 W) and a weaker co-propagating probe beam (up to 10 mW). Proper choice of the pump beam detuning permitted the observation of one-photon Raman gain and two-photon Raman gain. A wide variety of other gain and absorption features were also observed.

At the outset of these experiments, a decision had to be made about what alkali atoms would be optimum for two-photon Raman gain. Once a particular atom was selected, changing would be expensive because the choice of atom mandates which lasers to purchase. Since the required lasers cost around \$100,000, it was important to choose carefully and to keep flexibility in mind. To further complicate the decision, we wanted to be able to perform the experiment in a vapor cell or in an atomic beam. The possibility of using a vapor cell was very intriguing, because extremely large number densities can be realized, up the 10^3 times as larger than atomic beam systems.

Alkali atom	ground state splitting	wavelength of D1 line
	(MHz)	(nm)
lithium(6, 7)	228, 804	670.8
sodium(23)	1772	589.0
potassium(39(94%), 41(6%))	462, 254	769.9
rubidium(85(72%), 87(28%))	3036, 6835	794.8
cesium(133)	9193	852.1
francium	46700	817.2

Table 3.1: Ground state splittings and wavelengths of the D1 line for the alkali atoms. The numbers in parenthesis after the atom name are the isotopes and their relative abundance.

In searching for a suitable alkali atom, the ground state splitting must balance two opposing requirements. Smaller the ground state splittings have larger the scattering rates since the resonant enhancements are larger. However, if the ground state splitting is too small, it will be difficult to spectrally isolate the one-photon and the two-photon gain and the two-photon gain may actually sit on the wing of the one-photon gain, as was the case in the Oregon experiment [20]. Since the natural widths of these Raman transitions are typically on the order of a few MHz, the peak width will be set by the residual Doppler width, which was expected to be between 1 and 100 MHz depending on the exact experimental configuration. Thus we needed the one-photon Raman and two-photon Raman transitions to be separated by at least 100 MHz. We decided to use potassium 39 which has a ground state splitting of only 462 MHz as shown in Table 3.1 [89–92]. Potassium has the added advantage that the high power laser needed for the strong pump beam (generated by a Ti:Sapphire laser) can also be tuned to the rubidium resonances, thus providing a second possible alkali with which to work.

All of the work described in this thesis involves potassium 39. There was some work done with rubidium, but no two-photon gain was observed and once results were obtained with potassium, the rubidium experiment was discontinued.

There is one piece which has been overlooked in the description of Raman processes in potassium: how the population inversion is created. Looking at Figure 3.1, we see that since the two ground state levels are closely spaced, the populations of levels $|a\rangle$ and $|c\rangle$ are essentially equal at room temperature. In order to obtain any Raman type gain, inversion must be created between levels $|a\rangle$ and $|c\rangle$. An additional laser beam could be added at the $|b\rangle$ to $|c\rangle$ frequency, but by tuning the Raman pump laser near the $|b\rangle$ to $|c\rangle$ line (typically within a couple of GHz for the vapor cell experiment), it will preferentially optically pump the atoms from $|c\rangle$ to

$|a\rangle$. Experimentally this simplifies the apparatus since only two beams are needed. In addition, the large Rabi frequencies necessary to obtain significant two-photon Raman gain are sufficient to optically pump the entire Doppler distribution in the potassium vapor cell.

These pump-probe experiments were originally performed by Hope Concannon and are described in-depth in her doctoral thesis [57]. A typical spectra with a large two-photon gain of $\sim 30\%$ is presented in Figure 3.2. Additionally there is one-photon Raman gain 230 MHz to the left of the two-photon gain and three-photon Raman gain 77 MHz to the right. The small peak just to the left of the two-photon gain is one-photon Raman gain from ^{41}K which has a ground state splitting of 254 MHz. For this data the pump beam has a power of $\sim 900\text{mW}$ with a waist of $64\text{ }\mu\text{m}$ and the probe beam has a power of $\sim 2.5\text{mW}$ with a waist of $28\text{ }\mu\text{m}$.

These results were very impressive as they represent the largest two-photon gain observed to date. However these very high intensities in the pump and probe beams cause many other effects which compete with the two-photon Raman process. The very conditions which permit large two-photon Raman gain, namely the high density vapor cell and the nearly copropagating beams, enable other gain mechanisms such as dressed-state gain processes and multi-wave mixing effects. Figure 3.3 shows a spectrum at similar conditions to Figure 3.2, with a large probe frequency range. The large broad peak on the left (feature (I) in Figure 3.3) is thought to be due to a dressed-state type process, however it was not at the detuning we expected. Furthermore this feature could be moved in frequency by changing the pump-probe crossing angle, which is not expected from a dressed-state process (however I will continue to refer to it as the dressed-state feature). Feature (VII) (on the right side of the spectrum) is a four-wave mixing process enhanced by the small crossing angle between the two beams. Since these two features are very broad, any cavity

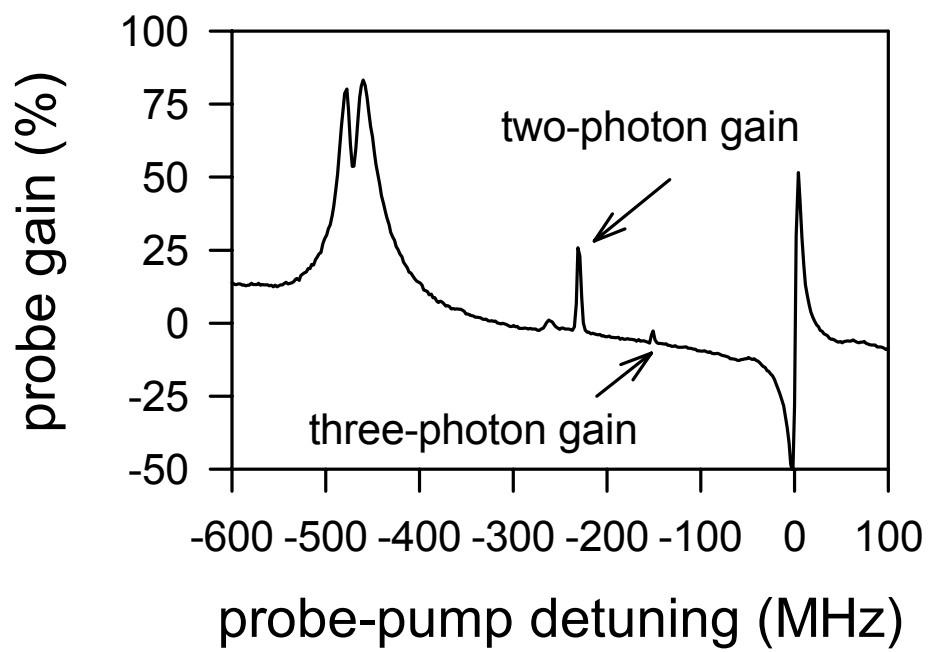


Figure 3.2: Experimental probe spectra from Hope Concannon’s work shows large ($\sim 30\%$) two-photon Raman gain. Also note significant three-photon Raman gain [21].

of reasonable length with a resonance at the two-photon Raman gain frequency will also have a longitudinal cavity resonance that is degenerate with one or the other of these broad features. Thus, if lasing is to be possible, these competing mechanisms must be suppressed.

In addition to these broad competing gain mechanisms, there were several other experimental difficulties with the vapor cell system. Self-focusing and defocusing due to the high density vapor cause the laser beams to expand, limiting beam intensities. This was a problem we routinely observed in the pump beam and for the probe beam at high powers. Similarly, channeling effects can occur where the probe beam is actually guided and focused (or defocused) by the strong pump beam [93]. Concannon suspected that this channeling effect was a problem in the buffer cell experiment, since small changes in the probe input angle would produce large jumps in the output angle, *i.e.* it appeared that the pump laser was steering the probe laser. This condition makes the optimum alignment extremely sensitive for the pump-probe experiments and would make laser cavity alignment very difficult. Finally, photoionization of the atoms via the absorption of multiple pump photons can destroy any inversion that has been created. Photoionization was observed for very high atomic number densities and pump powers and was characterized by the emission of a purple light from the vapor cell as the ions and electrons recombined. Ultimately, however, it was felt that a two-photon laser might be possible in a potassium vapor cell if the dressed-state feature and four-wave mixing feature could be suppressed.

3.3 Experimental results with buffer gases

Eliminating these broad features was the first priority when I took over the two-photon laser experiment. Even though we were not sure of the origin of these features, we knew that they involved the excited state of the D1 line since the widths were

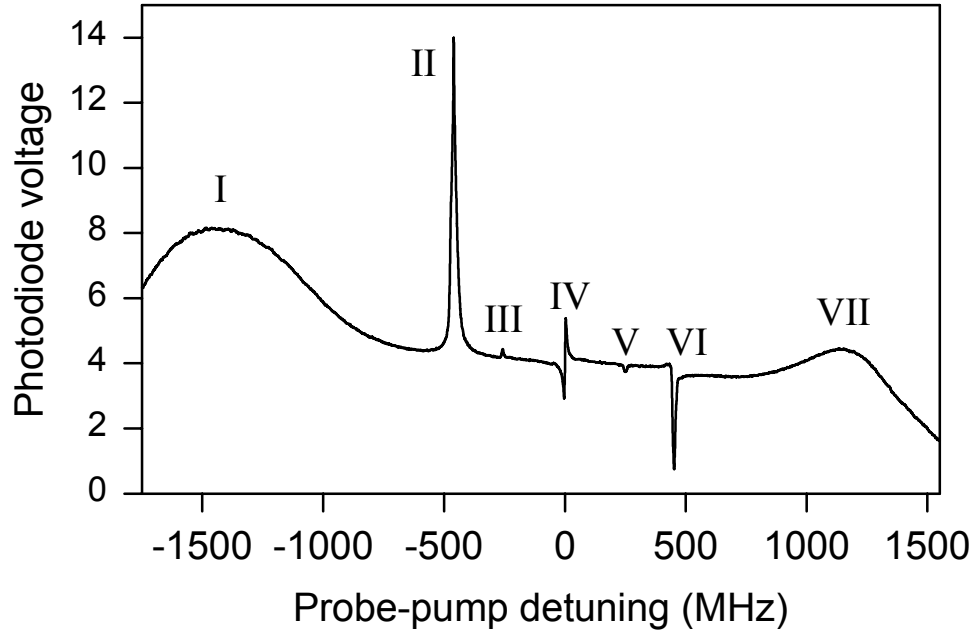


Figure 3.3: Experimental probe spectra from H. Concannon’s thesis, revealing some of the other mechanisms which compete with the two-photon gain (which is the small peak at -231 MHz). The features are (I) one-photon dressed-state gain, (II) one-photon Raman gain, (III) two-photon Raman gain, (IV) Rayleigh feature, (V) two-photon Raman absorption, (VI) one-photon Raman absorption, and (VII) multi-wave mixing feature.

on the order of the Doppler width of ~ 1 GHz. In addition, the self-defocusing was dependent on the population of the $4P_{1/2}$ excited state. Adding a buffer gas to the vapor cell alters the widths and strengths of the various features we observe since there will be collisions between the buffer gas and the potassium atoms. There are two types of collisions which can occur: collisions which perturb the electron due to the electric field of the colliding atom and resonant collisions where the electron changes energy levels. For an ensemble of driven alkali atoms, such as we have in the vapor cell experiment, perturbing collisions effectively dephase the coherence induced between the ground and excited states by the strong pump field. This increased dephasing rate will alter all processes occurring between the various levels, but may affect some more than others. There are also resonant collisions in which electrons change energy levels. In the vapor cell experiment this may be important since we can use gases that have resonances matching the $4P_{1/2}$ to $4P_{3/2}$ transition. This type of collisions should affect processes where the electron ends up in $4P_{1/2}$ level (*i.e.* dressed-state processes) and not alter processes involving the hyperfine levels of the $4S_{1/2}$ level (*i.e.* the Raman type processes).

Several gases were chosen for the buffer cell experiment. The noble gases helium and argon were used; these gases should have a substantial dephasing effect on potassium. Some theory suggests that the mass difference between helium and argon will have no effect on the dephasing rate [94], but experimentally there is a difference in the linewidth broadening due to helium versus argon [95]. For resonant collisions we used molecular nitrogen, since it has a vast number of vibrational and rotational energy states giving rise to many transitions which are nearly resonant with the $4P_{1/2}$ to $4P_{3/2}$ transition in potassium. Helium and argon also cause resonant transitions, but at a significantly lower rate than molecular nitrogen [96–98]. Other gases could have been tried, but we limited ourselves to these three because the literature indi-

cated that they were good choices, they were readily available, and they were safe to deal with in the lab.

My goal was to find a buffer gas and suitable gas pressure such that two conditions were met; that the maximum broad feature gain was significantly less than the maximum two-photon Raman gain and that the two-photon Raman gain was $> 10\%$. The broad feature gain must be significantly less than the two-photon gain so that there is no broad feature lasing, but only two-photon lasing. The 10% level for the two-photon Raman gain was a function of the expected losses in any laser cavity. Since this system used a buffer cell, there would be losses due to the cell windows. High anti-reflection coatings were not an option since potassium vapor tends to react with most coatings. It was expected that the losses would be at least 1% per surface. In addition, we would probably need other intracavity optics, such as polarizing beamsplitters, to introduce the strong pump beam into the cavity, adding further losses. The 10% number was not an absolute limit, but anything much below that looked impractical for the buffer cell configuration.

To test these ideas, I (in conjunction with Dr. Jeff Gardner) added small quantities of helium, argon, and nitrogen to the potassium buffer cell. These gases were tried individually, we did not try any mixtures of gases. A leak valve was used in conjunction with a Baratron gauge allowing us to accurately add gas pressures from a few milliTorr to several Torr. For the spectra presented, the pump power was ~ 780 mW and the probe power ranged from 1 to 6 mW. The temperature of the potassium was 170°C , which corresponds to a number density of 3.2×10^{13} atoms/cm³. The crossing angle was adjusted to maximize the two-photon gain. For these spectra the range was 4 to 8 milliradians. For comparison, a spectrum without any buffer gas is shown in Figure 3.4. There is substantial two-photon Raman gain (feature III), $\sim 50\%$, but the dressed state gain is $\sim 1200\%$ (feature I). The other features of

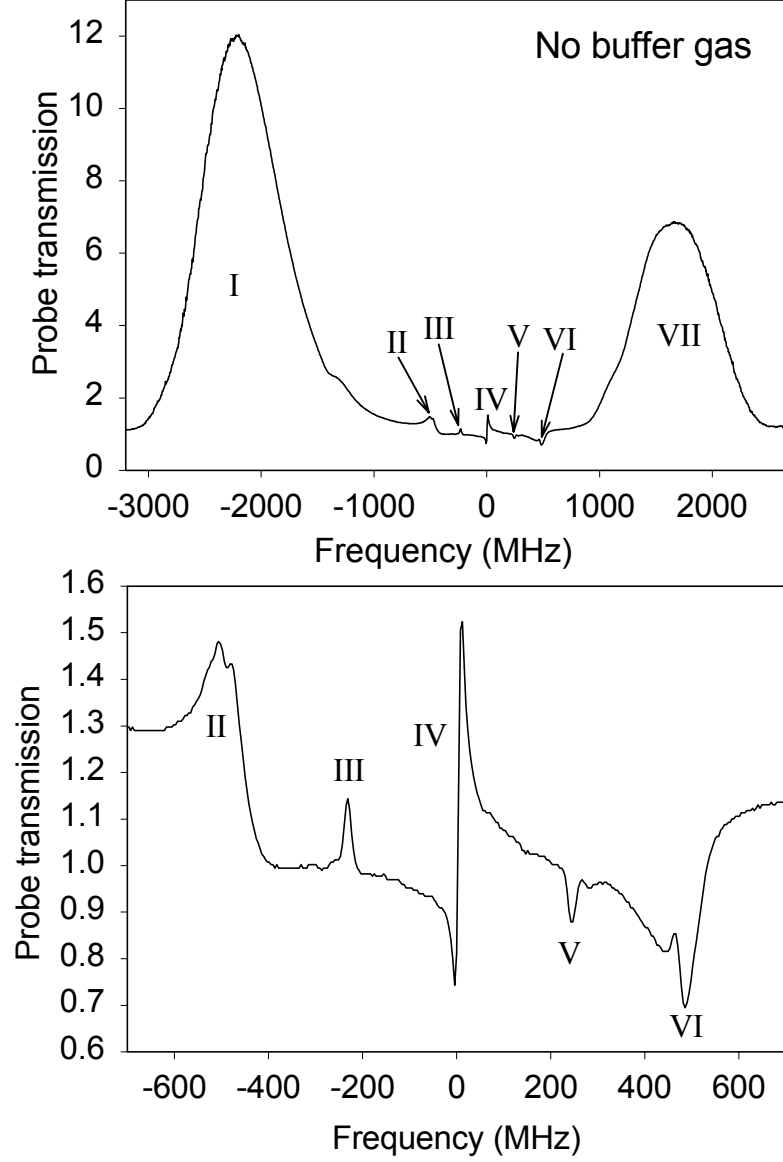


Figure 3.4: Experimental spectra with no buffer gas. The features are: (I) one-photon dressed-state gain, (II) one-photon Raman gain, (III) two-photon Raman gain, (IV) Raleigh resonance, (V) two-photon Raman absorption, (VI) one-photon Raman absorption, and (VII) multi-wave mixing..

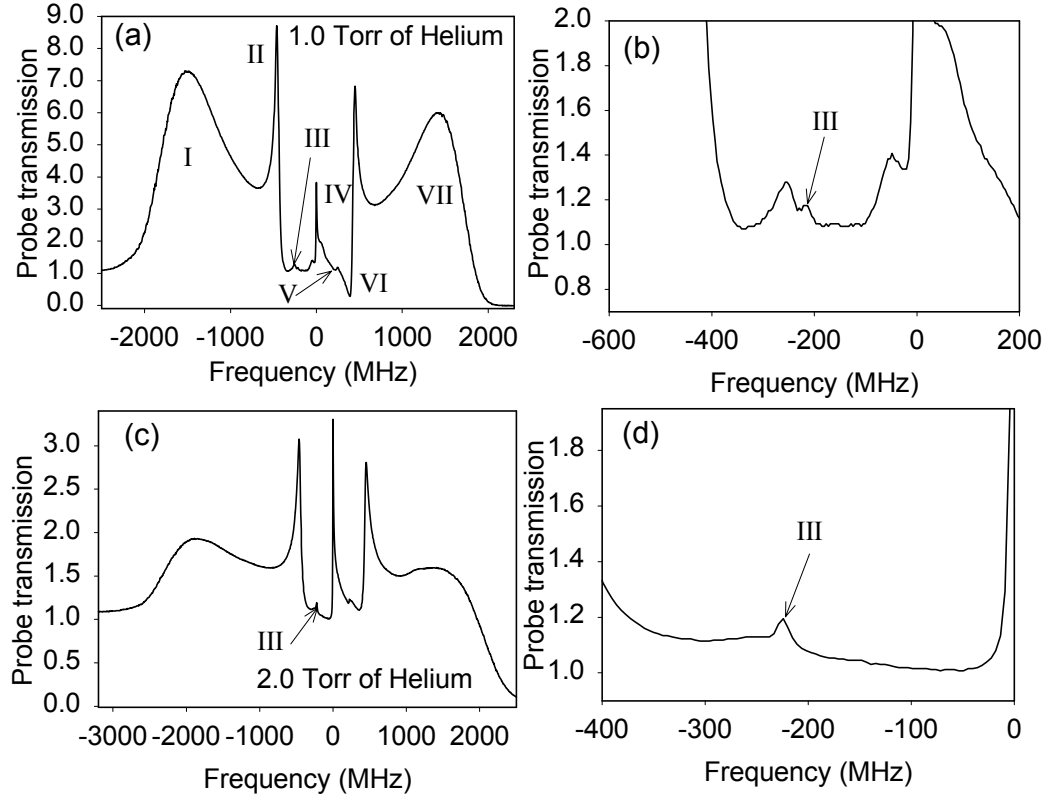


Figure 3.5: Experimental spectra with 1.0 Torr of Helium [(a) and (b)] and 2.0 Torr of Helium [(c) and (d)]. The numerals are the same as Figure 3.4

interest include the one-photon Raman gain (II), Raman absorption (V and VI), the Rayleigh resonance (IV), and the two broad features we wish to suppress (I and VII). For consistency I will continue to use these numbers to refer to these particular gain features throughout the rest of this section.

Our first choice was helium since it should have a high rate of dephasing collisions. Figure 3.5 shows spectra for helium buffer gas at pressures of 1.0 Torr [spectra (a) and (b)] and 2.0 Torr [spectra (c) and (d)]. There is a definite decrease in the gain

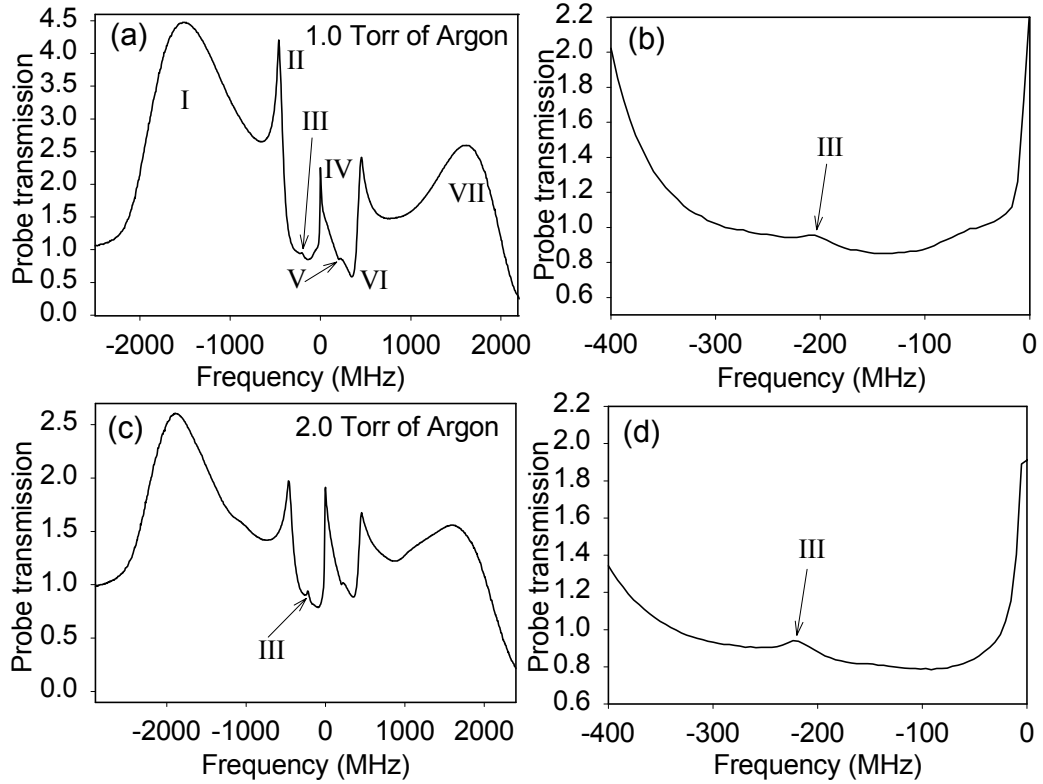


Figure 3.6: Experimental spectra showing 1.0 Torr of argon [(a) and (b)] and 2.0 Torr of argon [(c) and (d)].

for the broad features (I); for example, for 2.0 Torr of helium the broad feature has a gain of $\sim 80\%$, but the two-photon gain is now only about 10%. We have reached the lower limit for usable two-photon gain and the broad feature is still much larger. Thus, helium will not work as a buffer gas for our experimental conditions.

Argon is expected to have an effect similar to helium's [94]. The only significant difference is argon's larger size and mass. Figure 3.6 shows the results for 1.0 Torr [(a) and (b)] and 2.0 Torr [(c) and (d)] of argon. In comparison with helium, the

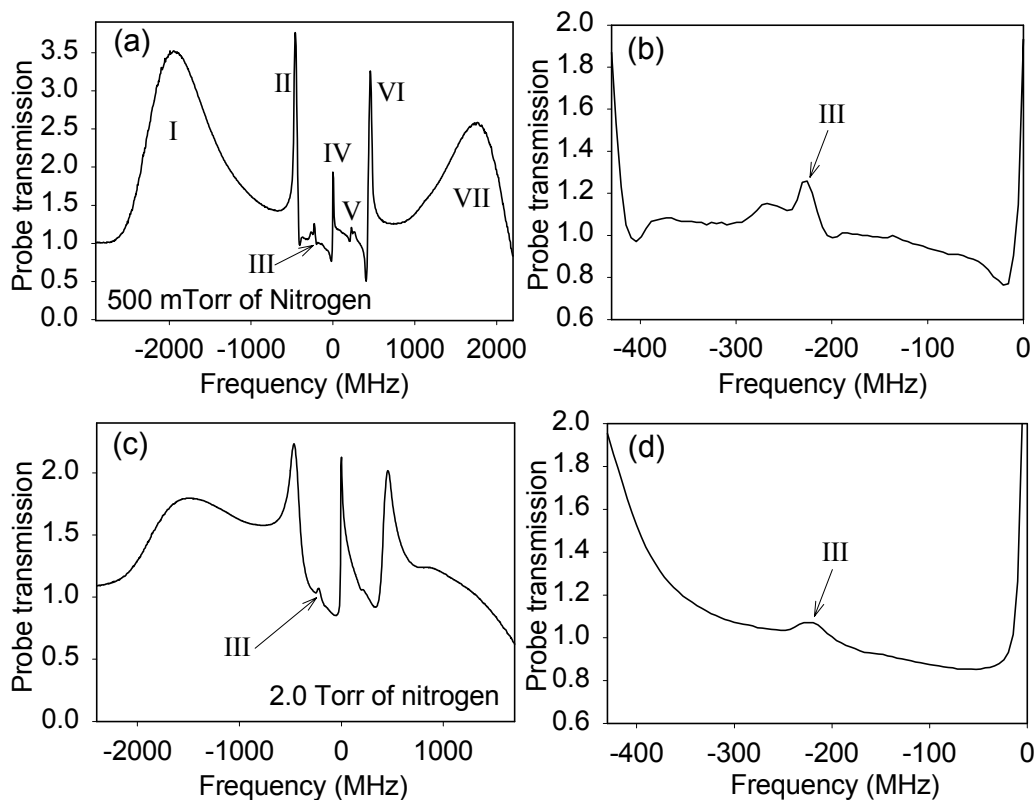


Figure 3.7: Experimental spectra showing 500 milliTorr of nitrogen [(a) and (b)] and 2.0 Torr of nitrogen [(c) and (d)].

argon is actually worse at meeting the criteria. There is a much larger impact on the one-photon and two-photon Raman features, while the broad feature is not suppressed as much as with helium.

The final gas we tested was molecular nitrogen, see Figure 3.7 for the experimental spectra. Gaseous nitrogen, unlike the noble gases, is a diatomic molecule and can undergo collisions which alter the vibrational and rotational energy of the molecule. Some of these energy spacings in nitrogen are close to transitions of potassium in-

cluding $4P_{1/2}$ to $4P_{3/2}$ transition. We would therefore expect nitrogen to have a significant impact on all the processes occurring in the potassium vapor. Figure 3.7 shows that there is a significant effect for even a small amount of nitrogen. Unfortunately, nitrogen seems to suppress and broaden all gain features, with no particular selectivity between the Raman and the dressed-state processes. Thus, nitrogen will not help with the problem of the broad features.

There are certainly many interesting questions to be asked and answered about the interplay of the various gain processes with the different buffer gases, but since none of the gases were useful in the two-photon laser, we decided to move on and consider some new experimental ideas.

3.4 Analysis of vapor cell experiments

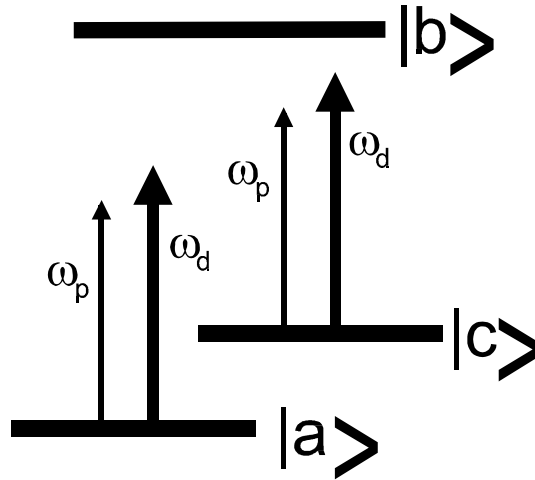


Figure 3.8: Three-level atom driven by two fields on each of the two transitions.

In thinking about what direction to take the experiments, we did some theoretical calculations which revealed several things that we had not considered previously.

First, we derived an effective two-photon Rabi frequency for the two-photon Raman type transition using time-dependent perturbation theory. Considering the three level atom from Figure 3.8, we let two fields of the form

$$\mathbf{E}^{\mathbf{p},\mathbf{d}}(\mathbf{t}) = \mathbf{E}_0^{\mathbf{p},\mathbf{d}} \mathbf{e}^{-i\omega_{\mathbf{p},\mathbf{d}}\mathbf{t}} + \text{c.c.} , \quad (3.1)$$

where $\mathbf{p}$ and $\mathbf{d}$ denote the probe and drive (or pump) fields respectively, interact with the $|a\rangle$ to $|b\rangle$ and $|c\rangle$ to $|b\rangle$ transitions. The one-photon Rabi frequencies for the interaction strengths of the fields with the transitions are

$$\Omega_{a,c}^{p,d} = \frac{2\boldsymbol{\mu}_{(\mathbf{a},\mathbf{c})\mathbf{b}} \cdot \mathbf{E}_0}{\hbar} , \quad (3.2)$$

where $\boldsymbol{\mu}$ is the dipole matrix element for the transition from $|a\rangle$ or $|c\rangle$ to $|b\rangle$. Employing these one-photon Rabi frequencies, an effective two-photon Raman Rabi frequency can be derived from the following relation

$$\Omega_{2\gamma} = \frac{\Omega_a^d \Omega_c^{p*}}{8\Delta(\Delta_{ac}/2)(\Delta + \Delta_{ac}/2)} [\Omega_c^{p*} \Omega_c^d - \Omega_a^d \Omega_a^{p*}] . \quad (3.3)$$

The detuning of the pump beam from the $|a\rangle \rightarrow |b\rangle$ transition is denoted by Δ and the probe-pump detuning is half the ground state splitting $\Delta_{ac}/2$, which is the resonance condition for the two-photon transition. This result is presented in our paper on the two-photon Raman gain in the buffer cell experiment [21]. The two-photon Raman Rabi frequency reveals that there is a destructive interference between two possible pathways as can be seen in Figure 3.9. One pathway involves two pump coherences on the $|a\rangle \rightarrow |b\rangle$ transition, one probe coherence on the $|a\rangle \rightarrow |b\rangle$ transition, and one probe coherence on the $|c\rangle \rightarrow |b\rangle$ transition. The other pathway involves one pump coherences on the $|a\rangle \rightarrow |b\rangle$ transition, one pump coherence on the $|c\rangle \rightarrow |b\rangle$ transition, and two probe coherences on the $|c\rangle \rightarrow |b\rangle$ transition. It is only the fact that the one-photon Rabi frequencies are slightly different for the $|a\rangle \rightarrow |b\rangle$

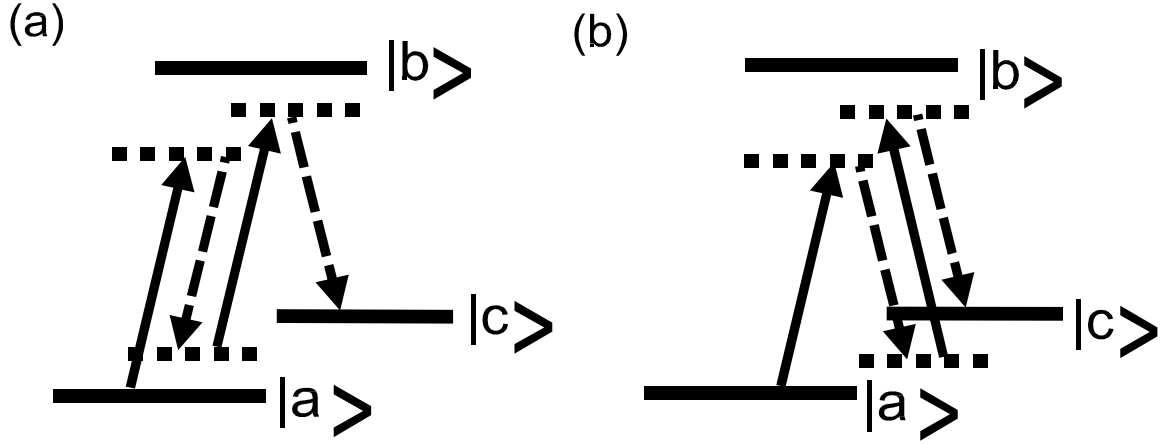


Figure 3.9: Illustration of multiple quantum pathways for two-photon Raman gain. The solid arrows are pump photons and the dashed arrow are probe photons. In the vapor cell experiment, these two pathways interfere destructively, reducing the observable two-photon gain.

transitions versus the $|c\rangle \rightarrow |b\rangle$ transitions that permits observation of any two-photon gain in the vapor cell setup. In considering the experimental results, this makes intuitive sense since the maximum one-photon Raman gain observed in the lab was $\sim 15,000\%$, the maximum two-photon gain was $\sim 30\%$ and the maximum three-photon gain was $\sim 5\%$. The two-photon gain is much closer to the three-photon gain, counter to the naive expectation that the ratio between the one- and two-photon gain and the two- and three-photon gain should be roughly the same. Thus, this configuration is handicapped from the start and may not be the ideal system for observing large two-photon gain. In looking for an alternative scheme, it is important to find something where there is either one quantum pathway or the multiple pathways interfere constructively rather than destructively.

Another difficulty with the vapor cell gain came to light when we began to think about what the cavity parameters would need to be for a two photon laser. Even

though we typically think of optical cavities as devices that build up and transmit light only for narrow resonances separated by the free spectral range, that is not true. Cavities transmit at all frequencies, but off resonance transmission is reduced by a factor on the order of the finesse as compared to the on resonance transmission. Thus if there is enough gain in the system, it will lase regardless of where the cavity resonances are tuned. If the cavity has just enough finesse (a measure of how long the photons remain in the cavity) for the two-photon feature to lase then the condition for the finesse is

$$F > \frac{g_{1-\gamma}}{g_{2-\gamma}} \quad (3.4)$$

where $g_{1-\gamma}$ is the maximum one-photon gain and $g_{2-\gamma}$ is the maximum two-photon gain. For the buffer cell this would indicate a finesse of approximately

$$F > \frac{15000}{30} = 500 \quad (3.5)$$

which is well in excess of the $F = 200$ cavity which had been designed for the vapor cell experiment. Finesses in excess of 200 would be extremely difficult to obtain since there are losses introduced by the vapor cell windows and the optics required to send the strong pump beam through the vapor cell.

3.5 New experimental concepts

In thinking about what direction to take the work, we reconsidered the idea of working with an atomic beam, but looking for Raman type processes instead of dressed-state processes, which were used for the previous two-photon laser in Oregon [20]. The problem with the barium atomic beam apparatus, in addition to the fact that the one-photon dressed state gain and two-photon gain were not spectrally resolved, was that the oven only functioned for about 6 hours at the temperatures necessary for reasonable atomic flux. After each run the entire system had to be disassembled,

cleaned and reloaded; a tedious and time consuming task. In the next chapter, I will describe how we designed a new atomic beam system, but for the moment, I shall consider the advantages of an atomic beam.

By using a collimated atomic beam, the Doppler broadening can be reduced from the ~ 800 MHz we experienced in the vapor cell to ~ 30 MHz. This is accomplished by having the laser beams orthogonal to the direction of the atom propagation, so that the Doppler broadening is only the transverse Doppler width which can be controlled by the oven aperture size. In addition, since the interaction length is set by the size of the atomic beam, there is no advantage to having the pump and probe beams colinear. These beams may be orthogonal as long as the Rabi frequencies are larger than the transverse Doppler width of the atomic beam and the pump beam is physically wider than the atomic beam. Orthogonal pump and probe beams eliminate any multi-wave mixing type processes that require a small pump-probe crossing angle. Furthermore, the feature widths can only be as large as the transverse Doppler width, so the broad features observed in the buffer cell experiment will be replaced by relatively narrow features. Proper choice of the cavity length will allow resonant enhancement for just the two-photon Raman process while avoiding these other narrow features. Finally, by using an atomic beam, the laser cavity can be built inside the vacuum chamber with no intracavity elements, making a cavity finesse on the order of several thousand or higher possible. This high finesse will overcome the loss in number density versus the buffer cell and enhance lasing of just the two-photon mechanism.

All of these ideas were already well known and were the rationale behind the two-photon dressed-state laser design in Oregon. The breakthrough was when we realized that there were other pathways for two-photon Raman process besides those we had previously considered. Specifically, new Raman type processes could occur by employing the magnetic sublevels of the hyperfine levels and different laser beam

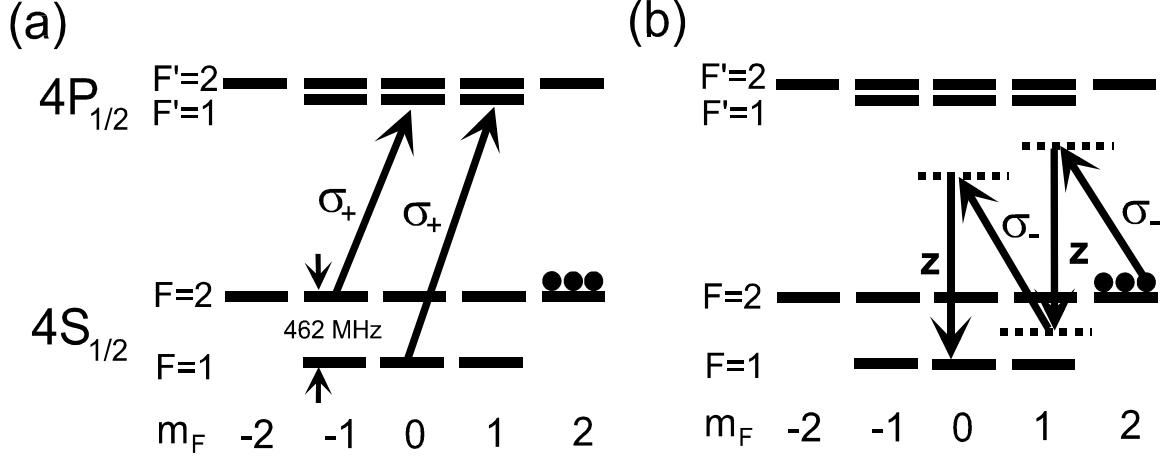


Figure 3.10: Diagrams showing the lasers used in creating the two-photon laser. In (a) the pump lasers that create the inversion via optical pumping are shown. These laser beams pump all the population into the $F=2, m_F = 2$ magnetic sublevel. From there the population is used for the two-photon transition as shown in (b). The σ_- photons come from the pump beam and the z photons come from the probe or lasing beam.

polarizations. The transition we decided to use is diagrammed in Figure 3.10. Two σ_+ polarized laser beams optically pump all of the atoms into the $F=2, m_F = 2$ magnetic sublevel. Once the atoms reach this level, they no longer interact with the optical pumping beams. If they undergo a Raman transition to another level, they are immediately pumped back to the $m_F = 2$ level for reuse. From the $F=2, m_F = 2$ level two σ_- pump photons and two linearly polarized z probe photons induce a two-photon stimulated Raman transition for the $F=1, m_F = 0$ level. This configuration minimizes the impact of quantum interferences since we sum over only a few Zeeman pathways instead of all of the Zeeman sublevels, as is the case in the vapor cell experiment. There are still quantum interferences, but they should be significantly reduced.

The actual geometry of the laser beams can be seen in Figure 3.11. The optical pumping beams with σ_+ polarization and the strong pump beam with σ_- polarization

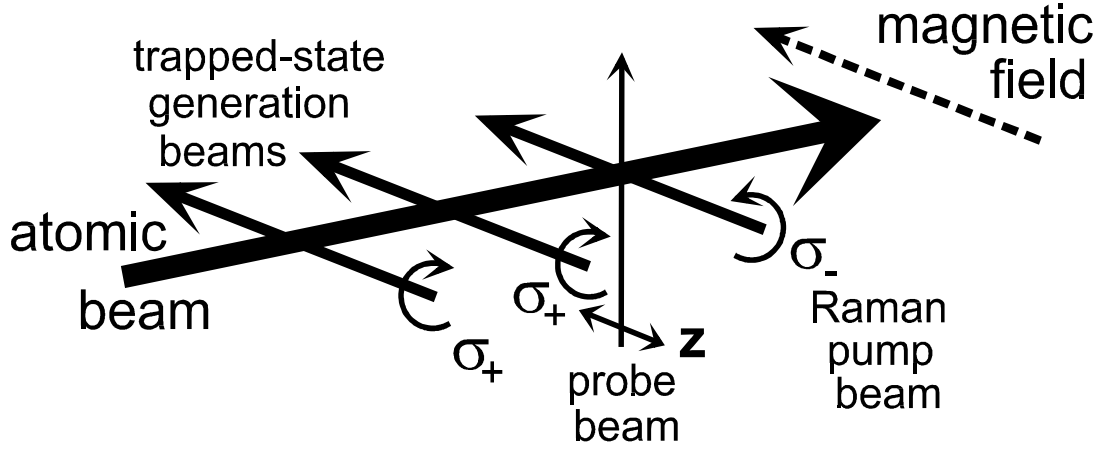


Figure 3.11: Experimental configuration consisting of a dense atomic potassium beam interacting with multiple laser beams in the presence of a weak, uniform magnetic field. The circularly polarized beams spatially overlap in the experiment; they are shown separated in the figure for clarity.

for the Raman transition are orthogonal to the atomic beam. Note that they are spatially separated in Figure 3.11 for clarity; in the actual experiment the optical pumping beams and the Raman pump overlap. The probe beam is aligned orthogonal to both the atomic beam and the pump beams. For the two-photon laser, the cavity is oriented along the line of the probe beam. The polarization axis for the laser beams is set by a weak magnetic field in the direction of the pump beams. This field is created by a set of Helmholtz coils and offsets any residual magnetic field from the earth or other laboratory equipment.

Experiments with this system occurred in two stages. The first stage consisted of gain measurements which were made with the pump beams and a probe beam. The details for this experiment are described in Chapter 4. Once the basic system was well understood and the gain features were mapped out, a high finesse laser cavity was added to achieve two-photon lasing. The two-photon laser experiment is covered in Chapter 6.

3.6 Theory for mechanisms in strongly driven potassium

Before considering the experimental results, I will present some simple theory which helps explain the experimentally observed features. First, consider the case of a weak pump beam and a weak probe beam, where we may use the bare atom basis and just add photons to make electronic transitions. There are three criteria that determine which processes are most likely to happen; in order of importance they are: 1) the fewer photons involved, the stronger the process, 2) the larger the atomic inversion the stronger the process, and 3) the closer the virtual intermediate levels are to real levels, the stronger the process.

Absorption is the strongest process since it only requires one photon. Since our system will have a relatively small Doppler width (smaller than the hyperfine splitting of the excited state) there are four resolvable absorption lines, as can be seen in Figure 3.12. To map out where these absorption lines are, let 0 be the frequency of the bluest line (*i.e.*, the $F = 1$ to $F' = 2$ transition), let blue detunings be positive, and let red detunings be negative. For potassium 39, the ground state splitting is $\Delta_g = 462$ MHz and the excited state splitting is $\Delta_e = 58$ MHz. With these definitions and parameters, the four absorption lines occur at 0, $-\Delta_e = -58$ MHz, $-\Delta_g = -462$ MHz, and $-\Delta_e - \Delta_g = -520$ MHz. In addition to the absorption lines from potassium 39, there are also absorption lines from potassium 41. The strength of those lines is reduced by $0.933/0.067 \approx 14$, which is the ratio of the relative abundance of the two isotopes in natural potassium.

The first multiple photon process is one-photon Raman gain, which requires one probe photon and one pump photon to make a transition between two hyperfine levels of the ground state or between two magnetic sublevels of a single hyperfine

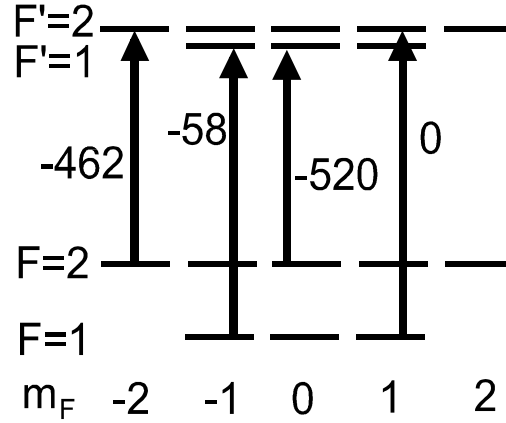


Figure 3.12: The four frequencies for absorption of the probe beam. All frequencies are in MHz and are relative to the $F=1$ to $F'=2$ line. Absorption happens for all the magnetic sublevels.

level. Since there are now pump and probe photons we let Δ_d be the detuning of the pump laser from the $F=1$ to $F'=2$ transition and Δ_p be the detuning of the probe laser from the pump laser. For one-photon Raman transitions among the sublevels of the ground states there will be three possible frequencies, $\Delta_p = 0$ (Figure 3.13 (a) & (c)), $\Delta_p = +\Delta_g$ (Figure 3.13(b)), and $\Delta_p = -\Delta_g$ (Figure 3.13(d)). Since the majority of the population will initially be in the $F=2$, $m_F = 2$ state, the processes in the Figure 3.13 (a) and (b) will dominate. However, as population is moved out of that level into the magnetic levels in the $F=1$ hyperfine levels, there exists the possibility of the creating an inversion for the processes (c) and (d) shown in the bottom of Figure 3.13.

One-photon dressed state gain is next in the hierarchy and will be the predominate competitor of two-photon Raman gain due to the wide variety of transitions that can be made and because they can be close in frequency to two-photon Raman gain. Since these processes can end up in either of the excited states, they show up in pairs separated in frequency by the excited state splitting. The predominate dressed state

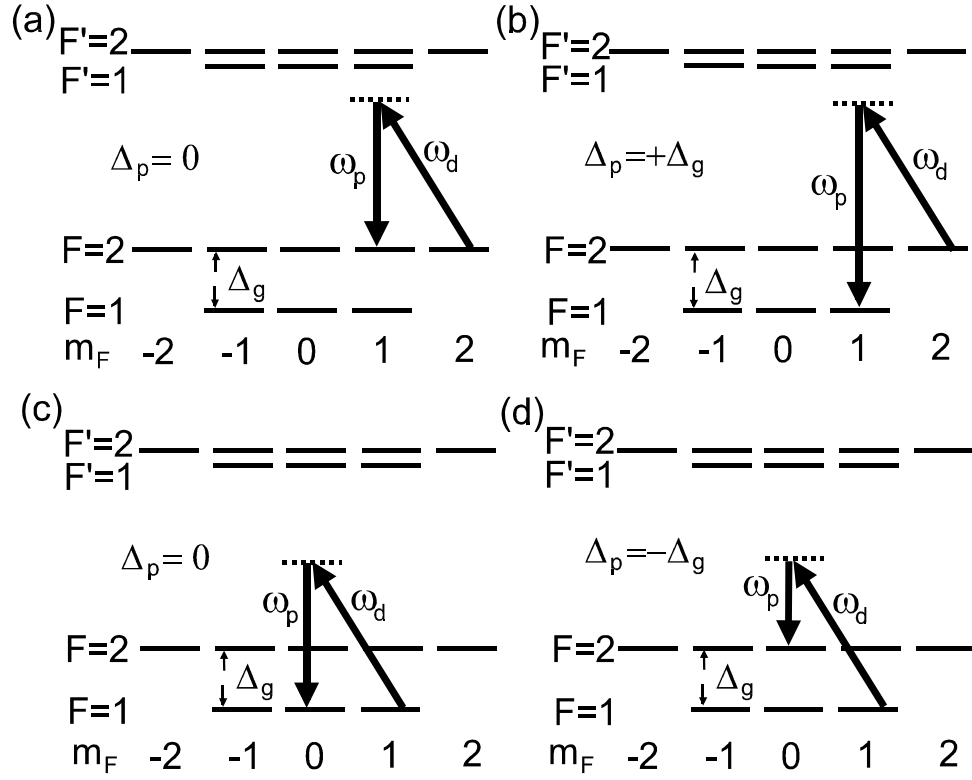


Figure 3.13: Possible pathways for one-photon Raman gain. Pathways (a) and (b) dominate when the population is pumped into the $F=2$, $m_F = 2$ sublevel.

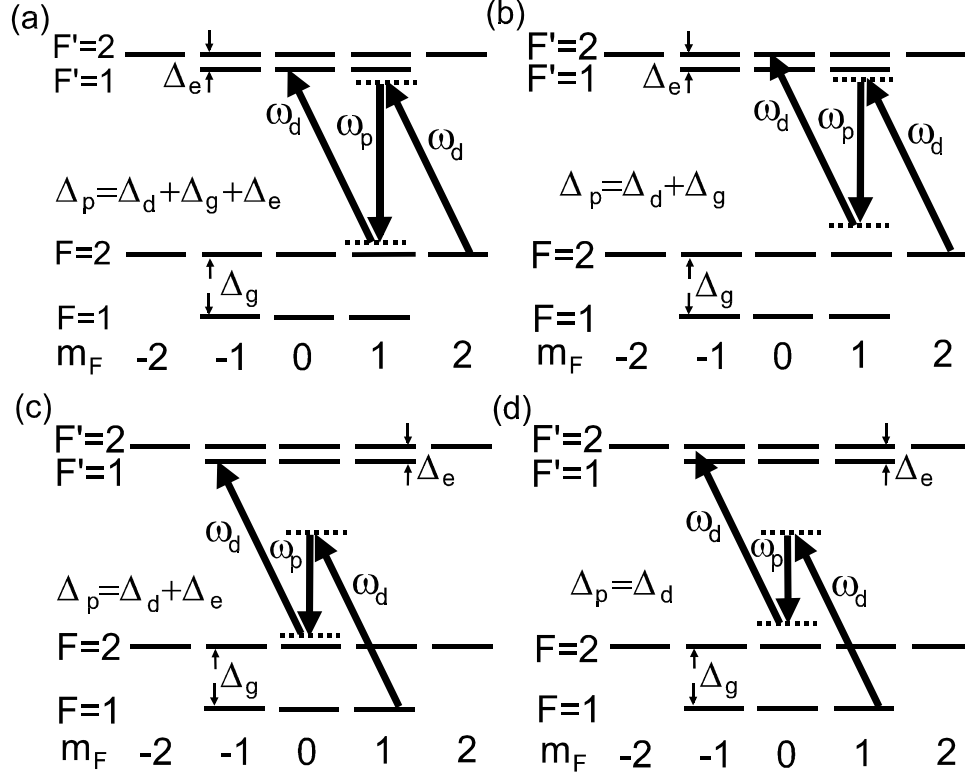


Figure 3.14: One-photon dressed-state processes occur between sublevels in the ground state and sublevels in the excited state, resulting in more possible resonant frequencies than for the one-photon Raman processes. Note that these processes typically occur in pairs [(a) & (b), and (c) & (d)] separated by the excited state splitting of 58 MHz.

transitions will be those starting at the $F=2$, $m_F = 2$ sublevel. Two possibilities are the dressed state processes with detunings of $\Delta_p = \Delta_d + \Delta_g + \Delta_e$ (Figure 3.14(a)) and $\Delta_p = \Delta_d + \Delta_g$ (Figure 3.14(b)). We also observe dressed-state pairs as the population moves throughout the sublevels; for example, the process with $\Delta_p = \Delta_d + \Delta_e$ (Figure 3.14(c)) and $\Delta_p = \Delta_d$ (Figure 3.14(d)).

After one-photon dressed-state gain comes two-photon Raman gain since four photons are involved. There will not be two-photon gain at ω_d (*i.e.* $\Delta_p = 0$) since it would be resonant with the one-photon gain which will preferentially occur since it

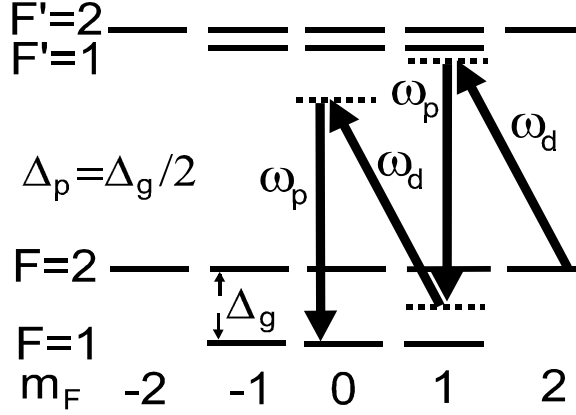


Figure 3.15: The two-photon Raman gain process that we use for the two-photon laser. In this process both probe photons are vertically polarized.

is a lower order process. The gain we are interested in occurs at $\Delta_p = \Delta_g/2$ as can be seen in Figure 3.15.

In Figure 3.15 the pump photons have circular polarization and the probe photons have vertical polarization. There is also the possibility of two-photon gain where one the probe photon has vertical polarization and the other probe photon has horizontal polarization. In our diagram, horizontal polarization is just a combination of left circular and right circular polarized light, as show in Figure 3.16. The resulting probe polarization would be 45° since one vertically polarized photon and one horizontally polarized photon are required for each stimulated emission event. In our system the gain for the 45° polarization is as large as for the vertical polarization.

Using five photons, two-photon dressed-state gain is next on the list of possible processes. Like the one-photon dressed-state gain, there are a variety of beginning and ending level possibilities for this process. Figure 3.17 just shows the one most likely to occur; the detuning is $\Delta_p = (\Delta_d + \Delta_g + \Delta_e)/2$. Again there are several two-photon dressed-state transitions that could be close in frequency to the two-photon Raman gain.

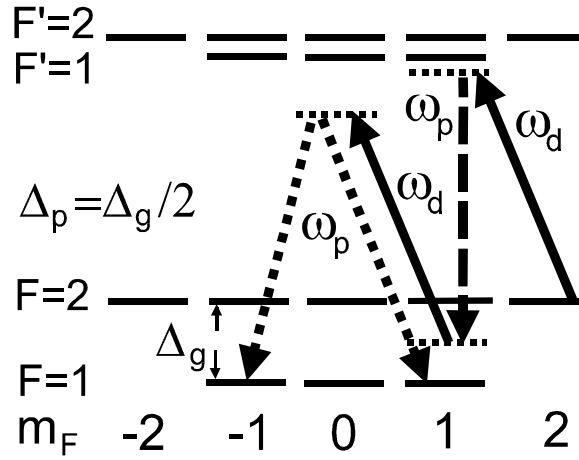


Figure 3.16: Another possible pathway for two-photon Raman gain in our experiment. Here one probe photon is vertically polarized (the dashed down arrow) and the second (the pair of dotted arrows) is horizontally polarized and thus appears as a superposition of left and right circular polarizations.

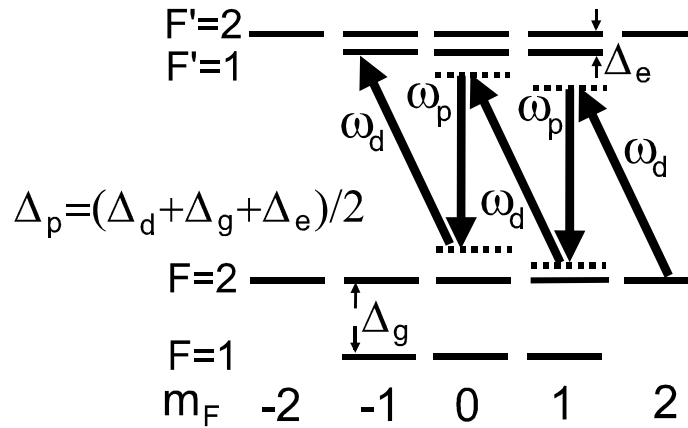


Figure 3.17: One of the many possible possible pathways for two-photon dressed state gain.

In order to determine exactly where these processes will occur, we must be able to account for the strong pump beam which is used in the actual experiment. The light shifts induced by the pump will substantially move the resonant frequencies for these processes. As a result, the location of a peak is dependent on the pump detuning AND the pump power. By performing a dressed-state analysis of the atom and pump field, the location of the energy levels can be determined, thus giving the exact resonant frequencies of the various processes.

3.6.1 Dressed-state analysis of Raman processes

This analysis was performed with the assistance of Olivier Pfister and follows the ideas first put forth by Cohen-Tannoudji [99]. The system consists of the atom interacting with (dressed by) a strong quantized pump field. The energy levels of the atom dressed by the pump are calculated and the probe beam is then added in as a perturbation to this system. This calculation is slightly more complicated because of all of the magnetic sublevels, but not unbearably so. The potassium energy levels are broken up into three set of levels (as can be seen in Figure 3.18), each set of levels being individually driven by the strong pump beam. The first set of levels (I) consists of the bare atom levels $F = 2, m_f = 2$; and $F' = 1, 2, m_f = 1$. The second set (II) consists of the bare atom levels $F = 1, 2, m_f = 1$ and $F' = 1, 2, m_f = 0$. Finally, the third set (III) consists of the bare atom levels $F = 1, 2, m_f = 0$ and $F' = 1, 2, m_f = -1$.

The Hamiltonian for this system is

$$H = H_o + V , \tag{3.6}$$

where

$$H_o = H_A + H_E , \tag{3.7}$$

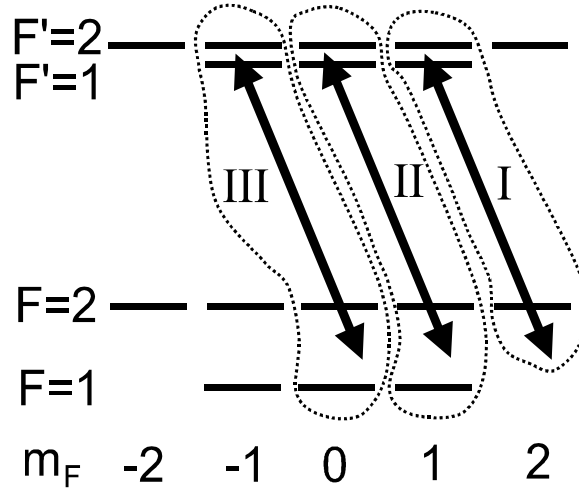


Figure 3.18: Three sets of levels are dressed by the strong pump beam. For simplicity these manifolds are referred to as I, II, and III.

H_A is the bare atom Hamiltonian, H_E is the Hamiltonian of the quantized pump field, and V is the electric dipole interaction between the pump and atom. The eigenstates of H_o are referred to as the uncoupled states and the eigenstates of H are the coupled states. We have ignored the interaction between the atom and the vacuum modes of the electromagnetic field, which give rise to spontaneous emission, since its effect on the resonance frequencies is negligible at our experimental resolution. The electric dipole interaction is given by

$$V = -\boldsymbol{\mu} \cdot \mathbf{E} , \quad (3.8)$$

where $\boldsymbol{\mu}$ is the atomic dipole and $\mathbf{E}$ is the pump electric field

$$\mathbf{E}(\mathbf{R}) = \sqrt{\frac{\hbar\omega_d}{2\epsilon_o V}} \boldsymbol{\epsilon}_d (\mathbf{a} + \mathbf{a}^+) , \quad (3.9)$$

where ω_d is the pump frequency and $\boldsymbol{\epsilon}_d$ is the polarization. If the pump field is in a coherent state $|\alpha \exp(-i\omega_d t)\rangle$ a classical field can be used in the interaction term [99].

One writes

$$\langle \alpha \exp(-i\omega_d t) | \mathbf{E}(\mathbf{R}) | \alpha \exp(-i\omega_d t) \rangle = \mathbf{E}_o e^{-i\omega_d t} + \text{c.c.} \quad (3.10)$$

where

$$\mathbf{E}_o = 2\varepsilon_d \sqrt{\frac{\hbar\omega_d}{2\varepsilon_o V}} \sqrt{\langle \mathbf{N} \rangle} . \quad (3.11)$$

$\langle N \rangle / V$ is a constant and $\mathbf{E}_o$ can be approximated by a classical field [99]. Using this classical field and the rotating wave approximation (which eliminates any quantities with optical frequencies in them) the Hamiltonian for set (I) is

$$H = H_o - \boldsymbol{\mu} \cdot \mathbf{E} = \quad (3.12)$$

$$\begin{bmatrix} E(a) + \hbar\omega_d & -\boldsymbol{\mu}_{(a) \rightarrow (c)} \cdot \mathbf{E}_o & -\boldsymbol{\mu}_{(a) \rightarrow (b)} \cdot \mathbf{E}_o \\ -\boldsymbol{\mu}_{(a) \rightarrow (c)} \cdot \mathbf{E}_o & E(c) & 0 \\ -\boldsymbol{\mu}_{(a) \rightarrow (b)} \cdot \mathbf{E}_o & 0 & E(b) \end{bmatrix}$$

where, (a) is the atomic level $F = 2, m_f = 2$, (b) is the atomic level $F' = 1, m_f = 1$, (c) is the atomic level $F' = 2, m_f = 1$, and $E()$ is the energy of that particular atomic level. To simplify, we use the Rabi frequency which is defined as

$$\Omega_{ij} = \frac{2\boldsymbol{\mu}_{ij} \cdot \mathbf{E}_o}{\hbar} . \quad (3.13)$$

If we now reference the pump detuning to the $F = 1$ to $F' = 2$ transition, as we did in the previous section, we may write the matrix as

$$H = H_o - \boldsymbol{\mu} \cdot \mathbf{E} = \quad (3.14)$$

$$\hbar \begin{bmatrix} \Delta_d + \Delta_g & -\Omega_{(a) \rightarrow (c)}/2 & -\Omega_{(a) \rightarrow (b)}/2 \\ -\Omega_{(a) \rightarrow (c)}/2 & 0 & 0 \\ -\Omega_{(a) \rightarrow (b)}/2 & 0 & -\Delta_e \end{bmatrix}$$

Using the experimental values for the pump detuning and the pump strength we diagonalize this matrix to obtain the energy levels of pump dressed atom. These new levels will be referred to as 1_I , 2_I , and 3_I , and are just linear combinations of the uncoupled atom plus laser levels.

To map out the light shifts of the two-photon resonance, we need the energy shifts for all three sets of levels. The dressed state levels consist of a periodic manifold

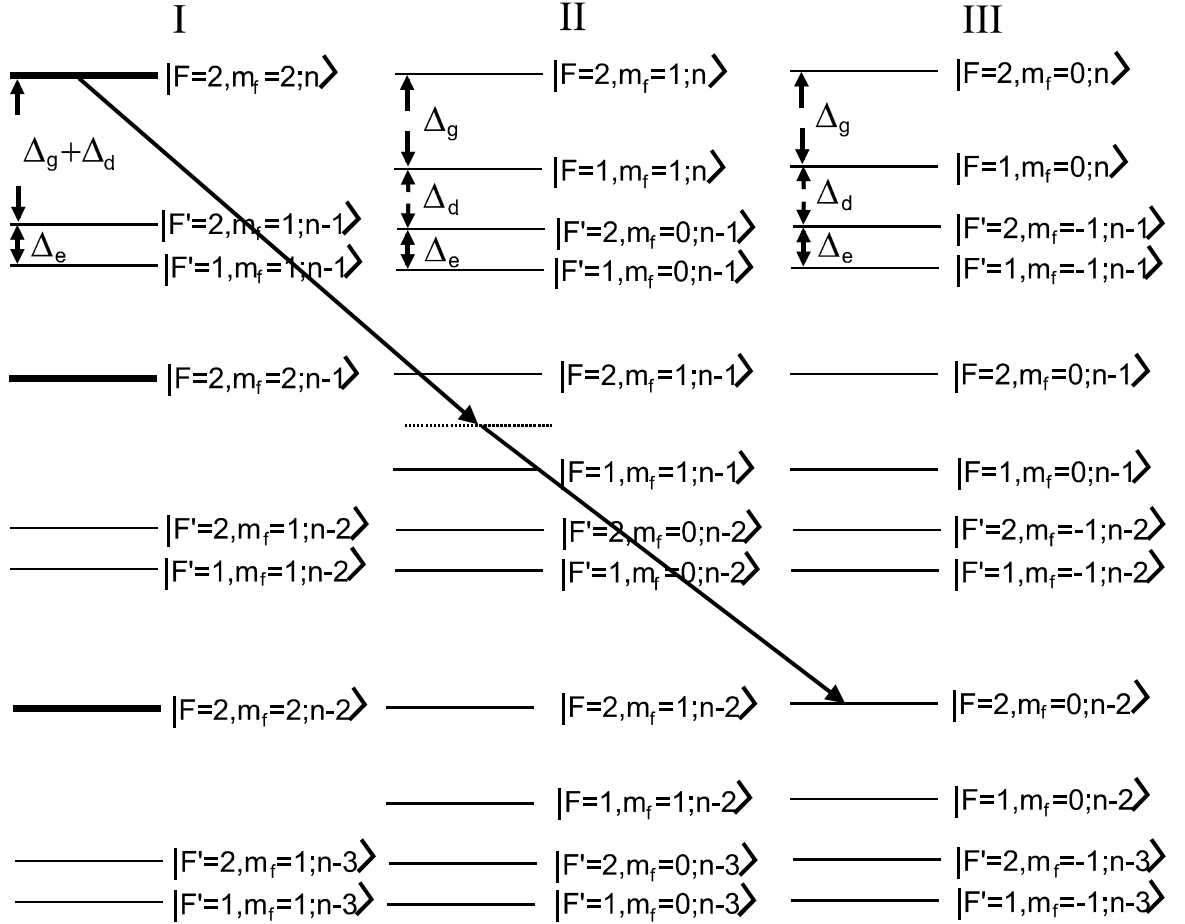


Figure 3.19: Uncoupled levels of the potassium atom and the circularly polarized pump beam. Each group of levels is nearly degenerate in energy; the label indicates the particular magnetic sublevel of potassium and the number of photons in the strong pump field. The dark lines indicate the levels with the majority of the electron population. Probe photons induce transitions between adjacent sets of levels. The transition shown is the two-photon Raman transition.

of energy levels, similar to the uncoupled levels in Figure 3.19. Moving up or down the ladder involves adding or subtracting one photon from the pump field. Probe processes appear, similarly to spontaneous emission, between groups of levels within a single set. In our situation the probe polarization prevents transitions within a set of levels; instead, the probe induces transitions between sets of levels, *i.e.*, the probe causes transitions between levels in set (I) and set (II) and transitions between (II) and (III), see Figure 3.19. (As an aside, a beam that is circularly polarized opposite to the probe can induce transitions between sets (I) and (III). This is the case of the optical pumping beams that bring population back from III to I.) Figure 3.19 displays our two-photon Raman transition from the $F=2, m_f = 2$ level to the $F=1, m_f = 0$ level, in the uncoupled dressed basis.

For simplicity I select the only three groups of interest out of three sets. Figure 3.20 shows how the various processes occur. The optical pumping moves the majority of the population into the $F=2, m_f = 2$, which is the top level in Figure 3.20. There is an inversion between that level and all sets of levels below it, thus there may be one-photon transitions to any of the levels in set (II) and two-photon transitions to any of the levels in set (III). As the pump power increases (*i.e.*, the Rabi frequency increases) there is shifting of the energy levels which thus shifts the resonances. It should also be noted that as the Rabi frequency becomes very large, there is significant mixing between the bare-atom levels. Most significantly, in set (I) the population effectively becomes spread amongst the three levels and hence gain processes can also start from the two other levels in set (I).

To determine the optimum Raman pump detuning, we plotted the differences in the energy levels (*i.e.*, the resonance frequencies) as a function of pump detuning for a specific pump power. Figure 3.21 shows a graph of the resonances for a pump Rabi frequency of 300 MHz. The various types of process are: (I) one-photon

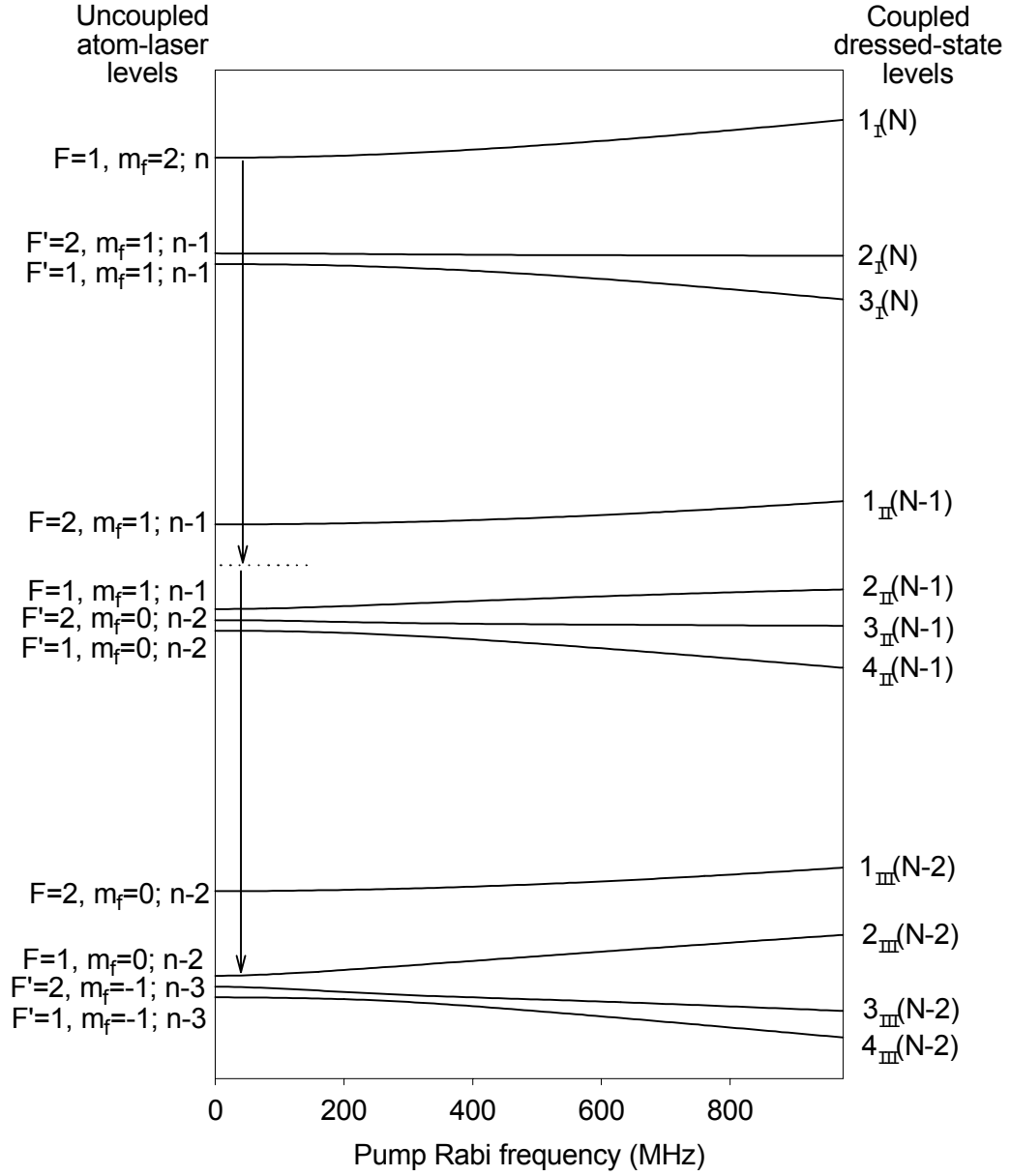


Figure 3.20: Plot showing the levels shifts as a function of the pump Rabi frequency. The labels on the left are the uncoupled levels corresponding to Figure 3.19 and the labels on the right are the dressed-state levels. For this diagram the pump detuning is +60MHz. The probe then induces transitions between the sets of levels, which become mixed for large pump Rabi frequencies.

Raman (occurs at the pump detuning), (II) one-photon Raman (+462 MHz from pump detuning), (III) one photon dressed-state, (IV) two-photon Raman, and (V) two-photon dressed state. The horizontal lines are the atomic absorption lines whose light shifts are neglected. The long dash lines are for ^{39}K and the short dash lines are for ^{41}K . We see that there are two sets of anticrossings, (A) and (B), which occur between -100 and 0 MHz. In (A), the blue one-photon resonance (II) anticrosses the two one-photon dressed-state resonances (III) and in (B), the two-photon Raman resonance (IV) anticrosses the two two-photon dressed state resonances (V). Within the circles, the mixing between the bare-atom levels is maximum and the three gain mechanisms are identical in nature to one another (*i.e.*, the distinction between “Raman” and “dressed-state” loses its meaning).

For this pump power we can see that the two-photon Raman frequency is very close to the absorption lines for negative pump detunings. When the pump detunings are positive, there are several features close by, (V) in Figure 3.21 but these are two-photon dressed state resonances. These are of little concern since they will be small (two-photon dressed state gain is a higher order process than two-photon Raman gain) and we do not care if another two-photon feature is near the two-photon Raman gain. Plots like Figure 3.21 were essential in our search for the optimum experimental parameters needed to obtain a large, well-isolated two-photon gain feature. In the next chapter I will present the spectra we observed for various pump detunings.

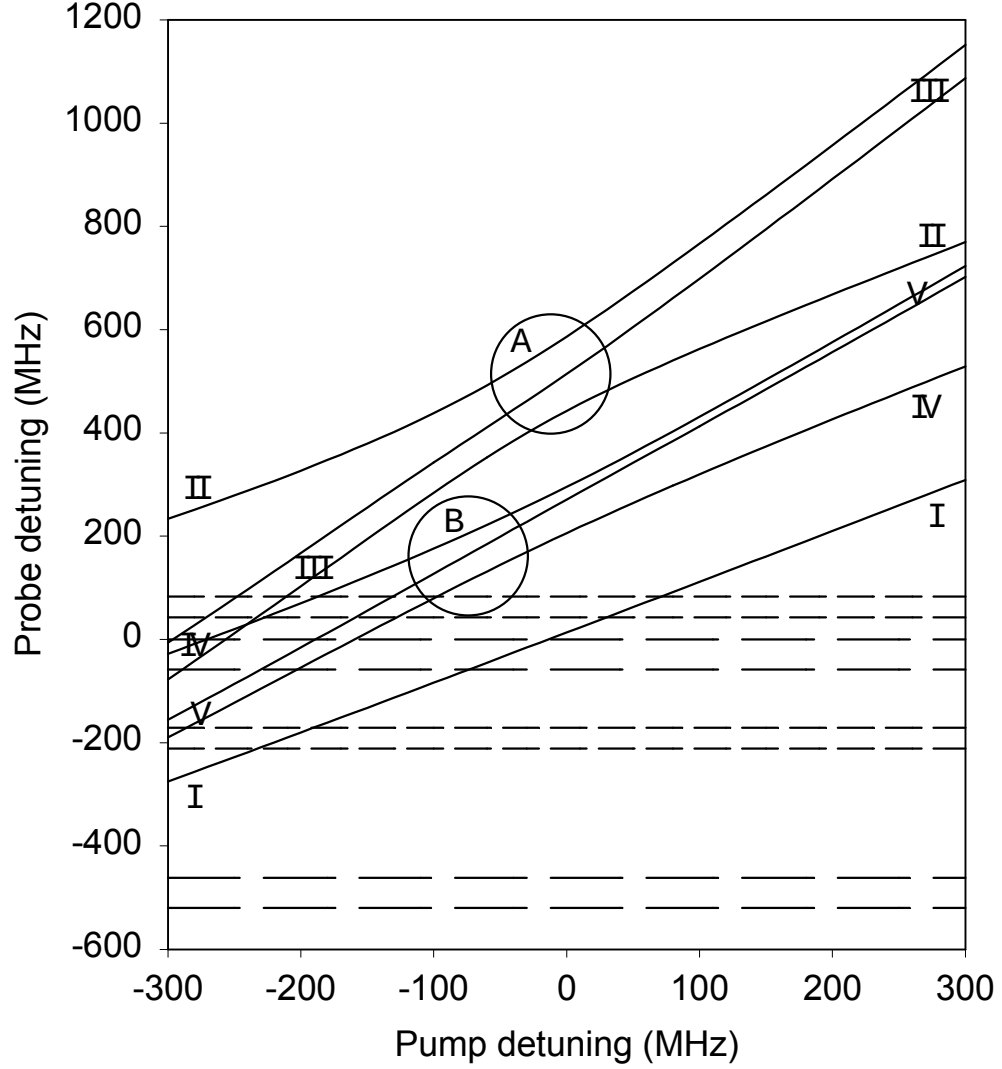


Figure 3.21: Graph showing the shifts in resonances as a function of the pump beam detuning. (I) The one-photon Raman feature which occurs at the pump detuning, (II) the blue one-photon Raman feature, (III) the two one-photon dressed state processes, (IV) the two-photon Raman process, and (V) the two two-photon dressed-state processes. The long dash horizontal lines are the ^{39}K absorption lines and the short dash horizontal lines are the ^{41}K absorption lines. Note that there are two anti-crossing (A) and (B).

Chapter 4

Two-photon Raman gain in a strongly driven potassium beam

4.1 Overview

This chapter gives the details of the pump-probe experiments performed using the potassium atomic beam system. The initial focus was on finding and maximizing the two-photon gain without any of the complications of the laser cavity. To this end we built a vacuum apparatus that used a six-way cross for the interaction region with an atomic beam apparatus attached to one side and a vacuum pump on the opposite side (see Figure 4.1). The optical pumping beams and the strong Raman pump were orthogonal to the atomic beam and the probe beam was orthogonal to the atomic beam and the pump beams. Using a subtracting detector setup to observe the probe beam allowed measurement of gains down to several parts in 10^5 .

Experimental spectra from this system are shown for a variety of pump detunings, pump powers, and probe powers. Two-photon gain was observed under a wide range of conditions where the optimum gain with the best spectral isolation occurred for pump detunings of +60 and +85 MHz. For these detunings the two-photon Raman gain as a function of probe power is measured, revealing the linear gain dependence on probe power.

Experimental apparatus overview

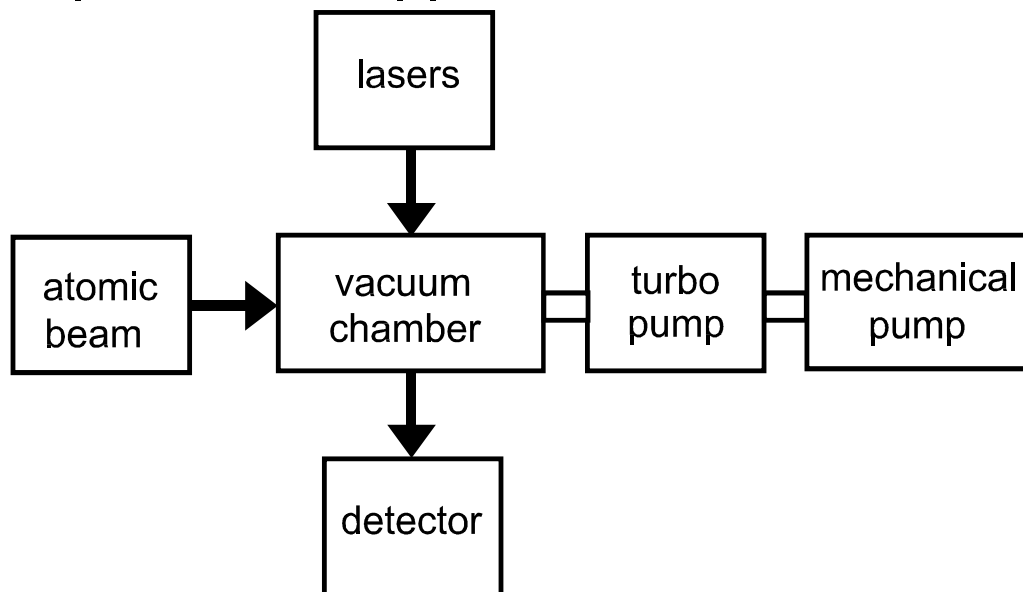


Figure 4.1: Block diagram showing the pieces of the two-photon gain measurement apparatus.

4.2 Experimental apparatus

4.2.1 Potassium atom

Since all of these experiments are dependent on the specific structure of the energy levels and transitions in potassium, I have included an overview of the relevant details. Potassium is a silvery metal and a member of the alkali family, which are favorites for scientific study due to their hydrogen-like nature since they only have one electron in their outer shells. Potassium, for example, has its outermost electron in the $4S_{1/2}$ level. We exploited the $4S_{1/2}$ to $4P_{1/2}$ transition which is referred to as the D1 line. Potassium melts at 63.5°C and boils at 758°C . For our work, the oven temperature is typically around 250°C . Potassium has two isotopes, ^{39}K and ^{41}K , which make up 93.26 % and 6.73 % of naturally occurring potassium, respectively. All of our work is done with ^{39}K , but the resonances in ^{41}K are close by and are present in the spectra.

The electronic energy level structure for potassium is illustrated in Figure 4.2. The ground state splitting in ^{39}K is 461.8 MHz and 254 MHz for ^{41}K [89]. The excited state splitting for the D1 line in ^{39}K is 57.7 MHz and in ^{41}K is 40 MHz. Also shown are the D2 lines with their 4 hyperfine levels.

Unlike previous work where only linear polarizations were used [21], this work also employs σ^+ and σ^- polarizations. Thus, it was necessary to calculate the transition strengths for all the $\Delta m_f = \pm 1$ transitions as well as the $\Delta m_f = 0$ transitions. The procedure used to determine the matrix elements is outlined in Hope Concannon's thesis, section 5.5.1 [57]. Figure 4.3 shows all the transition strengths for the D1 line in ^{39}K . These transition strengths are used for calculation the Rabi frequencies used in all of the theory work and for assessing the interference (constructive or destructive) between various quantum pathways.

In addition to the electronic energy level structure, we also need to know what

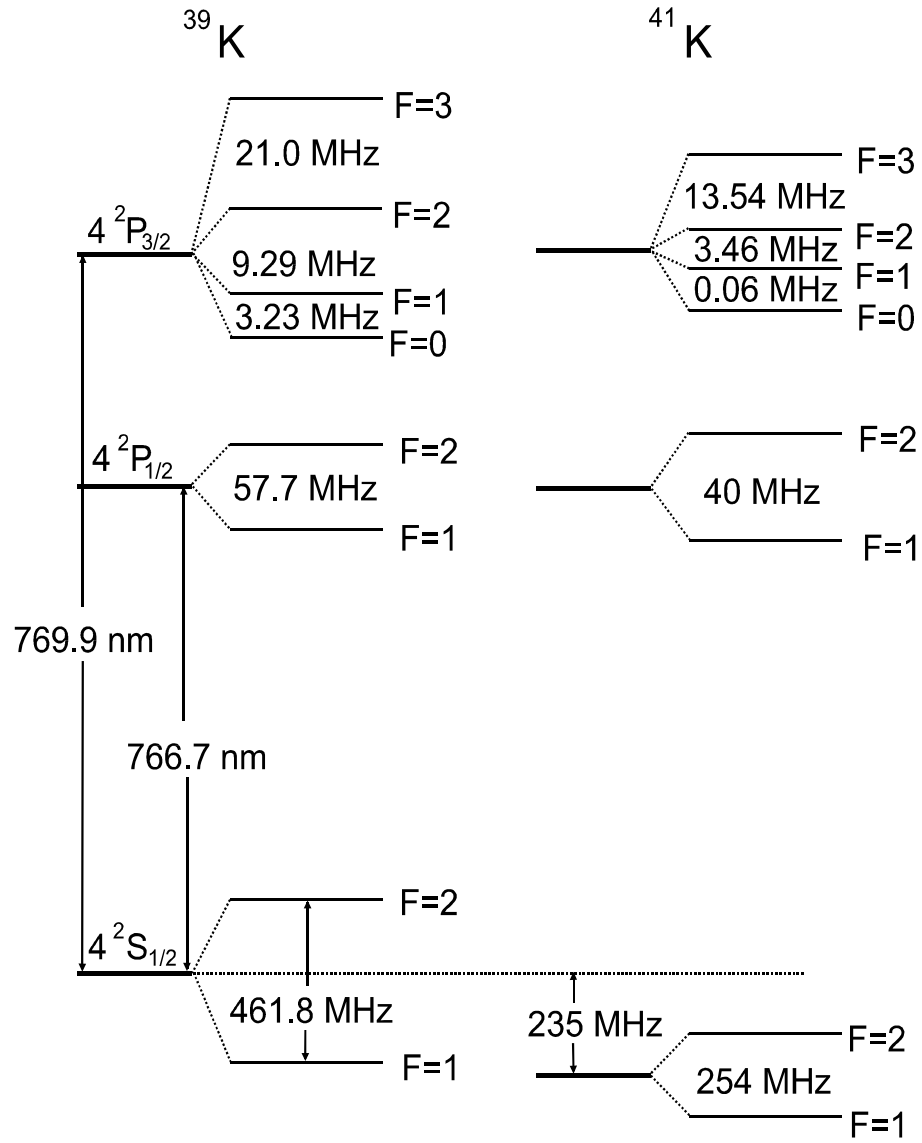


Figure 4.2: Level structure for the D1 and D2 lines of potassium 39 (the most abundant isotope at 93.3%) and potassium 41 (with 6.7% abundance).

D1 Line for Potassium 39

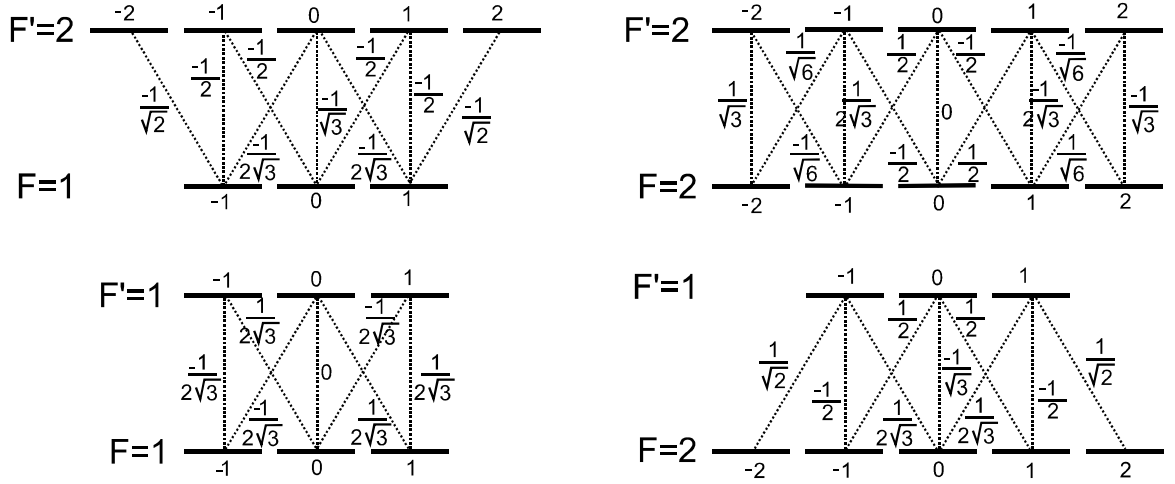


Figure 4.3: Matrix elements for the D1 line of potassium 39, normalized to $4.23ea_o$. F=1 & 2 are the ground states (the $4S_{1/2}$ level) and F'=1 & 2 are the excited states (the $4P_{1/2}$ level). Vertical lines are for transitions with π polarized light, while lines with negative slope are for light with σ^- polarization and lines with positive slope are for σ^+ polarization.

the properties of potassium are in the oven. Generally, atomic beams are formed by heating the material in an oven and allowing atoms to exit via a small hole. To know how many atoms will exit the hole in a given amount of time we need to know the number density inside the oven. This is a nontrivial function of the temperature; however, there are excellent empirical curves for the relationship between pressure (and hence number density) and temperature [100]. For potassium, the vapor pressure as a function of the temperature is given by

$$\log_{10}(P(T)) = 13.83642 - 4857.902/T - 0.00034940 * T - 2.21542 * \log_{10}(T) , \quad (4.1)$$

which may be converted to number density using the equation

$$n(T) = \frac{9.64 \times 10^{18} * P(T)}{t} , \quad (4.2)$$

where $n(T)$ is the number density in atoms per cubic centimeter and the temperature is in Kelvin. A graph of $n(T)$ versus temperature is shown in Figure 4.4. With this information, we are now prepared to design and build a high-flux atomic beam apparatus for potassium.

4.2.2 Atomic Beam

Background

In order to generate enough two-photon gain to build a viable laser, we need an atomic beam with significant flux. However, we also need a narrow divergence since the whole point of using an atomic beam is to limit the effects of Doppler-broadening by working in an essentially Doppler-free configuration. Atomic beam apparatuses come in many varieties, but most work on some variation of the same principle. An oven is used to heat the source of atoms which then pass out of a hole in the oven and are collimated by a second aperture some distance from the oven [101]. Naturally, this is a terribly inefficient way to build a source. In our particular system, for example,

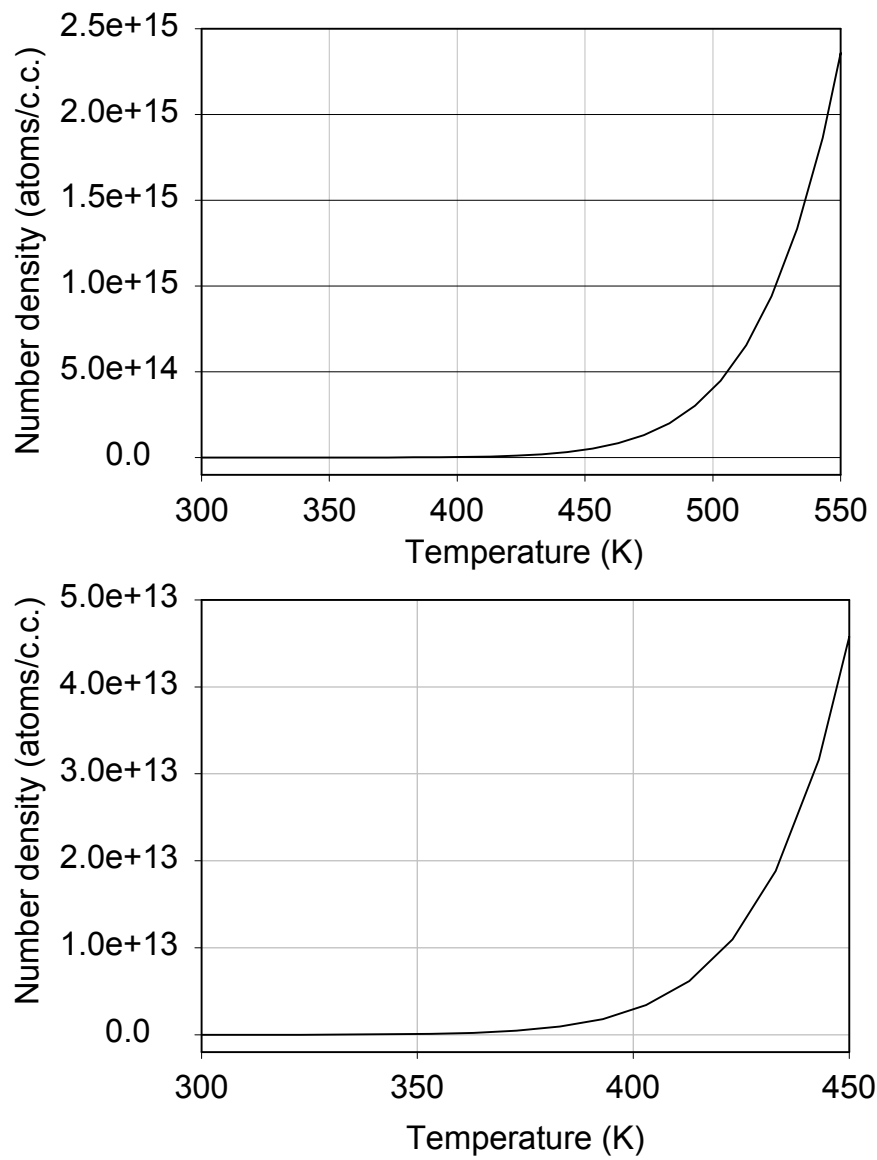


Figure 4.4: Graphs of the number density for potassium (in atoms/cm³) as a function of the temperature, for two different temperature ranges.

only 1 in 100,000 atoms leaving the oven actually makes it into the interaction region. Finding a balance between high flux and low divergence has been the challenge in creating the atomic beam for this experiment.

For the first design, we started by using a oven attached to a long tube that was lined with a fine metal screen, based on the ideas of Drullinger and coworkers [102]. This design overcomes the problem of atoms lost out of the beam by recycling the atoms back to the oven via the metal wick that lines the tube. Our system was similar to this, but we were unable to reach the fluxes we needed. Any time we got large fluxes by running the oven very hot, the beam broadened significantly. Evolving this design a step further along the line laid out by Swenumson [103], we built an atomic beam apparatus that employed several apertures down the length of the tube in an effort to block the atoms that were not in the beam. The walls of the tube were again lined with a metal screen that recycled the unused atoms back to the oven. Unfortunately this design did not work at the high oven temperatures required for the large atomic fluxes we needed.

After much experimentation, we discovered that if we cooled the tube (originally by pouring liquid nitrogen on it) the flux increased substantially while maintaining a narrow divergence angle. Apparently, all of the previous designs had fallen prey to the effect that atoms that hit the screen were bouncing instead of sticking. These atoms bounce back into the direction of the original beam, thereby scattering the straight atoms out of the beam and thus significantly reducing the flux. Chilling the walls caused these atoms to stick on the first bounce thus increasing the number of atoms that made it straight down the tube. Without cooling the tube wall, the high flux, large divergence beam had occurred when the tube was so hot that atoms either did not stick to the wall at all or were emitted by the tube wall, causing the entire tube to act as an atom source.

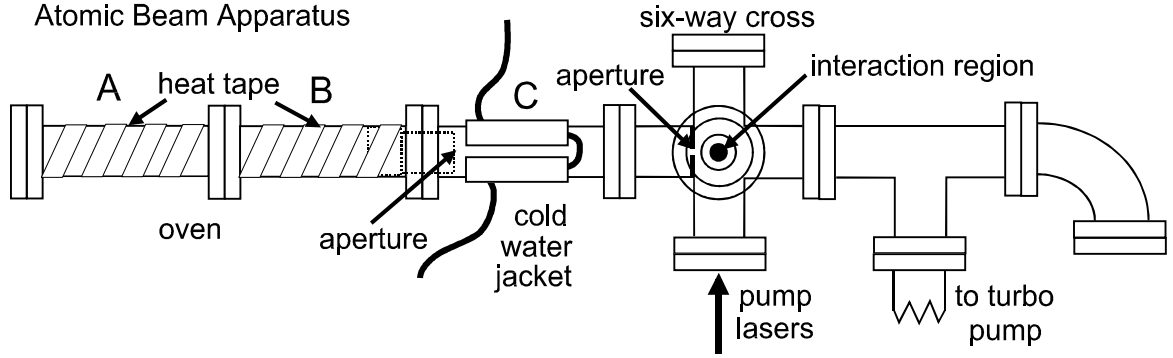


Figure 4.5: Schematic of atomic beam apparatus. Potassium is contained in the left two nipples which are heated. The two apertures collimate the atomic beam which then passes through the interaction region inside the six-way cross.

The next challenge was to design a source that could overcome these difficulties. The need for a cold tube and the desire to recycle the atoms seemed to be mutually exclusive until we came up with the idea of a two-stage atomic beam apparatus. Our idea was to have a cold tube while the beam was running, which would soon become coated with atoms. Once the source was used up, the tube would be heated while the oven was kept cool thus returning the unused atoms to the oven. To facilitate the process the tube would be mounted vertically with the oven below it so that the liquid atoms would naturally drain back into the oven when the tube was heated. We built an oven based on this design, using thermo-electric coolers to chill the cold tube. Unfortunately there was insufficient cooling and potassium escaped from the cold tube and formed a cloud in the vacuum chamber. In the end, we decided that even though the design was sound and the system could be made to work, it should be scrapped and we would build something simpler.

The system we built still employs bulk recycling of the atoms (versus the continuously recirculating ovens of Drullinger and Swenumson), but now the recycling is done by hand. In the new design, three $2\frac{3}{4}$ " nipples (*i.e.*, sections of vacuum tubing

with flanges on each end) are connected together. The first and the last are empty and the oven nozzle is press fit into the middle nipple, as illustrated in Figure 4.5. Several grams of potassium are added to the left nipple, which is heated along with the middle nipple to form the oven. The right nipple has a water cooling coil that reduces the temperature about 10° C below room temperature. A second aperture after the rightmost nipple collimates the atomic beam. As the beam is run, the potassium is moved from the oven on the left to the cold nipple on the right. Once the oven is emptied, the whole system is cooled and backfilled with argon. The nipples are then taken apart under argon and the left and right nipples are swapped, thus recycling the potassium back into the oven. Starting with 5 grams of potassium in the oven, the system may be run at maximum flux for approximately 20 hours before it is necessary to recycle the potassium. There is some loss associated with the recycling process so after four or five swaps, the potassium in the oven is essentially gone and must be replenished.

Theory for the atomic beam system

In order to determine if the oven is operating in a reasonable fashion, we developed a simple theory that gives a predicted optical absorption spectrum based on the physical aspects of the oven, the properties of potassium and the temperature of the oven. Linear absorption can be calculated using the expression

$$\frac{I_{out}}{I_{in}} = \exp(-\alpha z) \quad (4.3)$$

where I_{in} and I_{out} are the beam intensities before and after the beam respectively, α is the absorption coefficient, and z is the effective path length of the atomic beam. In order to calculate z , we use a model of the experimental apparatus as shown in Figure 4.6. In this diagram r_1 is the radius of the oven aperture, r_2 is the radius of the collimating aperture, d is the distance between the two apertures, and l is the

distance from the second aperture to the interaction region. Using this geometry, we can calculate the radius of the unclipped beam (*i.e.*, the region where the flux is uniform) and the radius of the entire beam (*i.e.*, including the region where the first aperture becomes eclipsed by the second aperture). These distances are

$$r_{inside} = r_2 + \frac{l}{d}(r_2 - r_1) , \quad (4.4)$$

$$r_{outside} = r_2 + \frac{l}{d}(r_2 + r_1) . \quad (4.5)$$

Provided that $r_1 \approx r_2$ and $d \gg r_2$, the atomic flux falls off linearly as the radius goes from r_{inside} to $r_{outside}$ (see the bottom of Figure 4.6). Thus, the effective path length z is given by

$$z = r_{inside} + r_{outside} = 2r_2(1 + \frac{l}{d}) , \quad (4.6)$$

which is then multiplied by the absorption coefficient for the whole Doppler distribution of the atomic beam.

The absorption coefficient is calculated using

$$\alpha = \frac{4}{3} \sqrt{\frac{\pi}{2}} \frac{n_p \omega_o}{\hbar c \sigma} \boldsymbol{\mu}^2 \mathbf{I}_o(\mathbf{T}, \boldsymbol{\omega}) \quad (4.7)$$

where n_p is the number density of the potassium in the interaction region, ω_o is the center frequency of the absorption line, σ is proportional to the Doppler width of the atomic beam, and $\boldsymbol{\mu}$ is the dipole matrix element. The lineshapes of the atomic beam and the natural lineshape of the resonance are combined in $I_o(T, \omega)$. The integral is given by

$$I_o(T, \omega) = \int_{-\infty}^{\infty} \frac{\exp \left[-\frac{1}{2} ((\omega - \omega_o)/\sigma(T) + (\gamma/\sigma(T))y)^2 \right]}{y^2 + 1} dy \quad (4.8)$$

where γ is the lifetime of the excited state. This integral has no closed form solution; to obtain a numerical result Mathematica was used [104]. Since there are four absorption lines, we calculate a lineshape for each line and then combine them to form

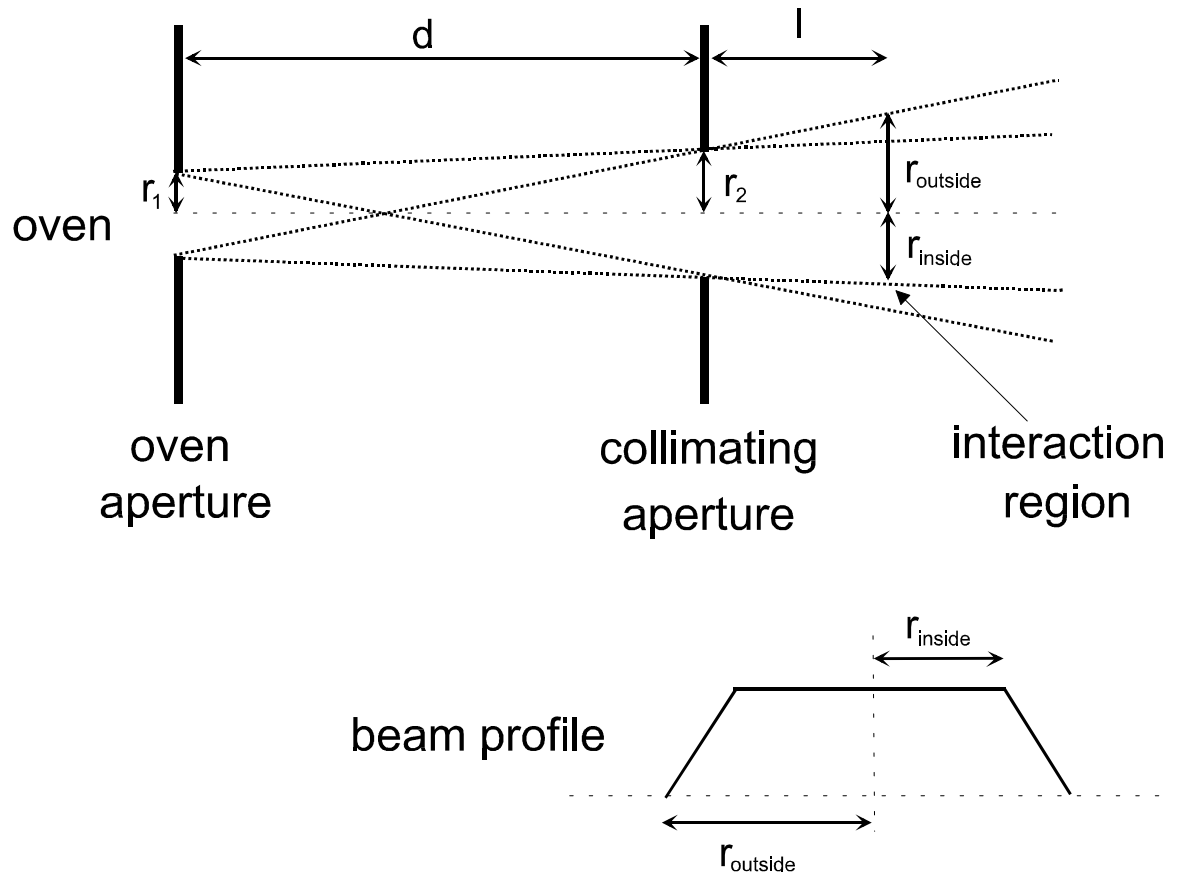


Figure 4.6: Schematic for the atomic beam. Two apertures collimate the atoms leaving the oven. In the interaction region there is a ring on the periphery of the beam where the flux falls off linearly as shown in the beam profile.

a theoretical spectrum. A theoretical result is presented in Figure 4.7 and shows excellent agreement with the experimental spectrum. These theoretical calculations provide a check of the temperature inside the oven. Since it would be difficult to place a thermocouple inside the oven we do not know this temperature exactly. The theoretical fits give a measure of the temperatures and are typically 20° C lower than the externally measured temperatures.

We can also calculate the atomic number density in the interaction region using this model of the experimental apparatus. The relationship between the two number densities is given by

$$n_{interaction}[t] = n_{oven}[t] \frac{r_1^2}{4(d+l)^2} \quad (4.9)$$

where $n_{oven}[t]$ is calculated using Equation 4.2, $r_1 = 1.5$ mm, $d = 15$ cm, and $l = 2$ cm. This gives a proportionality relationship of

$$n_{interaction}[t] = 1.95 \times 10^{-5} \cdot n_{oven}[t] . \quad (4.10)$$

Vacuum equipment used in atomic beam system

In order for the atomic beam to propagate >20 cm without being scattered via collisions with the background gas, the pressure in the chamber must be lower than 1×10^{-6} Torr. To obtain that kind of pressure in a large chamber, experimentalists use a diffusion pump backed by a rough pump. Recent technological process permits the use of turbo molecular pumps instead of the unwieldy and dirty diffusion pumps. For our experiment it is vital that the chamber is very clean, since any outside contaminates will coat the mirrors in the laser cavity and degrade the cavity finesse. Turbo pumps consist of small turbines spinning at molecular speeds; the Varian Turbo 70LP used in our lab runs at top speed of 75,000 revolutions per minute. A mechanical pump backs the turbo pump and provides a base pressure of ~ 1 mTorr. The turbo pump then takes the pressure down to $\sim 1 \times 10^{-7}$ Torr with all

of the cavity apparatus in the chamber. Since the turbo pump has a relatively low pumping speed (70 L/s compared to >200 L/s for a diffusion pump), it is crucial that everything in the chamber is very clean so as to minimize outgassing. Certain materials such as aluminum and many plastics and rubbers can not be used in the vacuum chamber because of their high outgassing rates. In addition, the turbo pump must be vibrationally isolated from the chamber and the optics table, because it will excite mechanical resonances in the two-photon cavity and the diode laser cavity. There are commercial vibration isolators available, but we just use a flexible bellows between the turbo pump and the chamber and mount the turbo pump on a separate table.

Potassium atomic beam operation

The potassium atomic beam operation commences by turning on the water chiller for the cold water jacket and then slowly heating up the oven. Currently, the oven is heated using two thermal tapes connected to two Variacs. The first thermal tape is wrapped around the nozzle end of the oven and is heated to $> 250^{\circ}$ C so that the nozzle temperature is always at least 20° C above the oven temperature. The second heat tape is wrapped around the back part of the oven and sets the actual temperature of the potassium vapor (since the vapor pressure is set by the coolest point in the oven). Once in thermal equilibrium, the oven temperature will remain within a degree without any adjustments to the Variac settings. This provides a stable atomic beam that will run at the same flux for many hours with minimal intervention.

A typical spectrum of the potassium absorption lines is shown in Figure 4.7. The dotted line is the experimental spectrum and the solid line is the theoretical fit. Here we have $\sim 50\%$ maximum absorption which corresponds to a number density in the

interaction region of 1.2×10^{10} atoms/cm³ and an oven temperature of $\sim 238^\circ$ K, as confirmed by the theory. The Doppler width (*i.e.*, the full width at half maximum) is ~ 30 MHz.

For the two-photon laser we need the maximum possible atomic flux. More atoms translates directly to more gain. Unfortunately, one cannot just increase the oven temperature forever and see increasing flux. Once the oven temperature reaches the point that the mean free path of the atoms is on the order of the oven aperture size, collisions in the beam start to broaden the Doppler width. In addition, running the beam at extremely high fluxes tends to create a background cloud in the interaction region. A spectrum showing this background cloud is given in Figure 4.8. A background cloud will result in absorption that will offset the gain and destroy any lasing. The maximum temperature we have been able to run at is $\sim 300^\circ$ C, which gives a line center absorption on the three strong lines of $\sim 95\%$.

As a final note, we still have one oven design that we would like to try. Developed by Hau *et al.* and called a “candlestick” oven [105], it has a chamber with a reservoir of material at the bottom and a heated tube on one side that wicks material up and ejects it out a hole into the chamber. A small collimating aperture opposite this hole provides the collimation. The authors have shown very high fluxes with reasonably small Doppler widths and excellent recirculating characteristics using this design.

4.2.3 Generation of laser beams

Argon ion laser

Generation of the fixed laser frequencies used in this experiment is done with a titanium doped sapphire crystal laser (Ti:Sa laser) which is pumped by an argon ion laser (Ar⁺ laser). The Ar⁺ laser is an Innova 310 built by Coherent and capable of generating powers in excess of 10 W and operating in the wavelength range 458

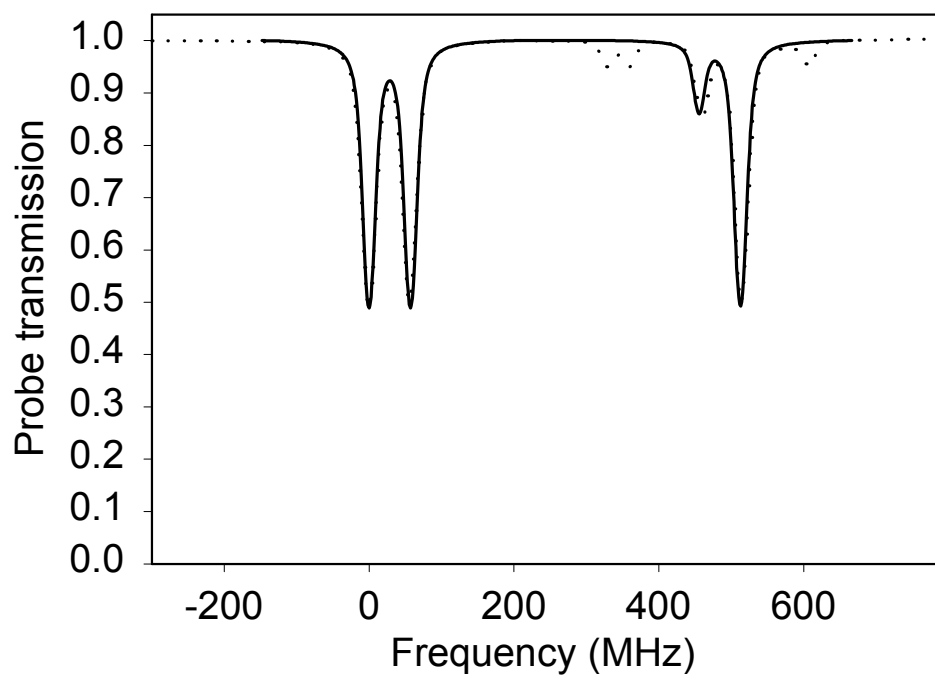


Figure 4.7: Absorption spectrum for potassium 39 with an oven temperature of 250°C. The dotted line is the experimental spectrum and the solid line is the fit from the theory. Note that the lines from potassium 41 are clearly visible in the experimental spectrum.

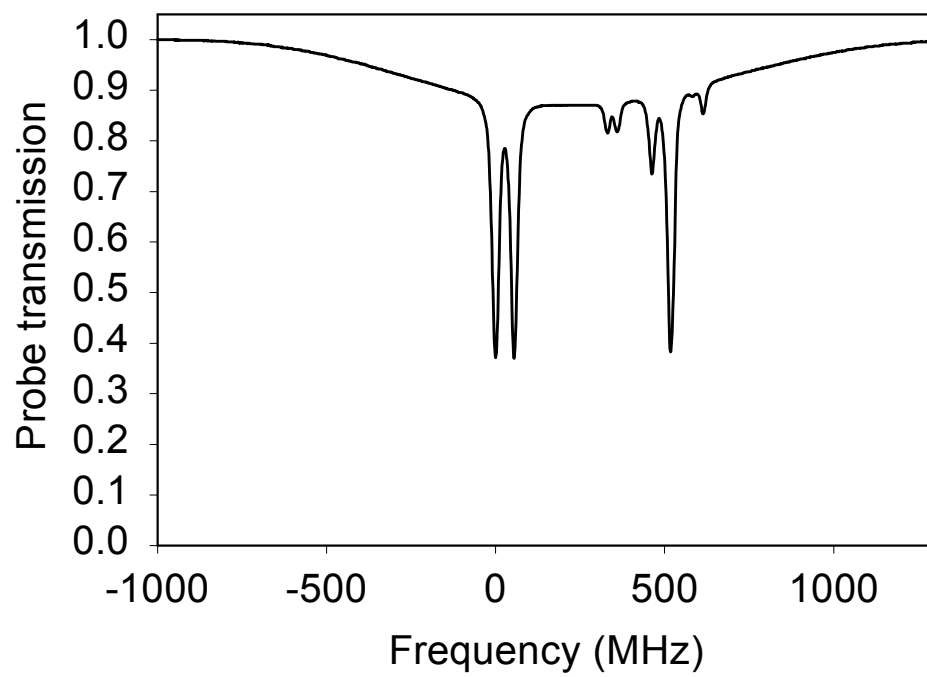


Figure 4.8: Sample spectrum showing a background cloud in the interaction region, which gives a Doppler width based on the temperature of the cloud, ≈ 1 GHz in this picture

nm to 514 nm [106]. The gain mechanism is excitation of the argon ions via the electric current which is passed through the plasma. For normal operation the current through the plasma is 55 A, the voltage drop is 227 V, and the output power in the TEM_{0,0} mode is ~ 6.7 W. Since the laser consumes roughly 12.5 kW of power, water cooling is required to remove the excess heat, with a typical water flow of 2.5-3.0 gallons/minute. Note that the overall efficiency of this laser is a rather dismal $\sim 1/2\%$.

Titanium:Sapphire laser

The Ti:Sa laser provides the high powered beam needed to drive the Raman transitions in potassium and the weak optical pumping beams needed to maintain the population inversion. The Ar⁺ laser is the pump source for the Ti:Sa laser, which consists of a four-mirror ring cavity with the crystal in the middle. The Ti:Sa laser may be tuned from 680 nm to 1025 nm. For the set of cavity optics we use, it is tunable from 700 nm to 825 nm [107]. A Faraday isolator in the cavity insures that the laser only lases in the forward direction since ring cavities are inherently bi-directional. Several elements are placed in the cavity to select and stabilize the frequency including a birefringent filter with a free spectral range of 400GHz, a thin etalon with a free spectral range of 225 GHz, and a thick etalon with a free spectral range of 10 GHz. Stabilization is provided by a piezo-electrically driven mirror and a rotating Brewster plate. Rotating the Brewster plate also allows the cavity to be scanned up to 30 GHz by varying the effective cavity length. The error signal used to correct the cavity length is generated by a thermally stabilized reference cavity keeping the overall drift of the laser frequency down to a few MHz per minute. The instantaneous linewidth is less than 1 MHz with the servos engaged as compared to ~ 40 MHz when the laser is operating in free run mode.

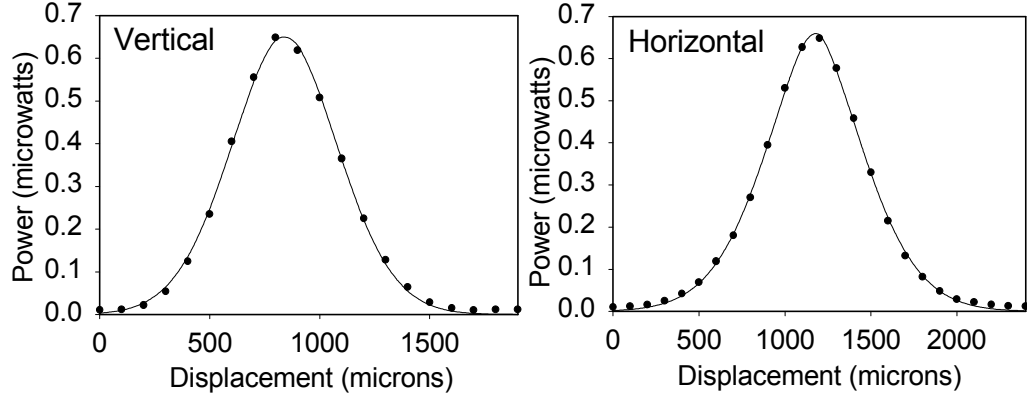


Figure 4.9: Beam profile for the Ti:Sapphire laser beam. Measurements were made using a 10 micron pinhole at 100 micron intervals. The solid line is a Gaussian fit to the data points.

With the 6.7 W of pump power from Ar^+ laser, the Ti:Sa can generate 1.0 W of power for an overall efficiency of $\sim 15\%$. Day to day operation involves adjusting the output coupler and the etalon alignment to restore maximum power. Every few months the mirrors and the sapphire crystal surfaces must be cleaned and realigned. This involves removing all of the intracavity elements and reinstalling them one by one, realigning the cavity for maximum power output at each step. The external reference cavity also requires periodic realignment when the laser can no longer scan the full 30 GHz. On rare occasions potentiometers in the feedback electronics must be adjusted to optimize the laser operation.

For these experiments it is important that the laser beam sizes match the size of interaction region. Thus, we can insure that the pump beam is larger than the probe beam and that the probe beam is much smaller than the atomic beam. Also, we need to know the laser beam sizes so that we can calculate the beam intensity in the

interaction region. To measure the beam sizes we moved a pinhole horizontally and vertically across the beam and measured the power after the pinhole at each step. Figure 4.9 shows the vertical and horizontal profiles for the Ti:Sa laser just after the laser. The points are measured powers and the line is a Gaussian fit. For the Ti:Sa we have a good qualitative fit, indicating a nearly TEM_{0,0} mode coming out of the laser. The waist size is 478 μm in the vertical direction and 531 μm in the horizontal direction. For the optical pumping beams, an expanding telescope expands the beam size in the interaction region to ~ 4.5 mm in both directions. The Raman pump beam is expanded using anamorphic prisms and cylindrical lens resulting in a beam size in the interaction region of 250 μm in the vertical direction and 3 mm in the horizontal direction. These dimensions match the size of interaction region, which is determined by the atomic beam size (~ 2.2 mm) and the probe laser or cavity waist (60 – 100 μm).

Diode lasers

The probe beam for the gain measurements and the injected pulses for the two-photon laser experiments come from a diode laser. The diode laser used in this experiment is an SDL-5401 which produces ~ 50 mW continuous-wave single-mode at a center frequencies ranging of 770 nm. Modifications to the diode were performed by Environmental Optical Sensors, Inc. (EOSI) and include anti-reflection coating the front surface of the diode laser. The EOSI system employs a Littman/Metcalf design [108] where the diode beam intersects a diffraction grating at grazing incidence and the first order diffracted beam hits a corner cube which retroreflects the beam back to the grating and then into the diode. The zeroth-order reflection provides the output beam. In this configuration, the grating remains fixed with respect to the diode and scanning is accomplished by moving the corner cube in an arc. The

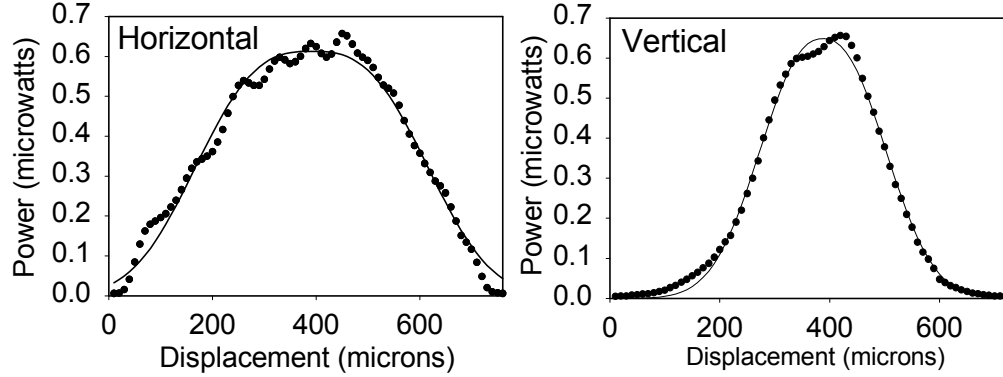


Figure 4.10: Beam profile for the EOSI diode laser. Data was taken with a 10 micron pinhole at 10 micron steps. The solid line is a Gaussian fit to the data points.

output power is 15 mW for a current of 75 mA. The laser can be scanned from 762 nm to 776 nm which covers both the D1 and D2 lines in potassium. Continuous scanning is accomplished using a PZT with a maximum scan width of ~ 150 GHz and a maximum scan rate of 1kHz. The corner cube eliminates the need to adjust the vertical alignment and a very stable cavity keeps longer term drift to a minimum. Current modulation up to 1 MHz is permitted via the control electronics and a bias-T directly connected to the diode permits modulation up to 1 GHz. The linewidth of the diode in the EOSI package is <1 MHz, compared with 10 MHz for the bare diode.

A significant shortcoming of diode lasers in general is poor beam quality. The configuration of the diode results in a beam that is elliptical (roughly 6:1), astigmatic, and multimode (*i.e.* not just $TEM_{0,0}$ as is the case for the Ti:Sa). As illustrated in Figure 4.10, even after sending the beam through a set of anamorphic prisms, the waist sizes are still significantly different. We measured the sizes to be $411 \mu\text{m}$

in the horizontal direction and $218\text{ }\mu\text{m}$ in the vertical direction. In addition, the beam is not very Gaussian; note the ripples in the power measurements compared to the Gaussian fit. Poor beam quality makes it more difficult to mode match the beam into the cavity for the two-photon laser experiments and to know the exact beam parameters for theoretical comparison in the gain experiment. For the gain experiments, the diode beam is focused to a spot size of $100\text{ }\mu\text{m}$ in the horizontal direction and $80\text{ }\mu\text{m}$ in the vertical direction in the interaction region.

Acousto-optic modulators

Acousto-optic modulators (AOM's) can shift the frequency of an incident laser beam anywhere from several MHz up to 2 GHz. Experimentally, they permit one laser to generate many optical frequencies separated by radio frequencies, making possible devices such as our two-photon laser, MOT's (magneto-optical traps) and optical scanners. In general, AOM's operate by propagating an RF acoustic wave through an optically transparent crystal and Bragg scattering a laser beam off of the grating formed by the RF wave. As can be seen in Figure 4.11, proper selection of the input angle permits the AOM to shift the light either to lower or higher frequencies. Conservation of momentum sets the angle between the straight through zero order beam and the first (second, third, etc) order shifted beam. This may also be thought of as the laser beam either absorbing or emitting phonons thus shifting the energy (and hence frequency) of the laser photons. The crystal is tellurium dioxide (TeO_2) for the AOM's used in our lab, and is driven by a lithium niobate piezoelectric transducer. One AOM used in the experiment is a Model 1206C from Isomet Corp. [109] and is tunable from 85 to 135 MHz. A second AOM used in the experiment is a Model ATM-2301A2 from IntraAction Corp. [110], which is tunable from 190 to 270 MHz. Both of these AOM's have an amplitude control as well as a frequency control

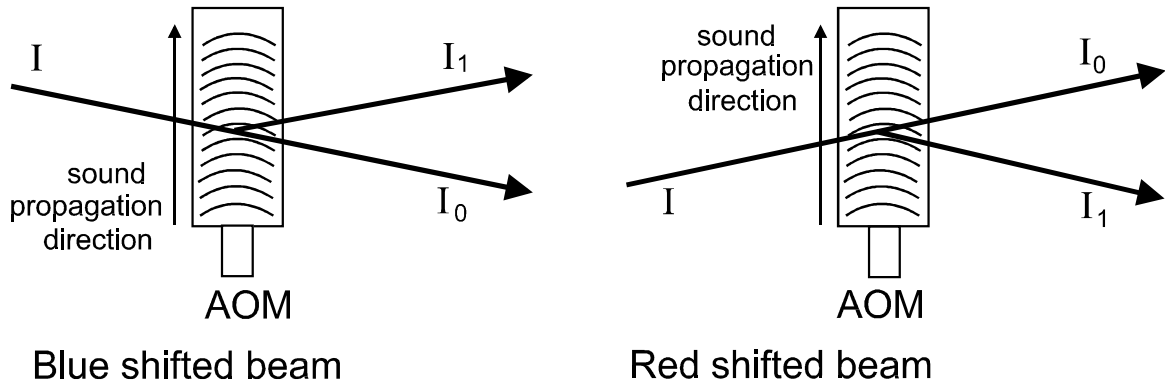


Figure 4.11: Beam shifts due to the AOM. I is the input beam, I_0 is the zero order output beam and I_1 is the first order or frequency shifted beam.

making it possible to control the amount of power shifted into the diffracted beam.

It is also possible to use AOM's in a double pass configuration as illustrated in Figure 4.12. By placing the AOM at the focus of a 1:1 telescope and retroreflecting the laser beam, the output laser beam does not move as the AOM is frequency tuned. A polarizing beam splitter and a quarter waveplate permit the separation of the input laser beam and the returning, shifted output laser beam. Thus, the entire system may be aligned for a particular AOM frequency and later the frequency can be changed without affecting the system alignment.

Saturation spectroscopy for absolute frequency reference

For any sort of spectroscopy where it is necessary to know the frequency of a feature, it is important to have an absolute frequency reference with respect to which measurements may be made. Atomic lines are very useful for this purpose as they are absolute, the natural linewidths are a few MHz (for potassium), and often these lines are very close to the frequencies that one is interested in measuring. Unfortunately, the natural linewidth is washed out by Doppler broadening, ranging from

AOM Double Pass Setup

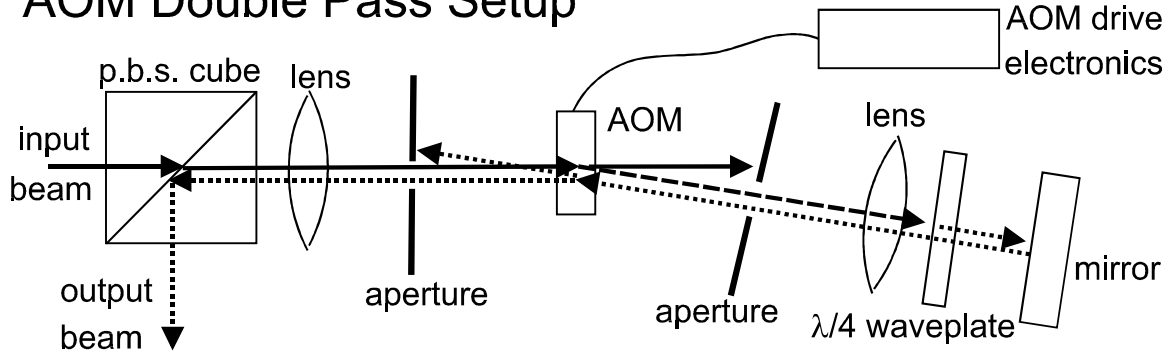


Figure 4.12: Double pass setup for an AOM. Double passing permits larger frequency shifts and has the important advantage that the frequency of the AOM may be tuned without moving the output beam. The P. B. S. is a polarizing beam splitter.

40 MHz in an atomic beam to ~ 2 GHz in a heated vapor. Several decades ago, a technique was developed called saturated absorption spectroscopy (informally called a Lamb dip) that permits observation of resonances at the natural linewidth even in a Doppler-broadened medium.

Saturation spectroscopy consists of a strong pump laser (strong compared to the saturation intensity) and a weak counterpropagating probe laser (well below the saturation intensity) with the same frequency. Experimentally, this can be achieved by passing a strong pump beam through the atomic vapor, attenuating the beam, and then retroreflecting the weak probe beam back through the vapor (see Figure 4.13). Using a polarizing beam splitter and a quarter-wave plate permits separation of the retroreflected probe beam from the strong pump beam. The probe beam is then sent to a photodiode detector. By scanning the laser across the resonance, one observes a broad dip with narrow spikes at the atomic transition frequencies, as illustrated in Figure 4.14. The probe beam is experiencing the saturation of the transitions induced by the pump beam. However since the beams are propagating in opposite directions

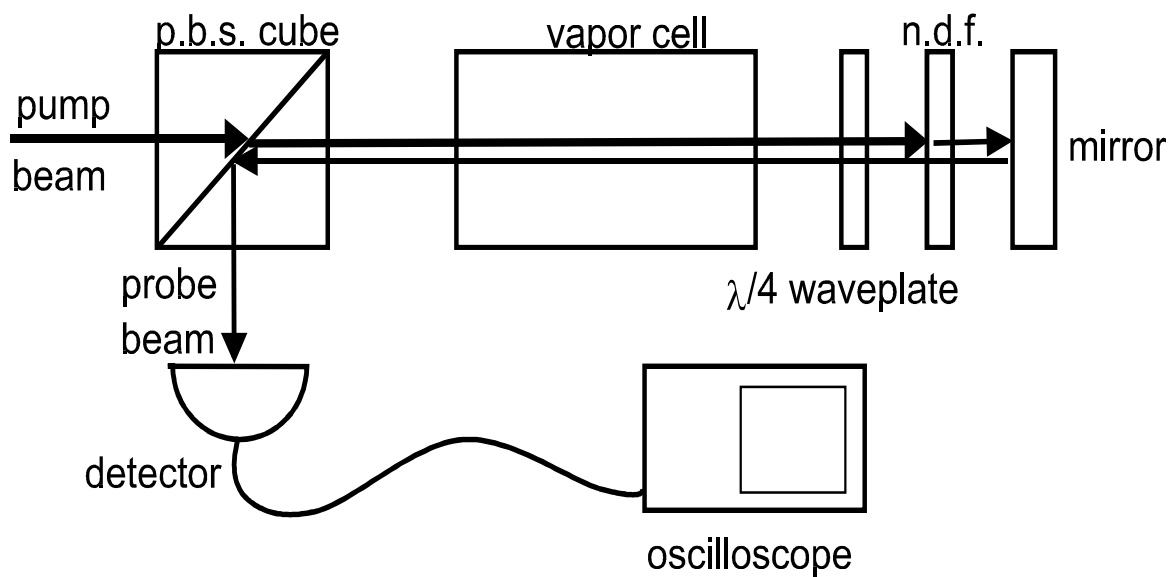


Figure 4.13: Saturated absorption spectroscopy setup used for establishing absolute frequencies using the atomic frequencies of the gas contained in the vapor cell. The p.b.s. is a polarizing beam splitter and the n.d.f. is a neutral density filter for attenuating the laser beam.

atoms, the probe beam is Doppler shifted in the opposite frequency direction from the pump laser. This means that for only the zero velocity (along the direction of the light propagation) atoms does the probe beam experience the saturation due to the pump beam. One minor complication is that for any pair of transitions there exists a velocity group such that when the lasers are tuned half way between the transitions, one laser beam is Doppler shifted to the red level and the other beam is Doppler shifted to the blue level, creating what is called a crossover resonance. In the sample saturated absorption spectrum from the D1 line of K, where there are two ground states 462 MHz apart and two excited state resonances 58 MHz apart, there are two groups of peaks separated by 462 MHz with two peaks within each group separated by 58 MHz. Midway between each pair of natural resonances is a crossover resonance and there is an additional set of crossover resonances in the middle of the scan. This middle set of crossover resonances actually experiences enhanced absorption and thus the resonances appear as dips on the background absorption instead of peaks. The best signals are obtained for a vapor temperature that gives a peak small single absorption between 50% and 75%.

Beat Note Generation

Our experiments generally involve the Ti:Sa and one or more diode lasers and it is extremely helpful to know the relative frequency of the different lasers. One method is to use multiple saturation spectroscopy setups and measure each laser frequency relative to the absolute frequency of the peaks in the saturated absorption spectrum. However with this method any nonlinearities in the laser scans must be accounted for and a reference vapor cell is needed for the saturation spectroscopy setups. A simpler and more accurate method is to use a beat note.

A beat note involves mixing a small amount of power from the two lasers on

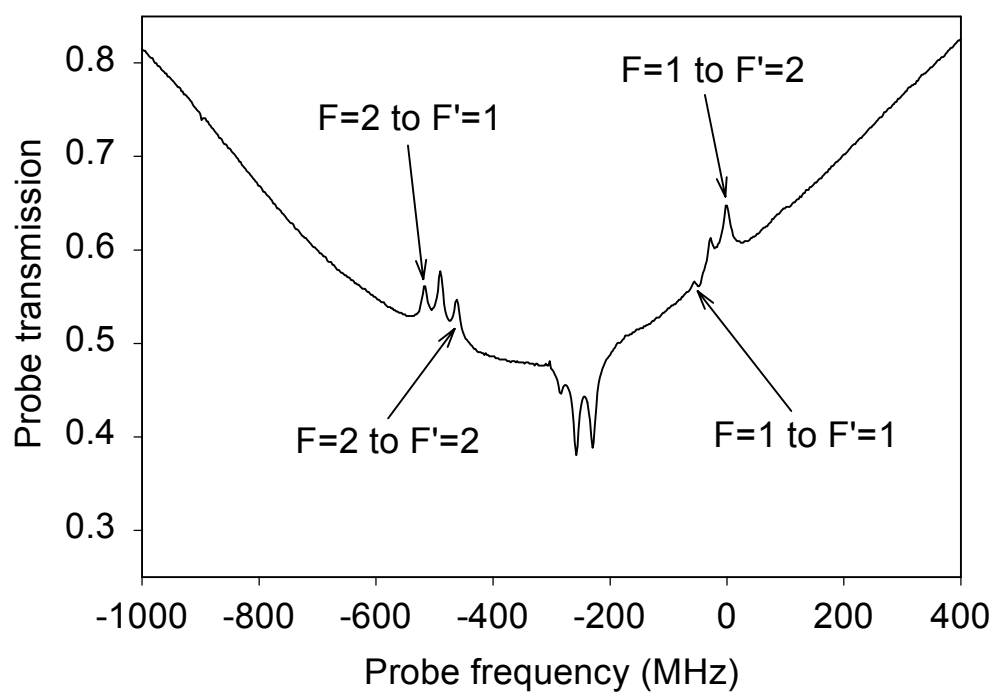


Figure 4.14: Saturated absorption spectrum. The probe transmission is normalized to 1 and the frequency zero is set at the $F=1$ to $F'=2$ hyperfine transition.

Beat Note System

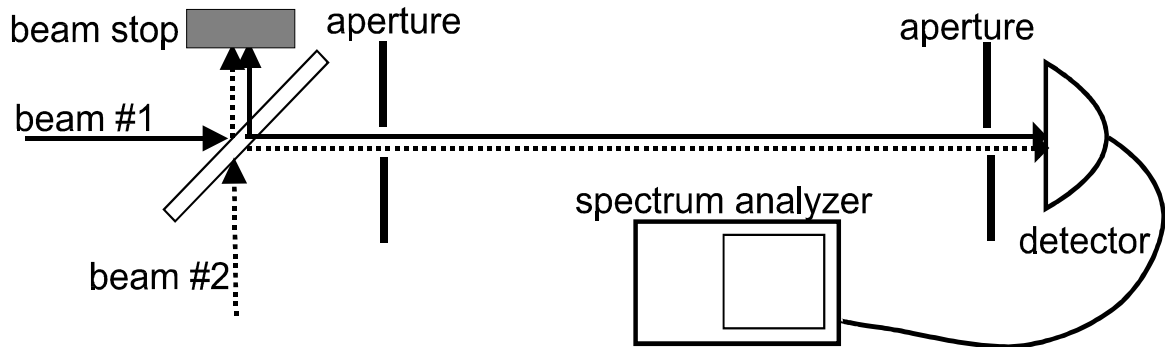


Figure 4.15: Beat note setup which allows accurate measurement of the relative frequency of two laser beams.

a high speed photodiode detector (high speed in this case being a detector with a high frequency cut-off between 100-300 MHz) and sending the detector output to a spectrum analyzer. The spectrum analyzer has a peak at the frequency corresponding to the difference in frequency between the two lasers. This system permits frequency difference measurements with an accuracy of 0.1 to 0.01 MHz. When setting up this system, it is crucial that the lasers overlap on the detector both in position and propagation direction, *i.e.* that the two beams have the same mode. As can be seen in Figure 4.15, the simplest way to align the beat note is to use a 50/50 beamsplitter to mix the two beams and then pass them through two apertures, separated by at least half a meter, before hitting the high speed detector. Also, one must be careful to avoid saturating the detector; an adequate beat note signal can be generated with only a few microwatts of power in each laser beam.

4.2.4 Laser beam detection methods and apparatuses

Direct detection using photodiodes

Detection of the laser beams is accomplished using high-quantum efficiency silicon photodiodes. The spectral response of the silicon photodiodes ranges from 400 nm to 1060 nm with quantum efficiencies in excess of 90%. These diodes are operated in photo-conductive mode and require a reverse bias, particularly for high frequency operation. Detector types used in the lab include the Hamamatsu S3994, the EG&G FFD-100, and the Hamamatsu S4751, all of which produce a current proportional to the input optical power. The Hamamatsu S3994 is a large area detector ($1\text{ cm} \times 1\text{ cm}$) with a bandwidth of $\sim 40\text{ MHz}$. Due to its large area, the S3994 is useful for measuring high powers (up to several mW) without saturating. The EG&G FFD-100 (and its close kin the FND-100) has an active area of 5.1 mm^2 and a bandwidth of $\sim 350\text{ MHz}$. These diodes are used as general, nonspecialized detectors in the lab. For higher bandwidths (*e.g.*, the beat note system) the Hamamatsu S4751 is employed, having a bandwidth greater than 500 MHz but a small active area of 0.5 mm^2 . When building a detector using these diodes, one can perform the current to voltage conversion using either a resistor or an op-amp (see Figure 4.16 for schematics). The high input impedance of the op-amp permits the use of a larger resistor for the actual I to V conversion and eliminates any problems with driving $50\ \Omega$ coaxial cables.

Subtracting Detector System

It is useful to have a detector system that can eliminate the constant background signal, when taking spectra where we are looking for small variations in the beam power (less than 1%). This can be accomplished electronically after the detector system, but can be accomplished with significantly less noise by using a second detector and directly subtracting the two currents [111]. In addition, this prevents saturation

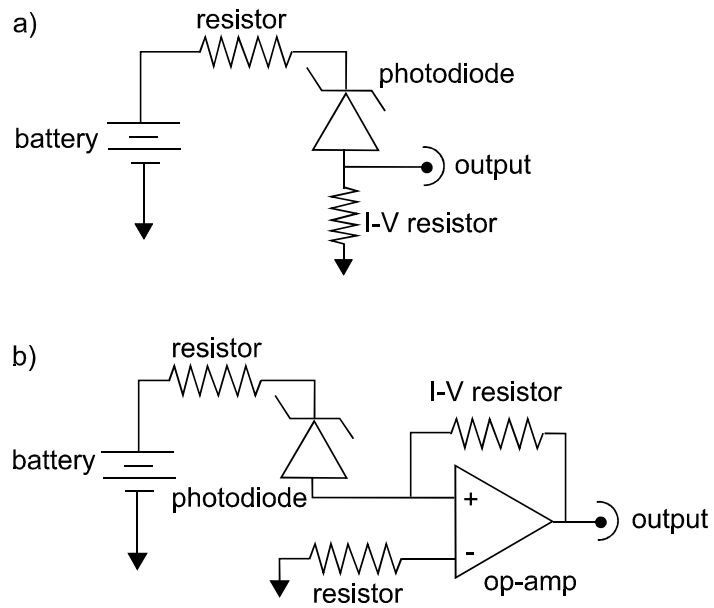


Figure 4.16: Electronic schematic for simple direct detection circuits. Current to voltage (I to V) conversion is performed either by a resistor (diagram a) or by a resistor in conjunction with an op-amp (diagram b).

Subtracting detector schematic

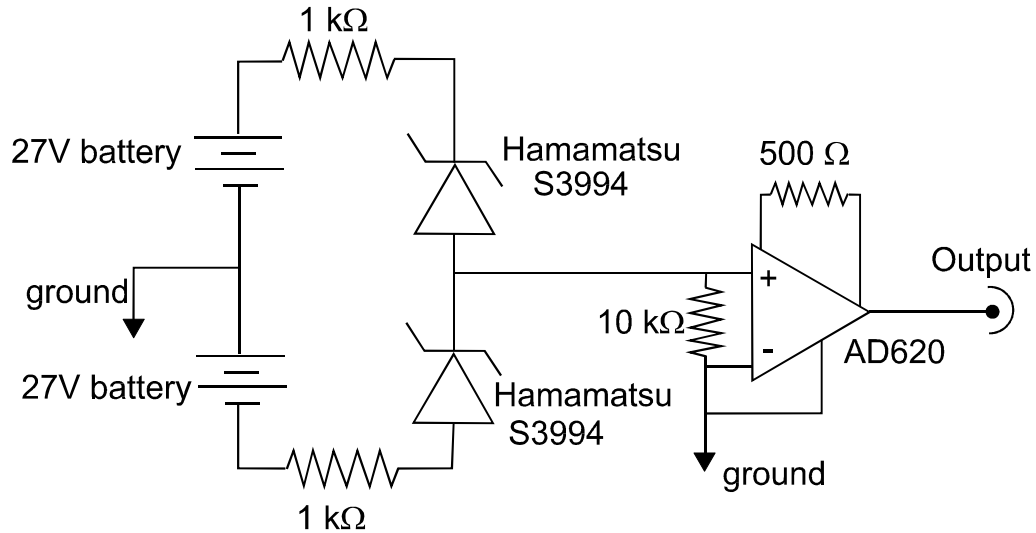


Figure 4.17: Electronic schematic for the subtracting detector setup. The two diodes provide direct current subtraction to a part in a thousand and the op-amp provides a gain of 1000 and the ability to drive downstream loads.

of the op-amp since the large background current is not amplified. Any noise on the ground can be subtracted out by using an instrumentation amplifier, such as the AD 620, with the minus side referenced to the local ground at the detectors. One other source of noise, line noise entering through the power supply lines, can be eliminated by using batteries to provide the reverse bias on the detectors and to power the instrumentation amplifier. Figure 4.17 shows a schematic of the subtracting detector system using an AD620 and two large area Hamamatsu detectors (part number S3994). This configuration gives a noise equivalent to 10 nW of optical power on the detectors with a bandwidth of less than 1 MHz. Since we expect to use ~ 1 mW of optical power for the two-photon gain measurements, this gives a sensitivity of 1×10^{-5} , which is substantially much better than the expected gain of $\sim 1 \times 10^{-3}$.

Computer aided data collection

In the lab, data is viewed using an oscilloscope, where the frequency of the laser is plotted on the x-axis and the response from the detector is plotted on the y-axis. The repetition rate for the laser is usually in the range of 15 to 30 Hz, making it possible to observe by eye what is happening as parameters in the experiment are changed. When we want to save a particular trace, an A/D card (which converts analog signals to digital signals) and a PC (personal computer) are used. The same signal that is fed into the oscilloscope also goes to the A/D card in the computer. A simple LabView program is then used to read in the data from the card, display it on the screen, and finally, to save the data to a disk file. This file has a standard ASCII format which permits any other programs to easily read in the data. Typically, we use a plotting program such as SigmaPlot to clean up the data, add axes, and produce graphs.

As an alternative to the oscilloscope-A/D card-computer data collection system, we also used a Tektronix 680B digitizing oscilloscope. This is a rather powerful and expensive oscilloscope with sample rates up to 5 GS/s [112]. Since this scope has very high sample rates, we were able to scan the diode laser faster than with the A/D card, thus reducing the jitter in the diode frequency scan. In addition, this scope has a feature that averages a user-specified number of scans for display. This permitted us to average out a great deal of the background noise and resolve features from the subtracting detector setup that are $<100 \mu\text{W}$ large. This corresponds to a power of $\sim 1 \text{ nW}$ on the subtracting detector. Furthermore, scans could be stored and saved to a disk drive in the oscilloscope and then sneaker-netted over to the PC and directly transferred to a data plotting program such as SigmaPlot. With its high resolution (both in time and voltage) and ease of storage and transfer, this digitizing scope greatly simplified the data collection process.

4.3 Experimental results for the two-photon Raman gain

The first experimental result was the optical pumping of the potassium atoms. We introduced two σ^- polarized beams into the interaction region to shift all of the population to the extreme $F=2, m_f = 2$ magnetic sublevel. The power in these beams is ~ 5 mW and the beam waists are 4.1 mm for the red and 5.0 mm for the blue optical pumping beams. In conjunction with the small magnetic field (~ 1 Gauss), these beams effectively pump the majority of the atoms. A spectrum with optical pumping is shown in Figure 4.18. The unpumped absorption is $\sim 50\%$ for this spectra; we see that the $F=2$ to $F'=2$ line now has $\sim 85\%$ absorption and the $F=2$ to $F'=1$ line has $\sim 10\%$ absorption left. Absorption from both of the $F=1$ lines is almost completely suppressed. Note that the remaining small absorption lines are due to the potassium 41.

Once the optical pumping is working, we know that there is a population inversion between the $F=2, m_f = 2$ level and the other magnetic sublevels. Thus, it should be possible to observe gain in the system. The dressed-state theory from the last section of Chapter 3 provides the roadmap for our search. We checked the frequencies where the two-photon gain appears isolated and that could be reached with the available AOM's. The two-photon gain is largest for small detuning and our best results were obtained for pump detunings ranging from +25 MHz to +85 MHz. To provide a framework I will present the results in order of the pump detuning, starting from the low frequency and going up in frequency. The pump detuning is measured from the bluest absorption line (the $4S_{1/2}$ ($F=1$) to the $4P_{1/2}$ ($F'=2$) line).

For all of this data, the probe beam waist is $90 \mu\text{m} \pm 10 \mu\text{m}$ and the probe power ranged from $1\mu\text{W}$ to $\sim 6\text{mW}$. For the Raman pump beam, a cylindrical lens is used

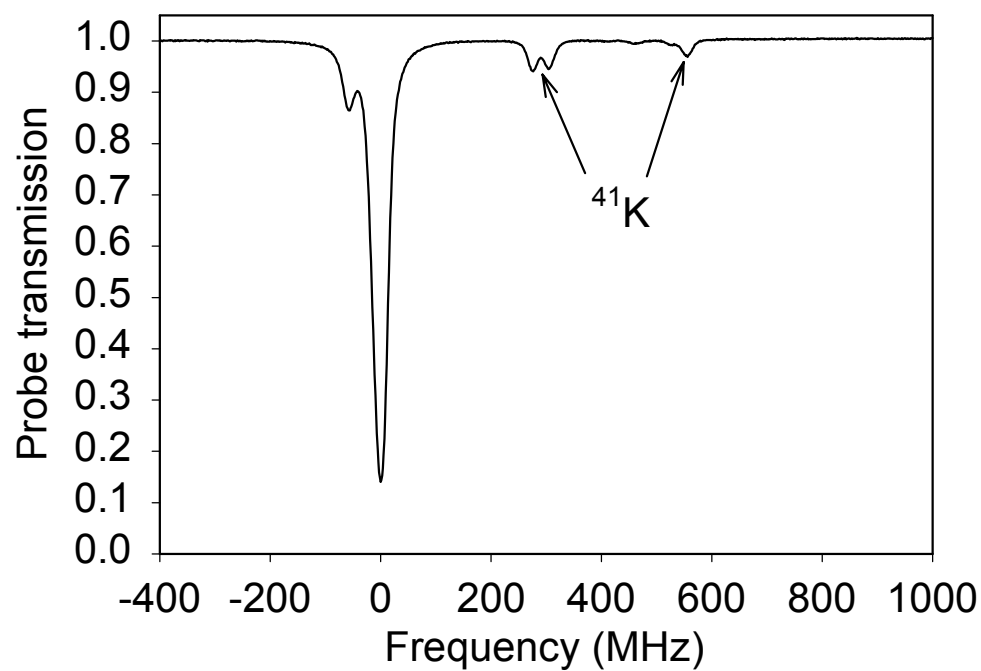


Figure 4.18: Spectrum showing optical pumping of potassium 39. The majority of the population is moved into the $F=2$, $m_F = 2$, magnetic sublevel, resulting in a single absorption line for potassium 39. Note that the potassium 41 lines are unperturbed.

to focus the beam to an elongated ellipse with a minor axis waist of 250 μm (more than twice the size of the probe waist) and the major axis waist of 3.0 mm (which is approximately the size of the atomic beam width). Thus, the pump beam covers the entire region where the probe beam overlaps with the atomic beam, but is not much larger so that power is not wasted. This was important since the two-photon gain requires large pump intensities, but we only have 1 W available from the Ti:Sa for the entire system. The power in the pump beam is varied from ~ 5 mW to a maximum somewhere between 200 and 300 mW, depending on the combination of AOM's used to shift the Raman pump beam frequency. The pump beam power is smoothly varied using a variable attenuator (Newport Model 925B).

The Rabi frequencies of the pump and probe beam can be calculated from the beam sizes and powers. Starting with the definition of the Rabi frequency given by

$$\Omega_{ij} = \frac{2\boldsymbol{\mu}_{ij} \cdot \mathbf{E}(\mathbf{r}, \mathbf{t})}{\hbar} , \quad (4.11)$$

and using the relations

$$I = \frac{c}{2\pi} |E(\mathbf{r}, \mathbf{t})|^2 , \quad (4.12)$$

and

$$I = \frac{2P}{w_o^2\pi} , \quad (4.13)$$

the resulting relation is

$$\frac{\Omega_{ij}}{2\pi} = 26.0 \left(\frac{d}{d_o}\right)_{ij} \sqrt{\frac{P(\text{mW})}{w_o^2(\text{mm}^2)}} \quad (4.14)$$

where Ω is the Rabi frequency, P is the beam power in milliwatts, w_o is the beam waist in millimeters, and $(d/d_o)_{ij}$ is the dipole matrix element for a particular transition ($\boldsymbol{\mu}_{ij}$), which may be obtained from Figure 4.3 and is normalized to $d_o = 4.23ea_o$. For the asymmetric pump beam replace w_o^2 with $(w_o)_{Vertical}(w_o)_{Horizontal}$.

4.3.1 Raman pump detuning of -597 MHz

The first choice was to use a pump detuning to the red of all the absorption lines, as had been the case for the experiments in the potassium buffer cell experiment. The indications were that detunings in-between the ground states would result in destructive interferences between the multiple quantum pathways thus reducing the gain. At that time we had not done any work with the detunings to the blue, since in the vapor cell blue detunings resulted in greater self-focusing of the pump beam. Figure 4.19 shows a spectra for a pump detuning of -597 MHz. We clearly see one-photon Raman gain at the pump detuning (0 MHz) and one-photon Raman gain detuned by the ground state splitting (462 MHz). The gain is $\sim 10\%$ which is substantial, without being large enough to cause problems for the two-photon laser (recall that there is a maximum ratio we can have between the one-photon and two-photon gain). However, we were unable to see any appreciable two-photon gain at this frequency. Using FM spectroscopy, we saw a feature that might have been two-photon Raman gain, but it was quite small and was on the shoulder of an absorption line. As we increase the power in the probe laser, which increases the two-photon gain, the large absorption line increases since the probe laser is now power broadening that line. Thus, the two-photon gain sits on the side of the absorption line and is not real gain, but merely reduced absorption. Also, the strong Raman pump laser is very close to the $F=2$ lines and hence is optically depumping the atoms, destroying the inversion. The signal to noise for this data is not very good because we were not yet using the subtracting detector setup, which was employed for the rest of the data. With these results we decided that it would be better to use pump detuning closer to 0 (the $F=1$ to $F'=2$ line).

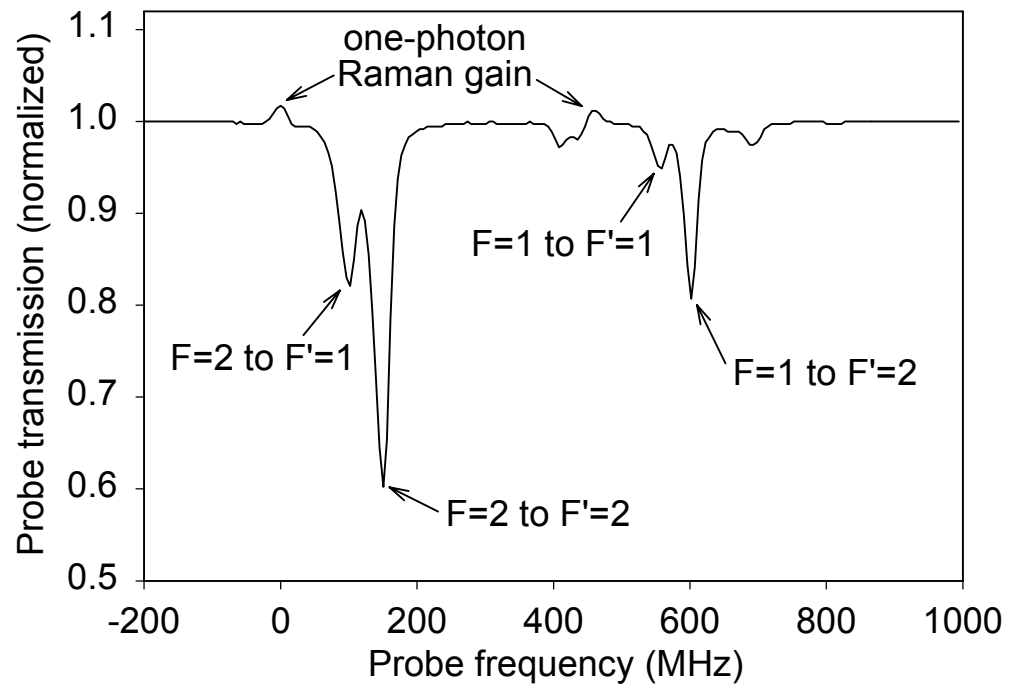


Figure 4.19: Probe spectrum for a Raman pump detuning of -597 MHz. Note the two one-photon Raman features, with pump-probe detunings of 0 MHz and +462 MHz.

4.3.2 Raman pump detuning of -60 MHz

The challenge for frequency detunings near 0 is to keep the detunings small while avoiding both the usual competing mechanisms and the absorption lines from the least abundant isotopes. Figure 4.20 shows a spectrum for a Raman pump detuning of -60MHz and a Raman pump power of 90 mW. The probe power is 5.85 mW. There are two large one-photon Raman peaks as usual. Slightly to the red of the blue Raman gain are two one-photon dressed state peaks that are also quite sizable at this pump detuning. In the middle are several small peaks and an absorption dip due to the least abundant isotope. Halfway between the two one-photon Raman peaks is a small peak, which is two-photon Raman gain. Given the nearby dips, it is difficult to tell if this is gain or just reduced absorption. In addition, the nearby one-photon dressed-state peaks are a source of concern.

4.3.3 Raman pump detuning of +25 MHz

The next pump detuning is +25 MHz. In Figure 4.21, we see that the two one-photon dressed-state gain features have now moved to the blue side of the blue one-photon Raman peak and thus are significantly farther away from the two-photon gain feature. The probe power for this spectrum is only 1.26mW, which is below saturation, so we do not see the maximum two-photon gain. However, a lower probe power reveals some of the nearby features that are saturated at high probe powers, such as the peak just to the red (the left) of the two-photon Raman gain feature. This additional peak is one-photon dressed-state gain starting from the $F=1, m_f=1$ level. Some population ends up in this level due to the strong Raman pump beam. The pump power for this spectrum is 146 mW.

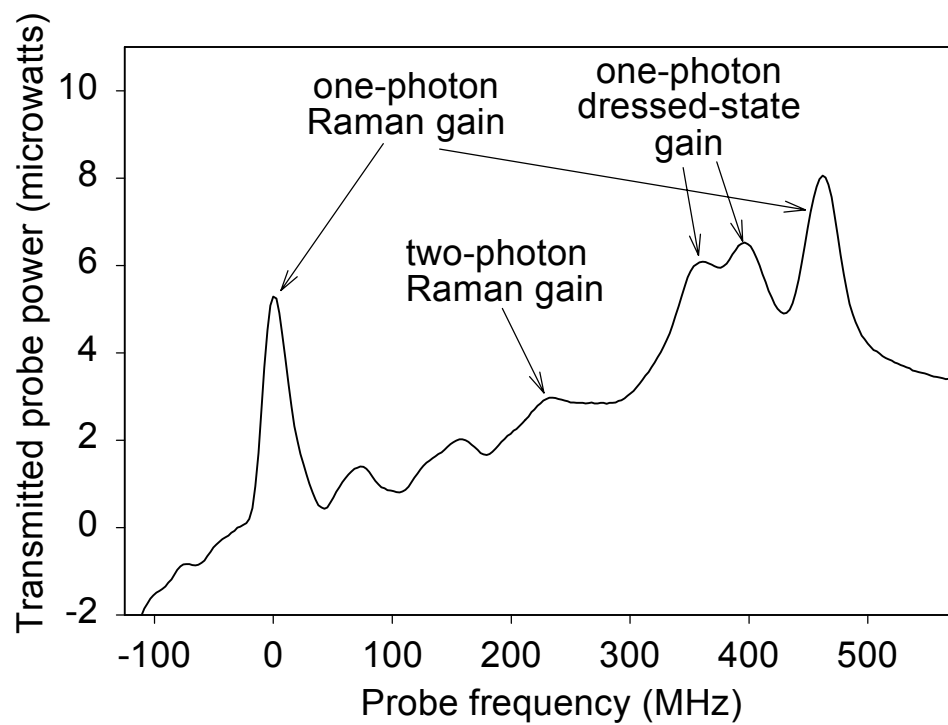


Figure 4.20: Probe spectrum for a pump detuning of -60MHz and a probe power of 5.85mW.

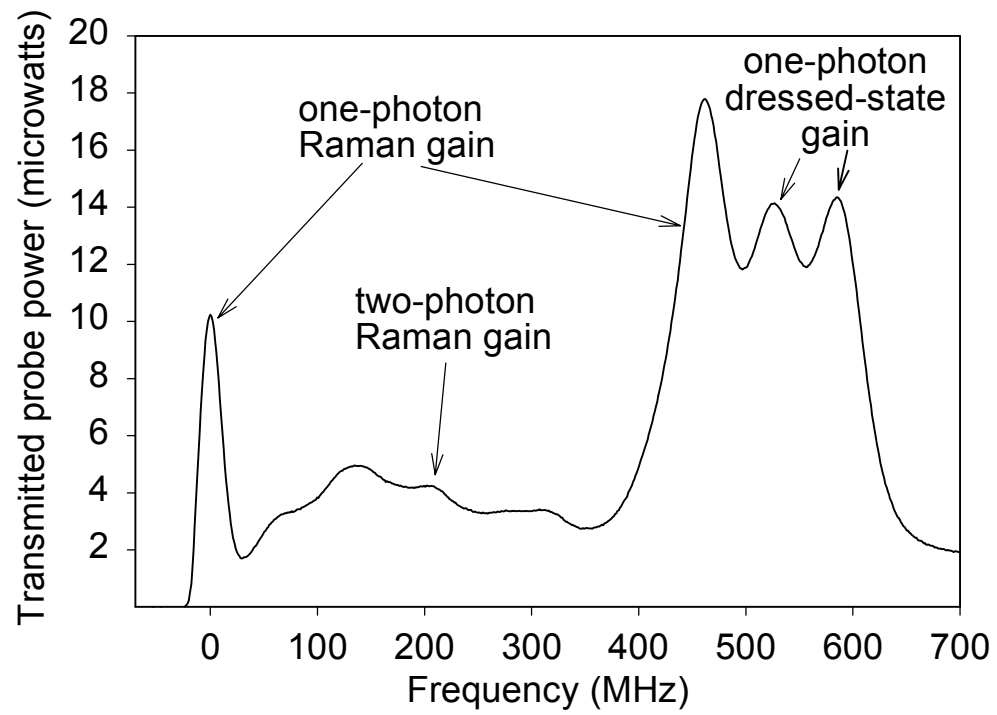


Figure 4.21: Probe spectrum for a pump detuning of +25MHz and a probe power of 1.26mW.

4.3.4 Raman pump detuning of +60 MHz

Some of our most promising results were obtained for a pump detuning of +60MHz. As can be seen in Figure 4.22, the probe spectrum clearly shows two one-photon Raman gain peaks and now the two one-photon dressed state gain peaks are well to the blue side of the bluest one-photon Raman gain peak. The probe power is 4.06 mW and the pump power is 300 mW for this spectrum. There are several features in-between the one-photon Raman peaks. The largest one, which is just slightly to the red of the middle, is the two-photon Raman gain peak. Note that there are no other features within at least one peak width. There is a feature to the blue of the two-photon Raman peak, which is due to one-photon Raman gain from the least abundant isotope. Originally we thought that it might be a two-photon process since it was in approximately the right spot and about the right size. Unfortunately it did not scale correctly with the probe power and thus revealed its origin as a one-photon feature.

This illustrates one of the problems we had with this experiment. Some features were very easy to recognize and assign to their mechanism of origin. However, there are many processes that can occur in this system, particularly at the large pump and probe powers required for two-photon gain. Thus, to be sure a gain feature is really a two-photon process, it is necessary to plot the gain as a function of the probe power and verify that there is a dependence of the gain on the probe power. Recall that for a one-photon process the gain is independent of the pump power until saturation. Once saturation is reached the gain slowly decreases to zero as the probe power is increased further. On the contrary, for a two-photon process, the gain increases linearly with increasing probe power until saturation is reached. After saturation, the gain levels off and then begins decreasing in an analogous fashion to the one-photon processes. Figure 4.23 shows a plot of the probe gain as a function

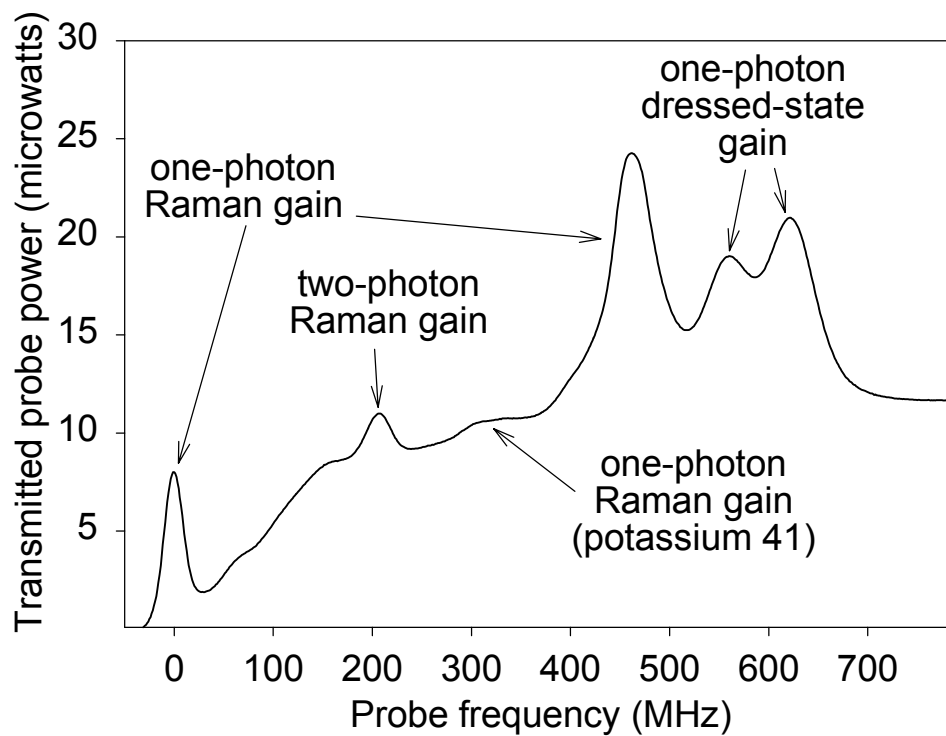


Figure 4.22: Probe spectrum for a pump detuning of +60 MHz, a pump power of 300 mW, and a probe power of 4.06 mW.

of probe power for a pump detuning of +60 MHz and a constant pump power of 300 mW. This data clearly shows a linear relationship with probe power and goes to zero for zero probe power. Saturation occurs slightly before 3 mW corresponding to an intensity of $\sim 20 \text{ W/cm}^2$. Using

$$I_{out} = I_{in}(1 + g^{(2)}I_{in}L)$$

we find $g^{(2)} = 8.9 \times 10^{-5} \text{ cm/W}$ from the data in the linear slope region. This linear dependence of the gain on the probe power is powerful support for our contention that this is indeed two-photon gain.

In order to obtain these large gains, the atomic beam flux must be maximized. As previously noted, the oven temperature can only be raised so high before the beam starts to broaden. Figure 4.24 shows a spectrum for weak probe absorption for an oven temperature of $\sim 280^\circ \text{ C}$. The maximum absorption (without optical pumping) is $\sim 95\%$ and the peak width is still approximately 30 MHz (note that the full width at half maximum occurs more than halfway down the absorption dip because of the exponential nature of absorption). Even the least abundant isotope has a line center absorption of $\sim 20\%$ at this beam flux.

4.3.5 Raman pump detuning +85 MHz

Since the results at +60 MHz were very encouraging, we wanted to check some nearby pump frequencies. We looked at 85 MHz after 60 MHz because it was the next closest frequency we could reach with the available AOM's. Figure 4.25 shows a probe spectrum at +85 MHz and it is quite similar to +60 MHz, except that the features are more isolated, *i.e.* there is more space between the two-photon Raman gain and it's closest neighbors. The probe power for this spectrum is 4.83 mW and the pump power is 197 mW.

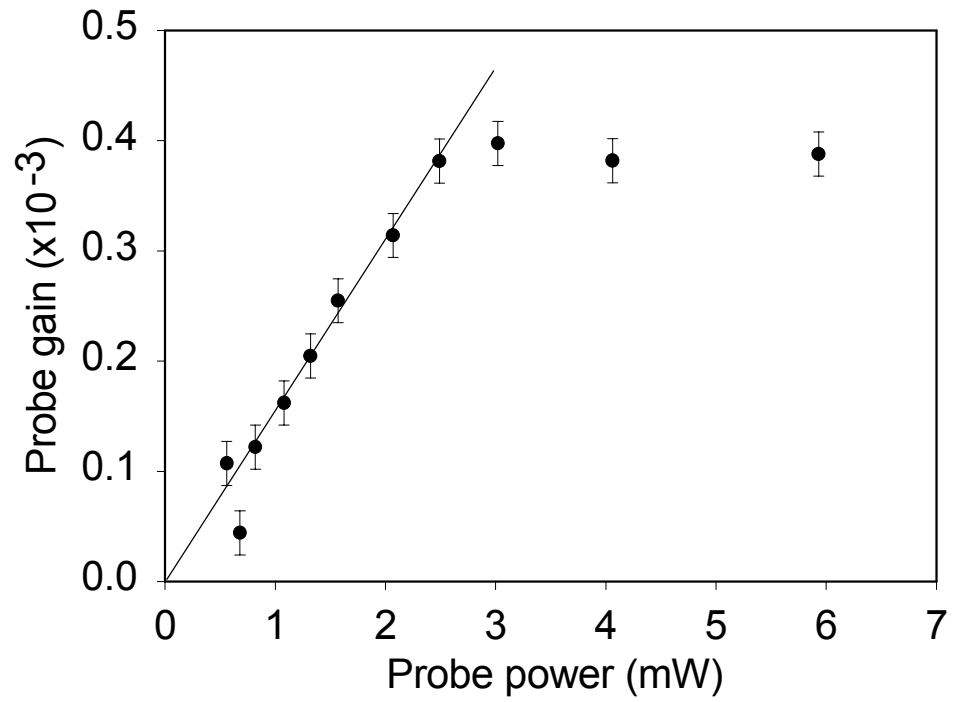


Figure 4.23: Gain versus probe power for a pump detuning of +60 MHz. This graph shows the linear relationship between probe power and gain before saturation. The gain slowly decreases with increasing probe power after saturation.

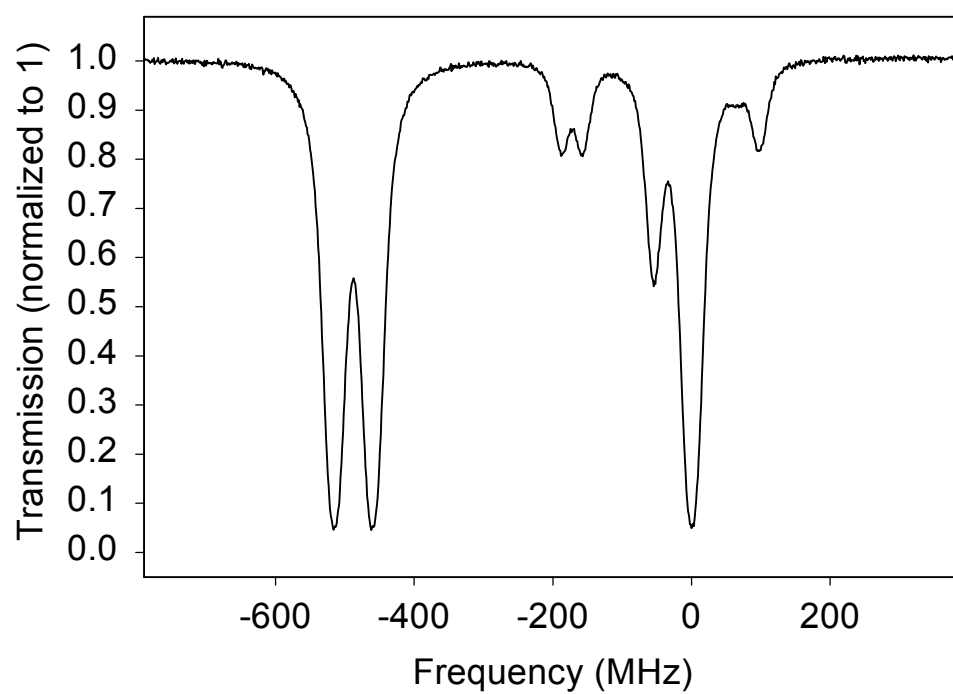


Figure 4.24: Spectrum showing absorption at very high atomic beam flux. The oven temperature is $\sim 280^\circ\text{C}$.

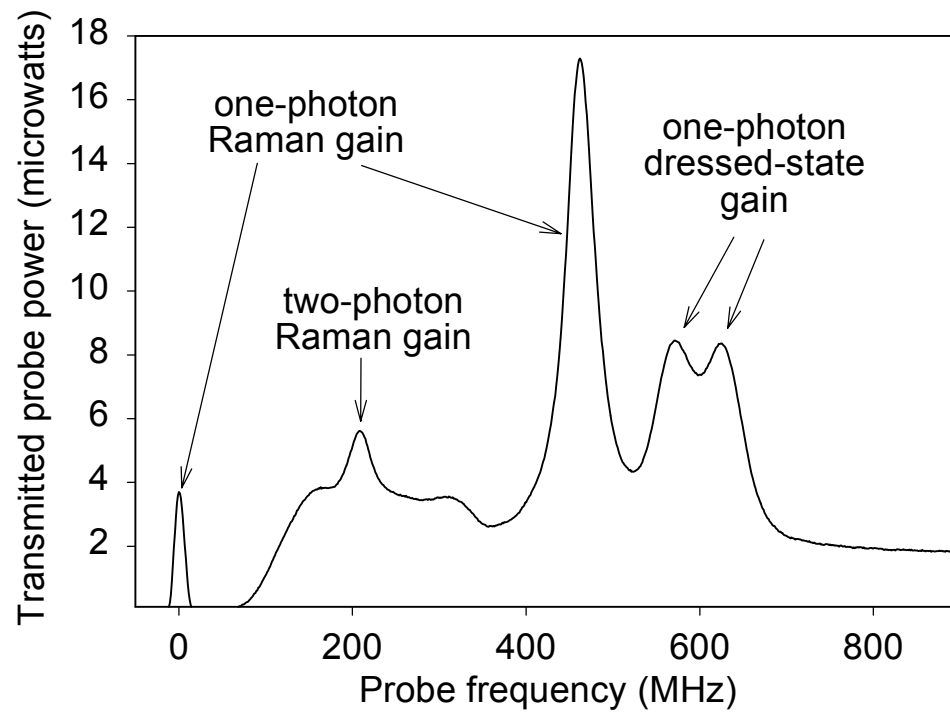


Figure 4.25: Probe spectrum for a pump detuning of +85MHz and a probe power of 4.83mW.

Figure 4.26 shows the relationship between gain and probe power for the two-photon Raman gain with a pump detuning of +85 MHz. Here again we see a linear relationship between probe gain and probe power until saturation. Note that the two-photon gain does not reach saturation until nearly 4 mW of pump power and the maximum gain is only 0.3×10^{-3} which is less than the maximum gain of nearly 0.4×10^{-3} observed at +60 MHz. The pump power is 197 mW for this detuning, which contributed to the reduced gain since the two-photon gain is dependent on the Raman pump power.

4.3.6 Raman pump detuning of +285 MHz

Since we would like to move the features farther from the absorption lines we tried a pump detuning of +285 MHz. From the analysis in Chapter 3 we know that all the Raman features tune with the pump frequency. At this detuning all of the features appear to be well isolated. In Figure 4.27 both of the one-photon Raman gain features are well away from the absorption lines. The one-photon dressed state features are now well to the blue of the bluest one-photon Raman gain peak. There is also a recognizable two-photon gain feature right between the one-photon Raman gain peaks. Unfortunately the maximum gain is considerably less than what we obtained at the lower frequencies. For a probe power of 4.96 mW and a pump power of 170 mW the two-photon Raman gain is only 0.22×10^{-3} , which is substantially less than what was observed at the smaller detunings. This, of course, makes sense given the dependence of the two-photon Raman Rabi frequency on the probe detuning.

When we plotted the probe gain as a function of probe power, we noticed that the gain is flat for low probe powers and then increases linearly for probe powers above 3 mW. This indicates that there is probably a one-photon feature which is nearly degenerate in frequency with the two-photon Raman gain. Interestingly enough, if

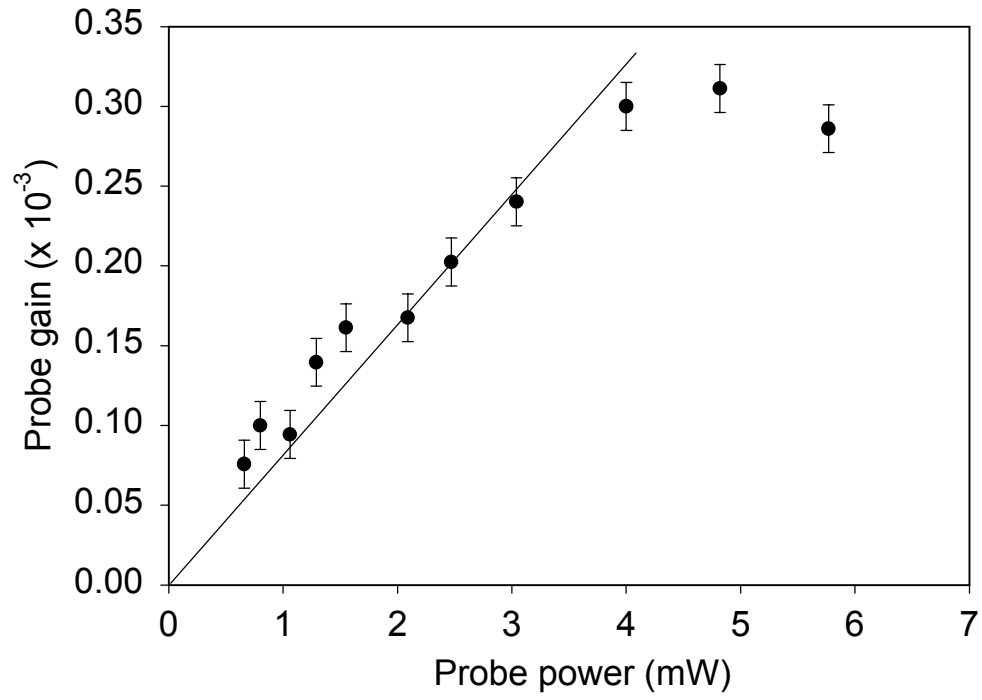


Figure 4.26: Probe gain as a function of probe power for a pump detuning of +85 MHz. The gain is linear with increasing probe power until saturation is reached at ~ 4 mW. The pump power is 197 mW for all data points.

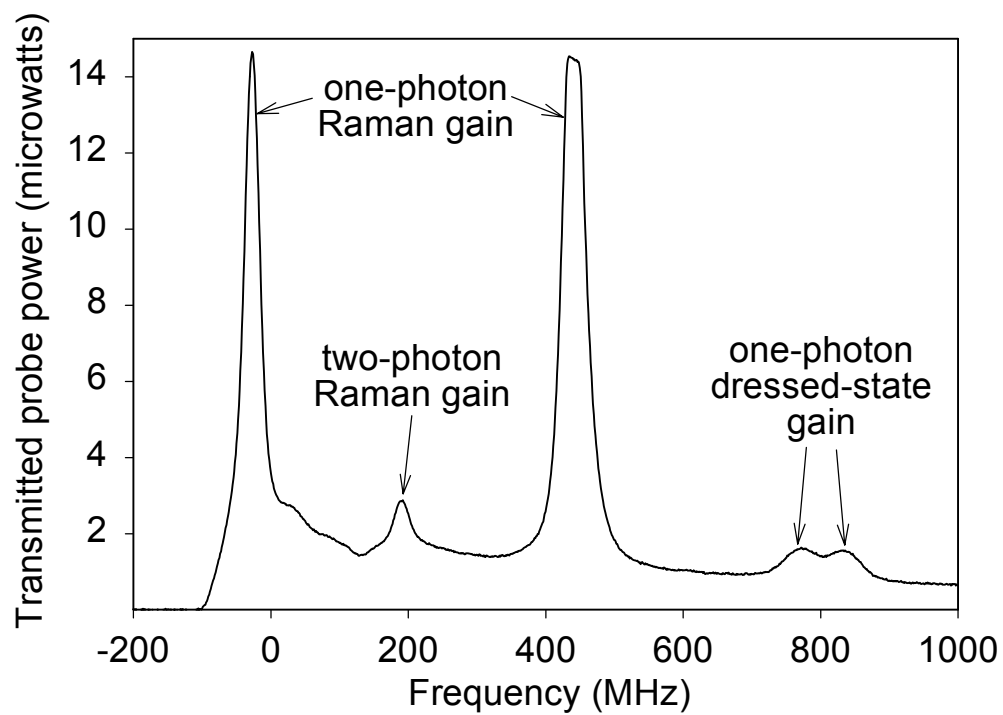


Figure 4.27: Probe spectrum for a pump detuning of +285 MHz. The probe power for this spectrum is 4.96 mW and the pump power is 170 mW.

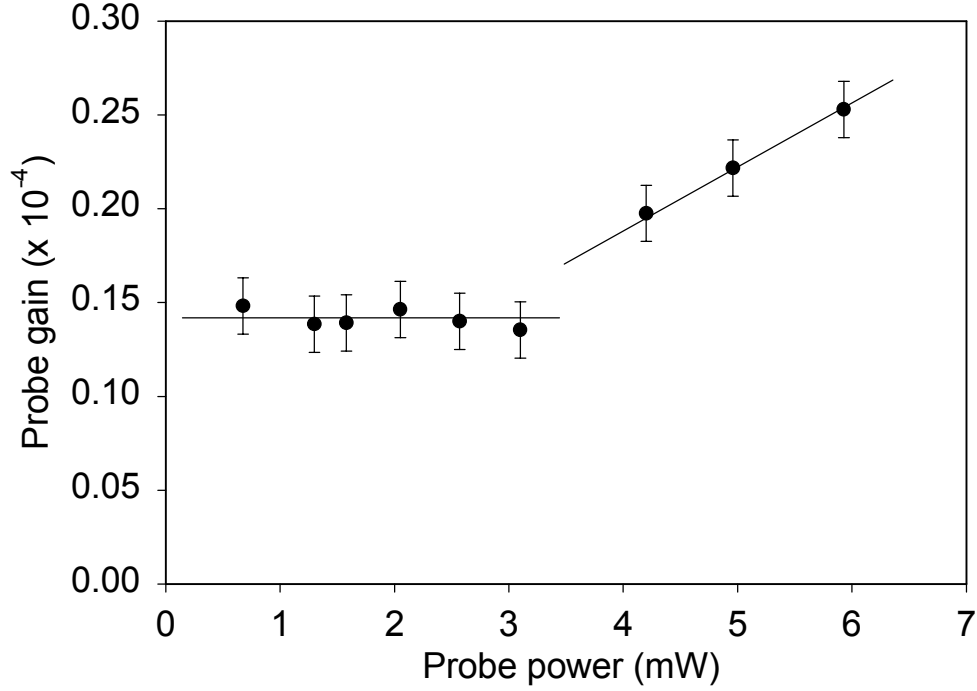


Figure 4.28: Graph of probe gain versus probe power for a pump detuning of +285MHz. Note the flat slope for low probe powers. This indicates that there is probably a one-photon gain feature which is degenerate with the two-photon gain.

this was an unsaturated one-photon feature, it would merely add a constant offset to the two-photon gain versus power graph. The fact that the total gain is actually flat for low powers indicates that the one-photon feature is being saturated as the probe power increases. Most likely it is just a coincidence that the saturation in the one-photon feature and the linear increase in the two-photon feature add to a constant total gain for low probe powers. The pump power for all of this data is 170 mW.

4.3.7 Alternate probe polarizations

The previous data was all taken before the two-photon laser experiment was set up. Once the two-photon laser was operating, we discovered that there are polarization instabilities in the output. These results suggested that there is probably significant two-photon gain for polarizations other than linear. To answer this question experimentally, we removed the laser cavity from the vacuum system and reassembled the pump-probe experiment. Data was collected for probe polarizations of vertical, 45° , and horizontal. Two-photon gain may occur for circular polarizations as well, but we have not checked this possibility yet.

Figure 4.29 shows the two-photon Raman gain as a function of probe power for vertical probe polarization (Figure 4.29(a)) and 45° probe polarization (Figure 4.29(b)). These graphs are virtually identical within the bounds of the experimental error. The pump detuning for this data is +80 MHz and the pump power is 420 mW.

We also checked the gain for a horizontally polarized probe beam and found that there was some gain present but it was significantly less (at least a factor of 4 less than for the vertically polarized probe), as illustrated in Figure 4.30.

4.4 Discussion of experimental results

There are several points to make in looking at all of the data from the atomic beam apparatus. The first item that bears mentioning is the degenerate one-photon Raman gain. Even though it is not directly related to the two-photon question at hand, it warrants some discussion. In the majority of pump-probe experiments (see H. Concannon's work [57] or the data in Chapter 3), there is a dispersive shaped feature where the probe and pump are degenerate. This feature is typically referred

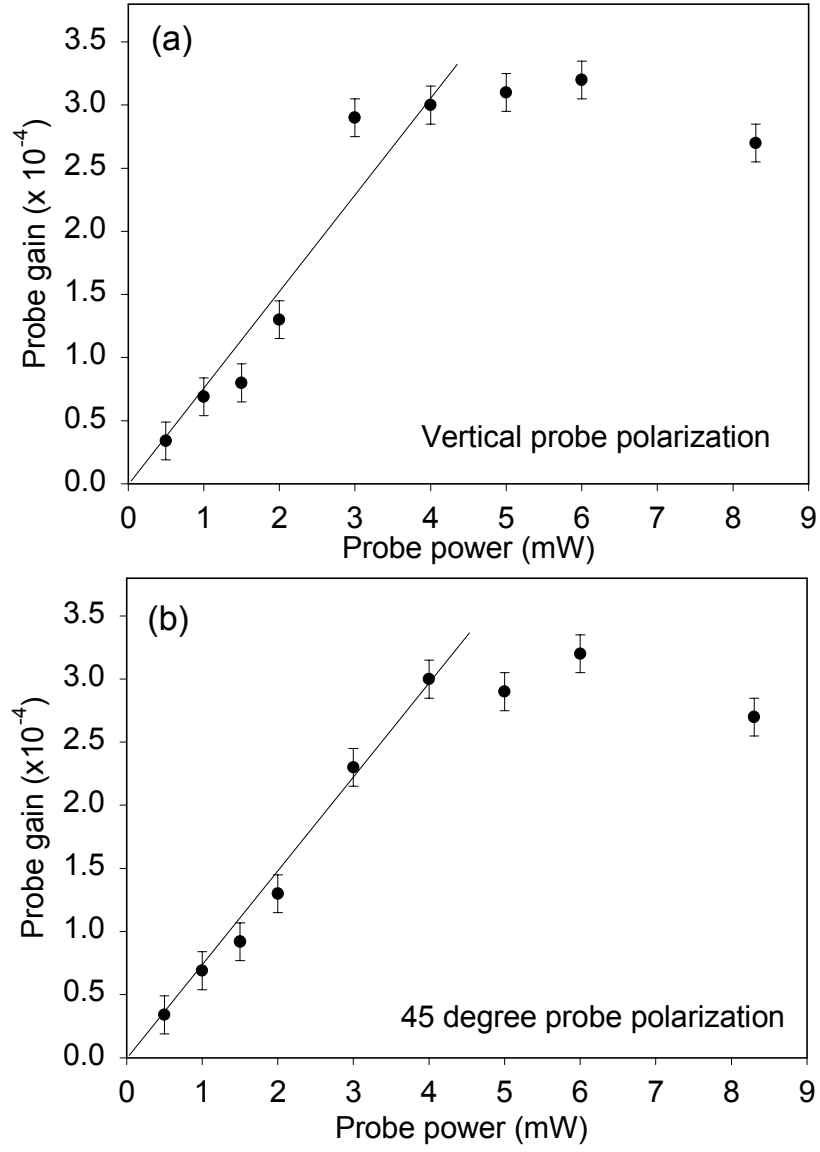


Figure 4.29: Two-photon Raman gain experienced by the probe as a function of the probe power. (a) Vertical probe polarization and (b) 45° polarization.

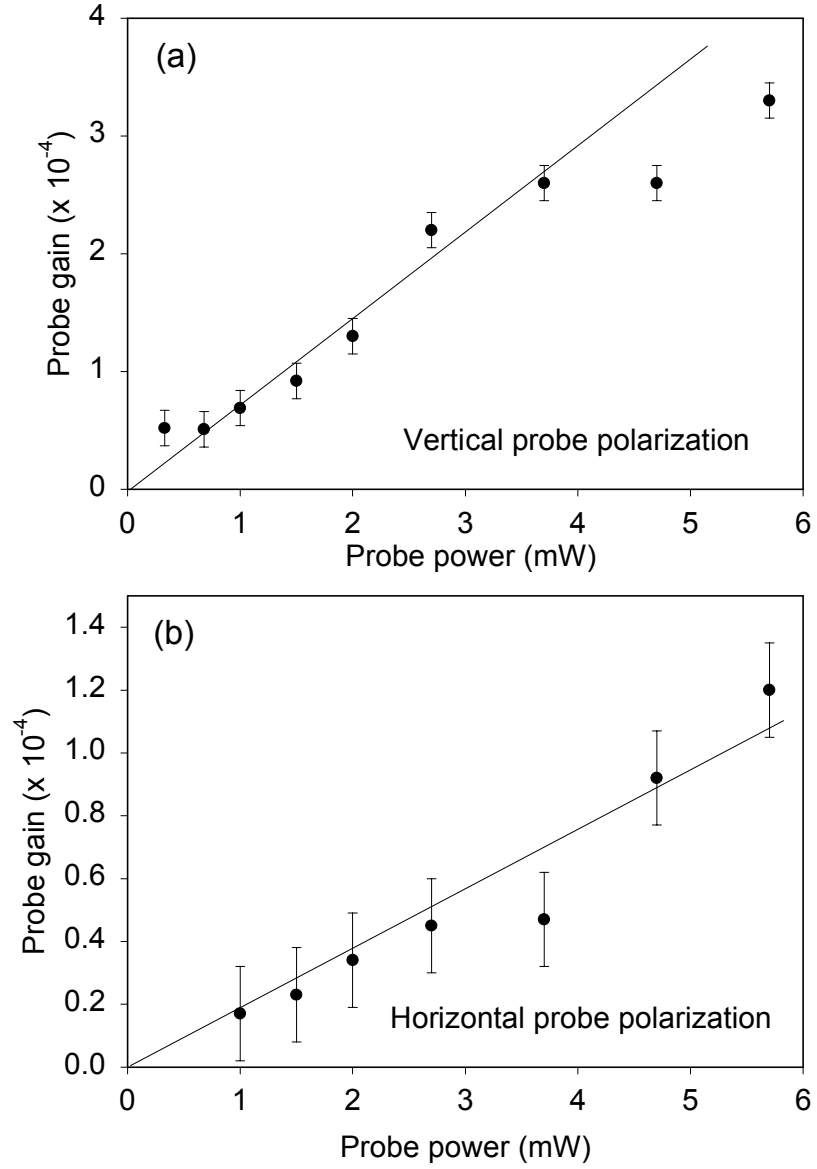


Figure 4.30: Two-photon Raman gain as a function of probe power. (a) Vertical probe polarization and (b) horizontal probe polarization.

to as a Rayleigh resonance and is due to population oscillations. However, for this experimental configuration we have gain when the pump and probe are degenerate. As a check, we performed some experiments using the beat note between the pump and the probe to confirm that this feature really occurs when the probe-pump detuning is zero. Recently, other scientists have suggested that atomic motion is required to have gain for degenerate pump and probe [113]. However, such is obviously not the case in our experiments.

Finally, from scrutinizing the data, the best place to try for two-photon lasing appears to be for a detuning somewhere between +25 MHz and +85 MHz. Here we have the largest two-photon gain by virtue of the small detuning. By careful selection of the frequency we should be able to find a spot where there are no competing mechanisms too close to the two-photon gain feature. With this information in mind, we move on to the next chapter where I will describe the optical cavity used for the two-photon Raman laser.

Chapter 5

Optical resonator for the two-photon laser

5.1 Overview

This chapter contains the description of the experimental apparatus used for the two-photon laser. An overview of the entire experimental apparatus is given followed by a detailed description of the cavity. A very high finesse cavity is required for the two-photon laser and working with such a cavity is not a trivial matter. Numerous factors were considered in the design and construction of the cavity, an explanation is given for each. Once the cavity was constructed, considerable time was spent aligning the cavity, the cavity apertures, and the probe beam and then measuring cavity parameters such as the throughput and the finesse.

5.2 Optical cavity basics

As noted in Chapter 1, any laser requires an optical resonator or optical cavity to select which spatial mode and frequency lases. The special requirements of the two-photon Raman laser necessitated that substantial time and effort be spent in the design and construction of the two-photon laser cavity. This section provides some background information on optical cavities and then describes in detail the cavity used for our two-photon Raman laser.

Figure 5.1 illustrates an optical cavity, which consists of two parallel mirrors. The mirrors are aligned such that an optical field propagating down the cavity axis and reflected by the mirrors, returns to the same place it started. This allows standing electromagnetic waves to exist between the mirrors, the requirement being that an

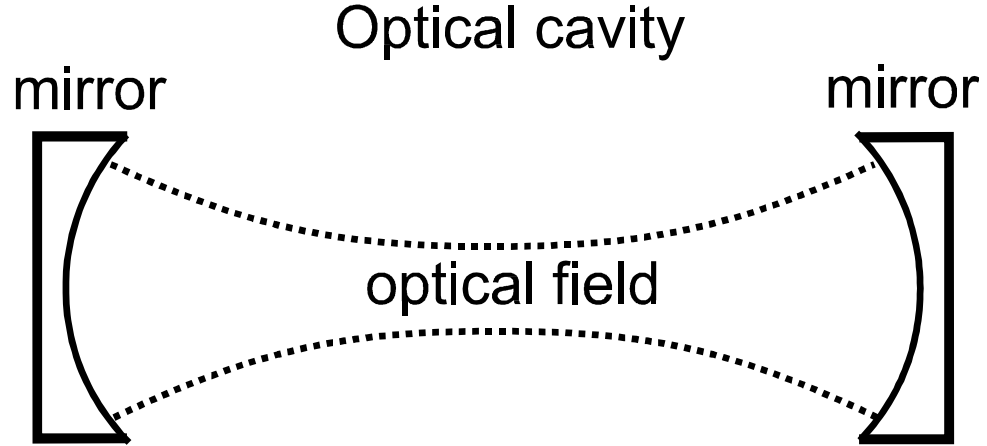


Figure 5.1: An optical cavity (also known as an optical resonator) consists of two mirrors aligned so that a standing electromagnetic wave can exist between the mirrors.

integral number of half wavelengths fits in the cavity or

$$d = q \frac{\lambda}{2} , \quad (5.1)$$

where d is the distance between the mirrors, q is an integer and λ is the wavelength of the light. For a typical optical cavity q is very large. For example, $q \approx 129,870$ for $d = 5$ cm and $\lambda = 770$ nm. Thus, the spacing between successive wavelengths that meet the resonance requirement is a small fraction of the wavelength. Typically, one measures this separation in frequency (instead of wavelength) and this quantity is called the free spectral range (FSR). The FSR is a property of the cavity and is equal to

$$\text{FSR} = \frac{c}{2d} . \quad (5.2)$$

For example, the FSR is 3.0 GHz for the 5 cm cavity. Each frequency that matches this resonance condition is referred to as a longitudinal mode of the cavity and the corresponding integer q is the longitudinal mode number.

Another important measure of the cavity is the narrowness of the cavity resonances. For the two-photon laser, we need to make sure that the cavity resonance

enhances only the two-photon process and not any other processes that occur at nearby frequencies. The width of the cavity resonances is related to the reflectivity of the mirrors. As an example, consider two slightly different frequencies of light in a cavity are in-phase and then begin propagating back and forth between the mirrors. The farther the distance traveled (or the greater the number of bounces) the larger the phase difference between the two frequencies until ultimately they will be out of phase. At this point the two frequencies will destructively interfere and will not be enhanced by the cavity. This quality of an optical cavity is measured using the finesse (F), which is the ratio of the free spectral range to the linewidth of the cavity (or how wide a resonance is) or

$$F = \frac{\text{FSR}}{\Delta\nu_{1/2}} , \quad (5.3)$$

where $\Delta\nu_{1/2}$ is the full width at half-maximum of the cavity resonance. The finesse can also be determined from the reflectivity of the mirrors and is given by

$$F = \frac{\pi(R_1 R_2)^{1/4}}{1 - (R_1 R_2)^{1/2}} , \quad (5.4)$$

where R_1 and R_2 are the reflectivities of the two mirrors [114]. In addition, the finesse provides an estimate of how much gain is required to initiate lasing. Recall that lasing will occur when the round trip losses (transmission through the mirrors and any absorptive or scattering losses in the cavity) equal the round trip gain. Assuming that there are no absorptive losses in the cavity, a single pass gain of

$$g > \frac{\pi}{F} \quad (5.5)$$

is necessary to support lasing. This can be thought of as the rate at which photons must be added to the beam inside the cavity to replace the photons lost through the mirrors or to intracavity losses. It is important to note that the linewidth of the

cavity and the buildup of the cavity (*i.e.*, how long the photons remain in the cavity) are both related to the reflectivity of the mirrors.

In addition to the longitudinal modes of the cavity, there are transverse spatial modes. These modes may be explored by considering the solutions to the wave equation in the cavity that meet the boundary condition $E = 0$ at the mirror surfaces. For two spherical mirrors in the paraxial approximation, the solutions are Hermite polynomials multiplied by a Gaussian and a phase term and are given by

$$E_{l,m}(x, y, z) = E_o \frac{w_o}{w(z)} H_l\left(\sqrt{2} \frac{x}{w(z)}\right) H_m\left(\sqrt{2} \frac{y}{w(z)}\right) \times \exp\left[-\frac{x^2 + y^2}{w^2(z)} - ik \frac{x^2 + y^2}{2R(z)} - ikz + i(l + m + 1)n\right]. \quad (5.6)$$

For this solution, z is the coordinate along the cavity axis ($z = 0$ at the center of the cavity) and x and y are the transverse coordinates ($x = 0$ and $y = 0$ on the cavity axis). The field amplitude is denoted by E_o , w_o is the spot size at the center of the cavity, $w(z)$ is the spot size at z , $R(z)$ is the radius of curvature of the beam at z , k is the wavenumber of the light, and n is the index of refraction of the material filling the cavity. The Hermite functions (H_l and H_m) are polynomials with order l and m (see Arfken [115]). These two indices determine the spatial profile of the optical beam and are referred to as the transverse mode numbers. These modes are referred to as TEM _{l,m} modes (transverse electro-magnetic modes). For a given q , the frequency of the transverse modes depends on the value of the sum of $l + m$. Thus, the TEM_{1,0} and TEM_{0,1} modes occur at the same frequency, which is different than the frequency for the TEM_{0,0} mode. For a more in-depth discussion and for plots and pictures of the spatial modes of a cavity, see Yariv's *Introduction to Optical Electronics*, sections 3.6 and 4.3 [116].

There are many possible mirror configurations for an optical resonator, the simplest to describe is a confocal resonator where two mirrors with the same radius of

curvature are separated by a distance equal to the radius of curvature. For a confocal cavity, the frequency for a mode q , $\text{TEM}_{1,1}$ (or $\text{TEM}_{2,0}$ or $\text{TEM}_{0,2}$) is degenerate with the frequency of the $q + 1$, $\text{TEM}_{0,0}$ as shown in Figure 5.2. When aligning a laser beam into a confocal cavity, one typically couples light into the higher order transverse modes as well as the $\text{TEM}_{0,0}$ mode. However if a $\text{TEM}_{0,0}$ laser beam is exactly aligned with the cavity axis, it will only couple to the $l + m = \text{even}$ modes. When this happens, every other peak in Figure 5.2 would begin to drop in amplitude as the exact alignment is reached.

There are other possible mirror configurations that result in a cavity where the transverse modes are frequency degenerate with the longitudinal modes. For example instead of have the $l + m = 2$ transverse mode match the $q + 1$ longitudinal mode, one could have the $l + m = 3$ or $l + m = 4$ transverse mode match the $q + 1$ mode. Each of this conditions occurs for a different mirror separation. These mirror separations may be calculated using the frequency separation between the transverse modes, which is given by

$$\Delta\nu_{l,m} = \frac{c}{2\pi n(z_2 - z_1)} \Delta(l + m) \left(\tan^{-1} \frac{z_2}{z_o} - \tan^{-1} \frac{z_1}{z_o} \right) \quad (5.7)$$

where $z_o = \pi w_o^2 n / \lambda$ and w_o is the beam waist [116]. By setting $FSR = p \cdot \Delta\nu_{l,m}$ (where p is an integer), the mirror separation $z_2 - z_1$ can be obtained. The first few mirror spacings are given in Table 5.1 for 5 cm radius of curvature mirrors. Shorter cavities have larger mode spacings which is advantageous since we do not want any nearby modes (transverse or longitudinal) that might lase on processes other than the two-photon Raman process. In addition, the finesse is constant regardless of the mirror separation so the larger free spectral ranges mean that the frequency width of the transmission peak is larger. This is important because we will be building a very high finesse cavity (necessary because the two-photon gain is so small) yet we want

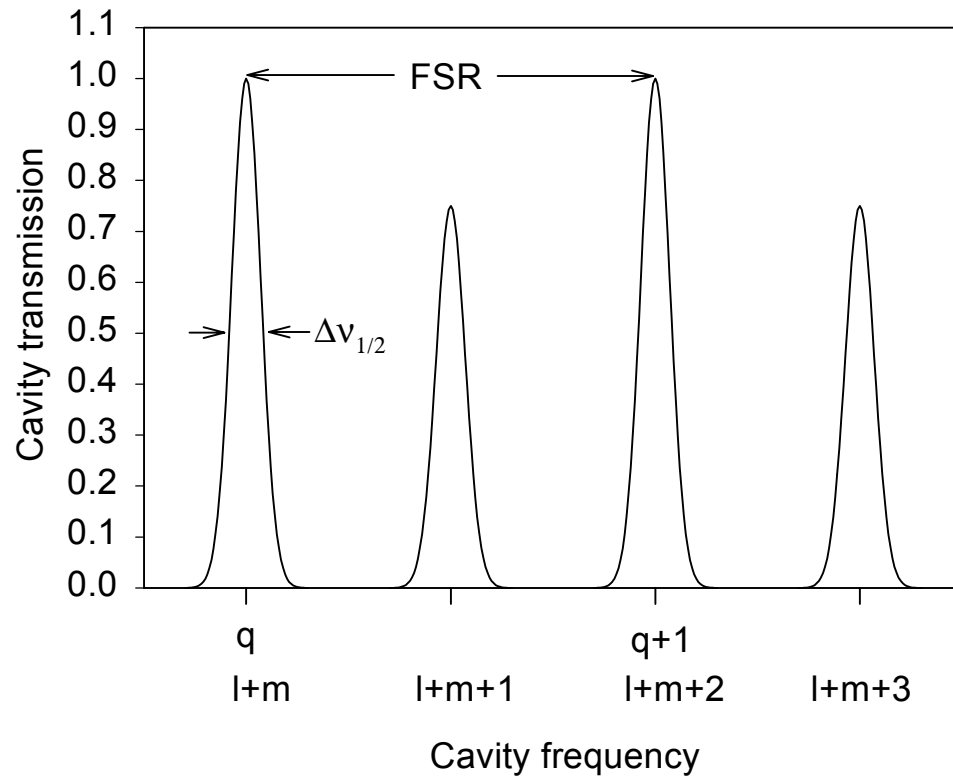


Figure 5.2: Frequency spectrum for the longitudinal and transverse mode frequencies for a confocal cavity

the transmission peak width to be on the order of the diode laser linewidth that we must couple into the cavity. The diode laser linewidth is ~ 1 MHz and hence it will be hard to couple any diode laser light into the two-photon laser cavity if the cavity linewidth is much smaller than this.

mirror separation (cm)	transverse mode spacing (GHz)	longitudinal mode spacing (GHz)
5	1.5	3
2.5	2.0	6
1.465	2.56	10.2
0.955	3.16	15.8

Table 5.1: Mirror separation, transverse mode spacing and longitudinal mode spacing for a cavity consisting of two 5 cm radius of curvature mirrors.

An example of a shorter (or sub-confocal) cavity transmission spectrum is given in Figure 5.3. For this spectrum, the $l + m = 4$ transverse mode is frequency degenerate with the $q + 1$ longitudinal mode. As in the case of the confocal cavity the intermediate transverse modes will be suppressed if the input laser is perfectly aligned with the cavity axis. That is, every fourth peak will become larger and the others will decrease in size.

There is one other problem that must be taken into consideration in the design of the cavity. The Hermite-Gaussian modes are actually the solutions for parabolic mirrors, NOT spherical mirrors. Typically, this is not a problem because the spherical surface is very close to the parabolic surface in the paraxial limit (*i.e.* for small transverse mode numbers). However, for very large transverse mode number this is not the case. For example, if the cavity length is set so that $l + m = 4$ mode is degenerate with the $q + 1$ mode, the $l + m = 100$ mode will not be degenerate with the $q + 25$ mode. This could be problematic for the two-photon Raman laser because the gain medium will be large enough to overlap with the high order transverse modes and thus gain features near the two-photon Raman feature could lase on very high

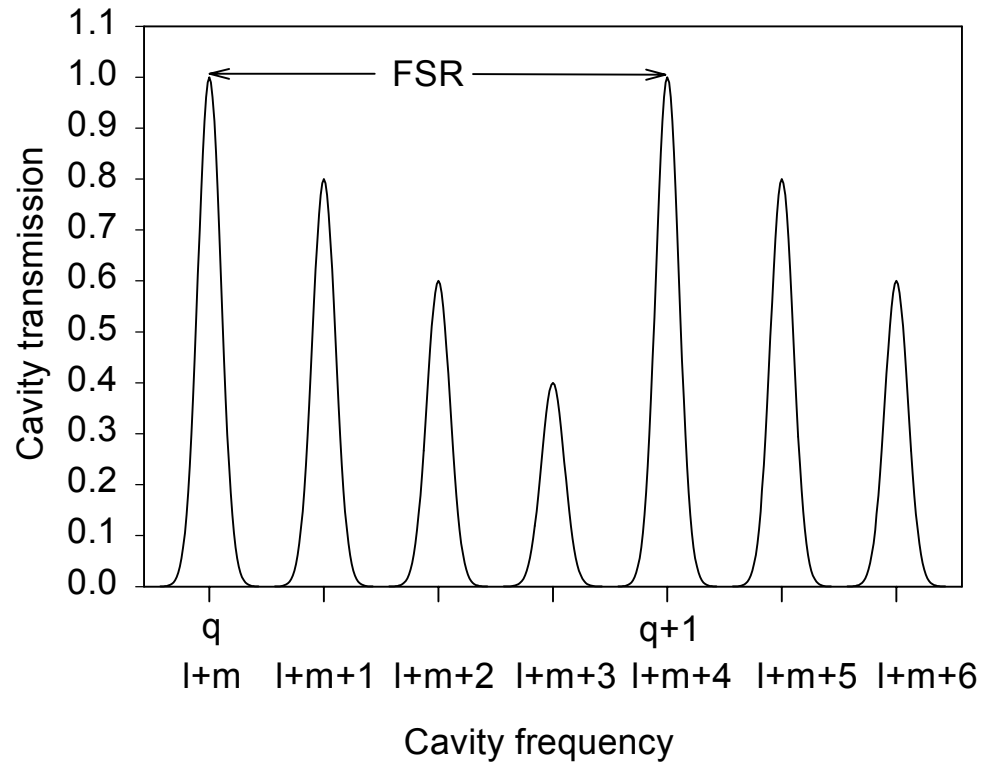


Figure 5.3: Cavity transmission spectrum for the case when the $l+m=4$ transverse mode is frequency degenerate with the $q+1$ longitudinal mode.

order transverse modes while the two-photon laser is lasing on the $\text{TEM}_{0,0}$ mode. One solution to this problem is to place apertures in the cavity that block part of the mirror surface. If chosen correctly, the apertures will not attenuate the $\text{TEM}_{0,0}$ mode but will attenuate the very high order transverse modes.

With these ideas in mind, I will move on to the specifics of the cavity actually used for the two-photon Raman laser.

5.3 Cavity design requirements

In designing the two-photon laser cavity there were a several requirements that needed to be met. First, the two-photon gain was expected to be less than 10^{-3} , necessitating very high reflectivity mirrors and no additional losses within the cavity. Next, the cavity must lase only on the two-photon Raman feature and not on any of the one-photon gain features. Fortunately, all of the other gain processes (one-photon Raman, one-photon dressed-state, and two-photon dressed state) are within 1 GHz of the two-photon Raman frequency. Thus, no cavity modes will be resonant with other gain features as long the free spectral range (FSR) is larger than ~ 1 GHz.

In order to match a cavity frequency to the two-photon lasing frequency, the length of the cavity must be tunable by at least one free spectral range. This tunability must be very accurate since the FSR will be > 10 GHz but the linewidth of the two-photon gain feature is ~ 10 MHz. In addition, the cavity length must be stable to thermal fluctuations. For example, a 5 cm length of stainless steel will undergo a change in length of 850 nm for a temperature change of 1° K, which would be just greater than two FSR's for 770 nm light. Thus, materials with low coefficients of thermal expansion must be used and the cavity must be isolated from thermal fluctuations. In addition to the fine cavity length adjustment, there must be a coarser cavity length adjustment since the cavity will initially be aligned in air, but will be used in

a vacuum chamber. For a 5 cm cavity, the difference in the index of refraction of air ($n = 1.000293$) [117] and the index of refraction of vacuum ($n = 1$) changes the effective cavity length by 14.6 microns. This length difference is sufficient to lift the mode degeneracy between the longitudinal and transverse modes of the cavity so we must be able to change the length to reestablish the mode degeneracy.

As noted in the previous section, we will also need apertures inside the cavity that are small enough to block the very high order transverse modes of the cavity but large enough as not to attenuate the TEM_{0,0} mode (thus acting as a source of loss in the cavity). Due to spherical aberrations, the higher order transverse modes have a different resonance frequency than the TEM_{0,0} mode. This allows the possibility that higher order transverse mode will lase on a large gain feature (*e.g.*, the one-photon Raman gain) when the TEM_{0,0} mode is resonant with the two-photon Raman process. The final constraint is that all of this apparatus must fit inside the vacuum chamber.

The solutions to these requirements included several state-of-the-art pieces of mechanical hardware. Very high reflectivity mirrors were used to provide sufficient buildup for the two-photon Raman gain to lase. Short cavity length adjustments were performed by a piezoelectric transducer. Commercial picomotor screws (from New Focus, Inc.) were used for the larger cavity length adjustments. To minimize thermal drift, Super-Invar steel was used in the construction of the cavity. The following paragraphs provide the details for each of these pieces of hardware.

The first set of mirrors we tried were special ordered from VLOC and were supposed to be very high reflectivity at 770 nm with a 5 cm radius of curvature. However, the finesse was only about 3000 (indicating a buildup of ~ 1000) when we set up the cavity with these mirrors. This was less than specified and insufficient for the two-photon laser, since the maximum gain we had observed was only 4×10^{-4} (see Chapter 4). This was a significant setback since these types of special orders

take several months and cost several thousand dollars. Fortunately, we were able to obtain another set of mirrors from REO (Research Electro-Optics Inc.) that were already coated for 770 nm. These mirrors have a diameter of 7 mm and a radius of curvature of 5 cm. The transmittivity of the mirrors is <300 ppm, the absorptive losses are <5 ppm, and the outside surface is flat with a anti-reflection coating of $R < 0.25\%$. In order to increase the cavity linewidth, we set the distance between the mirrors at 1.465 cm (see Table 5.1). This means that the fourth order transverse mode overlaps with the next longitudinal mode. For this configuration, the FSR is 10.24 GHz and the estimated finesse (using $R_1 = R_2 = 0.9997$) is $\sim 10,500$. Using these two numbers the estimated linewidth of the cavity is 0.97 MHz.

In order to translate linearly one of the mirrors by small lengths, a piezoelectric transducer (or PZT) is used - a length change of only 385 nm (one-half of the light wavelength) moves the cavity from one resonance to the next one. Piezoelectric materials have the properties that an applied mechanical stress induces an electric potential and vice versa. The material in our PZT's is lead zirconate titanate, a ceramic manufactured by EDO Corporation [118]. Since we want to use a single PZT to move a cavity mirror, a thin-wall tube poled from the inner curved surface to the outer curved surface is used, so an electric field applied across the radius of the tube changes the length of the tube. This length change is given by

$$\Delta L = \frac{2d_{31}V L}{OD - ID} , \quad (5.8)$$

where d_{31} is a property of the material ($d_{31} = -262 \times 10^{-12} \text{m/V}$ for our PZT's), V is the applied voltage, L is the length of the PZT, OD is the outer tube diameter, and ID is the inner tube diameter [119]. The tube used in the cavity is $0.5'' = 1.270$ cm long with an outer diameter of $0.5'' = 1.270$ cm and an inner diameter of $0.463'' = 1.176$ cm, with a maximum applied voltage of ~ 500 V. These dimensions

and maximum voltage give a maximum length change of $\Delta L = 3.5 \times 10^{-6}$ m, which is about ten FSR's.

While the PZT works well for scanning the cavity length over several free spectral ranges, it is insufficient for the 15 μ m length cavity needed to compensate for the transition from air to vacuum. It would be possible to obtain PZT's with larger length translations, but they would require larger voltages and would suffer from the fact that PZT's drift over time. This drift is not a problem if one is just scanning the cavity or using feedback to lock to a line, but it would be much better to have a stable method of changing the cavity length to compensate for index of refraction changes. A solution to this problem is the Picomotor from New Focus, Inc., which is a 1/4–80 screw that is turned by two jaws driven by a PZT [120]. By slowly expanding the PZT, the jaws do not slip and thus move the screw, but quickly contracting the PZT results in slippage between the jaws and the screw, allowing the jaws to move without turning the screw. Large distances may be moved with very fine resolution by repeating this slow expansion - fast contraction many times. The process may be reversed to turn the screw in the opposite direction. These devices are static so that they do not move unless the PZT is actuated. The three picomotors used in the cavity have a maximum range of 0.25" with a resolution < 40 nm and are installed at three corners of movable endplate, allowing the mirror to be translated linearly or tilted in the x or y direction.

The PZT's and the picomotors allow for sensitive adjustment of the cavity length, but they will be useless if the cavity length is changing wildly due to thermal drifts and acoustic vibrations. To this end, we built a very rigid cavity out of materials with low coefficients of thermal expansion. Stainless steel, for example, has a coefficient of thermal expansion of $17 \times 10^{-6}/^{\circ}$ C [121] so a 1° C temperature change for a 5 cm cavity will change the length by 850 nm, which is two FSR's. Our cavity is built

from Super-Invar steel, which has a coefficient of thermal expansion of $0.3 \times 10^{-6}/^{\circ}\text{C}$, 50 times smaller than stainless steel.

5.4 Mechanical design of the chamber and cavity

5.4.1 The Can

The entire apparatus for the two-photon laser cavity is contained inside a large vacuum chamber, nicknamed “The Can.” This chamber is fashioned from a section of 8” beam line (8” diameter stainless steel pipe) and is 18.5” long. There are three sets of radial ports, the first set contains eight ports equidistant apart, the second set contains four ports and the third set contains three ports, as shown in Figure 5.4. The two-photon laser cavity is situated horizontally, even with the first set of ports. The pump beams intersect the cavity vertically and one of the diagonal ports is used for electrical feedthroughs. Anti-reflection coated windows are used on the four ports where laser light enters or leaves the can. In the second set of ports the vertical ports also have windows, which are used for a weak probe laser which is used to check the atomic beam flux and verify the optical pumping. Vacuum is maintained with the same combination of pumps used in the beam apparatus, a turbo-molecular pump backed by a mechanical pump. The typical pressure in the can is less than 10^{-7} Torr. The oven apparatus is also the same as was used in the pump-probe experiments. The only change is that the second aperture is now inside the can and is mounted on an X-Y translation stage so that the atomic beam can be steered by a small amount, permitting accurate alignment of the atomic beam with the center of the laser cavity.

The Can

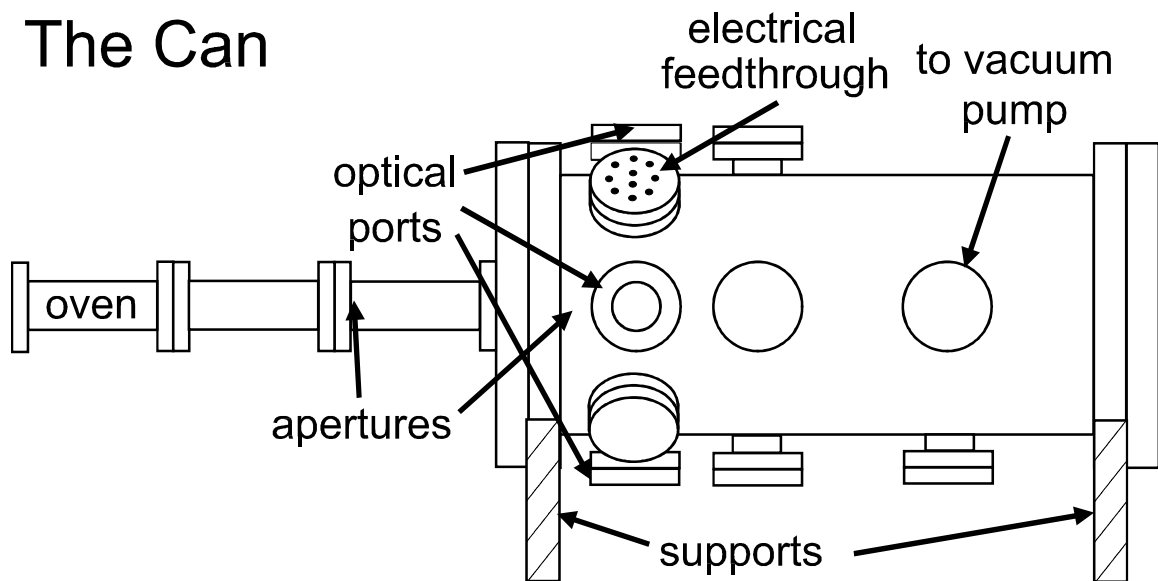


Figure 5.4: Schematic of the vacuum chamber used for the two-photon laser. The chamber is a piece of stainless steel pipe set on its side. All ports point radially outward. The baseplate on the left has a 1.5" diameter hole through which the atomic beam passes.

5.4.2 The optical cavity

Figure 5.5 shows a schematic of the two-photon laser cavity, which is formed by two $1/2''$ thick Super-Invar plates separated a distance of $2.692''$ by $0.685''$ diameter Super-Invar rods. Mounted in the left endplate is the $1/2''$ long cylindrical PZT with the 7 mm diameter high reflectivity mirror glued on top. On the right side, the mirror is mounted on a $1/4''$ Super-Invar plate which rests on three picomotor screws which are bolted into the endplate. The right mirror is glued in a $1/2''$ extender that is screwed into the small plate. The cavity is bolted to a superstructure that is attached to the plate closest to the atomic beam. The mounts for the cavity apertures hang from the side of the superstructure.

Figure 5.6 illustrates the view of the cavity from the end with the picomotors. The mirror is not attached to this end plate, but rather to a smaller plate that is held in place against the picomotors by two stiff springs. The four Super-Invar rods are located at the four corners and the three picomotors are set in an L. This L configuration allows easy adjustment in the x or y direction by using a single motor, or longitudinal translation by using all three motors.

To help with visualization of the cavity configuration, I have included two photographs of the apparatus. Figure 5.7 is a top down view of the cavity. One can clearly see the baseplate (at the bottom of the picture) and the superstructure that holds the cavity assembly and the aperture mounts. Also attached to the baseplate is the X-Y stage to which the movable atomic beam aperture is mounted. Directly in the middle of the picture, one can see a cavity mirror on the right side and the tops of two picomotors on the left side.

A closer view is given by Figure 5.8 where both cavity mirrors can be seen. In this picture all three picomotors are visible on the left as well as the PZT on the right.

Laser cavity - side view

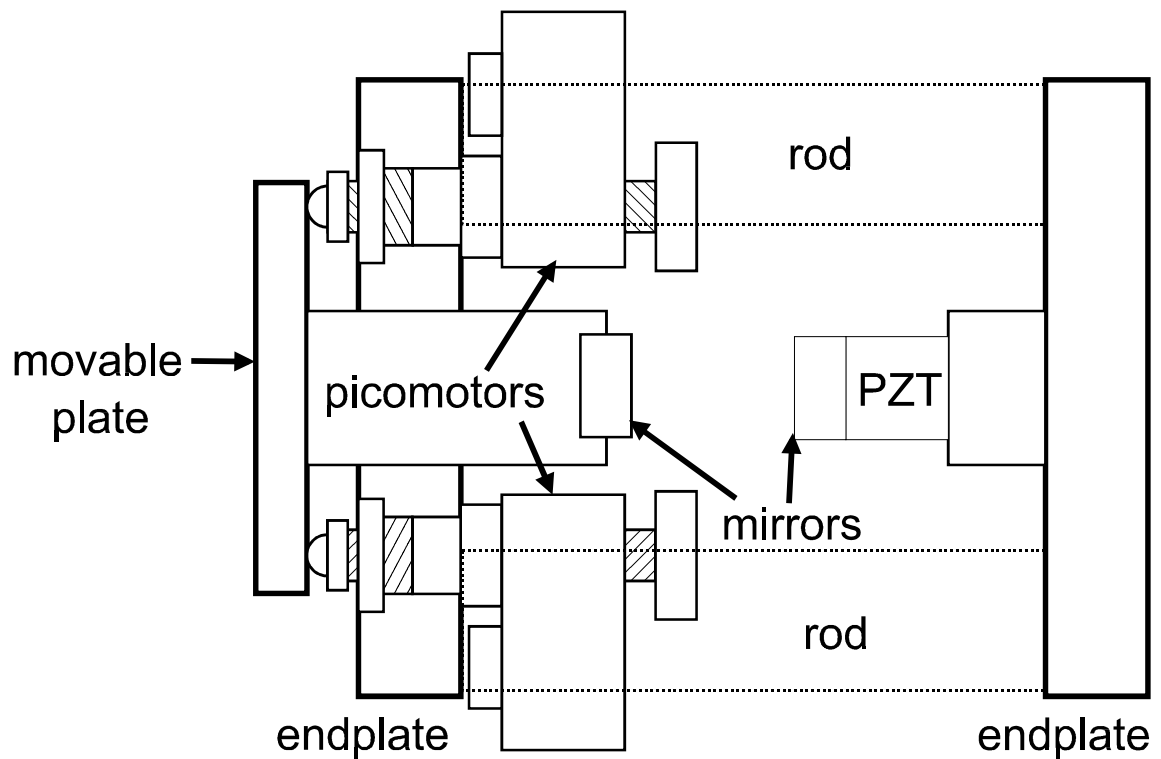


Figure 5.5: Side view of the two-photon laser cavity. The picomotors permit coarse translation of the right mirror, up to 0.25" with a resolution of 40 nm, while the PZT allows fine translation of the left mirror, up to $\sim 3.5 \mu\text{m}$.

Laser cavity - end view

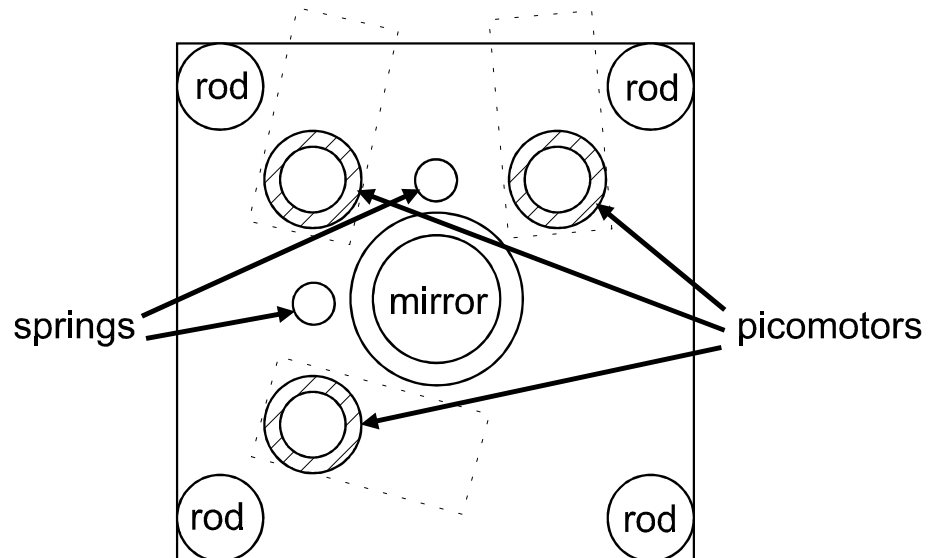


Figure 5.6: End view of the two-photon laser cavity. The large center hole is where the mirror mount screws into the endplate. The three picomotors are angled so that they do not interfere with the pump laser and atomic beam paths.

Figure 5.7: Top view looking down into the cavity. The cavity axis is horizontal, the atomic beam goes from the bottom to the top and the pump laser beams come out of the page. The apparatuses at the top of the picture are the X-Y stages used to align the cavity apertures.

Figure 5.8: Close view of the cavity clearly showing the two mirrors in the middle of the picture. Note the PZT behind the right mirror and the three picomotor screws around the left mirror. The two coils provide the magnetic field for the interaction region.

Coming down from the top are the two cavity apertures that will move directly in front of the mirrors. There is enough travel on the picomotor screws so that these apertures may be completely removed from the cavity. Finally, one can see the two coils that provide the magnetic field for the interaction region.

5.5 Cavity Operation

5.5.1 Cavity alignment

Injecting the probe laser beam into the high finesse cavity is not a simple task. In order to obtain maximum coupling and throughput, it is necessary that the probe beam is well aligned with the cavity axis and that the probe beam parameters, *i.e.*

the spot size and beam convergence, are matched to the cavity's $\text{TEM}_{0,0}$ mode. The most difficult part is the preliminary alignment. The technique we used was to simultaneously center the probe laser spot on the input mirror and overlap the incoming beam with the beam reflected from the front surface of the mirror. If the two mirrors are reasonably parallel, this technique generally couples some light into the cavity. Scanning the probe laser or the cavity length (to change the frequency) produces a cavity output similar to Figure 5.9 where the transmission peaks are roughly the same size. The next task is to maximize the peak heights by adjusting the position and direction of the input probe beam.

Once some light is coupled into the cavity, the higher order modes are suppressed by optimizing the probe alignment with the cavity axis. As noted before, this cavity is set up in a sub-confocal arrangement where the fourth order transverse mode is frequency degenerate with the next longitudinal mode. Thus, when the cavity is well aligned, every fourth peak becomes larger while the three intermediate peaks drop in size. To find where the cavity axis is located, we look at the output spots. When misaligned, there will be multiple output spots; once these multiple spots are observed we adjust the input beam alignment so that these spots move together and ultimately are on top of each other. In our case the initial transmission was so small that we used an infrared viewer and looked directly into the cavity. Once the intermediate peaks start to decrease in size it is usually possible to adjust the probe beam by just looking at the cavity transmission. At this stage it is important to optimize the input beam parameters, by adjusting the position or focal length of the lens used to focus the laser beam into the cavity. Figure 5.10 shows the cavity transmission for a probe beam that is very close to alignment with the cavity axis.

Once the cavity is well-aligned it is possible to check several parameters of the cavity. We measured the throughput of the cavity at a cavity resonance and found

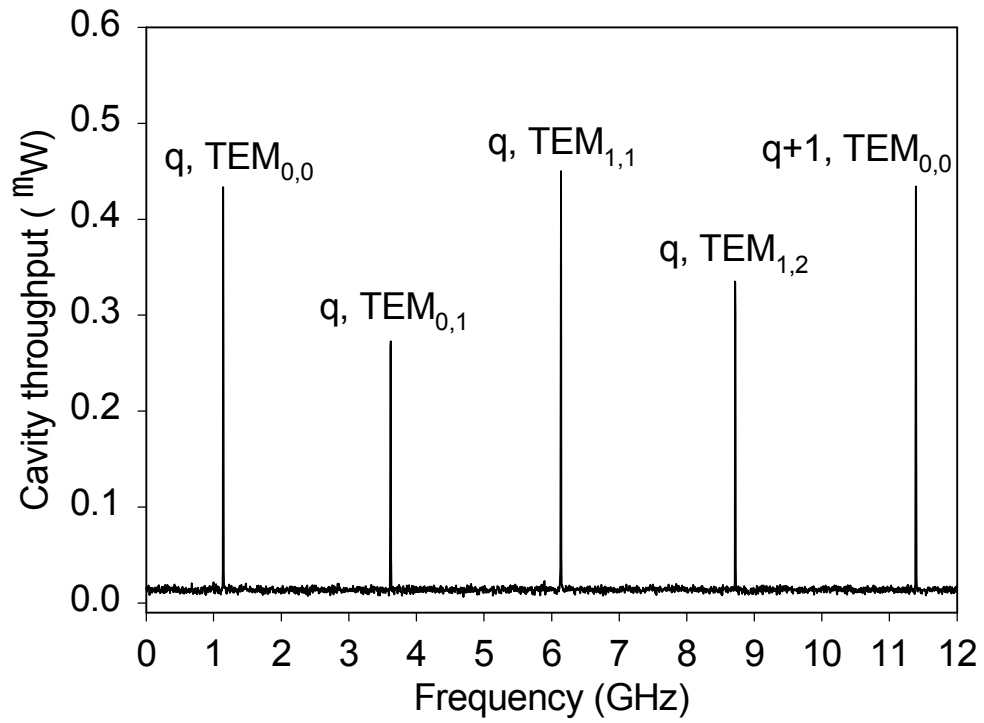


Figure 5.9: Cavity transmission as a function of frequency for a misaligned probe beam. All the peaks are roughly the same size.

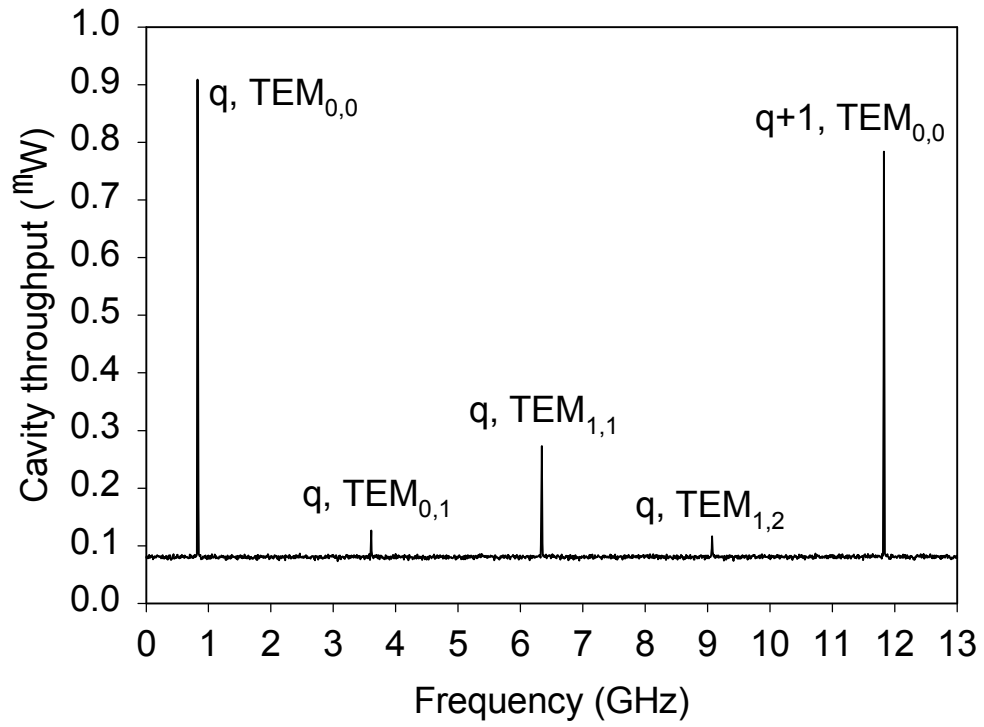


Figure 5.10: Cavity transmission for a nearly aligned probe beam. Every fourth peak increases in size and the intermediate peaks decrease in size.

it to be in excess of 50%. Ideally, 100% transmission should be possible, but losses in the cavity mirrors or less than perfect beam alignment will reduce the transmission. Figure 5.1 illustrates the decrease in the maximum cavity transmission as a function of the ratio of the losses due to absorption to the losses due to transmission. The expression for this graph is

$$\left(\frac{I_t}{I_i}\right)_{\max} = \frac{1}{(1 + \alpha/\beta)^2} ,$$

where I_i and I_t are the incident and transmitted intensity, α is the fractional intensity loss per pass and β is the transmitted loss per pass. The mirrors from REO were expected to have a transmittivity of <300 ppm and losses of <5 ppm, which would give a ratio of 0.016 and an expected maximum cavity transmission of 97%. The cavity is extremely sensitive, any misalignment of the input beam, small amount of contaminate on the mirrors, or mismatch between the cavity and laser beam modes will significantly reduce the measured maximum cavity throughput.

The next measurement we obtained was of the cavity finesse. One possible way to measure the finesse is to measure the ratio of the full width at half maximum of a transmission peak and the free spectral range (the distance between adjacent peaks). It was impossible to make an accurate measurement using this technique, because the linewidth of the diode laser used for the probe beam was on the same order as the cavity transmission peak width, making it difficult to exactly measure the cavity linewidth. Our preliminary measurement indicated that the cavity finesse was in excess of 3,000.

We also optimized the cavity length by making sure that the higher order transverse modes were degenerate with the TEM_{0,0} modes. This was necessary since the initial mirror alignment was performed outside the vacuum chamber and difference in the refraction indices of air and vacuum are significant enough at this level to

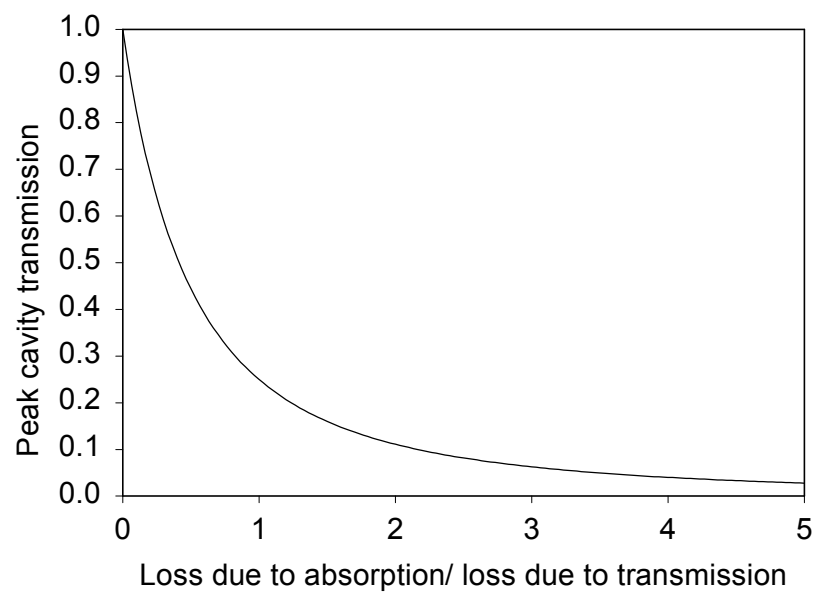


Figure 5.11: Maximum cavity transmission as a function of the ratio of the losses due to absorption or scatter to the losses due to transmission.

noticeably change the optimum cavity length.

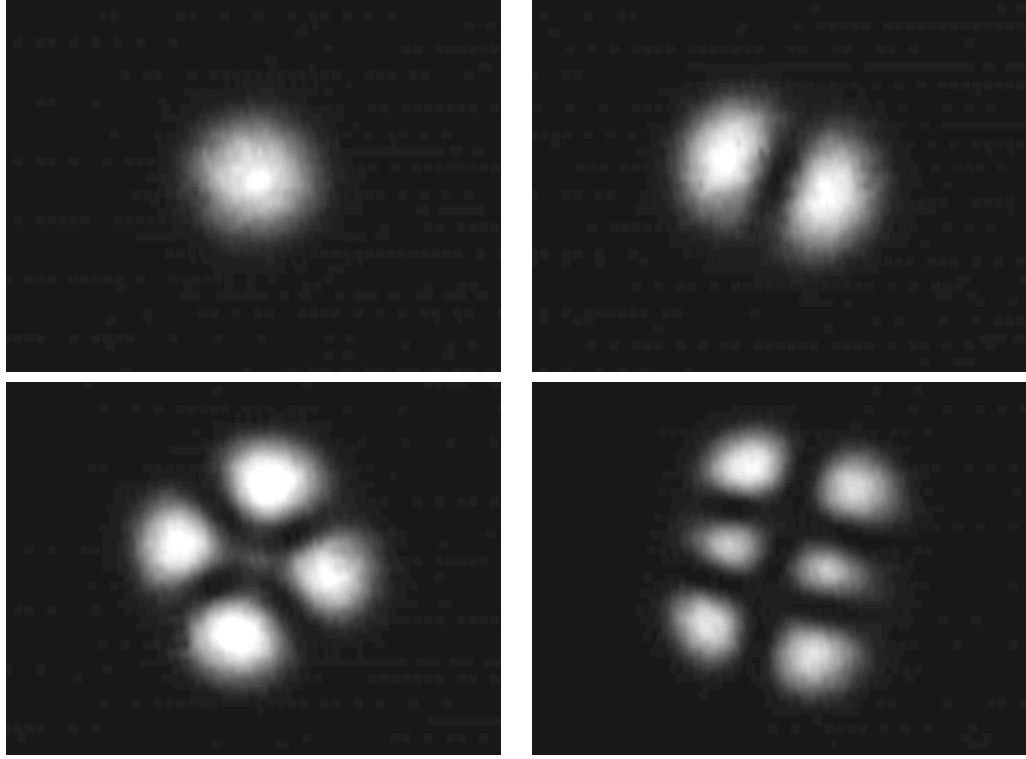


Table 5.2: Output spots for each of the four lowest order transverse modes.

To check that the input laser was aligned with the cavity, we photographed the output spot for various resonance frequencies. These are displayed in Table 5.2. The top left picture is a TEM_{0,0} mode, the top right picture is a TEM_{1,0} (or TEM_{0,1}), the bottom left picture is a TEM_{1,1}, and the bottom right picture is a TEM_{1,2}. Looking back at Figure 5.10 the left most peak is a TEM_{0,0} mode and the next three peaks are the TEM_{1,0}, TEM_{1,1}, and TEM_{1,2} modes respectively. The peaks then repeat starting again with the TEM_{0,0}. These pictures were obtained with a high sensitivity CCD camera which was connected to the computer via a Snappy Frame Grabber [122]. This permitted us to acquire pictures from the camera and turn them into a computer readable format. Our input laser beam was so well mode-matched with the TEM_{0,0} mode of the cavity that we had to misalign the input laser beam to

couple enough light into the transverse modes so we could take these pictures.

5.5.2 Cavity Finesse

Since we were unable to precisely measure the finesse using the width of the transmission peak, we needed an alternative method that is more suitable to high finesse cavities. One experimental method is to couple some light into the cavity and then quickly shut off the light source by using a shutter or by shifting the cavity resonance [123]. The larger the cavity finesse, the longer light will remain in the cavity. This is referred to as the ringdown time of the cavity. To make this measurement, it is important to shut off the input light much faster than the ringdown time. For our system we used an acousto-optic modulator (an AOM, see Chapter 4 for further explanation) to quickly shut off the diode beam. A high speed Hamamatsu S3994 detector was used to look at the ringdown from the cavity. This ringdown is just an exponential decay, so fitting a curve to the data provides an accurate measurement of the ringdown time and hence the finesse of the cavity.

The first step in this measurement is to place the high speed detector right after the AOM, (but in the front of the cavity) and capture a falling edge of the optical pulse. Such a feature is shown in Figure 5.12. This allows us to verify that the falling edge is indeed sharp and that the bandwidth of the detector and electronics is sufficient to capture the decay curve. For the edge feature in Figure 5.12, the 90% to 10% fall time is approximately 50 nanoseconds and the complete switch from on to off occurs in less than 100 nanoseconds. Once this has been verified, the detector is placed after the cavity and the experiment is repeated.

Figure 5.13 shows the decay curve as light leaves the cavity after the injected pulsed is turned off. The curve has been given a small offset and normalized to one

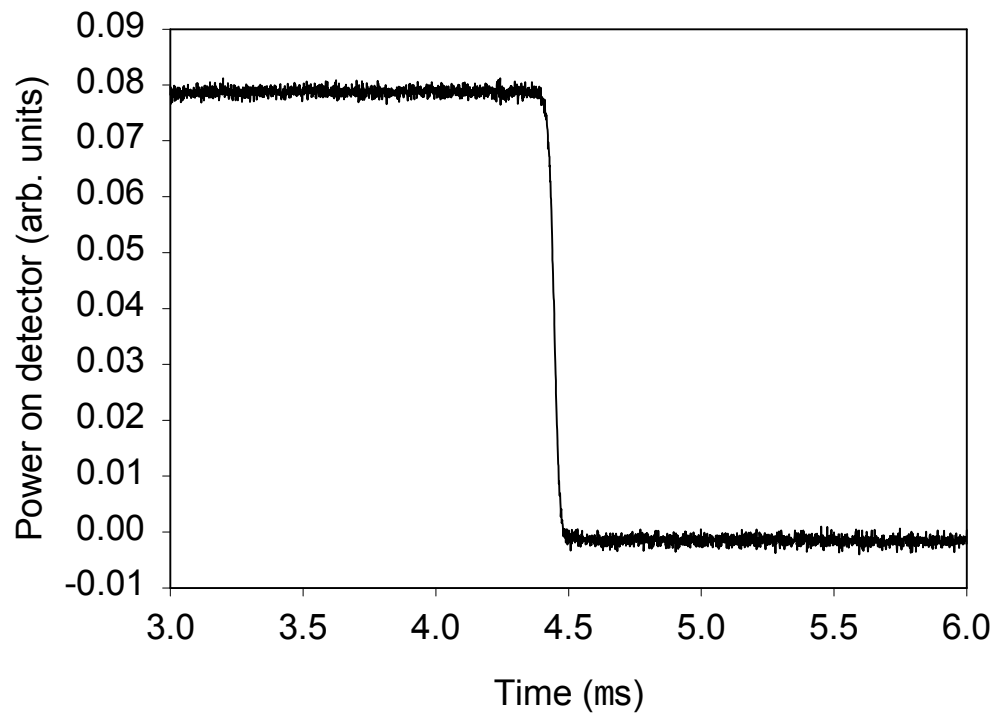


Figure 5.12: The falling edge of the optical pulse as seen by the detector in front of the cavity. The fall time is roughly 50 nanoseconds.

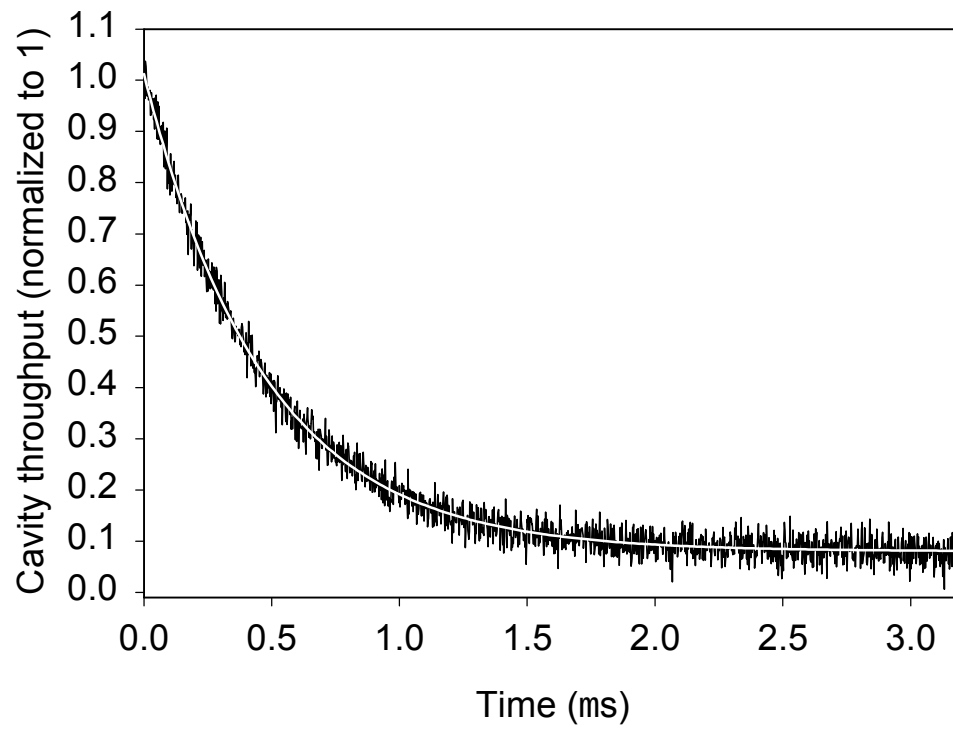


Figure 5.13: The falling edge of an optical pulse passing through the cavity as seen by a detector placed after the cavity. The $1/e$ decay time is $0.471 \pm 0.003 \mu\text{s}$.

to simplify the fitting process. An exponential decay curve of the form

$$y = y_o + a \cdot \exp(-bt) \quad (5.9)$$

is fit to the experimental data. For Figure 5.13, the fit values are $y_o = 0.0801 \pm 0.0009$, $a = 0.9323 \pm 0.0029$, and $b = 2.1248 \pm 0.0123$, and the fit curve is displayed in white. The 1/e cavity lifetime is the reciprocal of b, so for this data $\tau_{cavity} = 0.471 \pm 0.003 \mu s$. Once the cavity lifetime is calculated, the finesse may be obtained from the relation

$$F = \pi \cdot \tau_{cavity} \cdot FSR . \quad (5.10)$$

Since the free spectral range is 10.24 GHz, the cavity finesse is 15,140. The buildup due to the cavity is just the finesse divided by π , which for our cavity is 4,800. The gain necessary to sustain lasing is the reciprocal of the buildup, so we need a single pass gain of 2.1×10^{-4} , which is less than the maximum gain we observed in Chapter 4. We can also calculate the linewidth of the cavity since the finesse is just the ratio of the FSR to the cavity linewidth.

$$\Delta\nu_{1/2} = \frac{FSR}{F} = \frac{10.24 \text{ GHz}}{15,140} = 676 \text{ kHz} \quad (5.11)$$

Note that this measurement was made with the cavity apertures in place. Without the cavity apertures the finesse is 15,400.

5.5.3 Apertures in the cavity

In order to figure out what size apertures to try in the cavity, we calculated the expected size of the TEM_{0,0} mode in the cavity. Using simple Gaussian optics and the facts that the beam radius is infinite in the middle of the cavity and matches the mirror curvature on the each mirror surface, we can easily find that the beam waist in the middle of the cavity is 65.8 μm and 71.2 μm on each mirror. The TEM_{0,0} mode

is just a Gaussian so we can easily calculate the aperture size that blocks a given proportion of the beam power. For example, the aperture size that blocks 1 part in a million of the beam power for the TEM_{0,0} mode is 374 μm . Another method for estimating the aperture size we will need is given in Hercher's paper on spherical mirror cavities [124]. Using simple geometrical arguments, Hercher calculates the effects of spherical aberration on the beam path in the cavity. In other words, this is an estimate of when the off axis path difference is enough to change the resonance frequency by 1/2 of the cavity linewidth. The spot radius when this occurs is given by

$$p_s = \left(\frac{R^3 \lambda}{F}\right)^{1/4} \quad (5.12)$$

where R is the radius of curvature of the mirrors, λ is the wavelength of the light, F is the cavity finesse, and p_s is the spot radius. Using $R = 5 \text{ cm}$, $\lambda = 770 \text{ nm}$, and $F = 15,000$, we find that $p_s = 283 \mu\text{m}$, or the spot diameter is 566 μm . With this as a guide we ordered several apertures, with 300, 350 and 400 μm sized pinholes.

To gauge the effect of the apertures on the cavity, we placed each size aperture in front of a cavity mirror and measured the cavity throughput as a function of its horizontal position. This gave us a rough measure of the effect of the apertures on the TEM_{0,0} mode. A more accurate measurement was made by measuring the cavity finesse with the apertures in place. Any addition losses due to the apertures will show up as a reduction of the finesse. The results for apertures of sizes 300, 350 and 400 μm are show in Figure 5.14. The amount of light input is slightly different in each case, but it is still easy to see that the 300 μm aperture never lets all of the light through. The 350 micron aperture does let all of the light through and the 400 μm aperture appears to be too large. However, when we measured the finesse for two 350 μm apertures, it was 13,100, which is significantly less than the empty cavity finesse of 15,400. Next, we tried one 350 and one 400 μm aperture, but there was

still a reduction of the finesse to 14,840. In the end, we used two 400 μm apertures which decreases the finesse to 15,140. All of the data in Chapter 6 was taken with two 400 μm apertures in the cavity.

As a final check of the aperture sizes, we calculated the size of the transverse cavity modes. This allows us to determine which transverse modes are highly attenuated by the cavity apertures. The hope is that the apertures affect a relatively low order transverse mode. As an approximation I calculated the horizontal location of the last spot of the light of the $\text{TEM}_{0,m}$ mode for several values of m . Using Mathematica, I numerically obtained the location of the last peak for the Hermite-Gaussian's given in equation 5.6. This calculation assumes a confocal cavity, so the results are not completely accurate for our sub-confocal cavity, but will be close. The radial position of the last lobe for the first 8 transverse modes is given in Table 5.3. Doubling the radii to obtain a diameter, we see that a 400 μm aperture starts to significantly block the last lobe for a $\text{TEM}_{0,5}$ mode. This is a relatively low number, so high order transverse modes should not be a problem for our cavity with two 400 μm apertures in front of the mirrors.

m	last lobe position (μm)
0	0
1	78.2
2	123.8
3	159.2
4	189.2
5	215.7
6	239.7
7	261.8
8	282.3

Table 5.3: Approximate spatial size of the $\text{TEM}_{0,m}$ modes.

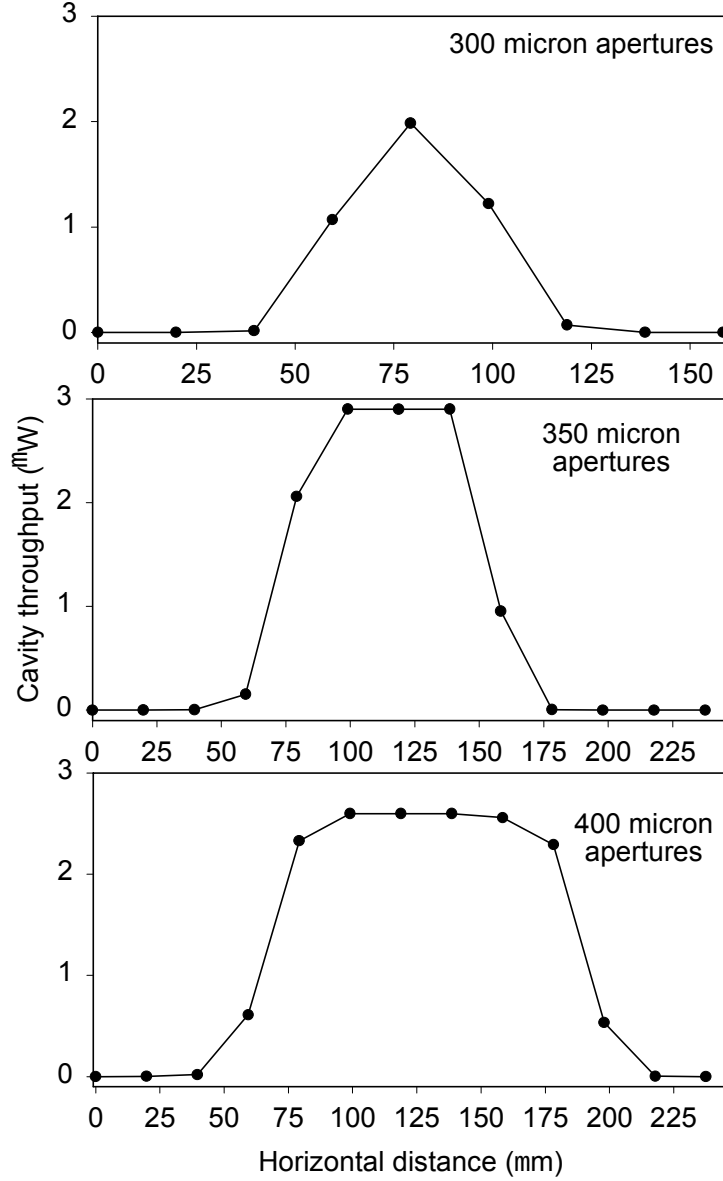


Figure 5.14: Cavity throughput as a function of the transverse location of the aperture. The 300 μm aperture does not allow maximum transmission for any aperture location, the 350 μm aperture permits maximum transmission over a distance of 40 μm , and the 400 μm aperture over a distance of 60 μm .

5.5.4 Two-photon lasing without cavity apertures

As an experimental check of the necessity of the apertures, I measured the power emitted by the cavity as a function of the cavity frequency without any apertures in front of the mirrors for a Raman pump detuning of +85 MHz. Figure 5.15 shows the output of the cavity as a function of the cavity frequency. The probe is not being injected, so the observed lasing is either one-photon lasing or two-photon lasing initiated by one-photon lasing. For large pump powers (a Raman pump power of 260 is illustrated in Figure 5.15(b)) lasing occurs for all cavity frequencies and we see a “tail” of higher order modes to the left of each of the low order modes. If the pump power is reduced to 32 mW, we are left with just the lowest order modes lasing on the two one-photon Raman features, as can be seen in Figure 5.15(a). The largest peak is the blue one-photon Raman feature lasing on a longitudinal mode (*i.e.* the transverse mode is TEM_{0,0}). The tail to the left of the largest peak consists of higher order transverse modes lasing as their frequency passes through the gain maximum of the one-photon Raman feature. Note that for both pump powers lasing occurs for all cavity frequencies. This data provides the experimental verification that cavity apertures are required for the two-photon Raman laser.

5.5.5 High speed detector

One additional piece of equipment is needed to observe the output of the two-photon laser. We require a detector system with high sensitivity AND high speed. The laser output is expected to be a maximum of a few microwatts and we want to be able to observe signals down to a few tens of nanowatts. In addition, the turn-on dynamics of the laser will probably occur in a time period on the order of the cavity decay time, which we have already measured to be $\sim 1/2$ a microsecond, so we would need a detector with a bandwidth of at least 10 MHz. There are numerous

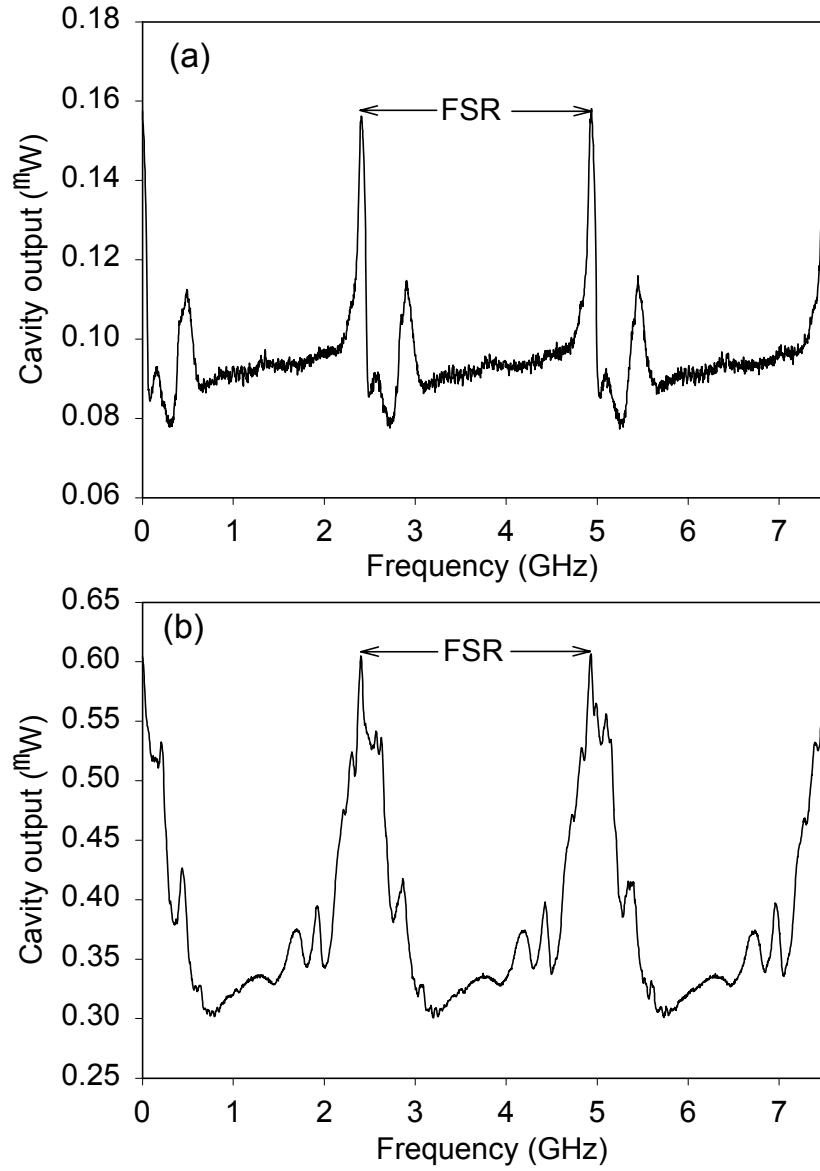


Figure 5.15: Cavity output as a function of cavity frequency with no apertures inside the cavity. Graph (a) is for a Raman pump power of 32 mW and graph (b) is for a pump power of 260 mW. Note that lasing now occurs at all frequencies.

photodiode detectors which are high speed and very quiet. However, it is difficult to find amplifiers which have the combination of speed and low noise necessary to turn the current output of the photodiodes into an easily measured voltage signal. Typically, the input noise of the amplifier is a sizable fraction of the photodiode signal for small amounts of light. The solution to this problem is an avalanche photodiode. This is a photodiode that uses an electronic avalanche system as the first stage of gain and then uses a transimpedance amplifier. The avalanche stage boosts the signal well above the noise of the transimpedance amplifier, thus vastly improving the signal to noise of the whole system, while maintaining the high speed of the device. This system is more advantageous than a photomultiplier because it has a high quantum efficiency and it is small, easy to use device. The APD (avalanche photodiode) we obtained is a Hamamatsu C5460 and has a response of $1.31 \text{ V}/\mu\text{W}$ and a bandwidth of 10 MHz [125]. The entire detector and amplifier apparatus is the size of a credit card, facilitating deployment in the lab.

5.6 Conclusion

In this chapter I have described in-depth the optical resonator used for our two-photon laser experiment. This cavity is sub-confocal and the length is set so that the fourth order transverse mode is frequency degenerate with the next longitudinal mode. Apertures were added to the cavity to suppress high order transverse cavity modes, without affecting the $\text{TEM}_{0,0}$ mode which will be used for the two-photon laser. The FSR for this cavity is 10.24 GHz and the finesse is $F=15,140$, giving a cavity linewidth of 676 kHz. In the next chapter I will describe how we used this cavity observe two-photon Raman lasing in the strongly pumped potassium atomic beam.

Chapter 6

Experimental realization of a two-photon Raman laser

6.1 Overview of experimental results

This chapter contains my experimental results from the two-photon Raman laser. I observed two-photon lasing based on a two-photon Raman process in strongly driven potassium atoms. As noted in Chapter 2, an external source of photons is necessary to initiate two-photon lasing. I was able to initiate two-photon Raman lasing using two distinct mechanisms: an externally injected pulse from another laser and a frequency degenerate one-photon process. This is the first experimental observation of two-photon lasing initiated by a frequency degenerate one-photon process.

To characterize the two-photon Raman laser, I measured the lasing threshold as a function of the atomic beam number density. At the upper limit of the atomic beam number density (which is set by the oven configuration) the round trip two-photon Raman gain is approximately twice the round trip loss, so the two-photon laser can operate a factor of two above threshold. I also measured the external probe power required to initiate the two-photon Raman laser. For the two-photon laser initiated by a one-photon process, I measured the two-photon threshold as a function of the atomic beam number density. As the number density is increased, one observes a sudden onset of two-photon lasing.

Once threshold conditions were understood, I observed the output properties of the laser light; most significantly, the polarization of the laser light. I observed large polarization instabilities in the laser light even though the output intensity

was relatively constant. These polarization instabilities change with the strength of the magnetic field in the interaction region. The instabilities are present for all the magnetic field strengths, number densities and pump powers that I tried. In addition I observed intensity fluctuations for an input pulse that is below the two-photon laser threshold.

Further results are given for larger Raman pump detunings. In addition, I present results for two-photon dressed-state lasing in our strongly driven potassium system.

6.2 Two-photon Raman laser initiated by an external pulse

I have observed two-photon Raman lasing initiated by an external laser pulse. As noted in Chapter 2, there are two conditions that must be met for two-photon lasing to occur: a sufficient number of inverted atoms and enough photons to support lasing. Since the two-photon laser does not produce enough photons from spontaneous emission to initiate two-photon lasing, these photons must come from another source. A pulse of light generated by an external laser tuned to the two-photon frequency can be injected into the cavity to initiate two-photon lasing. This technique was used to initiate the two-photon dressed-state laser built in Oregon [20].

The best results for two-photon Raman lasing initiated by an external pulse were obtained for a Raman pump detuning of +25 MHz. At this detuning the two-photon Raman gain is quite large since the intermediate state detunings are small. Figure 6.1 shows the output of the laser cavity as a function of the cavity frequency for a Raman pump power of 213 mW. The feature to the far left is the red one-photon Raman gain. It is quite small since it is in the wings of the absorption of the $F=1$ to $F'=2$ hyperfine line; this one-photon Raman feature is much larger at higher Raman pump detunings. On the right hand side is a combination of blue one-photon

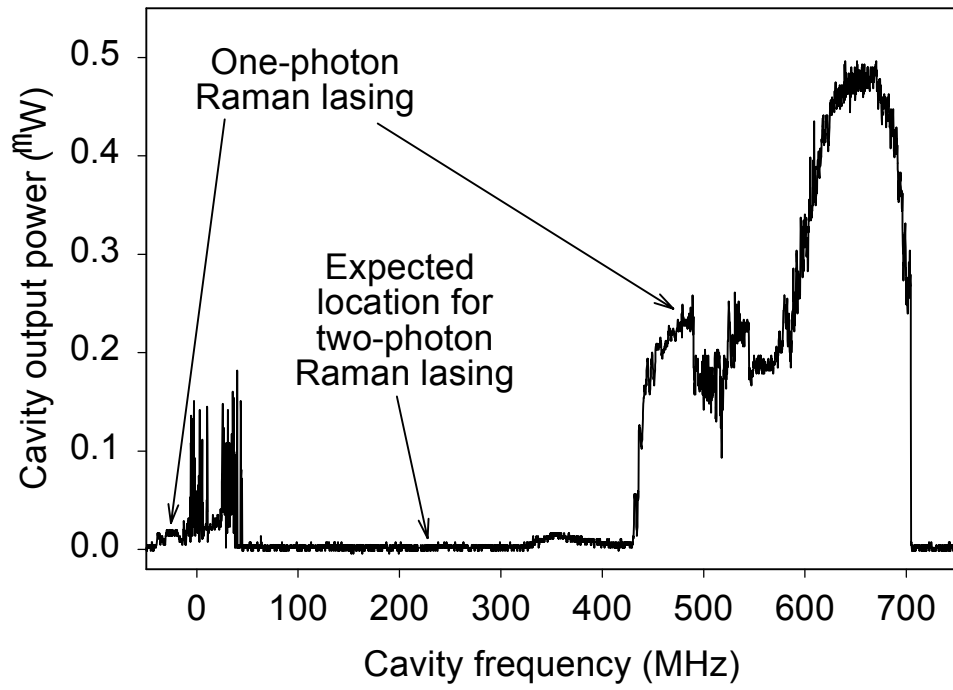


Figure 6.1: Output of the cavity versus the cavity frequency for a pump detuning of +25 MHz. The frequency increases from left to right. The arrow indicates the expected position of the two-photon Raman process.

Raman lasing, one-photon dressed-state lasing, two-photon dressed-state lasing, and possibly other features. The arrow in the middle indicates the expected frequency location for the two-photon Raman process. Since there are no photons to initiate the two-photon Raman laser, we do not see any light generated by the cavity at this frequency as the cavity is scanned. For this Raman pump detuning there are no other processes that occur at the same frequency as the two-photon Raman process. At higher pump powers there is some one-photon gain that occurs ~ 30 MHz away from the two-photon Raman process; however, there is enough frequency separation so that this one-photon process does not affect the two-photon process at all.

Once we had verified that there are no other processes occurring at the two-photon Raman lasing frequency, we initiated the two-photon Raman laser with an external pulse. This was accomplished by fixing the cavity frequency and the external laser frequency at the two-photon Raman frequency and injecting pulses of laser light into the cavity. An acousto-optic modulator (or AOM, see Chapter 4 for further explanation) was used to pulse the probe beam. This permitted very sharp edges (approximately 50 ns) and a convenient reference with which to trigger the fast digitizing scope and capture the temporal output of the two-photon laser. Using a crude method (*i.e.*, chopping the probe beam by hand), we observed two-photon lasing times of several seconds. Figure 6.2 shows a long time scan captured by the digitizing scope. The cavity output is zero before the laser is initiated. The two-photon laser is triggered by the external pulse and remains on for more than one fifth of a second. Once the two-photon laser turns off, it remains off until a new trigger pulse is injected into the cavity. The time that the laser stays on is limited by jitter, either in the cavity frequency (which was tuned by hand) or the Raman pump frequency. It is my expectation that a reasonable feedback loop and locking device should permit long term operation of the laser (many minutes), however this

is beyond the realm of the current experiments. This longer time scale operation will permit accurate measurements of the noise properties of the two-photon laser.

In addition to proving that the two-photon laser will remain on for an extended period of time, we also wanted to observe the short time scale behavior of the two-photon laser. In the Oregon experiment, there was a spiking of the output intensity as the two-photon dressed-state laser turned on. Figure 6.3 shows initiation of the two-photon Raman laser with a very short pulse (pulse width $\sim 0.12 \mu\text{sec.}$). This short pulse is sufficient to initiate two-photon lasing, but we do not observe the spiking which was seen in the two-photon dressed-state laser. Instead, the laser smoothly increases in power over several cavity lifetimes (the cavity lifetime is $0.47 \mu\text{s}$). This smooth increase in power suggests that the two-photon laser is operating well above threshold since the laser power increases after the external pulse is turned off. Once the maximum power is reached, there are fluctuations in the output intensity, but they are relatively small. Qualitatively, Figure 6.3(b) is very similar to the turn-on behavior in the rate equation model, specifically Figure 2.4(c). The power injected into the cavity is less than the maximum two-photon laser output power and it is seen that the two-photon laser output power builds up after the pulse is turned off.

These results are strongly indicative of a two-photon process. The two-photon Raman lasing occurred halfway between the frequencies of the two one-photon Raman processes, as expected. Furthermore, at no time did we observe any output of the cavity at the two-photon Raman frequency without an input laser pulse.

To further characterize the two-photon laser, I measured the maximum laser output power as a function of number density, as illustrated in Figure 6.4. More accurately, this figure shows the output laser power as a function of the measured temperature of the atomic beam oven. The number density values are then obtained from our model of the oven and previous calibrations with the two-photon gain mea-

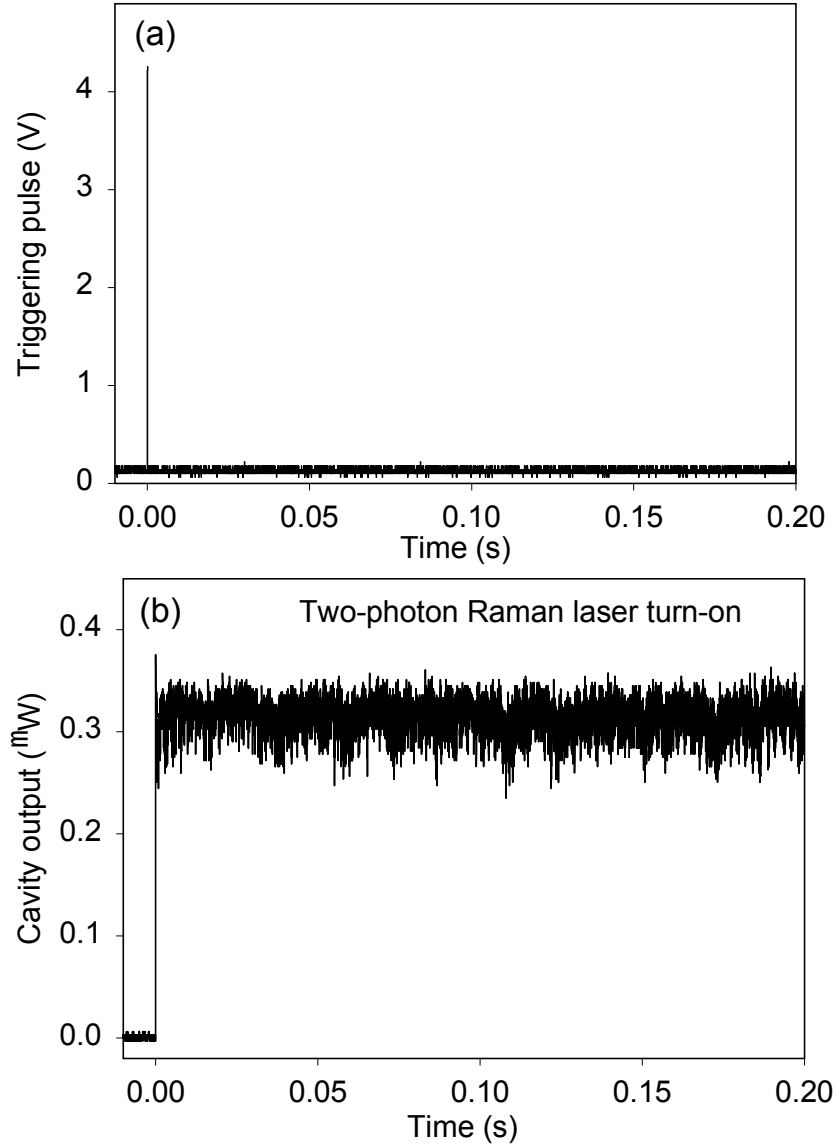


Figure 6.2: Two-photon Raman laser turn on. (a) The electronic pulse that switches the AOM on and off creating the light pulse. (b) The two-photon cavity output as a function of time. The two-photon laser requires an external pulse for initiation and then remains on for more than 1/5th of a second.

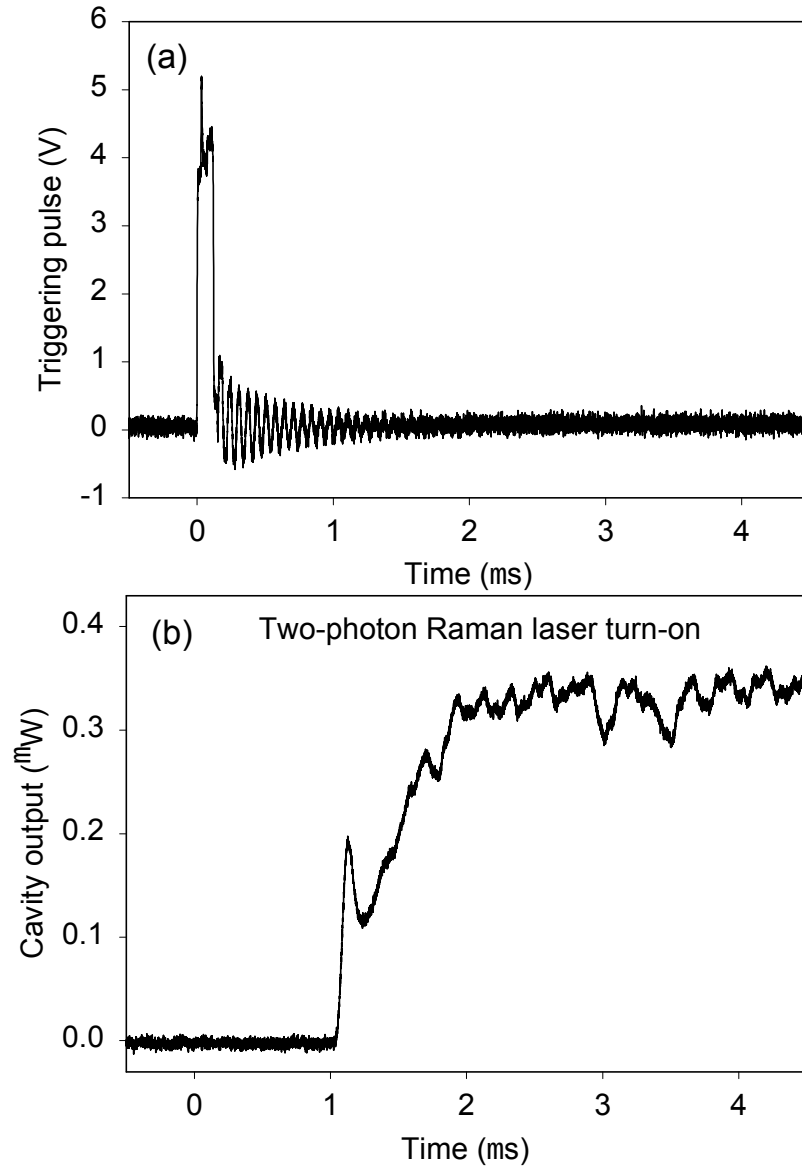


Figure 6.3: Turn-on behavior of the two-photon laser on the short time scale. (a) The electronic pulse which triggers the AOM. (b) The cavity output versus time. There is a delay in the electronics of $\sim 1 \mu\text{s}$ between the two signals and the ringing in the electronic pulse is not present in the optical pulse. In this case the input pulse power is actually below the full two-photon laser power and the output quickly builds to full power after the pulse is turned off.

surement setup. We know that the oven suffered from a saturation effect, *i.e.*, the atomic beam number density did not accurately reflect the oven number density for the very highest temperatures used. Thus, the saturation observed above 2×10^{10} atoms/cm³ is probably not a two-photon laser effect but rather that the atomic beam density is not increasing with increasing temperature. Looking at lower number densities, we see that threshold occurs at approximately 1×10^{10} atoms/cm³. From the rate equation theory in Section 2.1 we expect a square root relationship between the output power and the atomic beam number density just above threshold. Once the two-photon laser is well above threshold the output power should increase linearly with the atomic beam number density. It is difficult to make any assessment concerning the relationship between output power and the number density from the few data points in Figure 6.4, except that there is some saturation mechanism for high oven temperatures. Assuming that the number densities are correct up to 2×10^{10} atoms/cm³, the two-photon laser is operating approximately a factor of 2 above threshold. That is, the cavity round trip gain is twice the round trip losses due to the cavity (the other loss is photons leaving through the mirrors). This agrees with the results from Chapter 4 and Chapter 5. In Chapter 4, the maximum observed two-photon gain was 4×10^{-4} . In Chapter 5, the cavity finesses was measured and found to be $\sim 15,000$. This gives a cavity buildup of ~ 5000 , indicating that a gain of 2×10^{-4} is required to overcome cavity losses. Thus the maximum gain observed was twice the level of gain required for laser threshold.

In addition to a number density threshold, we also expect to see a threshold in the input probe beam power. The previous results show that some probe power is required to start the two-photon laser, but we want to know exactly how much probe power is needed. It is expected that the input probe power necessary to initiate two-photon lasing will be a function of the atomic beam number density.

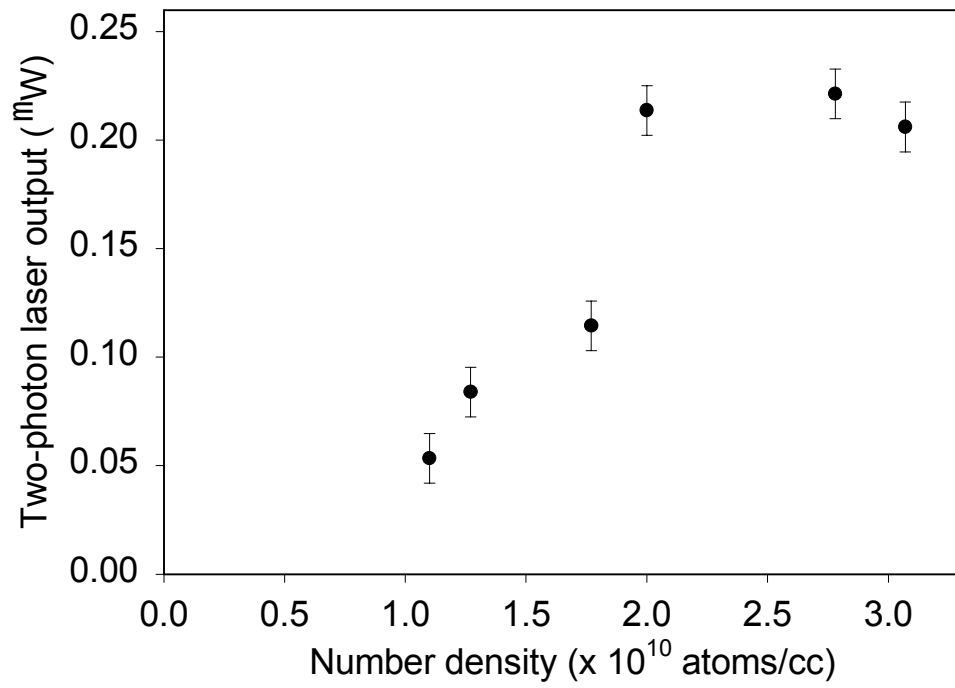


Figure 6.4: Peak two-photon laser output as a function of the atomic beam number density for a pump detuning of +25 MHz. The saturation is probably due to the atomic beam, not the two-photon laser.

For larger number densities fewer photons should be needed to initiate two-photon lasing. Unfortunately, this is a rather difficult measurement to obtain with the current experimental setup because it is hard to consistently couple the input probe beam into the cavity. Locking the diode laser frequency to the cavity frequency would significantly simplify this measurement.

Figure 6.5 shows the laser output for three different input probe powers. The Raman pump power is 280 mW and the oven temperature is 258° C for all three graphs. These graphs show that 10 μ W is insufficient to initiate two-photon lasing, 20 μ W is just above threshold, and 30 μ W of probe power is well above threshold. The vertical dotted lines represent the beginning and end of the input light pulses. Looking back at the rate equation model in section 2.1.2, we see a correspondence between Figure 6.5(a) and Figure 2.4(a), Figure 6.5(b) and Figure 2.4(b), and Figure 6.5(c) and Figure 2.4(d). While not an exact model of the two-photon laser, the rate equation model provides excellent qualitative predictions of the two-photon laser turn-on dynamics. The rate equation model does not predict any of the intensity oscillations observed in the output while the input pulse is on, particularly visible in Figure 6.5(a).

It bears repeating that this is a difficult measurement since the two-photon laser threshold is dependent on the optimization of all of the experimental parameters. Thus these results only indicate that I was unable to initiate two-photon lasing with 10 μ W of probe power. Due to experimental constraints, I was unable to measure the probe threshold for other atomic beam number densities. Future workers will be able to more accurately map the probe threshold power as a function of the atomic beam number density.

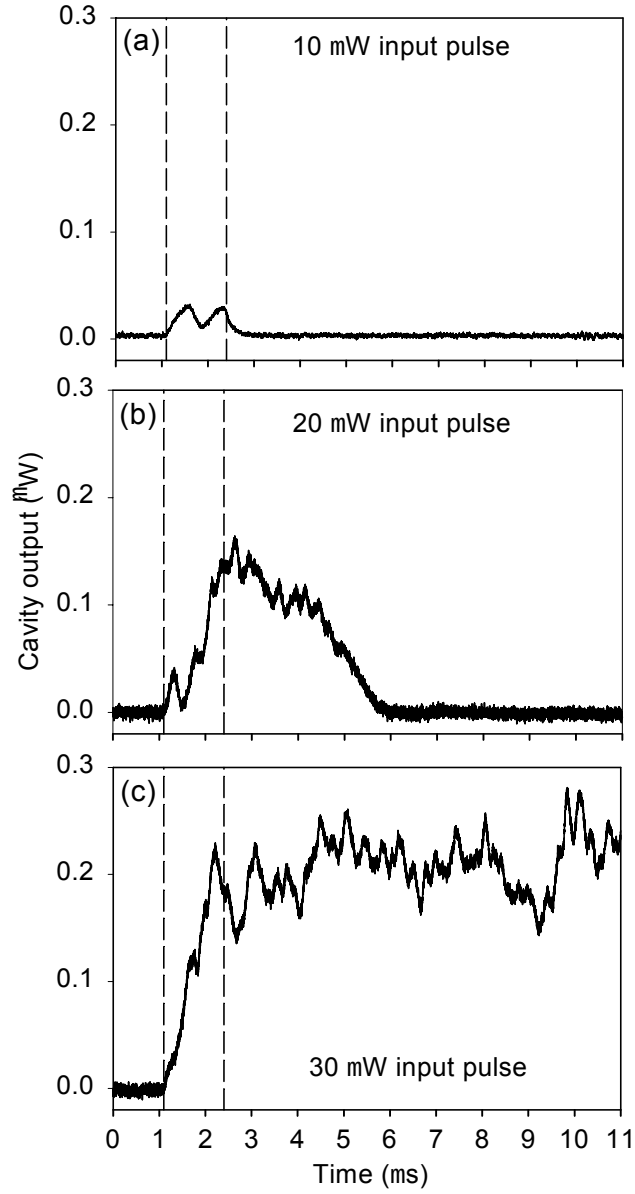


Figure 6.5: Two-photon laser threshold as a function of input probe power. The vertical dashed lines are the beginning and end of the externally injected laser pulse. (a) For $10\ \mu\text{W}$ the two-photon laser does not turn on, even though there is gain (note the oscillations in the pulse intensity). (b) At $20\ \mu\text{W}$ the two-photon laser almost turns on, but not quite. (c) For $30\ \mu\text{W}$ and above the two-photon laser turns-on.

6.3 Two-photon laser initiated by a one-photon process

In addition to two-photon lasing initiated by an external pulse, I have also observed initiation of the two-photon Raman laser by a frequency degenerate one-photon process. This is the first experimental observation of a such a two-photon laser initiation. In my experiment it is possible to overlap the frequency of a one-photon process (usually a one-photon dressed-state process) with the two-photon Raman process. This one-photon process can then provide sufficient photons to initiate the two-photon Raman lasing. For our experimental configuration this situation can occur for several pump detunings; however, the best results were obtained for a pump detuning of +85 MHz.

Figure 6.6 shows the cavity output power as a function of cavity frequency for three different potassium atomic beam number densities. Figure 6.6(a) illustrates the cavity output for a number density just below the two-photon laser threshold. As this atomic beam number density there is either insufficient two-photon gain or too few photons from the one-photon process to support two-photon lasing. Figure 6.6(b) shows the output for an atomic beam number density just above the two-photon Raman laser threshold and the two-photon lasing appears as a narrow spike sticking out of a broad one-photon feature. In Figure 6.6(c), the number density is well above threshold for two-photon lasing and the two-photon peak is the second largest feature after the one-photon dressed state feature on the right hand side of the graph. Note the difference in the vertical scales for the three graphs. There is a temperature difference of only 4° C between Figure 6.6(a) and Figure 6.6(b) versus a temperature difference of 40° C between Figure 6.6(b) and Figure 6.6(c). This would indicate that there is a large jump in the laser output as the two-photon

Raman process reaches threshold. These experimental results mirror the conclusions from the simple rate equation model from Chapter 3, where I showed that there is a threshold for the two-photon laser as the pump rate is increased.

To find the two-photon Raman laser threshold, I measured the peak output power of the two-photon laser as a function of the atomic beam number density; the results are given in Figure 6.7. There are more data points for this graph than for Figure 6.4 and it does appear that there is a square root relationship between the output power and the atomic beam number density near threshold. For number densities between 0.7×10^{10} atoms/cm³ and 2.0×10^{10} atoms/cm³ the relationship is linear. For the highest number densities (above 2.0×10^{10}) the output power appears to saturate. As noted before, this saturation is probably due to the potassium atomic beam and not to an effect in the two-photon laser itself.

In comparison to Figure 6.4, the number density threshold is much lower for the two-photon Raman laser initiated by a frequency degenerate one-photon process. There are two possible (but not mutually exclusive) explanations for this: 1) since the two-photon laser self-initiates in this case, it may be easier to optimize all experimental parameters thereby lowering threshold, and 2) the one-photon gain may reduce the amount of two-photon gain necessary to overcome cavity losses, thus lowering the threshold.

The next observation was the initiation behavior for the two-photon laser triggered by the one-photon mechanism. In order to see what was occurring at laser turn-on, we set the cavity frequency at the two-photon laser frequency and chopped the Raman pump beam. Figure 6.8 shows the two-photon initiation as the Raman pump beam is turned on. As in the case of the external pulse injection, the full Raman pump beam turn-on happens in less than 50 nanoseconds. However the two-photon laser takes almost $2 \mu\text{s}$ to completely turn-on, compared to the cavity lifetime of 0.47

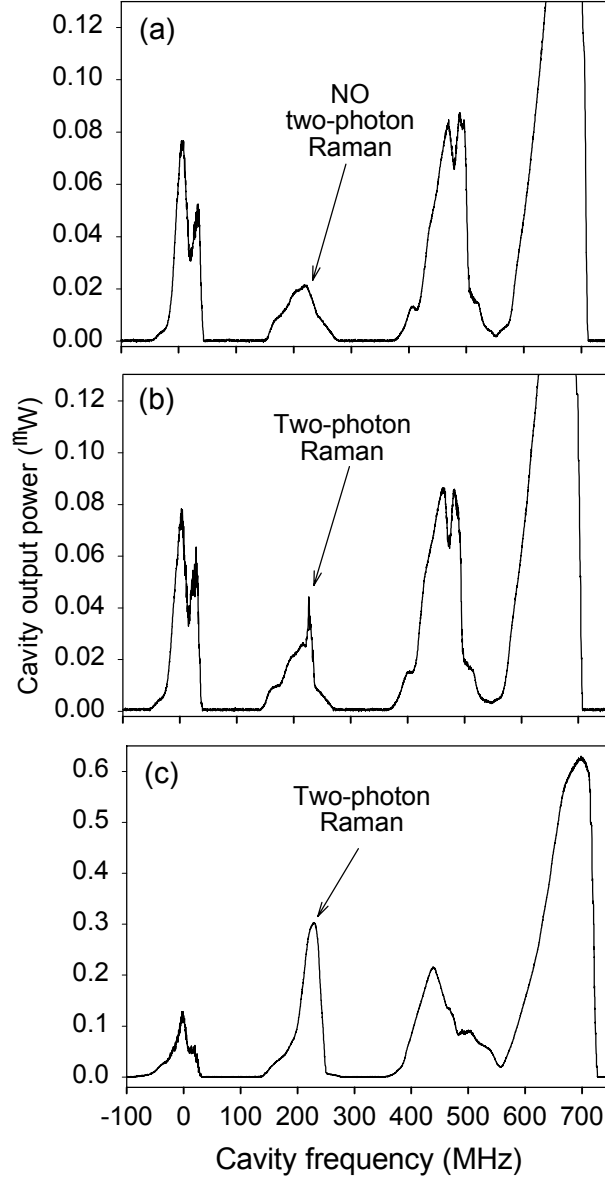


Figure 6.6: Laser cavity output as a function of cavity frequency for three atomic beam number densities. Note the change in scales. The pump beam detuning is +85 MHz. (a) The cavity output just below the two-photon laser threshold. (b) The two-photon laser just above threshold. (c) The two-photon laser well above threshold. In (b) and (c) the two-photon laser is initiated by a frequency degenerate one-photon process.

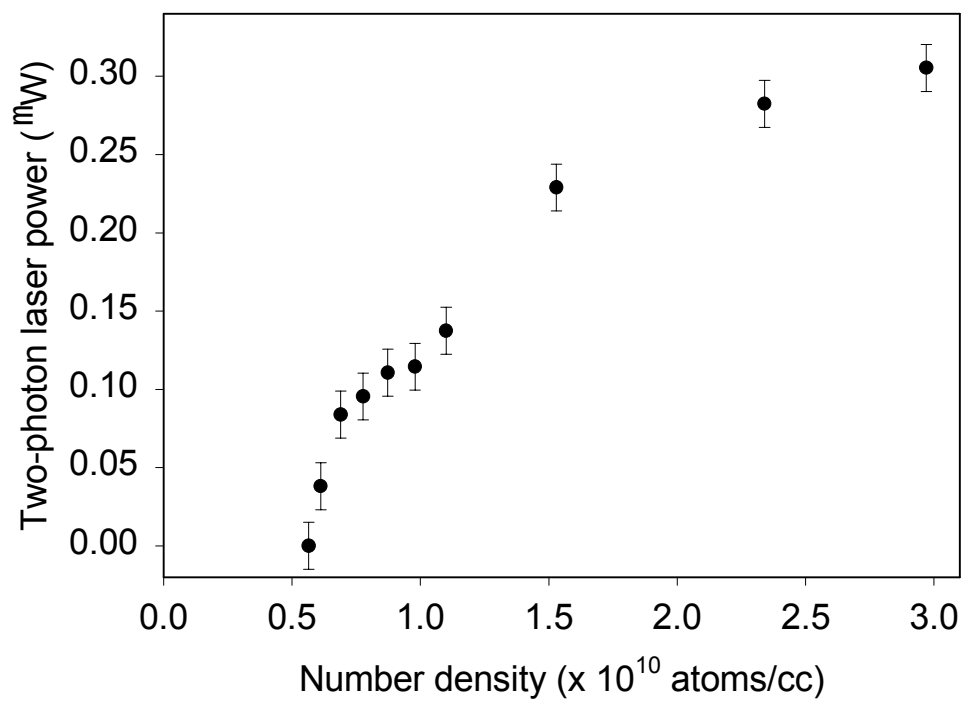


Figure 6.7: Maximum two-photon Raman laser output as a function of the atomic beam number density. The saturation at the highest number densities is due to the atomic beam, not the two-photon laser.

μs . The turn-on dynamics are very smooth, there is no spiking (as was the case in the dressed-state two-photon laser in Oregon) or even small oscillations (as observed for the two-photon Raman laser initiated by an injected pulse).

Comparing this result to the rate equation model in section 2.1.4, we can see a qualitative resemblance between Figure 6.8 and Figure 2.6(c). This just indicates that the one-photon gain is much larger than the two-photon gain, but we already knew that since Figure 6.6 shows that the one-photon feature lases for a much lower number density than the two-photon feature.

6.4 Polarization instabilities in strongly driven potassium

After studying the turn-on behavior of the two-photon laser, I focused on measuring the polarization of the light output by the two-photon laser. Figure 6.9 shows the output of the laser as a function time with a high quality linear polarizer in front of the high speed detector. Three different polarizer angles, for three different initiation events, are shown: vertical (Figure 6.9(a)), horizontal (Figure 6.9(b)), and 45° (Figure 6.9(c)). For all three plots the input probe beam is vertically polarized. In all cases the intensity after the polarizer is oscillating in time with a period of approximately $0.11 \mu\text{sec}$ and an amplitude of $\sim 50 \%$. The oscillations suggest that part of the output polarization is oscillating in time. The frequency of these observed oscillations is $\sim 9.3 \text{ MHz}$. The constant power portion of the output is either linearly polarized output or circularly polarized output. Linear polarization does not seem as likely since the constant power is nearly the same for all three polarizer orientations. Circular polarization is a definite possibility since any polarizer orientation would just reduce the output power by a factor of two since the detector would average the oscillations of the circular polarization passed by the polarizer. The Raman pump

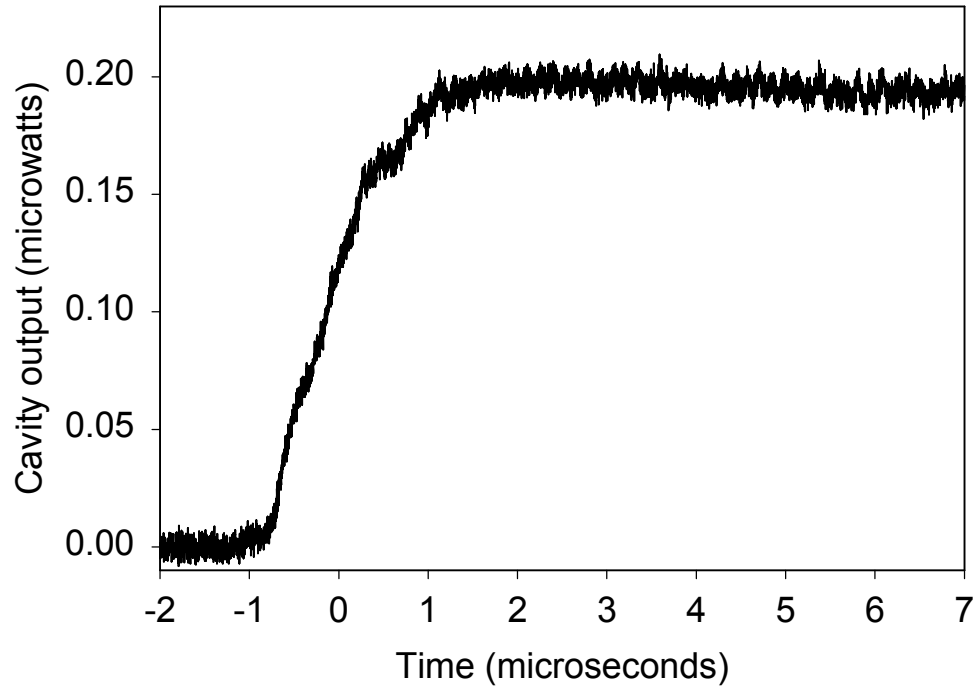


Figure 6.8: Turn-on dynamics for the two-photon laser initiated by the frequency degenerate one-photon process. The Raman pump beam was quickly turned on and the cavity output captured. Note that there is a slow, smooth turn-on unlike the sharp discontinuous turn-on with the injected pulse.

detuning for this data is +25 MHz, the Raman pump power is 280 mW, and the oven temperature is 258° C.

One possibility is that these polarization instabilities are due to the atoms interaction with two strong counter-propagating beams. These results are similar to those observed by Gauthier [76,77] in a sodium vapor cell. Gauthier's results were dependent on the magnetic field strength, so I acquired the polarization instabilities as a function of the magnetic field strength. It was suggested that there may be a magnetic field strength or two-photon gain level where these polarization oscillations do not occur. Figure 6.10 shows the laser output for three different magnetic coil currents. For our magnets, the field strength is 1.4 gauss/amp. In these plots the input probe pulse is present for the first 1.5 μ s of laser oscillation. The periodicity of the oscillation changes with increasing field strength, although it is difficult to quantify accurately. An excellent future experiment would be to send the detector output to a spectrum analyzer and see if there are several dominate frequencies present, once the laser is running for long time periods. I attempted this measurement but was unsuccessful as the laser drifted too fast for the spectrum analyzer to acquire sufficient data for a spectra. The Raman pump detuning is +25 MHz, the Raman pump power is 290 mW, and the oven temperature is 246° C.

Further data was collected for magnetic field strengths up to 5.6 Gauss and for the opposite current polarity. Polarization oscillations were present for all field strengths, although the nature of the oscillations certainly varied with the field strength. Furthermore, oscillations were observed for all probe input powers, Raman pump powers, and atomic beam number densities that we considered. Most likely we need a magnetic field strength of zero in the gain medium to eliminate these polarization instabilities. This is not achievable with the current experimental setup, as we only have one Helmholtz coil which produces a vertical magnetic field and thus cannot

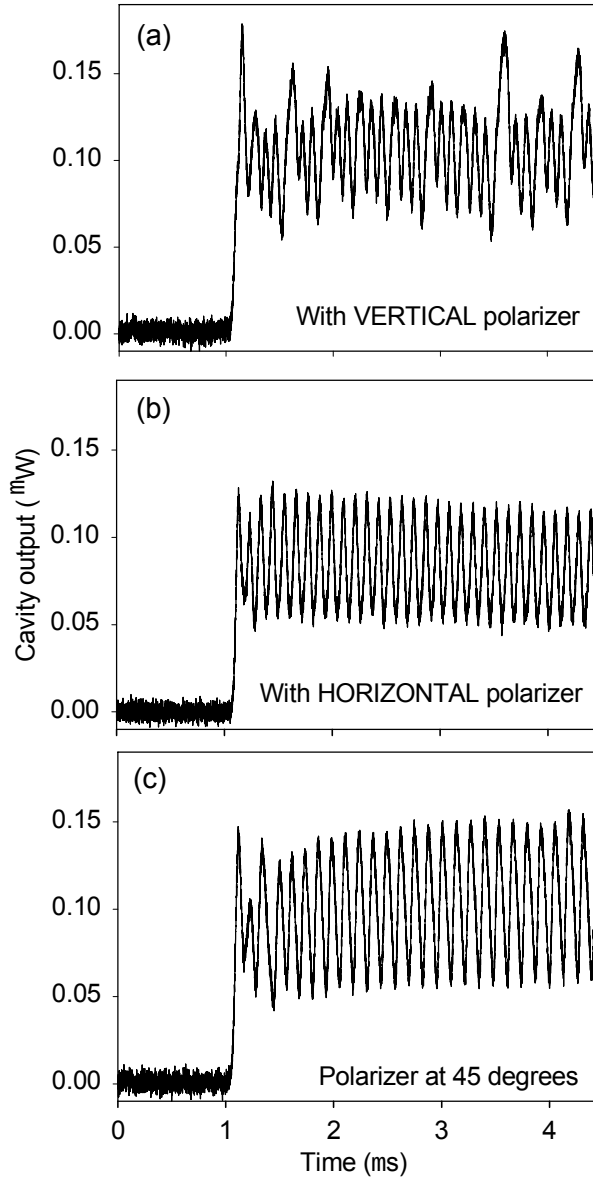


Figure 6.9: Two-photon laser output as a function of time for three different polarizer angles. The linear polarizer is placed between the two-photon laser and the detector. This data is for three different (not simultaneous) turn-on sequences. Note that the oscillation size is roughly 50% of the full intensity.

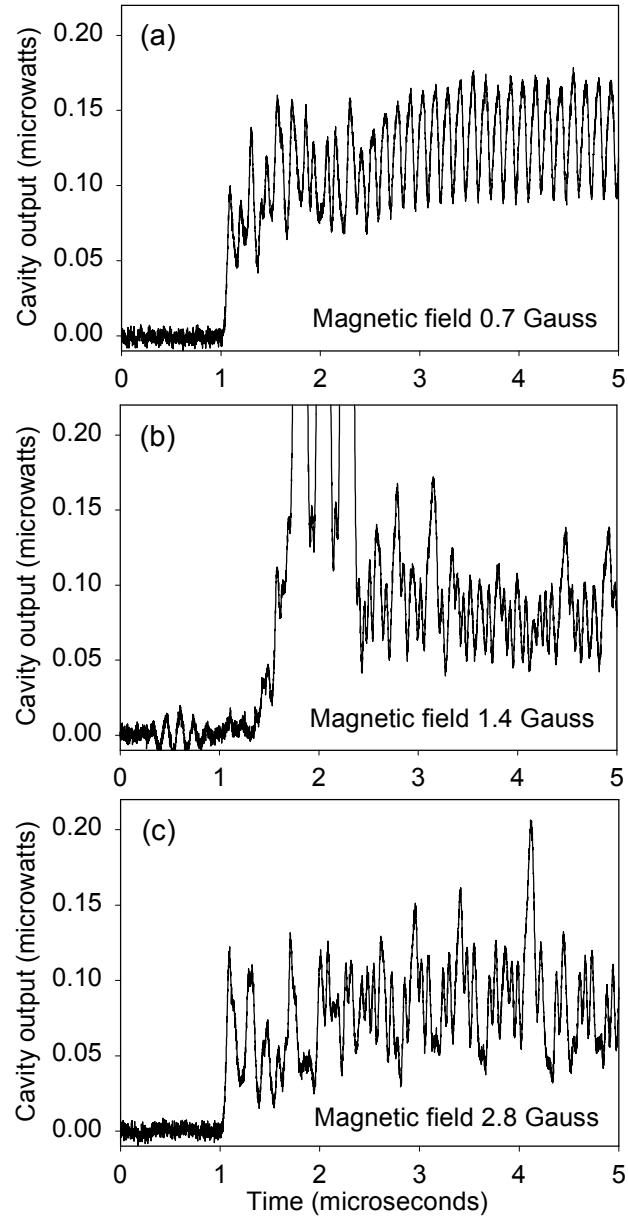


Figure 6.10: Output intensity after a vertical polarizer for three different magnetic field strengths. Oscillations were observed for all magnetic field strengths. At low fields the oscillations were periodic, but became less so for higher field strengths.

eliminate the magnetic field of the earth. A set of three orthogonal Helmholtz coils or shielding with mu metal should permit cancelation of the earth's magnetic field and thus eliminate these polarization instabilities.

Another intriguing piece of instability data was obtained by injecting an external pulse when the atomic beam number density is below threshold for the two-photon laser. Figure 6.11 illustrates what happens to the probe pulse with and without the Raman pump beam. We see intensity oscillations in the probe pulse with a period of $0.13 \mu\text{sec}$ or an oscillation frequency of 8 MHz. The Raman pump detuning is +25 MHz, the Raman pump power is 280 mW, and the oven temperature is 235°C . It is worth repeating that these are intensity oscillations, NOT polarization oscillations. There is no polarizer in front of the detector. Looking back to Figure 6.5 we see that there are also intensity oscillations for the $10 \mu\text{W}$ pulse injected into the cavity.

I also measured the polarization oscillations for a Raman pump detuning of +85 MHz. As before, the measurements were obtained by placing a high quality polarizer in front of the detector and recording the light intensity on the detector. Figure 6.12 shows the cavity output as a function of cavity frequency for no polarizer, a vertical polarizer and a horizontal polarizer. The dark areas in the plots indicate very fast oscillations in the intensity, oscillations substantially faster than the cavity scan rate which is just a few Hertz. This allows us to easily observe which cavity features experience polarization oscillations and which do not. For those features that exhibit oscillations, we can then take time series pictures with the cavity frequency fixed on one particular feature at a time. For the case of no polarizer (Figure 6.12(a)), there are some intensity oscillations on the red one-photon Raman gain (the feature on the far left). We suspect that these oscillations are due to some two or possibly even three photon processes, which happen very close to the pump frequency. These features were observed at several pump detunings and always exhibited oscillations

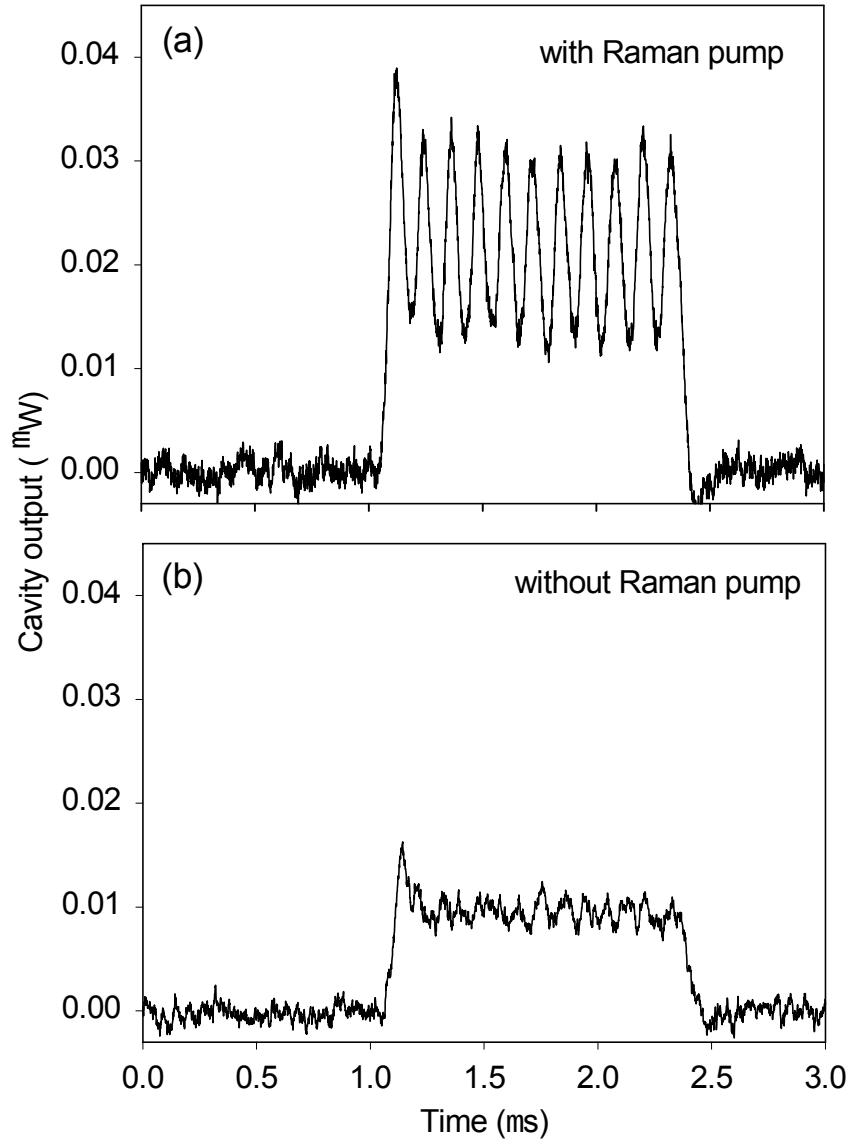


Figure 6.11: Oscillations in the intensity of the input pulse for pulse powers below two-photon lasing threshold. The top graph shows the cavity output with the Raman pump beam and the bottom graph shows the output without the Raman pump beam.

in the output intensity. With the vertical polarizer (Figure 6.12(b)), we can clearly see that the red and blue one-photon Raman features are vertically polarized with no oscillations. The two-photon Raman and the one-photon dressed state feature both exhibit polarization oscillations. There is also a feature to the far blue side of the one-photon dressed state gain which is completely vertically polarized and does not oscillate. Finally we have the cavity output with a horizontal polarizer (Figure 6.12(c)), which suppresses all output from the one-photon Raman features, but the two-photon Raman and the one-photon dressed-state features still oscillate. The Raman pump power is 300 mW and the oven temperature is 260° C.

Next, I measured the output of the individual features as a function of time. Looking first at the two-photon Raman lasing feature, I set the cavity frequency at the two-photon Raman resonance frequency and recorded the light passing through the polarizer as a function of time. As can be seen in Figure 6.13, the polarization instabilities have a period of approximately 0.25 μ s, or a frequency of 4 MHz. The size of the oscillations is much larger than in the time spectra in Figure 6.9. There is also a constant background for the output with the vertical polarizer while the output for the horizontal polarizer has no offset. This would suggest that the output polarization has a linear component in the vertical direction, perhaps due to the frequency degenerate one-photon process. For Figure 6.13, the Raman pump power is 300 mW and the oven temperature is 260° C.

I also acquired the output of the one-photon dressed-state feature after a polarizer. The one-photon dressed-state feature experiences polarization oscillations as illustrated in Figure 6.14. The period for these oscillations is 0.64 μ s, giving a frequency of 1.6 MHz, while the amplitude is still 100%. There is no constant background for either plot, indicating that there is no linearly polarized output light at all.

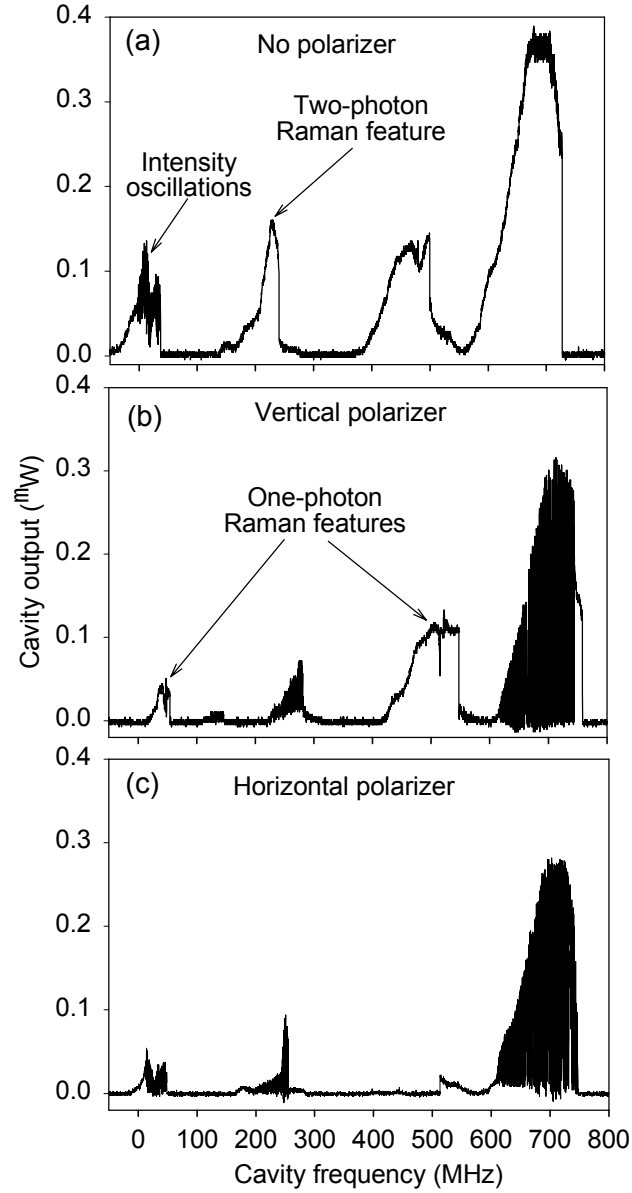


Figure 6.12: Output of the cavity as a function of cavity frequency for (a) no polarizer, (b) vertical polarizer, and (c) horizontal polarizer. The dark regions indicate intensity oscillations.

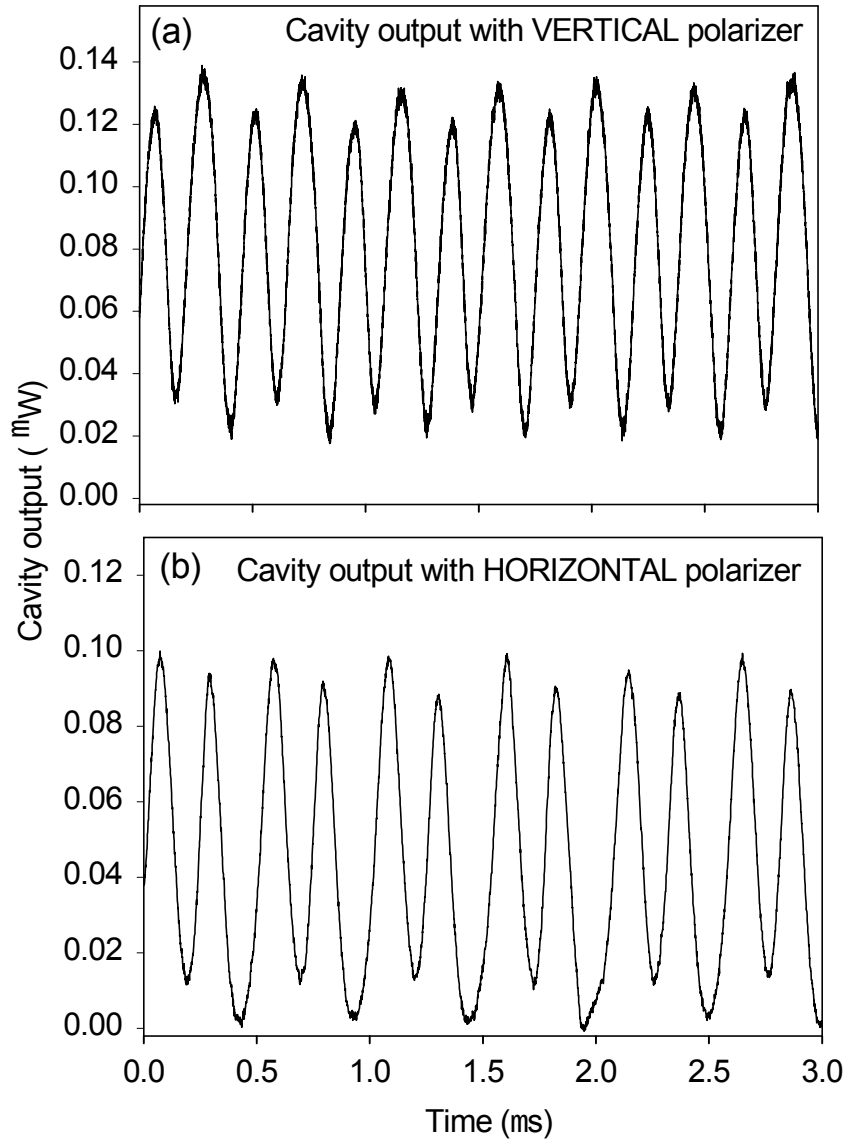


Figure 6.13: Intensity oscillations of the two-photon Raman laser as a function of time for (a) vertical polarization and (b) horizontal polarizations. In this case the oscillations are almost 100%.

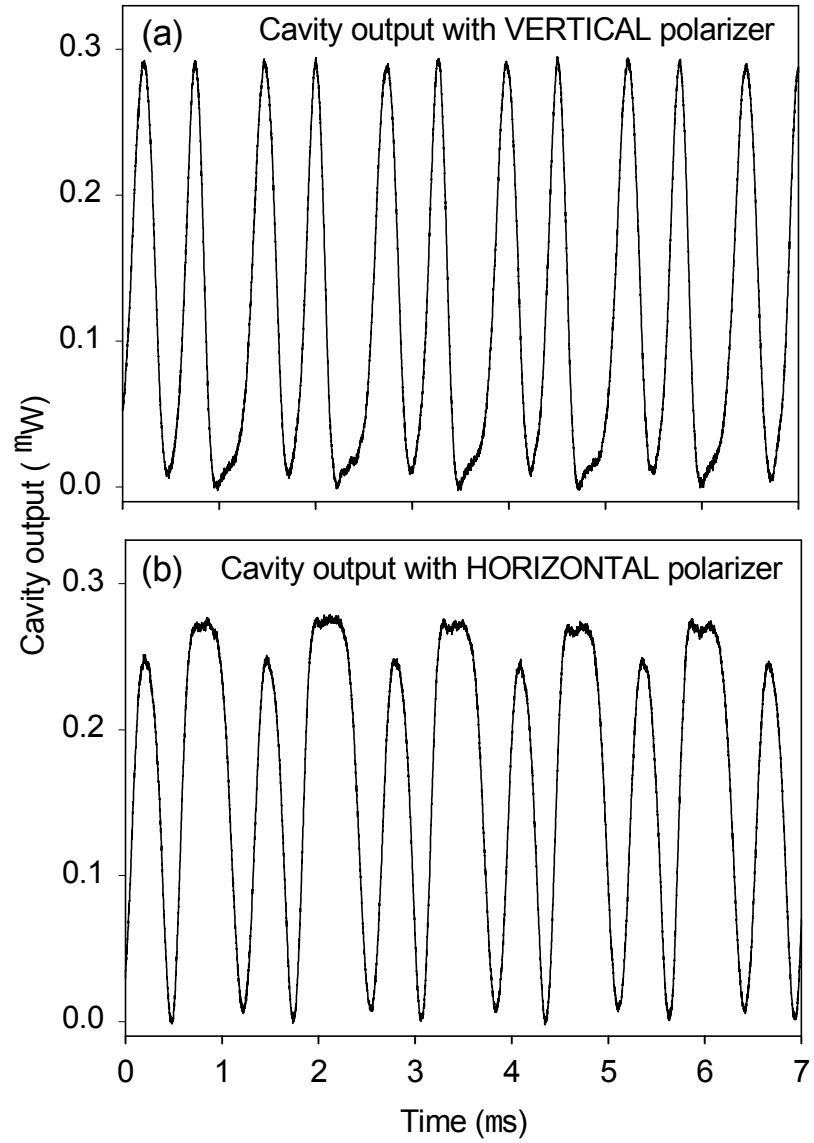


Figure 6.14: Intensity oscillations of the one-photon dressed-state laser as a function of time for (a) vertical polarization and (b) horizontal polarization. The oscillations are almost 100%, however the period is different that the two-photon Raman laser.

6.5 Other experimental results

One of the concerns with the two-photon laser is to verify that it really is two-photon lasing and not some high order transverse mode lasing on a one-photon mechanism. External triggering goes a long way in supporting our assertion that this is two-photon lasing, but other checks are also helpful. One of these checks is to observe the output mode of the laser to insure that it is indeed lasing on the $\text{TEM}_{0,0}$ mode and not some other higher order mode. A higher order mode would be at another frequency thus possibly fooling one into thinking one was seeing two-photon lasing at a given frequency when in fact it was one-photon lasing at another frequency. We were constantly viewing the output spot with an IR viewer and it always appeared to be $\text{TEM}_{0,0}$. Figure 6.15 shows an example the output spot of the two-photon laser. This picture was taken with a CCD camera and a frame grabber card which converted the video image into a computer readable format. We also placed a polarizer in front of the camera, but the polarization instabilities happen far to quickly for the video camera to see. To completely verify that the output spot is not oscillating between higher order modes would require a high speed camera or perhaps multiple detectors with apertures so that each detector views a different part of the output mode. Any variations between the response of the different detectors would indicate that the laser is rapidly jumping between high order modes.

For the sake of completeness, we recorded the cavity output power for various Raman pump detunings. The dressed-state theory from Chapter 3 led us to believe that larger Raman pump detunings would provide better spectral isolation of the two-photon Raman feature. The only concern was whether there would be sufficient two-photon gain at the larger Raman pump detunings to support lasing. Experimental results were obtained for Raman pump detunings of +135 MHz, +200 MHz, +285



Figure 6.15: Output spot for the two-photon Raman laser. The spot looks much like the $\text{TEM}_{0,0}$ spot acquired previously for an empty cavity, as shown in Table 5.2.

MHz, and +335 MHz. We were unable to observe any spectrally isolated two-photon Raman lasing for these Raman pump detuning. We were able to spectrally isolate a two-photon dressed-state feature at a detuning of +135 MHz. Results showing the initiation of two-photon dressed-state lasing with an external pulse are presented. One possible reason that we did not observe two-photon Raman lasing at the larger pump detuning was that the cavity finesse was degraded over the course of taking these measurements. By the time the experiments were concluded, the cavity finesse had dropped below 6,000, a substantial reduction from the initial finesse of 15,000. This problem has been solved and a new cavity will soon be built. The new cavity should have a finesse in excess of 20,000, thus facilitating the search for two-photon Raman lasing at larger pump detunings.

6.5.1 Raman pump detuning of +135 MHz

The results for +135 MHz are basically the same as +85 MHz, except that one of the two-photon dressed-state mechanisms is spectrally isolated. For smaller detunings

the two largest two-photon dressed-state features overlap with the blue one-photon Raman feature. Even though this is not the focus of this thesis, two-photon dressed-state lasing in this system is still an important observation. Since both two-photon Raman and two-photon dressed-state lasing are present in the system, it will be possible to sort out which effects are common to two-photon lasers and which are a function of the underlying mechanism. Figure 6.16 shows the turn-on of the two-photon dressed-state laser; the dashed vertical lines are the beginning and the end of the injected laser pulse. There are oscillations in the output intensity while the pulse is present, however these quickly die away once the pulse is turned off. The Raman pump power is 300 mW and the oven temperature is 255° C. No further observations were conducted on the two-photon dressed-state laser, however this will be an important feature for future experimentalists to study.

6.5.2 Raman pump detuning of +200 MHz

At a pump detuning of +200 MHz, all the features are easily resolved, as can be seen in Figure 6.17. The Raman pump power is 425 mW and the oven temperature is 263° C. There is some two-photon Raman lasing, but it is on the side of a broad one-photon dressed-state feature. This two-photon Raman is initiated by the broad one-photon process, as was the case for a pump detuning of +85 MHz. Higher detuning may spectrally isolate the one-photon feature from the two-photon Raman since the two-photon Raman tunes as the pump detuning while the one-photon dressed-state tunes as twice the pump detuning.

6.5.3 Raman pump detuning of +285 MHz

At +285 MHz pump detuning, we observed one-photon red and one photon blue Raman lasing (the feature on the left and the feature in the middle of Figure 6.18,

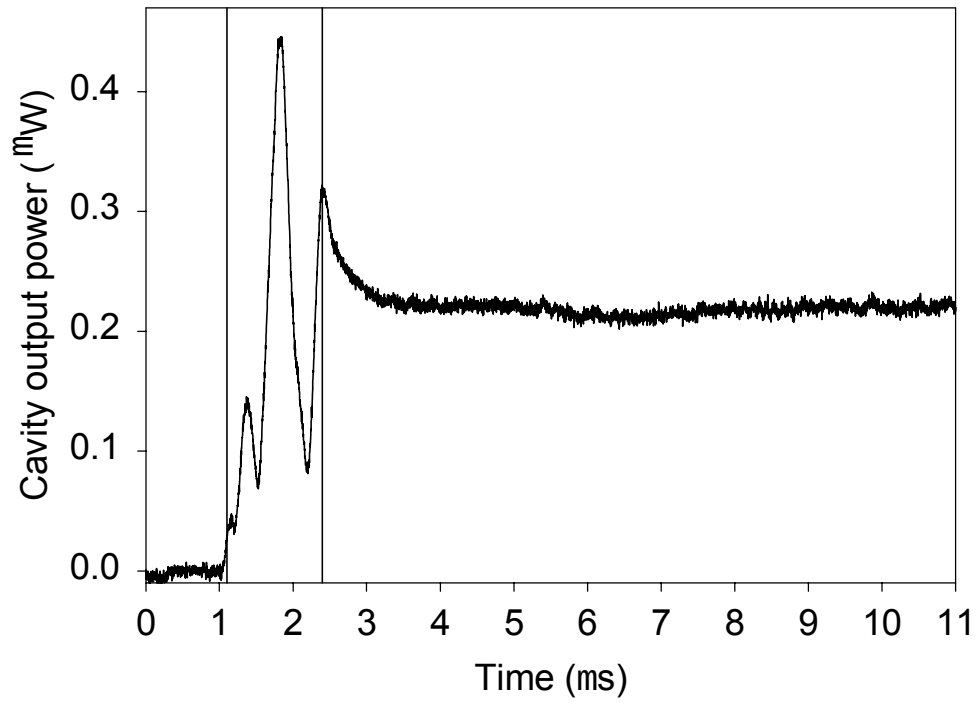


Figure 6.16: Turn-on of the two-photon dressed-state laser for a Raman pump detuning of +135 MHz. The two vertical lines represent the beginning and end of the externally injected optical pulse.

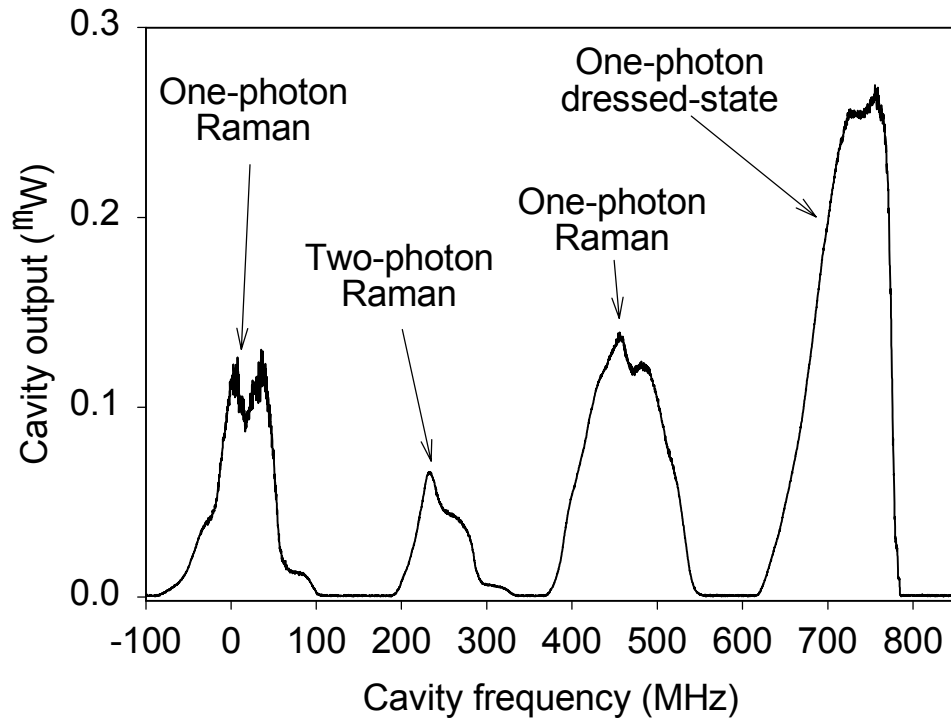


Figure 6.17: Cavity output versus cavity frequency for a pump detuning of +200 MHz. Two-photon Raman lasing is present, but it is overlapping with a one-photon feature.

respectively) as well as some one-photon dressed-state gain (the feature to the far right). However, no two-photon Raman lasing was observed. From Figure 6.18, one can see that there are no other features between the two one-photon Raman features, so there can not be two-photon lasing initiated by one-photon lasing as there is at +200 MHz pump detuning. I also tried injecting probe pulses of various powers and detunings without any success. It should be noted that this was attempted after the mirrors had begun to degrade. For the spectra in Figure 6.18 the cavity finesse was about 6,500. The Raman pump power is 450 mW and the oven temperature is 270° C.

6.5.4 Raman pump detuning of +335 MHz

The results for a pump detuning of +335 MHz are similar to those for +285 MHz. The one-photon Raman features are clearly present as well as the one-photon dressed-state gain to the blue, as can be seen in Figure 6.19. There are no competing mechanisms in the area where the two-photon Raman lasing should be, however I was unable to initiate any two-photon lasing with an external pulse. As in the case of +285 MHz, the cavity finesse is only 6,500. The Raman pump power is 450 mW and the oven temperature is 237° C.

6.6 Discussion of experimental results

The strongly driven potassium beam system is extremely rich in terms of mechanisms and behaviors. We have observed one-photon Raman lasing, two-photon Raman lasing, one-photon dressed-state lasing, two-photon dressed-state lasing, and possibly other mechanisms. Verification of the two-photon Raman lasing rests on a multitude of measurements. The two-photon Raman lasing occurs midway in frequency between the two one-photon Raman features as shown in the amplification

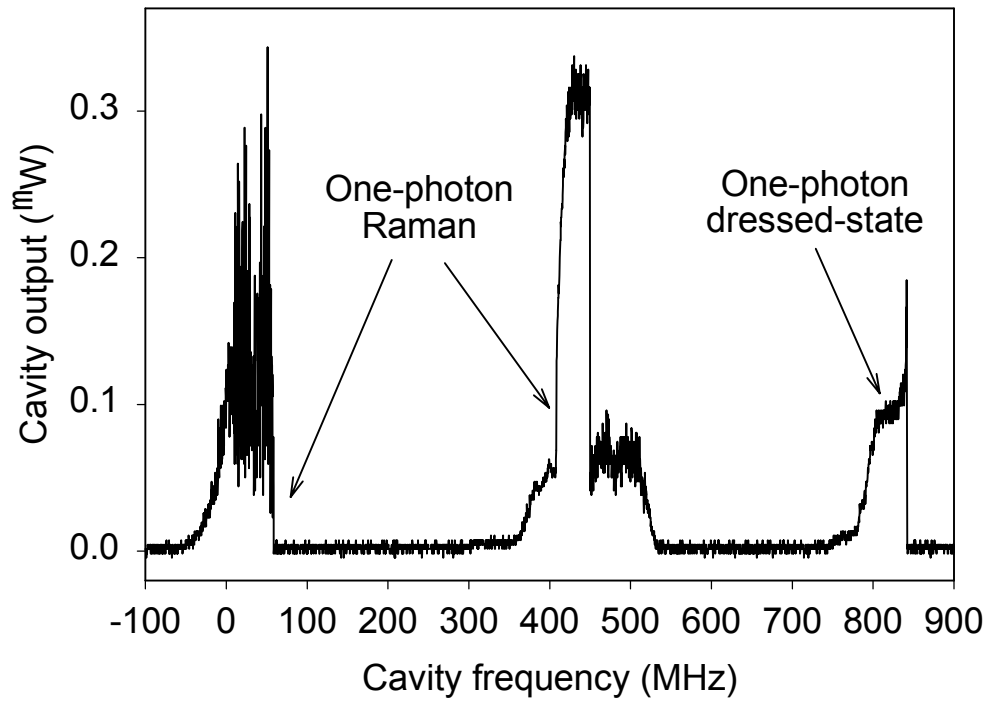


Figure 6.18: Cavity output versus cavity frequency for a pump detuning of +285 MHz. No two-photon Raman lasing was observed at this pump detuning.

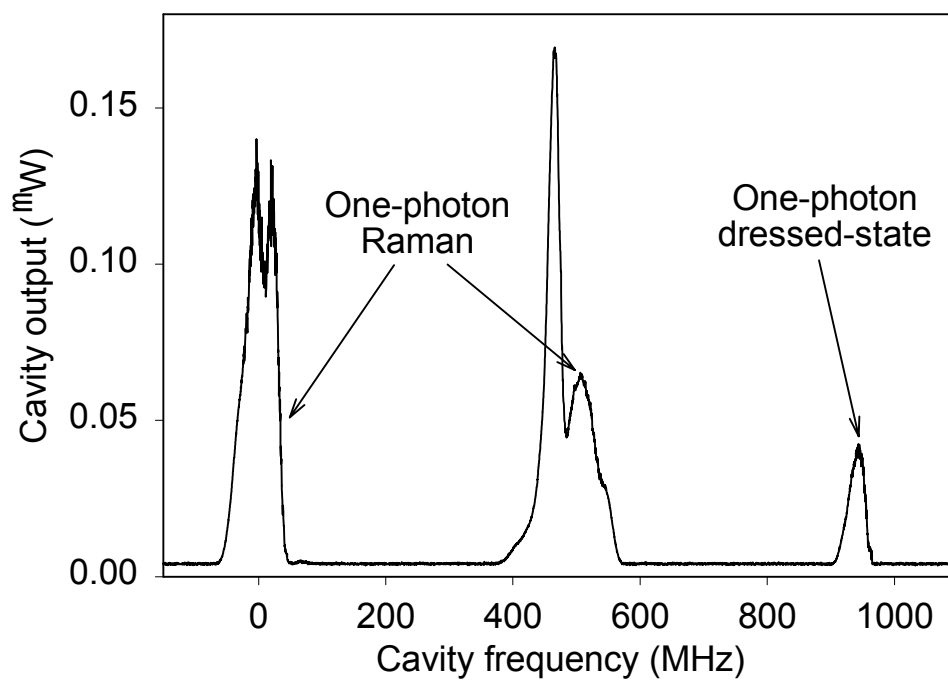


Figure 6.19: Cavity output as a function of cavity frequency for a pump detuning of +335 MHz. Once again, no two-photon Raman lasing was observed.

measurement in Chapter 4 and the dressed-state theory discussed in Chapter 3. The number density threshold for observing two-photon Raman lasing is consistent with gain measurements performed in Chapter 4 and the known cavity finesse. The two-photon Raman laser requires photons from another source to turn-on, either from an external laser or from a frequency degenerate one-photon mechanism. There is a definite photon number threshold for the two-photon laser, *i.e.*, the external source must be above a certain intensity level or the one-photon mechanism must produce sufficient photons to start the two-photon Raman laser.

Once we had established that this was indeed two-photon Raman lasing, we conducted further exploration of the turn-on behavior and polarization instabilities of the two-photon Raman laser. We observed intensity oscillations in the output whenever an external pulse is introduced into the two-photon Raman laser. However few spikes or oscillations were observed once the external source was extinguished. For the case of the initiation via a frequency degenerate one-photon process, the two-photon laser turns on smoothly when the Raman pump beam is chopped.

One of the more interesting observations was the polarization instabilities in the output light from the two-photon Raman laser. The output polarization appears to be oscillating at a frequency that is on the order of the cavity lifetime (at least for low magnetic field strengths). These instabilities were present for all pump powers, atomic beam number densities, and magnetic field strengths that we experimentally checked. As the magnetic field strength was increased the oscillations increased in frequency and became more aperiodic. Intensity oscillations were also observed in some of the other lasing features, including the one-photon dressed-state laser.

As stated before, this is an extremely rich system and I have only scratched the surface of what experiments and results can be obtained. Further experiments should be able to uncover the origin of the polarization instabilities and begin to explore the

noise properties and photon statistics of the two-photon laser.

Chapter 7

Conclusion and Future directions

7.1 Conclusions

This thesis has covered my successful efforts in realizing a two-photon laser in strongly driven potassium atoms. Two-photon lasing based on two distinct two-photon gain mechanisms was observed and characterized. The first was two-photon Raman lasing, based on two-photon Raman gain. In two-photon Raman scattering, two probe photons and two pump photons interact with the atom resulting in an atomic transition and the emission of four identical photons. The other mechanism was two-photon dressed-state gain. This process is similar to the Raman scattering, except that the scattering process uses three pump photons and the result is an atomic transition from the ground to excited state and the emission of four identical photons.

I built up the evidence for two-photon lasing by first conducting extensive experiments measuring the two-photon gain in the potassium beam apparatus. Probe beam spectra were obtained for a variety of probe powers, pump powers, and pump detunings. By comparing with theory, it was possible to identify which gain features corresponded to which scattering processes. Furthermore, by mapping the relationship between the probe power and the gain, I showed that the gain was due to a two-photon process.

Once I understood what processes can occur in the strongly pumped potassium system, a cavity was added. Lasing was observed on many features including several one-photon Raman gain mechanisms, one-photon dressed-state mechanisms, two-

photon Raman mechanisms and two-photon dressed-state mechanisms. The difficult task was then to gather the necessary data to support our contention that we were indeed observing two-photon lasing. This was done by several techniques including mapping the lasing threshold as a function of atomic number density, pump power, and probe power. I also showed that an external source of photons was required to initiate two-photon lasing, as is expected.

This work was dependent on the previous work done by others. In particular, Mossberg and Gauthier's realization of two-photon dressed-state laser in barium showed that a two-photon optical laser could be built. Hope Concannon's initial work in potassium revealed the possibilities for two-photon Raman gain and set a new record for the largest two-photon gain. My experiments were a natural extension of the earlier work and were only successful because of what had been learned in the previous experiments.

7.2 Future experiments

The future experiments for this apparatus fall into two categories. First is the systematic optimization of the two-photon laser via a comprehensive exploration of the parameter space. The other, and much more important, category is the list of ground-breaking experiments that can be performed with a working two-photon laser.

7.2.1 Optimization of the two-photon laser

The optimization list is presented as a reminder to future students who work on the experiment, so they will have a reference for things to try. The first task will be to redesign the atomic beam so that potassium can not escape and coat the cavity mirrors. The current design allowed potassium from the cold region after the oven

aperture to leak into the main chamber. Some of these atoms were deposited on the cavity mirrors. When the chamber was opened these atoms reacted with the air to form potassium hydroxide, which then attacked the mirror coating. We observed a stepwise fall in the cavity throughput and finesse each time the chamber was opened. To prevent any reoccurrence of this problem, a liquid nitrogen cold trap has been designed to replace the water cooled nipple right after the atomic beam oven. This liquid nitrogen trap should prevent any potassium that is not in the atomic beam from reaching the vacuum chamber. Once this trap is built and tested with the current setup, new mirrors can be installed in the cavity without fear of coating them with potassium.

With the cavity apparatus back in optimum condition, one should look for two-photon lasing at the higher pump detuning, particularly +285 MHz. There should be sufficient buildup in the cavity to permit lasing at this pump detuning and it appears that there are no competing gain mechanisms nearby. This may be an excellent location to study the two-photon laser without having to deal with all of the gain mechanisms which can lase at +25 MHz.

There are a variety of other parameters changes that should be explored. Changes to the pump beam include other pump detunings, pump powers, and pump spot sizes in the two-photon laser. The apertures in the cavity may need to be smaller, or perhaps only one aperture is necessary instead of two. Other polarizations should be explored, since the only input pulse polarization we tried was vertical.

If the experiment apparatus were to be redesigned, there are several things that might be tried. Changing the length of the cavity would change the linewidth: a longer cavity would have a narrower linewidth. This might in turn narrow the linewidth of the lasing features, although it seems most likely at this point that the frequency spread over which most features lase is already larger than the cavity

linewidth. As noted before in Chapter 4, there may be other atomic beam designs which would combine the high flux required for these experiments with the simple continuous operation of a recirculating oven. Along these lines, dropping the temperature of the cold region after the oven may be advantageous. One might even go so far as to consider the addition of a liquid nitrogen trap in the system, which would chill out atoms which are not in the atomic beam.

7.2.2 Experiments with a two-photon laser

The real excitement behind this project is the experiments that can be performed with a two-photon laser. An optical two-photon laser has been built before, so this is not ground-breaking in that respect. However, it is the first real opportunity to study the properties of a two photon laser in-depth.

One of the first additional experiments would be to look at the correlations in the photons emitted by the two-photon laser. In the current configuration the two mirrors are identical so light is emitted from both ends of the laser. It would be possible to place a detector at each output and then add and subtract the detectors responses to look at the photon correlations. Also, a 50/50 beamsplitter could be placed at one output, splitting the light into two beams which would then fall on detectors. The outputs of these detectors could also be added and subtracted to observe any correlations. This experiment should be relatively simple to perform. We already have the detectors and electronics available in the lab, so the only piece that is needed is a stable output from the laser for a long enough period to perform the sampling measurements. Several minutes of stable laser output should be sufficient to perform the correlation experiments.

An intriguing possibility offered by this system is three-photon lasing. No three-photon features were observed in the gain spectra taken without the cavity, but there

were hints of three-photon lasing features when the cavity was added. Three photon gain may merely require more atoms or more pump power.

Farther down the road is a study of the bistable properties of the amplifier. In Chapter 6, I showed that the two-photon laser needs an external photon source (from either an injected pulse or a nearby one-photon process) to start lasing. However, once lasing is initiated, the external source is no longer needed. Thus the optical two-photon laser naturally exhibits bistable behavior. There are only a few optical systems which show bistable behavior. Bistable behavior opens the door to the possibility of an optical switch, which would be a tremendous breakthrough. Both faster optical communications and optical computers need a robust optical switch before they will become reality.

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Biography

William J. Brown was born in Cincinnati, Ohio on May the 16th in the year 1971. The son of William C. and Marilyn Brown, he grew up in various towns in the eastern part of Kentucky. After graduating first in his class from Jackson City High School in 1989, he headed to Lexington, Kentucky to study at Transylvania University. There he completed a B.A. in physics, mathematics, and computer science and graduated summa cum laude in 1993. Still feeling uneducated he entered Duke University in the fall of 1993 as a Townes Fellow with the goal of completing a Ph.D. in physics. In 1996 he received an A.M. in physics while working with Dr. Daniel Gauthier. In 1997 he received a Walter Gordy Fellowship for excellent research. During his time at Duke he participated in numerous intramural teams and had the opportunity to be a Resident Advisor for an undergraduate dorm for two years. He met a fellow R.A., Jane, who became the significant other in his life. After Duke, he plans to move to Washington, D.C. and see what the “real world” has to offer.

LIST OF PUBLICATIONS

‘Amplification of laser beams counter propagating through a potassium vapor: The effects of atomic coherence’, W. J. Brown, J. R. Gardner, D. J. Gauthier, and R. Vilaseca, *Physical Review A* 56, 3255 (1997).

‘Observation of large continuous-wave two-photon optical amplification’, H. M. Concannon, W. J. Brown, J. R. Gardner, and D. J. Gauthier, *Physical Review A* 56, 1519 (1997).

‘Amplification of laser beams propagating through a collection of strongly driven, Doppler-broadened, two-level atoms’, W. J. Brown, J. R. Gardner, D. J. Gauthier, and R. Vilaseca, *Physical Review A* 55, 1601 (1997).

LIST OF PRESENTATIONS

‘Experimental realization of a two-photon laser in strongly driven potassium

atoms', O. Pfister, W. J. Brown, and D. J. Gauthier, invited talk at the Quantum Electronics and Laser Science Conference, May 1999.

'A new mechanism for continuous wave two photon amplification', O. Pfister, W. J. Brown, and D. J. Gauthier, contributed talk at the International Quantum Electronics Conference, San Francisco, California, May 1998.

'Quantum noise properties of a saturated amplifier', W. J. Brown, J. R. Gardner, and D. J. Gauthier, contributed talk at the Quantum Electronics and Laser Science Conference, Baltimore, Maryland, May 1997.

'Collective atomic recoil and dressed-state resonances', J. R. Gardner, W. J. Brown, D. J. Gauthier, and R. Vileseca, contributed talk at the Quantum Electronics and Laser Science Conference, Baltimore, Maryland, May 1997.

'Laser beam amplification resulting from collective atomic recoil', D. J. Gauthier, W. J. Brown, J. R. Gardner, and R. Vilaseca, contributed poster at the Quantum Electronics and Laser Science Conference, Baltimore, Maryland, May 1997.

'Observation of large continuous-wave two-photon optical amplification', J. R. Gardner, W. J. Brown, and D. J. Gauthier, contributed talk at Division of Atomic, Molecular, and Optical Physics, Ann Arbor, Michigan, May 1996.